SELECTION AND OPTIMIZATION OF MARINE OIL SPILL RESPONSE OPERATIONS USING ARTIFICIAL INTELLIGENCE AND SOFT COMPUTING TECHNIQUES

by

Saeed Mohammadiun

Ph.D., Civil Engineering, Tarbiat Modares University, Iran, 2018

M.A.Sc., Civil Engineering, Tarbiat Modares University, Iran, 2013

B.Sc., Civil Engineering, Shahrood University of Technology, Iran, 2011

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The following individuals certify that they have read, and recommend to the College of Graduate Studies for acceptance, a thesis/dissertation entitled:

Selection and optimization of marine oil spill response operations using artificial intelligence and soft computing techniques

submitted by Saeed Mohammadiun in partial fulfillment of the requirements of

the degree of Doctor of Philosophy.

Dr. Rehan Sadiq, School of Engineering, UBC Okanagan

Supervisor

Dr. Kasun Hewage, School of Engineering, UBC Okanagan

Co-Supervisor

Dr. Nicolas Peleato, School of Engineering, UBC Okanagan

Supervisory Committee Member

Dr. Zheng Liu, School of Engineering, UBC Okanagan

Supervisory Committee Member

Dr. Anas Chaaban, School of Engineering, UBC Okanagan

University Examiner

Dr. Bishnu Acharya, College of Engineering, University of Saskatchewan

External Examiner

Abstract

Marine oil spill incidents are detrimental to both natural environment and human health. Water quality, marine ecosystems, and shoreline conditions can be deteriorated by the spilt oil. Swift and efficient response to an oil spill is crucial to minimize the adverse consequences. However, the oily waste generated from response operations may also become a challenge, requiring careful waste management strategies. Widely used oil spill response methods (OSRMs) include mechanical containment and recovery (MCR), in-situ burning, and the use of chemical dispersants. Choosing the most suitable method is a complex process depending on various factors. Among OSRMs, MCR is the most effective in removal of spilt oil from the marine environment. The management of oily wastewater generated during MCR requires careful attention, as it comprises a significant portion of overall oily waste.

This study developed multiple tools to aid selecting OSRMs in harsh and remote offshore waters. These selection tools employ machine learning techniques and historical response data to predict appropriate OSRMs for new incidents. The tools were developed in MATLAB^{TM} using various artificial intelligence and soft computing techniques, such as fuzzy decision tree (FDT), Gaussian process regression (GPR), and artificial neural network, individually or in combination. FDT-based tools were also integrated with regression analysis techniques and an optimization algorithm to enhance their performance. Optimized FDTs integrated with regression analysis and GPR were found to be the most effective techniques based on the prediction power.

Furthermore, this study developed an integrated optimization tool to efficiently manage the mechanical response process. This tool aims to minimize the time and cost associated with MCR and oily wastewater management (OWM) and the volume of weathered oil during the operation. The tool encompasses three components of multi-objective optimization, oil weathering process, and MCR and OWM operational agents, simulating detailed response procedures. Applying the tool to a case study in Canada led to a notable reduction in the time and cost of the entire response, and a considerable increase in the volume of recovered oil. It provides an effective approach to manage response process, and significantly reduces the environmental and socio-economic impacts of oil spill incidents.

Lay Summary

The increasing oil and gas explorations and shipping in remote offshore waters (e.g., Arctic regions) have raised the risk of severe economic, social, and environmental impacts from oil spill incidents. Several decision support tools have been developed in this study, identifying the most appropriate response strategy to mitigate these negative impacts. The tools facilitate the timely selection of the best method to respond to an oil spill and optimize the response process based on multiple environmental and operational factors, such as wind speed, wave height, and response capabilities. The machine learning techniques enable response method selection tools to learn from past oil spills by identifying subtle patterns among decision-making factors to be effectively used for managing potential future incidents. Furthermore, the response process optimization tool can schedule available resources (e.g., response vessels) to reduce response time and cost, and enhance oil recovery, thereby maximizing operational efficacy and minimizing environmental damage.

Preface

Saeed Mohammadiun prepared the research concepts and contents of this thesis under the supervision of Dr. Rehan Sadiq and Dr. Kasun Hewage at the University of British Columbia, Okanagan Campus. This study has produced five journal articles, one conference oral presentation, one podcast, one web-based media release, and a part of a scientific newsletter. Four journal articles have been published, and one is currently under review. Chapters 3 to 7 and a part of Chapter 1 of this thesis were used as materials for the journal and conference articles. Saeed Mohammadiun was the main author, responsible for conceptualization, investigation, methodology development, implementation, writing, and editing of the articles. Dr. Rehan Sadiq, Dr. Kasun Hewage, Dr. Guangji Hu (University of British Columbia), Dr. Jianbing Li (University of Northern British Columbia), Dr. Reza Mirshahi (Iran University of Medical Sciences), Mr. Abdorreza Alavai Gharahbagh (Azad University – Shahrood Branch), and Dr. Ezzedin Bakhtavar (University of British Columbia) collaborated on the articles, especially in reviewing and editing papers. Ms. Arezoo Rahimi aided with the graphical presentations in this research, and the necessary permission for artwork usage is included as an appendix (Appendix E) to this dissertation. Dr. Zheng Liu and Dr. Nicolas Peleato served as members of the supervisory committee, providing critical and constructive feedback to improve the quality of the research.

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- **S. Mohammadiun**, G. Hu, A. Alavi Gharahbagh, J. Li, K. Hewage, and R. Sadiq, "*Intelligent Computational Techniques in Marine Oil Spill Management: A Critical Review*" (2021) Journal of Hazardous Materials, Elsevier, Volume 419, 126425, DOI: doi.org/10.1016/j.jhazmat.2021.126425 (Chapter 3)

- **S. Mohammadiun**, G. Hu, A. Alavi Gharahbagh, R. Mirshahi, J. Li, K. Hewage, and R. Sadiq, "*Optimization of Integrated Fuzzy Decision Tree and Regression Models for Selection of Oil Spill Response Method in the Arctic*" (2021) Knowledge-Based Systems, Elsevier, Volume 213, 106676, DOI: doi.org/10.1016/j.knosys.2020.106676 (Chapter 5)
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- "*When It Comes to Oil Spills, Experts Should Turn to Artificial Intelligence*", Web-based Media Release, University of British Columbia, October 2021, URL: https://news.ok.ubc.ca/2021/10/13/when-it-comes-to-oil-spills-experts-should-turn-to-artificial-intelligence/
- "*Can AI Help to Mitigate the Effects of Catastrophic Oil Spills?*" Ask AI Podcast, Episode 32, December 2021, URL: https://www.askai.org/post/e32-how-can-ai-help-to-mitigate-catastrophic-oil-spills

Table of Contents

Abstract ..... **iii**
Lay Summary ..... **iv**
Preface ..... **v**
Table of Contents ..... **vii**
List of Tables ..... **xi**
List of Figures ..... **xii**
List of Abbreviations ..... **xiv**
Acknowledgements ..... **xvi**
Dedication ..... **xvii**
Chapter 1 – Introduction **1**
1.1 Background 1
1.2 Research Motivation 6
1.2.1 Multi Partner Research Initiative 8
1.3 Research Questions and Objectives 8
1.4 Thesis Structure 10
Chapter 2 – Proposed Framework and Research Methodology **13**
2.1 Introduction 13
2.2 Research Framework 13
2.3 Literature Review and Content Analysis 14
2.4 FDT-based OSRM Selection Tool 15
2.5 Optimization-Based FDTR Selection Tool 15
2.6 Performance Evaluation of MLs for OSRM Selection 16
2.7 Integrated Optimization of MCR and OWM Operations 16
Chapter 3 – Literature Review and Content Analysis **17**
3.1 Introduction 17
3.2 Marine Oil Spill Management (MOSM) 18
3.2.1 Major Components of MOSM 18
3.2.2 Oil Spill Response Methods 20
3.2.3 Computational Techniques used in MOSM 23
3.2.4 Numerical Modeling in MOSM 26
3.2.4.1 Physical, Chemical, and Biological Processes 26
3.2.4.2 Numerical Simulators 30
3.3 Statistical Evaluation of Relevant Studies 31

3.4 Descriptive Analysis of Oil Spill Detection, Characterization, and Monitoring 34
3.4.1 Hyperspectral and Multi-Spectral Imagery 35
3.4.2 Synthetic-Aperture Radar Imagery 35
3.4.3 Computational Techniques in OSDCM 37
3.4.3.1 Technique Evolution in OSDCM 37
3.4.3.2 Image Segmentation Process using AI/SC-Based Tools 38
3.4.3.3 Deep Learning Application to Enhance OSDCM 41
3.5 Descriptive Analysis of Spill Management and Response Optimization 43
3.5.1 Oil Spill Response Selection and Risk Analysis 45
3.5.1.1 Response Selection 45
3.5.1.2 Risk and Uncertainty Analyses 48
3.5.2 Oil Spill Response Optimization 50
3.6 Summary 53
Chapter 4 – Oil Spill Response Method Selection using Fuzzy Decision Tree 55
4.1 Introduction 55
4.2 Method 56
4.2.1 Model for OSRM Selection Tool 56
4.2.2 Fuzzification of Attributes 59
4.2.3 Oil Spill Data Acquisition 61
4.2.4 Building Fuzzy Decision Trees 62
4.2.4.1 Fuzzy Interactive Dichotomizer 3 Algorithm 63
4.2.4.2 Building Decision Trees 66
4.2.4.3 Decision Tree Pruning 67
4.3 Results and Discussion 68
4.3.1 Visualization of Decision Trees used in OSRM Selection 70
4.3.2 Evaluation of Decision Trees 74
4.4 Summary 77
Chapter 5 – Optimization of Fuzzy Decision Tree-Based Tools 79
5.1 Introduction 79
5.2 Method 81
5.2.1 Model for Optimization-Based FDTR Selection Tool 81
5.2.2 Oil Spill Dataset Preparation 82
5.2.3 FDTR Development and Optimization 83
5.2.3.1 Optimization Iterative Process 85

5.2.3.2 Fuzzy Decision Tree with various Regression Analyses 88
5.2.4 Statistical Evaluation and Response Method Selection 92
5.2.4.1 Hyper Volume Indicator Calculation 92
5.2.4.2 Statistical Evaluation 93
5.3 Results and Discussion 94
5.3.1 Tool Enhancement by Optimization 94
5.3.2 Pareto Fronts of Optimized Tools 96
5.3.3 Visualization of Optimized FDTRs 101
5.4 Summary 103
Chapter 6 – Comparison of Different Machine Learning Techniques for OSRM Selection 104
6.1 Introduction 104
6.2 Method 105
6.2.1 Model for Performance Evaluation of MLs for OSRM Selection 105
6.2.2 Regression-Based Machine Learning Techniques 108
6.3 Results and Discussion 112
6.3.1 Comparison of Prediction Accuracies 113
6.3.2 Error Evaluation in the Regression-Based Techniques 116
6.3.3 Sensitivity Analysis to the Number of Cases 119
6.4 Summary 120
Chapter 7 – Integrated optimization of oil spill response and oily waste management operations 122
7.1 Introduction 122
7.2 Method 123
7.2.1 Model for Integrated Optimization of MCR and OWM 125
7.2.1.1 Multi-Objective Optimization Agent 126
7.2.1.2 Mathematical Model Agent 127
7.2.1.3 Oil Weathering Process Agent 135
7.3 Hypothetical Case Study 137
7.4 Results and Discussion 139
7.4.1 Two-Objective Optimization 139
7.4.2 Three-Objective Optimization 146
7.5 Summary 152
Chapter 8 – Conclusions 154

8.1 Introduction 154
8.2 Originality and Contributions 155
8.3 Limitations 157
8.4 Recommendations 157
Bibliography 160
Appendices 174
Appendix A – Summary of Reviewed Articles in ODCM 174
Appendix B – Summary of Reviewed Articles in Spill Management and Response Optimization 183
Appendix C – Hypothetical Dataset of Oil Spill Incidents used in OSRM Selection Tools 193
Appendix D – Pseudocode Developed for Optimizing MCR and OWM Operations 194
Appendix E – Input Parameters and Assumptions of Case Study of MCR and OWM Optimization 197
Appendix F – Artwork Usage Consent 202

List of Tables

Table 1-1. Challenging oil spill incidents since 2002 2
Table 3-1. Brief descriptions of computational techniques commonly used in MOSM 25
Table 3-2. Decision variables, objective functions, and constraints used in OSRM optimization 52
Table 4-1. Important attributes and conditions favored by different oil spill response methods . 58
Table 4-2. Definitions and linguistic variables of attributes 61
Table 4-3. Summary statistics of hypothetical oil spill data..... 62
Table 4-4. Results of the simple FDT-based tool applied to multiple datasets 69
Table 4-5. Prediction performance of decision trees developed through two approaches 74
Table 4-6. Distribution of attributes identified in different hierarchies of decision trees (10,000 runs)..... 76
Table 5-1. The setting of parameters for NSDE..... 86
Table 5-2. Different information measures to construct a decision tree..... 90
Table 5-3. Comparison of IG-LR (from previous chapter) and the optimization-based FDTRs .. 95
Table 5-4. Statistical parameters of normalized HVIs..... 98
Table 5-5. Evaluation of normalized HVIs based on the analysis of each regression analysis .. 99
Table 5-6. Evaluation of normalized HVIs based on the analysis of each information measure 99
Table 6-1. Average prediction accuracies and computation time for different techniques 114
Table 6-2. Average absolute correlation coefficients and prediction errors in regression-based ML techniques..... 117
Table 6-3. Performance of regression-based tools on a specific spill incident in testing dataset 118
Table 6-4. Variations of statistical metrics of prediction accuracies with the size of the training dataset 119
Table 7-1. List of time related variables 129
Table 7-2. List of binary decision variables..... 129
Table 7-3. Constraints of the MCR and OWM optimization problem 132
Table 7-4. Overall comparison of SDA and selected OPA-II solutions 141
Table 7-5. Comparison of operational cost and workload of vessels in SDA and OPA-II 141
Table 7-6. Comparison of different mini barges in SDA and OPA-II..... 142
Table 7-7. Comparison of different trucks in SDA and OPA-II..... 143
Table 7-8. Treatment operation comparison in SDA and OPA-II..... 143
Table 7-9. Completion time (hr) for operations related to each component in different approaches 151

List of Figures

Figure 2-1. Proposed research framework 14
Figure 3-1. Schematic presentation of oil spill management 19
Figure 3-2. Schematic presentation of the most common offshore OSRMs 22
Figure 3-3. Statistical evaluation of reviewed studies applying computational techniques in oil spill management 32
Figure 3-4. Origin countries distribution of the reviewed papers 33
Figure 3-5. Simplified depiction of identification and response team mobilization processes in Western Canada (WCMRC, 2021a) 44
Figure 4-1. The model of FDT-based tool for selection of suitable oil spill response method 57
Figure 4-2. Effects of attributes on the operational suitability of oil spill response methods 59
Figure 4-3. Graphic illustration of linguistic variables presented in three fuzzy sets 60
Figure 4-4. Structure of a decision tree 63
Figure 4-5. Decision tree (FDT-AP1) for the selection of suitable oil spill response methods: (a) unpruned and (b) pruned versions 71
Figure 4-6. Fuzzy decision trees (pruned) for (a) mechanical containment and recovery (MCR), (b) use of chemical dispersant (UCD), and (c) in-situ burning (ISB) 73
Figure 4-7. Prediction performance of decision trees developed through two approaches 75
Figure 4-8. Distribution of terminal nodes reached at different hierarchies of decision trees (a total of 10,000 iterations) 77
Figure 5-1. The developed optimization-based FDTR model for oil spill response selection 82
Figure 5-2. A typical FDT with various regression analyses (FDTR) and information measures 89
Figure 5-3. The area considered in the calculation of HVI (Yazdi et al., 2018) 93
Figure 5-4. Pareto fronts of different generations in one of the iterations of the GPR-GINI tool 96
Figure 5-5. Final Pareto fronts (generation No. 200) of different iterations in GPR-GINI 97
Figure 5-6. Typical normalized HVI variations for one of the GPR iterations 100
Figure 5-7. Final Pareto front for GPR-GINI tool with the highest HVI 101
Figure 5-8. One set of FDTRs developed based on GINI information measure and different regression analyses 102
Figure 6-1. The model to evaluate the performance of different ML techniques in the OSRM selection 106
Figure 6-2. Sensitivity analysis to the number of (a) neurons and (b) layers of ANNs 113
Figure 6-3. Boxplots of obtained testing accuracies through all iterations of tools 116

Figure 6-4. Distribution of testing correlation coefficients for regression-based ML techniques 117
Figure 7-1. Network of marine oil spill response and liquid oily waste management operations 124
Figure 7-2. Model for optimization of oil spill response and oily waste management operations 125
Figure 7-3. An illustration of the hypothetical oil spill incident considered for optimization 139
Figure 7-4. Variation of time and cost in the final Pareto front of OPA-II and SDA..... 140
Figure 7-5. Variations in remaining volume of oil slicks in SDA and OPA-II 144
Figure 7-6. Volume of stored oily wastewater in big barges in SDA and OPA-II 146
Figure 7-7. Final Pareto front of three-objective optimization (OPA-III) 147
Figure 7-8. Volume of weathered oil in three different approaches 148
Figure 7-9. Oil slicks removal in OPA-III 150
Figure 7-10. Variations of skimmed and remaining oil volumes in Slick 3 in different approaches 150

List of Abbreviations

AI	artificial intelligence
ANN	artificial neural network
ANN-BR	artificial neural network - Bayesian regularization
ANN-GD	artificial neural network - gradient descent
ANN-LM	artificial neural network - Levenberg Marquardt
BAT	Best Available Technology
BLOSOM	Blowout and Spill Occurrence Model
BN	Bayesian network
CART	Classification and Regression Trees
CBR	case-based reasoning
CCG	Canadian Coast Guard
CNN	convolutional neural network
COCO	Common Objects in Context
CPU	central processing unit
DEV	deviance
DL	deep learning
EU	European Union
FDT	fuzzy decision tree
FDT-AP1	fuzzy decision tree - Approach 1
FDTR	fuzzy decision tree with regression analysis
FPN	feature pyramid network
GA	genetic algorithm
GCS	growing cell structures
GEP	genetic expression programming
GINI	GINI impurity
GMDH	group method of data handling
GNOME	General National Oceanic and Atmospheric Administration Operational Modeling Environment
GoM	Gulf of Mexico
GP	Gaussian process
GPR	Gaussian process regression
GPU	graphics processing unit
H	high
HVI	hyper volume indicator
ID3	Interactive Dichotomizer 3
IG	information gain
ISB	in-situ burning

L	low
LCML	Life Cycle Management Laboratory
LR	linear regression
M	Moderate
MC	Monte Carlo
MCR	mechanical containment and recovery
MEDESS-4MS	Mediterranean Decision Support System for Marine Safety
ME	misclassification error
MIP	mixed integer programming
ML	machine learning
MOSM	marine oil spill management
MPRI	Multi Partner Research Initiative
NEBA	Net Environmental Benefit Analysis
NLR	non-linear regression
NN	neural network
NP	non-deterministic polynomial-time
NRDAM/CMR	Natural Resource Damage Assessment Model for Coastal and Marine Resources
NSDE	non-dominated sorting differential evolution
ODE	ordinary differential equation
OFDT	optimized fuzzy decision tree
OPA	optimized approach
OSCAR	Oil Spill Contingency and Response
OSDCM	oil spill detection, characterization, and monitoring
OSRM	oil spill response method
OWM	oily wastewater management
RGB	red, green, and blue
ROC	receiver operating characteristic
RS	remote sensing
SAR	synthetic-aperture radar
SC	soft computing
SDA	shortest distance approach
SLIQ	Supervised Learning in Quest
STD	standard deviation
SVR	support vector regression
UBCO	University of British Columbia - Okanagan
UCD	use of chemical dispersants
US	United States
VV	vertical - vertical

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Dedication

To my parents, for their love, support, and encouragement

Chapter 1 – Introduction

1.1 Background

Growing marine oil exploration and transportation activities have increased the risk of oil spills in offshore waters. An oil spill is the incidental release of crude oil or petroleum products into natural environments (Li et al., 2016; Uche-Soria and Rodríguez-Monroy, 2019). It is estimated that about two million tons of oil enter the marine environment every year (Ivshina et al., 2015). Oil extraction, transportation, consumption, and natural seep are the primary oil release sources (Hu et al., 2019b; NRC, 2003). Large oil spills can have catastrophic environmental, human health, and socio-economic consequences (Li et al., 2016; Wan et al., 2022; Yekeen et al., 2020). There is a long-recorded history of response to large oil spills and related concerns since the Torrey Canyon incident in 1967. This oil tanker released about 120,000 tons of crude oil near the UK Western coasts (Huang et al., 2020). One recent infamous incident was the 2010 Deepwater Horizon spill in the Gulf of Mexico (GoM), US, which led to 700,000 tons of spilt oil during about 12 weeks of a wellhead leakage. As one of the most catastrophic oil spills in history, this blowout incident caused more than US$20 billion of economic loss, including US$14 billion of clean-up costs (Ivshina et al., 2015; Ye et al., 2019). This incident resulted in a Clean Water Act penalty of US$5.5 billion for its responsible industry entity and approximately US$8.8 billion in damages to natural resources (EPA, 2023). Vessel incident is also a major cause of marine oil spills. For example, in 2018, the Sanchi oil tanker incident spilt about 136,000 tons of natural-gas condensate and 2000 tons of bunker fuel into the East China Sea. This spill has significantly impacted vulnerable species in a large area (Chen et al., 2020). As one of the most recent vessel incidents, the 2020 Wakashio vessel incident in Pointe d'Esny, Mauritius, has resulted in the release of approximately 4,000 tons of fuel oil in the Indian Ocean (Rajendran et al., 2021). Although the frequency of medium/large-scale oil spills from tank vessels has been decreased in recent years, small-scale incidents may also concern the local stakeholders and communities due to the significant adverse

environmental and socio-economic impacts. For example, in 2016, Nathan E. Stewart tugboat ran aground in Bella Bella, Canada caused a small spill (110 m^3 of diesel oil), which has greatly affected the socio-economic activities of the indigenous community in the impacted area (TSB, 2016). Table 1-1 summarizes several other challenging oil spill incidents within the last two decades.

Table 1-1. Challenging oil spill incidents since 2002

Incident	Location	Year	Amount of oil spill (ton)	The main type of spilt oil	Reference
Prestige	Coast of Galicia, Spain	2002	63,000	Heavy fuel oil	(Laffon et al., 2006)
Tasman Spirit	Port of Karachi, Pakistan	2004	30,000	Light crude oil	(Janjua et al., 2013)
Hebei Spirit	West coast of Korea, Republic of Korea	2007	10,900	Mixed crude oil	(Kim et al., 2010)
California pipeline	Southern California, US	2021	about 150	Medium crude oil	(Brown et al., 2021)

As climate change is rapidly reducing sea ice coverage in the Arctic Basin, there is a growing interest in exploring new maritime shipping routes in Arctic waters. For example, ship traffic in the Canadian Arctic nearly tripled during 1990-2015, and further offshore exploration is either ongoing or proposed in other Arctic areas (Carter et al., 2018b; Potter et al., 2012). This increasing interest has also triggered concerns over the risk of oil spills in the cold marine waters. Like response operations in temporal waters, a response in remote cold areas also aims to rapidly eliminate the spilt oil from seawater and minimize the environmental impacts. Unlike the oil spill response in temperate climates, the response in fragile and harsh cold environments requires special disaster-prevention measures to be in place. The spilt oil may lead to more detrimental impacts in cold waters due to the lower biodegradation rate of contaminants in low-temperature environments (Ivshina et al., 2015). Additionally, there are logistical constraints in Arctic and Sub-Arctic waters to respond oil spills (Polaris, 2009). Due to the low population density in these regions, the required workforce and facilities to handle a large oil spill are limited (Hu et al., 2020; Polaris,

2009). The absence of essential infrastructure, such as ports and roads, in the Arctic region can also impede the transportation of both response personnel and equipment to potential locations affected by oil spills (Nuka Research Group, 2007).

To minimize the adverse impacts of an oil spill incident, it is critical to clean up spilt oil in a well-organized and timely manner. Both short-term and long-term contingency plans should be developed for this purpose. In contingency plans, appropriate response methods required before, during, and after incidents should be identified through effective marine oil spill management (MOSM) (Li et al., 2016; Liu and Wirtz, 2006). MOSM can broadly cover multiple components, including but not limited to, oil spill detection, characterization, and monitoring (OSDCM); data acquisition for a spill area; risk assessment; response method selection and process optimization; clean-up operations; and monitoring response progress to make complementary decisions if required (Hu et al., 2020; Huang et al., 2020; Moroni et al., 2019; Ye et al., 2020). Oil spill response method (OSRM) selection and process optimization are the principal components of MOSM.

In Canadian waters, Transport Canada is the primary federal agency, setting guidelines and regulations for responding to oil spills. Transport Canada oversees the entire response regime, enforcing standards and implementing legislation. In the Canadian response regime, industry entities also have specific responsibilities in oil spill management. They operate through government-certified response organizations, including Western Canada Marine Response Corporation, Eastern Canada Response Corporation, Point Tupper Marine Services, and Atlantic Emergency Response Team, all of which comply with the Canadian Response Organization Standards. Each response organization maintains the necessary plans and logistics for spill response. To operate in Canada, industry entities must have agreements with the response organizations to handle spills on their behalf, and these entities pay annual fees to these organizations to maintain the required level of preparedness. Vessels in Canadian waters must also have shipboard oil pollution emergency plans, and oil facilities are obliged to have both

emergency plans and response equipment on-site. In the event of a spill, the responsible industry entity must oversee the response operation (WCMRC, 2018, 2017).

The Canadian Coast Guard (CCG) also oversees the industry's operational responsibilities, ensuring they respond appropriately to marine pollution incidents. Furthermore, several other federal agencies play crucial roles within this response regime. Environment and Climate Change Canada contributes scientific, technical, environmental, and wildlife expertise to spill responses. The Department of Fisheries and Oceans provides essential scientific expertise and information to ensure effective spill responses while also issuing the necessary permits to safeguard wildlife and aquatic resources (WCMRC, 2018, 2017).

Alternatively, the primary entity responsible for spill response in the Canadian Arctic is CCG. To enhance response preparedness and capability in the Arctic, basic equipment for handling small near-shore spills were positioned in Resolute, Arctic Bay, and Pond Inlet in Nunavut, as well as in Kugluktuk and Ulukhaktok within the Beaufort region. Also, there are stockpiles of response equipment prepositioned in Iqaluit, Nunavut and Tuktoyaktuk, Northwest Territories. The Mary River Mine on Baffin Island, Nunavut also has its own reserve of response resources. Furthermore, the CCG base in Hay River, Yellowknife, Northwest Territories provides access to further oil spill response resources (WWF-Canada, 2017).

An oil spill response in offshore waters (i.e., both temperate and cold waters) is generally conducted by one or more of three major OSRMs, including mechanical containment and recovery (MCR), use of chemical dispersants (UCD), and in-situ burning (ISB). The suitability of response methods for a specific oil spill incident varies depending on many attributes (factors) including, but not limited to, type and volume of the spilt oil, characteristics of the oil slick, ice coverage, and various environmental factors such as wave height (Mohammadiun et al., 2021a; Polaris, 2009). These response methods or a combination of them should be meticulously selected and optimized based on these attributes. The characteristics of spilt oil change constantly under the effects of weathering processes, including evaporation, drift, spreading,

emulsification, dispersion, biodegradation, photo-oxidation, and sedimentation (Ivshina et al., 2015; You and Leyffer, 2011; Zhong and You, 2011). The dynamic change of the characteristics of spilt oil could affect the efficacy of applied response methods. The work arrangement of response fleets can also affect the response efficacy; for instance, different arrangements of response vessels and skimmers dispatched for the same oil slick could result in different response duration and collection rate (Ye et al., 2019). The above-mentioned weathering processes lead to a continuous variation of spilt oil characteristics over time, making OSRM more complicated and challenging. Alternatively, some response operations may lead to the generation of a considerable amount of oily waste. Generally, the volume of oily waste generated from response operations can be up to ten times greater than the original volume of spilt oil due to the emulsification process and shoreline sediment contamination (IPIECA, 2014). The large volume of oily waste could easily exceed the capacity of locally available waste handling facilities, resulting in a great challenge for its effective management. Therefore, in addition to the OSRM selection, the management of oily waste is another important aspect that needs to be carefully considered in the development of contingency plans.

Since MOSM is a very complicated undertaking affected by a variety of factors (e.g., oil weathering processes, response methods, and environmental conditions), there is a dire need for intelligent computational techniques to assist decision-making in various complex MOSM tasks (Ye et al., 2019). Moreover, the recent expert panel report of the Royal Society of Canada emphasizes the use of knowledge discovery techniques to enhance decision-making in complex oil spill management (Lee et al., 2015). In this direction, decision support tools are essential to help MOSM stakeholders make transparent decisions based on reliable data and techniques. In the context of MOSM, an effective decision support tool should be able to systematically find the most appropriate management practices for each component of MOSM. It can be a computation-based platform connecting MOSM components to find efficient solutions (Moroni et al., 2019). The first step for developing such a tool is to collect required oil spill data from different sources,

such as synthetic-aperture radar (SAR) image data, datasets of previous incidents, and simulation-based models (Angelliaume et al., 2017; Chen et al., 2017; Janati et al., 2020). By evaluating the collected data, an effective decision support tool can suggest both proactive or reactive practices to prevent oil spill incidents or mitigate adverse impacts of an oil spill if it occurs (Moroni et al., 2019). Intelligent computational techniques have a great potential to be used for developing decision support tools for the most essential components of MOSM. Oil spill response practitioners and decision-makers can utilize these tools to effectively coordinate response operations, enhance response training process, and improve contingency planning.

1.2 Research Motivation

As mentioned, it is widely accepted that responding to an oil spill should be conducted in a well-organized and timely manner. To achieve this, effective decision support tools can be employed to efficiently coordinate the entire response process and enhance decision-making. Artificial intelligence (AI) and soft-computing (SC) are among the most effective available computational techniques for supporting decision-making (Mohammadiun et al., 2021a; Moroni et al., 2019). AI aims to develop automated models computationally trained to imitate human intelligence capability in order to find precise or deterministic solutions (Kok et al., 2010). AI encompasses cognitive computing, machine learning (ML), digital image processing, artificial neural network (ANN), and deep learning (DL) etc. (Duch and Mandziuk, 2007). The SC techniques consist of intelligent or evolutionary approaches that aim to find sufficiently accurate approximations of exact solutions for NP-hard (non-deterministic polynomial-time hard) class problems (Ibrahim, 2016). NP-hard class problems have no analytical or exact numerical answers obtainable within a reasonable computation time. Some of the major SC techniques can be enumerated as fuzzy logic, different evolutionary and metaheuristic optimization algorithms (such as genetic algorithm (GA) and differential evolution), probabilistic reasoning, and some uncertainty assessment techniques (Duch and Mandziuk, 2007; Ibrahim, 2016; Sadiq et al., 2004). SC and AI have a

degree of overlapping regarding the combined application of techniques to find solutions more efficiently.

AI/SC techniques have the capacity to identify subtle patterns among input parameters (e.g., decision-making factors) that may not be readily discernible by humans or modeled effectively using parametric and deterministic techniques. SC techniques can also discover efficient solutions, considering the multifaceted aspects of complex phenomena (i.e., NP-hard class problems) that are beyond the scope of conventional mathematical techniques. In this regard, AI/SC techniques can be used in a wide range of problems for developing robust decision support tools to ultimately facilitate the decision-making process. Therefore, computational techniques, particularly AI and SC techniques, can also be used to develop tools for effective MOSM, such as ANN, fuzzy decision trees (FDTs), evolutionary algorithms, and stochastic-based tools (Guo and Zhang, 2014; Khanna et al., 2013; Ye et al., 2019; Zeng and Wang, 2020). Recent progress in computational techniques, such as the novel flexible evolutionary and uncertainty analysis techniques, facilitates the development of robust and accurate decision support tools in MOSM (Jiao et al., 2019; Mohammadiun et al., 2021a; Yekeen et al., 2020).

These tools can aid oil spill response practitioners and decision-makers to timely select the most effective and efficient strategy for responding to oil spills, which are time-sensitive operations. These tools can also be valuable during training and contingency planning stages to enhance response preparedness. Such tools could potentially alleviate the burden on experienced decision-makers who might not always be readily available and could make errors in the high-pressure aftermath of an oil spill. Furthermore, considering all influential decision-making factors, environmental and economic aspects, and the limitation in available resources for a response is a highly complex task, resulting in an NP-hard class problem. This highlights the essential role of comprehensive AI/SC-based decision support tools in tackling such challenges effectively.

In this direction, supported by the department of Fisheries and Oceans Canada through the Multi Partner Research Initiative (MPRI), the present research thoroughly investigated the capacity of advanced computational techniques in facilitating effective MOSM.

1.2.1 Multi Partner Research Initiative

Fisheries and Oceans Canada developed MPRI under the National Oceans Protection Plan to enhance the response preparedness for oil spill incidents in Canadian waters while advancing science-based decision making in oil spill response. By supporting over 220 researchers, the initiative allocated approximately CA$45.5 million over a five-year period to harness the knowledge and insights of oil spill experts, both within Canada and internationally.

At the University of British Columbia - Okanagan (UBCO), we were actively involved in one of the MPRI sub-projects entitled "Improved Decanting and Oily Waste Management Strategies for Marine Oil Spill Response." In this sub-project, UBCO collaborated with four other Canadian universities, including the University of Northern British Columbia, Memorial University, and Dalhousie University. Our research progress and outcomes were also overseen and guided by the MPRI technical advisory committee, which consisted of multiple advisors from diverse industries and government entities.

Given UBCO's research potential, robust computational capabilities, and available expertise, our work mainly focused on enhancing oil spill response performance and improving oily waste management operation through the utilization of novel computational tools. We intended to develop these tools using proficient AI/SC techniques.

1.3 Research Questions and Objectives

The potential of advanced computational techniques in the field of MOSM requires further investigation. In this direction, the sub-field of response method selection faces several notable research gaps. The existing methodologies tend to consider only a limited set of decision-making factors when choosing the most appropriate OSRM, often overlooking crucial variables that could

significantly impact the outcome. Additionally, while the environmental implications of OSRM selection are often assessed, the operational suitability of these methods has not received adequate attention within the decision-making process. Evaluation of this operational aspects is essential, as it directly affects the efficiency and effectiveness of response efforts in real-world scenarios. Furthermore, the use of ML-based tools for OSRM selection is hindered by the lack of sufficient and high-quality data to develop these tools.

Alternatively, there are also several critical research gaps in the sub-filed of response process optimization. In this regard, there is a strong connection between oil spill response and oily waste management operations, yet this vital linkage is often overlooked in terms of comprehensive modeling. Incorporating this connection into a holistic model is crucial to ensure the efficient coordination of the entire process. Also, in order to efficiently coordinate the repones process, there is a dire need for multi-objective optimization that considers not only the environmental aspects but also the operational details of both oil spill response and oily waste management. The process optimization while accounting for real-world constraints is a complex task that requires advanced optimization techniques used within appropriate decision support systems. It is also needed to develop a comprehensive case study encompassing various aspects of oil spill response and oily waste management operations. Such case studies serve as testing grounds for potential optimization tools, allowing researchers to assess the effectiveness and robustness of their approaches under realistic conditions.

Concerning the research gaps mentioned above, the following main questions are answered through different phases of this study:

- What is the significance and potential of utilizing advanced computational techniques in MOSM, and how can their performance be improved?
- How can response practitioners transparently select appropriate response strategies using a comprehensive decision-making model that considers operational suitability and all influential decision-making factors?

- How can decision-makers effectively manage the oil spill response process and the resulting oily waste to minimize environmental impacts and associated costs?

To answer above questions, this research study aimed to evaluate the potential of applying advanced AI/SC techniques in MOSM and develop practical decision support tools to intelligently select appropriate response strategies and optimize response operations in order to reduce environmental impacts and response costs. Different AI and SC techniques are used to develop intelligent decision support tools to obtain efficient strategies for MOSM. These techniques include evolutionary optimization, fuzzy set theory, and classification techniques such as decision trees. The tools are expected to facilitate the selection of the most suitable oil response techniques and optimize the response process for tackling offshore oil spills. The intelligent techniques used in the tools imitate oil spill response experts' cognition and knowledge to make accurate response decisions rapidly.

To develop the proposed tools following sub-objectives are defined:

1) Identify potential approaches to enhance the performance of computational techniques in MOSM and suggest computation-based MOSM frameworks;
2) Develop intelligent decision support tools for selecting OSRM in Arctic offshore waters, taking into account practical considerations and limitations;
3) Improve the predication power and applicability of FDT-based OSRM selection tools working based on AI/SC techniques; and
4) Perform a comprehensive integrated optimization of oil spill response and oily wastewater management operations to minimize environmental impacts and operational costs.

1.4 Thesis Structure

This thesis consists of eight chapters summarized as follows:

- Chapter 1 serves as an introduction to the research, outlining the problem, motivation, and objectives. In particular, this chapter highlights the potential of applying intelligent

computational techniques in the field of MOSM. By emphasizing the significance of these techniques, the chapter sets the stage for the rest of the research.

- Chapter 2 proposes a framework for the present study to meet the sub-objectives laid out in Chapter 1. This chapter briefly discusses the various phases of the proposed research framework.
- Chapter 3 presents a comprehensive literature review and content analysis of essential components of MOSM and available techniques, particularly intelligent computational techniques, for effective management of oil spill incidents. It examines how intelligent computational techniques can be used for effective MOSM, offering insights and recommendations for improving the overall response strategy.
- Chapter 4 outlines a comprehensive methodology for creating an FDT-based decision support tool for facilitating OSRM selection, along with presenting the outcomes of its application. The decision support tool, developed using fuzzy theory, decision tree technique, and linear regression (LR) analysis, was trained and tested on a hypothetical oil spill dataset. This chapter elaborates the development process of the FDT-based OSRM selection tool, enabling readers to understand the underlying techniques and methods involved. FDT used in Chapter 4 was initially assumed as a capable intelligent technique for a transparent decision-making in OSRM selection. Accordingly, we enhanced its prediction accuracy and robustness while comparing its performance with other promising ML techniques in the next two chapters.
- Chapter 5 provides a detailed explanation and outcomes of the model that was used to improve the prediction power of the decision support tool developed in Chapter 4. This chapter describes the use of multiple regression analyses and tree splitting criteria, along with a metaheuristic optimization algorithm, to identify optimal training and testing datasets for constructing the decision trees. The improved decision support tool can achieve a

higher level of accuracy and robustness in selecting OSRM for a given set of environmental and operational factors.

- Chapter 6 demonstrates the proposed methodology for comparing the performance of various ML techniques used for OSRM selection. The chapter provides detailed results on the performance evaluation of multiple ANNs, support vector regression (SVR), Gaussian process regression (GPR), and the previously developed FDTs. The performance evaluation was based on prediction accuracies and several statistical metrics, which allows for the selection of the most reliable and accurate techniques using a small-sized dataset.
- Chapter 7 presents an innovative tool to simultaneously optimize oil spill response process and oily waste management operations. The tool was proposed using a metaheuristic optimization algorithm through a multi-agent system to minimize response time, cost, and volume of weathered oil. This tool was used to optimize the scheduling of available response assets in a hypothetical oil spill incident considering the detailed operational procedures of response and waste management.
- Chapter 8 summarizes the research reporting outcomes of various tools developed for OSRM selection and response process optimization. Additionally, this chapter presents the originality and limitations of this study and offers recommendations for future works.

Chapter 2 – Proposed Framework and Research Methodology

2.1 Introduction

This chapter provides an overview of the proposed framework for developing intelligent decision support tools for oil spill response selection and optimization. Details of the developed tools within the thesis framework are presented in the Method sections of the corresponding chapters for each decision support tool (Chapters 4 to 7).

The tools developed within this framework are particularly designed to facilitate the decision-making process for OSRM selection and response process optimization. They aim to decrease the human role in decision-making, subsequently mitigating the risk of errors while considering a wider range of influential decision-making factors. The developed tools founded on state-of-the-art models, effectively incorporating critical factors into the decision-making process for a timely and efficient oil spill response. These decisive factors encompass aspects such as the operational suitability of OSRMs, operational considerations within response and oily waste management, as well as environmental and economic features. These models can also serve as fundamental guidelines for the creation of more comprehensive decision support tools for MOSM, should better data become accessible. Moreover, they can be adapted to suit specific case studies based on local requirements and regulations. The used techniques through ML models can also provide a reference for ML studies in other fields struggling with similar data issues.

2.2 Research Framework

Figure 2-1 depicts the overall framework of this research, outlining the consecutive steps taken to achieve the proposed sub-objectives. Firstly, a comprehensive literature review was conducted to understand the recent advancements and identify knowledge gaps in the application of intelligent computational techniques in MOSM. Subsequently, OSRM selection tools were developed using FDT to enable a transparent decision-making process in selecting the most appropriate response method. FDT was selected as the main technique to develop the OSRM

selection tools due to its notable capacity in transparent decision-making. The FDT-based tools were then optimized to enhance their prediction accuracy and robustness. In the next step, the performance of tools developed based on multiple widely used ML techniques as well as the previously developed FDT-based tools was compared to not only investigate the relative prediction power of FDTs but also suggest the best performing tools for working with relatively small-sized datasets. The ML techniques were primarily chosen for OSRM selection in this study due to their potentially favorable performance, as observed in small-sized datasets within other research domains. Finally, the operational process of MCR was optimized alongside oily wastewater management (OWM) operation to reduce overall time and cost, as well as the volume of weathered oil.

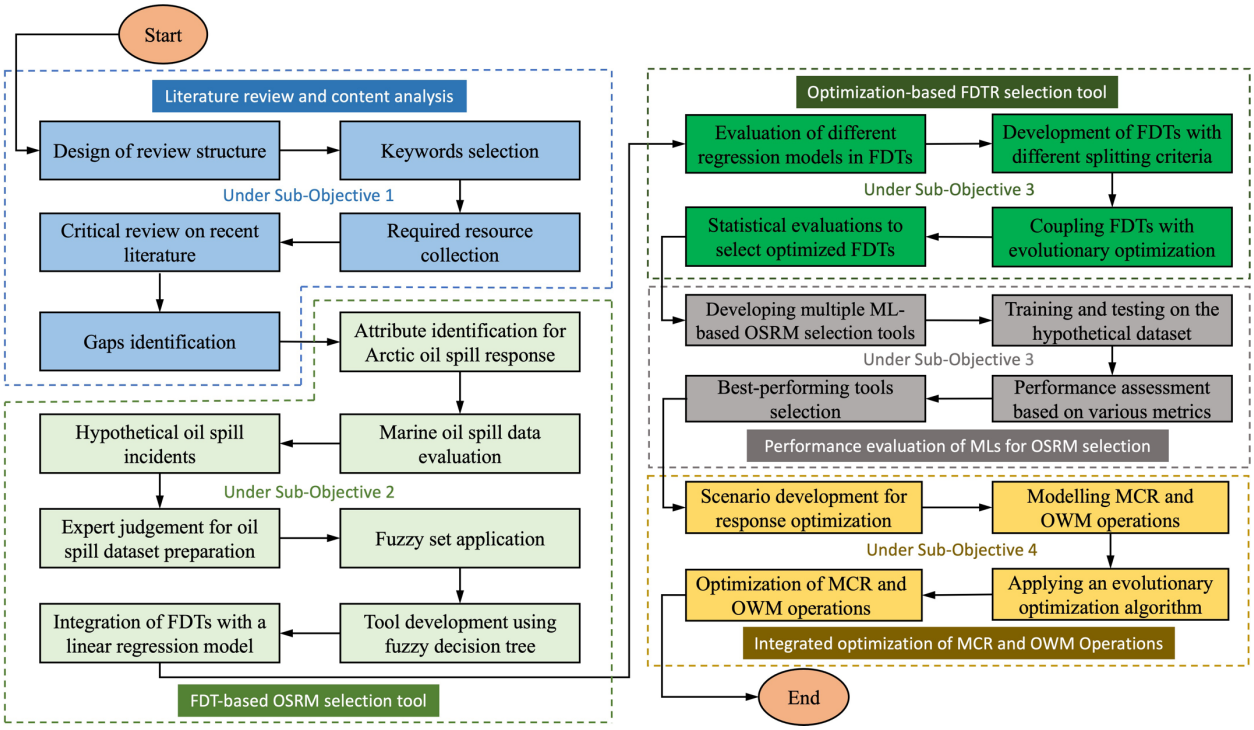


Figure 2-1. Proposed research framework

2.3 Literature Review and Content Analysis

A comprehensive literature review and content analysis was conducted to evaluate the role of intelligent computational techniques in oil spill detection and monitoring as well as OSRM

selection and process optimization. Moreover, this critical review covered various response methods, the adverse effects of oil spills on marine ecosystems, and the prominent challenges in MOSM. This literature review was performed under the first research sub-objective outlined in Chapter 1.

2.4 FDT-based OSRM Selection Tool

An FDT-based decision support tool was developed to facilitate OSRM selection in a practical and transparent way. The tool was developed based on a decision tree technique which is appropriate for transparent decision-making especially in case of limited amount of data. This decision support tool (briefly called “OSRM selection tool”) is capable of promptly generating suitability indices, in terms of field application feasibility, for commonly used OSRMs under specific environmental conditions. The tool can consider various influential attributes in OSRM selection, including environmental attributes and spilt oil characteristics. To this end, a hypothetical, yet realistic, dataset of oil spills in Arctic waters was generated based on experts’ judgment, and the dataset was used to demonstrate decision support tool’s applicability. Also, fuzzy logic was incorporated into the decision-making process to account for the epistemic uncertainties originated from human judgments on influential attributes because of incomplete information and subjective perception of spill information. This model defined under the second research sub-objective outlined in Chapter 1.

2.5 Optimization-Based FDTR Selection Tool

A comprehensive model was designed to enhance the prediction accuracy and robustness of the intelligent OSRM selection tool. For this purpose, a metaheuristic evolutionary optimization algorithm was integrated with the OSRM selection tool to find the optimal combination of training and testing datasets. Appropriate objective functions concerning training and testing prediction accuracy and number of generated classification rules were defined to enhance the prediction power of OSRM selection tool. In addition, effects of different tree splitting criteria and embedded

regression analyses were evaluated in this model. This model was defined under the third research sub-objective outlined in Chapter 1.

2.6 Performance Evaluation of MLs for OSRM Selection

A model was developed aiming to identify the most suitable ML techniques for the selection of oil spill response methods under data constraints. Several ML techniques were compared on the performance of predicting suitable OSRM in cold waters with given environmental information. Multiple regression-based ML techniques were used to develop predictive tools with a relatively small hypothetical oil spill dataset. The performance of these tools and previous FDT-based tools was compared based on prediction accuracy and other important statistical metrics. This model was defined under the third research sub-objective outlined in Chapter 1.

2.7 Integrated Optimization of MCR and OWM Operations

MCR is an ideal response method to oil spill incidents. However, its effectiveness can be compromised by response capacity limitations in the affected areas, such as available transportation and storage capacities of oily wastewater and the availability of fleets of response vessels. Management of waste generated from an oil spill response is a crucial step in MOSM as it is usually a bottleneck of any response operations. To this end, the essential steps in MCR were identified, and practical variables were optimized based on the consideration of available resources and constraints such as the number of response vessels, storage capacity, response time, and cost. To demonstrate the integrated optimization, a hypothetical dataset of an oil spill incident information, response and storage capacities, waste treatment facilities, estimated unit costs, and other operational requirements were developed based on the consultation with oil spill response experts. An optimization tool was developed to help oil spill response practitioners efficiently manage the oil spill response process and the generated oily wastewater in an integrated manner. This model was defined under the fourth research sub-objective outlined in Chapter 1.

Chapter 3 – Literature Review and Content Analysis

A version of this chapter has been published in Journal of Hazardous Materials, entitled "Intelligent Computational Techniques in Marine Oil Spill Management: A Critical Review." https://doi.org/10.1016/j.jhazmat.2021.126425.

3.1 Introduction

This chapter initially discusses MOSM with an emphasize on the application of intelligent computational techniques in this field. It also includes a discussion on OSRMs as a critical part of oil spill management. Additionally, commonly used computational techniques and numerical models in MOSM, as well as influential processes affecting the fate and transport of spilt oil, are introduced.

Subsequently, this chapter provides a detailed review of literature published since 2010 on the application of computational techniques, particularly AI/SC techniques, in MOSM. Google Scholar and Science Direct search engines were used to find relevant publications using the combinations of keywords in oil spill management and computation-based research domains. The keywords in the oil spill management domain included oil spill waste management, OSRM selection, and OSDCM; the keywords in the AI/SC-based computation domain included AI, ML, optimization algorithms, metaheuristic techniques, SC, data mining, uncertainty analysis, and decision support systems. An example of combination could be "oil spill detection, AI."

Following that, statistical analysis and descriptive analysis (or meta-analysis) were carried out to evaluate the collected literature. Statistical analysis was conducted to determine the temporal distribution of the studies, publishers' engagement in this field of research, various research sub-fields, geographical locations of the studies, and available case studies. Descriptive analysis was conducted by evaluating each paper based on the review structure to identify its objectives, techniques and models used, strength/novelty of the research, and limitations and potential recommendations for enhancement.

The collected studies generally fall into two categories: (i) the combined use of remote sensing (RS) and AI/SC-based tools in OSDCM; (ii) selection and optimization of OSRMs using SC and advanced mathematical techniques. Representative studies in the two categories were summarized in two separate review tables presented in Appendices A and B of this thesis.

3.2 Marine Oil Spill Management (MOSM)

Various components of MOSM and major computational techniques and numerical models frequently used in this field are introduced in the following sections. Timely response is a crucial aspect of spill management; thus, this chapter also includes a review of OSRMs that are commonly used in major oil spill incidents.

3.2.1 Major Components of MOSM

OSDCM, risk evaluation, response selection and process optimization, and waste management are the major components of an MOSM system. A hypothetical MOSM is schematically shown in Figure 3-1. OSDCM tools, working based on data obtained from aerial and satellite images and the analysis of possible samples collected by vessels, are needful in MOSM. These tools can be used to detect oil spill incidents and predict the trajectory of spilt oil slicks (Cantorna et al., 2019; Fingas and Brown, 2014; Xu et al., 2014; Zodiatis et al., 2016). Another popular device used in oil spill monitoring is sensor-equipped buoys, which can detect and trace petroleum hydrocarbons on the water surface (Lassalle et al., 2020; Moroni et al., 2016; Sai et al., 2020). As illustrated in Figure 3-1, a buoy can also be equipped with different instruments to obtain accurate metrological, oceanic, and geographical data. An intelligent buoy can be used for oil spill tracing in ice-infested waters (Sai et al., 2020).

As depicted in Figure 3-1, there are some highly vulnerable and sensitive (high-risk) species to oil spills, such as sea birds and mangrove forests/saltmarshes, that should be considered within oil spill risk assessment in MOSM (Kakalis and Ventikos, 2008; Ozigis et al., 2019). The presence of these vulnerable and sensitive species in the ecosystem also stresses the necessity of a timely

and suitable response to reduce the environmental risks (Balogun et al., 2020; Giri et al., 2011; Ozigis et al., 2019).

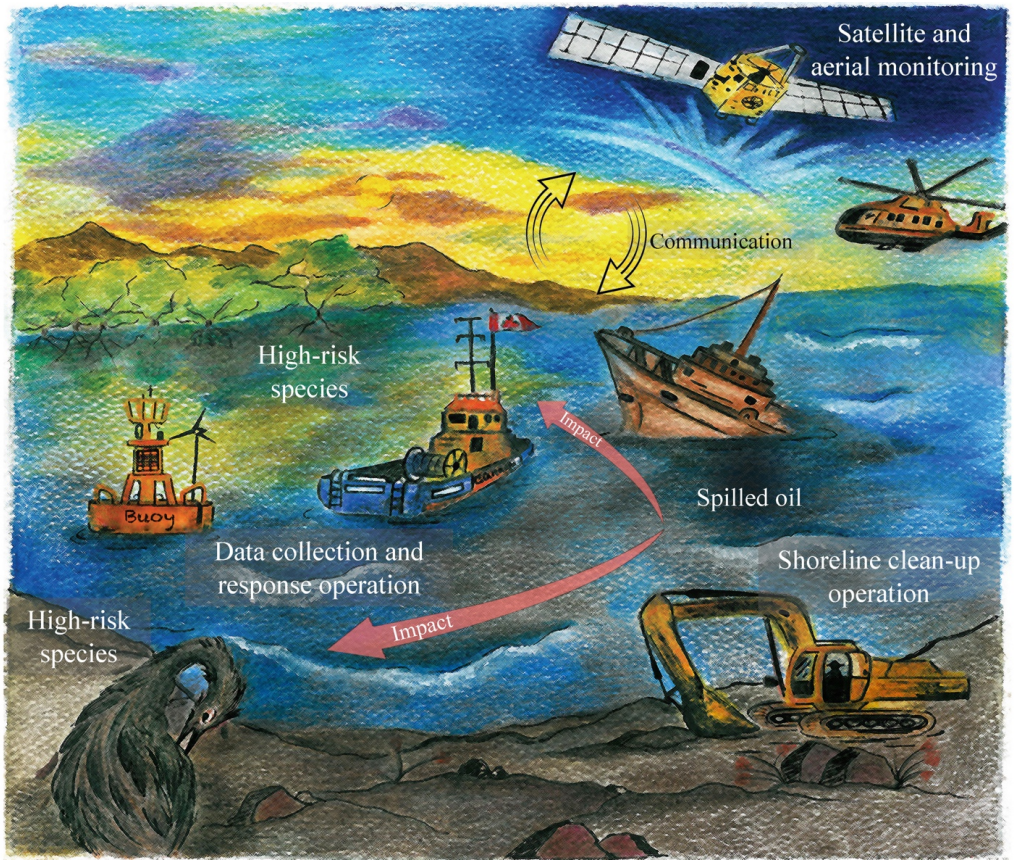


Figure 3-1. Schematic presentation of oil spill management

Once an oil spill is identified, response actions should be taken immediately to control the spread of the spilt oil. While offshore oil spill response often involves the use of MCR, ISB, and UCD, the cleanup of oil-contaminated shorelines requires extensive labor and machinery, such as excavators (Ivshina et al., 2015; Mohammadiun et al., 2021a; Rongsayamanont et al., 2017). Management of response-generated oily waste, including its storage, transportation, treatment, and disposal/discharge, can also be a challenging task in MOSM (IPIECA, 2014; POSOW, 2016). Although a substantial portion of solid oily wastes is sent to incineration plants and landfill sites for disposal, it is necessary to have a waste management plan to reduce, reuse, and recycle the wastes (IPIECA, 2014; POSOW, 2016). The collected oil-water mixture may also be transported to oil-water separators or decanting facilities for the recovery process (IPIECA, 2014). It is also

necessary to estimate the quantity of wastes generated from oil spill response for effective waste management. Typically, the volume of waste generation varies from 40% to 200% of the volume of spilt oil, and in extreme cases it may exceed ten times the initial spill volume (IPIECA, 2014).

3.2.2 Oil Spill Response Methods

Preventing spilt oil from reaching shorelines is important because significant amounts of oily waste can be generated during shoreline cleanup (IPIECA, 2014). Figure 3-2 depicts the widely used offshore OSRMs. Generally, offshore cleanup methods fall into three categories: 1) mechanical response methods including the use of booms, skimmers, and sorbents; 2) chemical response methods including burning, dispersant chemicals, and solidifiers; and 3) biological response methods including the application of fertilizers or some microorganisms to enhance biodegradation of spilt oil (IPIECA, 2014; Ivshina et al., 2015; Li et al., 2016; Zhong and You, 2011). One or a combination of these response methods should be carefully selected and optimized based on individual oil spill conditions, such as variable properties of spilt oil and environmental conditions (Hu et al., 2020).

The most ideal method for oil spill management is the MCR, which involves using booms to contain the spilt oil, followed by skimmers to collect the oil from the water surface. The collected oil-water mixture is then transported to oil-water separators or decanting facilities for the recovery process and treatment (IPIECA, 2014). There are various types of booms, including above-water freeboard, below-water skirt, floating device, fire-resistant etc. Their performance mostly depends on the height and velocity of waves and the response vessel's towing speed (Hu et al., 2020; Ivshina et al., 2015; Mohammadiun et al., 2021a). A typical MCR operation starts with deploying large-scale booms to contain an oil spill, and then continues with pumping the oil into preliminary (temporary) storage vessels by the use of skimmers passably separating oil and water. Finally, the remaining residues of the spilt oil can also be collected using absorbing blankets or diluted by the application of dispersants (Li et al., 2019). The operational suitability of MCR can be

determined based on the location of an oil spill, an oil spill thickness, and weathering conditions (Hu et al., 2020; Mohammadiun et al., 2021a; Polaris, 2009). However, MCR can be effectively performed within a narrow window of opportunity due to the oil weathering process, and its application may be impeded by ice-infested waters and remote response locations in Arctic and Sub-Arctic areas. The efficacy of some skimmers can also be reduced by spilt oil with a high viscosity (Mohammadiun et al., 2022).

ISB is one of the primary and emergency methods of oil spill response, with its efficacy significantly depending on the oil slick thickness. Fire-resistant booms may be used in ISB to provide the minimum required thickness (2 to 3 mm) for successful combustion, which can be initiated using a handhold igniter or a Heli-Torch (Hu et al., 2020; IPIECA, 2014). However, ISB application should be restricted to the regions located at least 1 to 2 km away from a residential area due to the release of toxic smoke produced by burning volatile components of oil (Ivshina et al., 2015). ISB is an effective response method for oil spill incidents in remote locations like Arctic and Sub-Arctic waters (Mohammadiun et al., 2022). The remoteness of the response location, thickness and continuity of oil slicks, and weathering conditions are the most influential attributes defining the efficacy of ISB (Hu et al., 2020; Mohammadiun et al., 2021a; Polaris, 2009).

UCD is suited for the application in adverse weather conditions, deep-water incidents, and areas near vulnerable coastal (Rongsayamanont et al., 2017). Equipped aircraft or ships can spray a dispersant chemical, such as a type of Corexit®, to break up the oil in water (Mohammadiun et al., 2022). Enough time should be provided for the chemical compound to penetrate the oil and disperse it into small droplets inside the water column in UCD (Ivshina et al., 2015). The application of this method may be restricted by local regulations pertaining to the oil spill location due to the temporary elevation of eco-toxicity in the impacted area (Potter et al., 2012). The suitability and effectiveness of UCD are influenced by the viscosity of the oil, location of the spill, thickness and continuity of oil slicks, and weathering conditions (Hu et al., 2020; Ivshina et al., 2015; Mohammadiun et al., 2021a; Polaris, 2009). The presence of floating sea ice and weak

waves may reduce the effectiveness of dispersants in cold climates; however, emulsification may be impeded in these environments (Potter et al., 2012). In the lack of sufficient flow advection such as in some ice-infused waters, appropriate propellers can be used to achieve the desirable mixing energy for UCD.

The bioremediation technique increases the biodegradation rate of hydrocarbon contaminants, reducing their persistence time. As a rich source of required nutrients of phosphorus and nitrogen, fertilizers can be used to improve the biodegradation rate in seawaters. This response method requires a significant flow advection and/or dispersion inside sufficiently deep waters. Prior to applying this response method, a human and environmental risk assessment should be performed while considering local regulations (Ivshina et al., 2015).



Figure 3-2. Schematic presentation of the most common offshore OSRMs

As mentioned earlier, the characteristics of spilt oil would change under the effects of weathering processes (Ivshina et al., 2015; You and Leyffer, 2011). This variation in oil spill characteristics over time may have a considerable influence on different components of MOSM. The dynamic change of the characteristics of spilt oil and work arrangement of response fleets could affect the efficacy of applied response methods and the performance of OSDCM tools (Mera et al., 2017; Ye et al., 2019). The oil-water emulsion may also increase the total volume of response-generated waste due to a rise in water content substantially hindering the waste management in MOSM

(Zhong and You, 2011). The oil characteristics and response efficacy can also be affected by harsh environments, especially in Arctic and Sub-Arctic areas (Carter et al., 2018a; Hu et al., 2020; Potter et al., 2012). The application of computational techniques in MOSM may significantly facilitate oil spill management by intelligently considering effects of the abovementioned complex processes in decision-making.

3.2.3 Computational Techniques used in MOSM

The following sub-section provides an overview of computational techniques commonly used in MOSM to enhance the decision-making process. The scope of computational techniques in this thesis, encompasses AI/SC and conventional mathematical techniques used to facilitate decision-making in MOSM. For example, ML techniques (e.g., ANNs and FDTs) and optimization algorithms (e.g., metaheuristic algorithm) used to detect oil spills, select the most suitable OSRM, and optimize response processes are reviewed. Major inputs for these techniques include: (i) aerial/satellite image data for OSDCM; (ii) datasets of influential attributes in decision-making for OSRM selection; and (iii) response resources availability and capacity, local regulations, the cleanup target, and properties of spilt oil for OSRM process optimization.

ANNs are a widely used AI/SC technique in MOSM, and they function as learning algorithms that mimic the neural networks found in the human brain. These networks consist of interconnected artificial neurons that are arranged into multiple layers, including input, hidden, and output layers, with different transformations (i.e., activation functions). The output of neurons is a weighted summation of the inputs combined with a linear/non-linear transformation. The ANN weights and biases are computed in a training process aiming to minimize the difference between training data and the ANN outputs by evaluating loss functions (Priddy and Keller, 2005; Shanmuganathan, 2016).

Metaheuristic evolutionary optimization is another useful computational technique used in MOSM. It includes a group of algorithms that can find sufficiently accurate estimation of optimal solutions

within an iterative process (e.g., a GA working based on natural evolution). Metaheuristic algorithms are the enhanced version of heuristics, and are capable of escaping from local optima and estimating a global optimal value. They can be applied to a wide range of optimization problems as they require neither explicit form of the objective functions nor gradient calculation of functions. These algorithms use different operators, which are various mathematical processes applied to candidate optimal solutions (variables), to enhance objective function values (fitness) through the iterative process (generations). Exploration and exploitation capacity can be defined for the operators of the algorithms. Exploration is the ability of an algorithm to perform a better global search by randomizing some parameters to find new solutions within a search space. Exploitation is the ability to perform a better local search by tuning solutions that are already found (Dan and Simon, 2013; Mohammadiun et al., 2018; Yazdi et al., 2018).

Case-based reasoning (CBR) is another technique used in OSRM selection based on experience and records of previously happened oil spill incidents (Huang et al., 2020; Mata et al., 2011). The CBR technique is designed to solve problems in a way that mimics how human experts would solve them, by reasoning from prior experiences and adapting those solutions to new situations. In this technique, solutions of previous incidents are stored and evaluated for their potential of being used in future similar oil spill responses concerning influential attributes such as oil slick characteristics and oceanic data (Mata et al., 2011). In CBR, the stored solutions for past incidents, are used as a knowledge base to guide decision-making for new incidents. When a new oil spill incident occurs, the CBR system retrieves and analyzes past cases that are similar in key attributes to the current situation. The system then adapts the most relevant solution to the new case, trying to provide a rapid and effective response to the incident.

Other commonly used computational techniques are briefly introduced in Table 3-1.

Table 3-1. Brief descriptions of computational techniques commonly used in MOSM

Technique/Model	Description	Additional References
Machine Learning	It is the development of a computational tool identifying meaningful patterns among influential attributes in a dataset that can be used to predict intended output values without being directly programmed for it. The ML's weights and parameters can be determined (or tuned) based on a training dataset of cases; the trained tool can be used to predict outputs for any new cases (e.g., a testing dataset).	(Ibrahim, 2016; Mehryar Mohri et al., 2013)
Binary Classification	It is the classification of elements in a dataset into two disjoint sets based on the predefined classification rule(s). For example, as the output of a binary decision support tool, a response method can be either appropriate or inappropriate for a specific oil spill incident based on values of inputs such as wave height, spill volume, and remoteness of spill location.	(Koyejo et al., 2014; Ozigis et al., 2019)
Semantic Segmentation	It is the binary classification of image objects into their corresponding classes through image processing; For example, a group of dark pixels in SAR image data can be classified into oil slick or non-oil slick classes.	(Garcia-Garcia et al., 2017; Guo et al., 2018)
Instance Segmentation	It is a more advanced image segmentation technique than semantic segmentation by differentiating between objects belonging to the same class. For example, it may be possible to differentiate various types of oil slicks (e.g., with different thicknesses) in instance segmentation of SAR images.	(Arnab and Torr, 2016; He et al., 2020)
Deep Learning	DL models are sophisticated and recently developed types of ML, performing similar to the human cognitive process. DL models usually have significantly more layers and neurons than conventional neural networks (NNs), and thus perform better in solving complex problems. DL models have been broadly applied in image processing for both semantic and instance segmentations.	(Aghdam and Heravi, 2017; Schmidhuber, 2015)
Feature Selection	These techniques select a limited number of more significant features extracted from image data, to enhance OSDCM models' robustness (e.g., a spill detection ML tool). Influential features can be selected based on the general patterns, such as their correlations or influence on an OSDCM tool's detection accuracy.	(Ma et al., 2014; Mera et al., 2017)
Clustering Algorithms	These algorithms divide a set of data into a number of distinct subsets (clusters) in which elements are similar to each other. Clustering can be used as a preprocessing step to enhance the developed ML tool's prediction power because the tools can be trained on a data cluster with more discernible patterns.	(Kamble, 1999; Xu and Wunsch, 2005)
Mixed Integer (linear/non-linear) Programing	A mathematical technique used to model optimization problems with both continuous and discrete (integer) decision variables. Discrete decision variables through optimization could be decision-making about whether an oil spill response should be performed. The technique can be applied to both linear and non-linear objective functions and constraints.	(P. Li et al., 2014b; Wolsey, 2008)
Multi-Agent System	It is usually an intelligent system consists of different agents (or modules), such as optimization and simulation modules, interacting to solve a highly complicated problem. In this system, an agent provides others with their required input data to perform a task that may be impossible to be conducted by a single agent.	(Deloach et al., 2001; Ye et al., 2019)
Bayesian Network (BN)	BN and its generalized version-influence diagram were also used in decision-making and uncertainty analysis in OSRM selection. By changing in one independent variable (e.g., influential attributes), the state of other dependent variables (e.g., response method suitability scores) can be estimated based on the probability distributions in BNs.	(Helle et al., 2015, 2011)
Monte Carlo (MC) simulation	MC is a robust uncertainty analysis technique. MC simulation works based on sampling from probability density functions of input variables that will be used in the simulation process to estimate probabilities of a model's outputs. MC can obtain probability distributions of desirable outputs even in highly complicated and non-linear systems.	(Kroese et al., 2014)

3.2.4 Numerical Modeling in MOSM

Oil spill numerical simulators can be used in different MOSM components from OSDCM to OSRM selection/optimization. These simulators are mostly used for fate and transport modeling of spilt oil in the marine environment (Abdallah and Chantsev, 2022). Influential physical, chemical, and biological processes in fate and transport of spilt oil are discussed in the following alongside widely-used numerical simulators for MOSM (ASCE, 1996; Zhong and You, 2011).

3.2.4.1 Physical, Chemical, and Biological Processes

Various physical, chemical, and biological processes affect a spilt oil in the marine environment within the three main routes of oil spill transport, fate, and ice and shoreline interactions. The transport of spilt oil in seawater is mainly affected by advection and spreading, while the fate of the oil is governed by other complex processes, such as evaporation, emulsification, dispersion, dissolution, biodegradation, and photo-oxidation. In cold regions and near shorelines, ice and shoreline interactions such as ice drifting and encapsulation as well as sedimentation and re-floatation can also alter the fate and transport of spilt oil (ASCE, 1996; Zhong and You, 2011).

Transport of spilt oil:

The spilt oil in marine waters creates a thin layer called oil slick expanding over the water surface. As a result of advection, which is an influential factor in the determination of the oil spill trajectory, the oil slick starts to move in the water. This movement is caused by the exerted momentum from a field of wave, current, and wind. Most of the relevant studies applied the superposition technique on different vectors of mean flow, wind, current, tide, and turbulent diffusion to predict oil slick location over time by considering a suitable time step. A comprehensive prediction method is to use complex two/three-dimensional numerical simulators to solve governing equations simultaneously, requiring significant computational resources (ASCE, 1996; Huang, 2005).

Additionally, the moving oil slick expands horizontally (spread) due to an imbalance in mechanical forces, including gravity, inertia, viscosity, surface tension, and interfacial tension between the

water and oil. The combined effect of spreading and influential fate processes can lead to a gradual reduction in oil slick thickness. Beyond a certain critical threshold, a further reduction in thickness can cause the oil slick to break up into smaller fragments. Spreading is a crucial process from an environmental perspective since it can significantly decrease oil slick density and mass over time. During the early stages of an oil spill incident, the location of the oil slick is heavily influenced by spreading, although this effect decreases as time progresses (ASCE, 1996; Huang, 2005).

Fate of spilt oil:

While the oil slick is being transported because of flow advection, its volatile and light components (e.g., diesel) can evaporate, and its heavier components (e.g., bunker oil) can form water-oil emulsions (i.e., emulsification) or precipitate into the sediments (i.e., sedimentation). In addition to these processes, biodegradation and photo-oxidation may also affect spilt oil during a long-term period compared to other processes. The entire process is considerably controlled by oil properties, meteorological, and oceanic conditions (Tkalich et al., 2003). However, some of these processes, such as dissolution, biodegradation, photo-oxidation, and sedimentation, are often neglected in marine oil spill studies due to their negligible effects or complexity.

Evaporation is a critical process that affects oil spill fate by changing oil density and viscosity. It removes oil components with low-boiling points and reduces the volume of spilt oil over time. Oil slick volume is especially affected by the evaporation process during the initial 12 hours after an incident. The evaporation rate of heavy to light oils varies from 10% to 75% of the oil mass, respectively. Evaporation can also occur in the emulsified or even dissolved oil with a variable evaporation rate. Generally, a logarithmic variation can be assumed for the evaporation rate in a multi-phase fluid of spilt oil in seawater. This rate particularly depends on the oil temperature, wind and current speed, oil slick area, and percentage of volatile components. The evaporation rate increases significantly with higher temperatures, high-velocity winds, and rough seawaters (Zhong and You, 2011). Ordinary differential equation (ODE) were used to model the evaporated

fraction of oil over time based on these influential parameters and some empirical constants (Li et al., 2016; Ye et al., 2019; Zhong and You, 2011).

Oil emulsification in water is the dispersion of oil droplets inside the water column, producing an immiscible mixture referred to as “mousse” or “chocolate mousse” (Fingas, 2014). This process has a significant impact on the characteristics of spilt oil, potentially increasing the total volume of the oil-water mixture up to five times the volume of the initial spill (Lassalle et al., 2020). Emulsification may cause the spilt oil specific gravity to increase by up to 30% of its initial value, typically occurring within the first day of an incident. The effects of emulsification on mass and specific gravity of oil slicks can be considerable, even more than a week after an oil spill (ASCE, 1996; Fingas, 1995). Additionally, an oil-water emulsion can be an unstable mixture, with a tendency to lose some of its water content over time. The most significant effect of emulsification is on oil viscosity, potentially forming a heavy and semi-solid oily waste. This process can also reduce the evaporation rate and oil spreading (ASCE, 1996). To model the effects of emulsification, Mackay suggested a widely used formulation, defining an ODE problem to estimate the variation of fractional water content over time using empirical constants, wind velocity, and an exponential variation for oil density with time (Fingas, 1995).

Oil dispersion is another influential process whereby small droplets of oil are formed and dispersed throughout the water column, causing a reduction in oil slick continuity. The size and buoyancy of the oil droplets as well as the water turbulence intensity, are governing factors of this process. Although dispersion usually increases the spilt oil biodegradation rate, it causes a decrease in evaporation. It is important to note that larger and unstable dispersed droplets may rise again towards the water surface, coalescing into an oil slick. Turbulence kinetic energy is an influential parameter determining the degree of stability of dispersed oil in the water column (ASCE, 1996). The dispersion rate can be estimated based on temporal distribution of wind velocity, oil spill area, oil spill volume, interfacial tension of oil and water, and oil spill viscosity (Zhong and You, 2011).

The dissolution of oil in water is a process that can be crucial in the toxicity assessment of oil spills, particularly in shallow waters or spills with significant amounts of light aromatic components. However, the volume of dissolved oil is usually negligible in oil spill incidents. There is a notable correlation between evaporation and dissolution processes in oil spill incidents. During the initial 12 hours after oil spills, evaporation rates are typically higher, while dissolution rates are low. However, as oil slicks spread, the interfacial contact between oil and water increases, leading to an increase in dissolution rates. Higher temperature values may also cause a slight increase in oil dissolution (ASCE, 1996; Fingas, 2014).

Biodegradation is a natural process that reduces the persistence of hydrocarbon contaminants and alters the properties of oil in water over an extended period. In this process, marine micro-organisms (e.g., bacteria) break down the oil's toxic compounds and eliminate them. An approximate estimation of the biodegradation rate can be derived from the growth and decay rates of microbial populations. While the natural biodegradation rate of spilt oil is insignificant, the process can be accelerated by the application of bioremediation response techniques. As mentioned, fertilizers can be utilized to improve the biodegradation rate and it is necessary to have a significant flow advection and/or dispersion within adequately deep waters (Huang, 2005; Ivshina et al., 2015).

Photo-oxidation is a process in which petroleum hydrocarbons undergo oxidation in the presence of sunlight, producing soluble compounds in water. Although this process can change the characteristics of spilt oil, its overall effects on oil slicks are negligible. However, from a biological and toxicity perspective, the process can be significant due to its potential toxic by-products for marine environments. Simplified ODEs have been used to model this process based on the radiation angle of the sun, slick area and thickness, and cloud coverage (Fingas, 2014; Huang, 2005).

Ice and shoreline interactions:

In general, there is a lack of sufficient study and data on the complex interaction between spilt oil, ice, and shorelines. Different forms of sea ice have varying interactions with oil spills, such as fast ice, which is frozen along the coast or sea, and pack ice, which floats freely on the water surface. While oil slicks are unlikely to drift considerably either over or under the fast ice, they can easily drift with pack ice. An oil slick may also be encapsulated by ice until the spring thaw when the released oil returns to the water surface and undergoes weathering processes at a low temperature. The encapsulated oil slicks can also encounter with a vessel when they lie in a ship traffic route (Hu et al., 2020; Polaris, 2009).

The interaction between oil and shorelines is also highly complex and involves sediment deposition, incorporation of suspended sediments, penetration into shoreline sediment, and re-floatation. Surface and capillary adsorption are two primary components of this interaction, depending on the size of the sediment particles (Chao et al., 2003). Sedimented oil can be confined to the seabed by other sediments, posing a serious threat to biota. The sedimented oil can also be re-suspended (re-float) in the water column (Fingas, 2014). Sedimentation effects on oil spills, particularly on the oil slick location and distribution, may become increasingly important over a long period (ASCE, 1996). The rate of oil decrease in water due to sedimentation can be estimated using a simplified ODE based on the sediment load and mixture salinity (Huang, 2005).

3.2.4.2 Numerical Simulators

Numerical simulators are developed to incorporate the effects of the influential processes mentioned earlier in MOSM. However, many marine oil spill studies tend to overlook certain processes such as ice and shoreline interactions due to limited data availability or the complexity of these processes.

Various oil spill simulators have been developed to monitor and predict oil slick transport using Lagrangian, Eulerian, and hybrid Lagrangian-Eulerian models. Those models consider advection,

spreading, and hydrodynamic conditions in oil slick simulation in the marine environment. As mentioned, the advection governs the movement of spilt oil in the water, while the spreading mainly affects the fate of spilt oil and may lead to breaking up the oil slick into smaller fragments (ASCE, 1996; Huang, 2005). Hydrodynamic effects can be investigated by numerically solving three-dimensional Navier-Stokes equations or its depth-averaged, two-dimensional form of shallow water equations (Janati et al., 2020; Mohammadiun et al., 2016, 2015). As mentioned, the processes affecting the fate of spilt oil are usually modeled using differential and empirical equations based on the relevant influential parameters and empirical constants (Li et al., 2016; Ye et al., 2019).

Some commonly used oil spill simulators can be enumerated as the Natural Resource Damage Assessment Models for Coastal and Marine Resources (NRDAM/CMR), General National Oceanic and Atmospheric Administration Operational Modeling Environment (GNOME), Blowout and Spill Occurrence Model (BLOSOM), Oil Spill Contingency and Response (OSCAR), MEDSLIK-II, MIKE 21/3, and SIMAP (Abdallah and Chantsev, 2022; De Dominicis et al., 2013; Grubesic et al., 2017). Those simulators are effective models in MOSM for predicting the location of oil slicks and properties of spilt oil over time.

3.3 Statistical Evaluation of Relevant Studies

Figure 3-3 shows the statistical evaluation of the major reviewed studies, found using the combinations of the keywords mentioned earlier. These studies apply computational techniques to different components of MOSM. About 45% of the reviewed papers are relevant to OSDCM using various integrated RS and AI/SC-based tools. In this category, the automatic detection of oil spill incidents, intelligent determination of spilt oil properties, and monitoring oil spill effects on natural environments were investigated using different aerial and satellite image data. The number of publications in this research category has been increasing in recent years due to the increased satellite image data access (e.g., RADARSAT-2, TerraSAR-X, Sentinel 1&2, and

Landsat data) and considerable progress in image segmentation and processing techniques, such as DL-based tools (Bianchi et al., 2020; Marghany, 2015; Yekeen et al., 2020). Moreover, about 40% of the studies focused on selection and/or optimization of OSRMs. The enhanced availability of meteorological and oceanic data, advancements in comprehensive models for oil spill trajectory simulation, and progress in knowledge discovery have heightened interest in this component of MOSM in recent years.

About 10% of reviewed papers were pertaining to literature review while lacking detailed consideration of computational techniques in this field. More than half of the evaluated literatures were published by Elsevier, and most of them were found in Marine Pollution Bulletin (32%) and Journal of Hazardous Materials (16%) among a total of 19 Elsevier journals. About 29% of reviewed papers were published before 2015, and 71% of them were papers published during the recent five years.

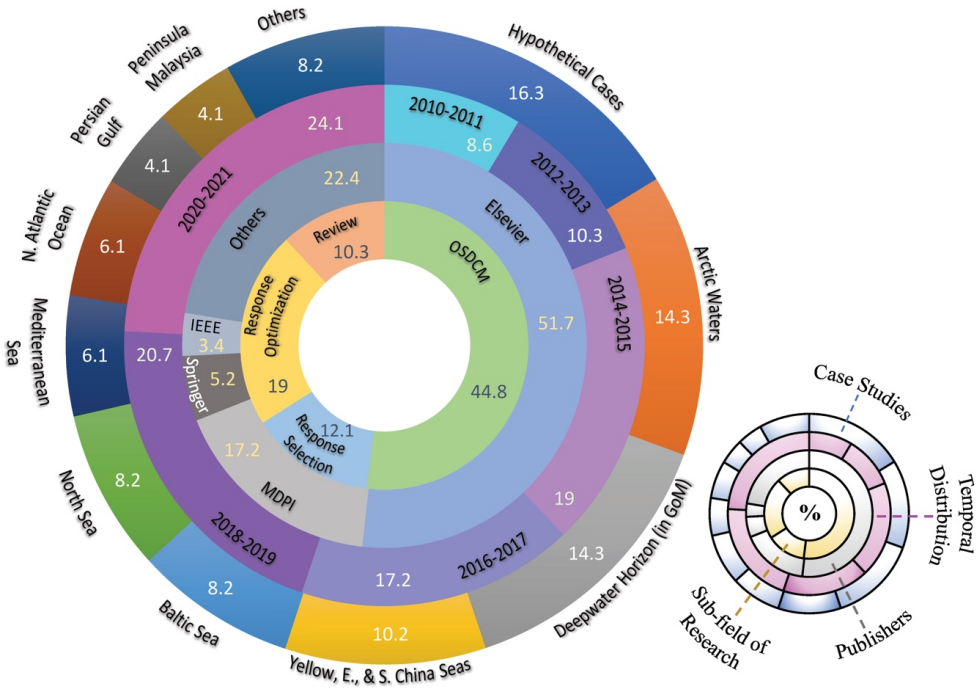


Figure 3-3. Statistical evaluation of reviewed studies applying computational techniques in oil spill management

In terms of geographical distribution of the reviewed studies, 40% of articles evaluated oil spill case studies in the Arctic (i.e., the Norwegian Sea, the Barents Sea, the Greenland Sea, and the Arctic Ocean), the Deepwater Horizon oil spill in the GoM, and incidents in China's open seas. Figure 3-4 displays the countries of origin of the evaluated papers and significant international waters. As depicted, Canada, China, and the United States were the top three countries accounting for more than half of the reported studies on the development of computation-based tools in oil spill management. Both Canada and the United States have extensive coastlines on their eastern and western sides. Also, their territory covers vast Arctic/Sub-Arctic areas. Global warming has reduced ice-covered areas in Arctic and Sub-Arctic regions, offering new potential marine transportation routes in these environments, so as oil spill risks (Carter et al., 2018a; Potter et al., 2012).

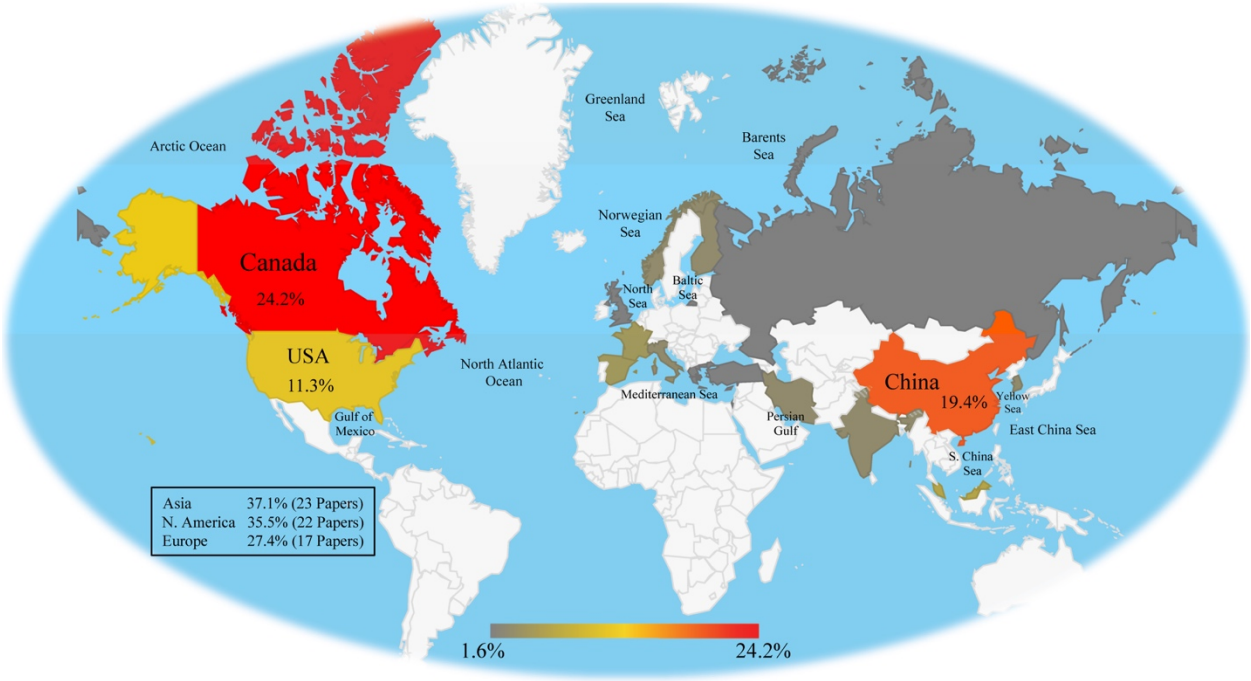


Figure 3-4. Origin countries distribution of the reviewed papers

Additionally, the East China Sea, the South China Sea, and the Yellow Sea were the extensively studied regions due to considerable marine transportation activities and recent utilization of advanced RS tools for oil spill detection in these areas (Moroni et al., 2019; Zeng and Wang,

2020). European countries adjacent to the Mediterranean Sea, the Norwegian Sea, the North Sea, and the Baltic Sea also published extensive studies (27.4% in total) in the area of using computational techniques in MOSM. The mentioned seas were frequently considered in studies due to significant ship traffic in these waters and the availability of sufficient data, such as RS data provided by the CleanSeaNet service of the EU maritime safety agency (Moroni et al., 2019; Zodiatis et al., 2016).

The Deepwater Horizon oil spill in USA was the most investigated real-world case study due to its catastrophic environmental impacts, significant financial burden, data availability, and media exposure. However, hypothetical oil spill incidents were the most frequently considered case studies, revealing a lack of sufficient real-world datasets for studying oil spills. Particularly, a significant deficiency in real-world data was noticed for oil spill incidents in harsh environments (Hu et al., 2020).

3.4 Descriptive Analysis of Oil Spill Detection, Characterization, and Monitoring

RS can provide response practitioners with essential data for OSDCM and for determination of the recovery rate of contaminated natural environments by evaluating long-term images. AI/SC-based tools can be applied in combination with RS to intelligently evaluate image data and identify influential features for defining various indices in OSDCM (Balogun et al., 2020; Ozigis et al., 2019; Yekeen et al., 2020). For instance, these tools can identify and classify different objects of interest (e.g., different types of oil slicks) based on subtle features (e.g., reflectance in a specific wavelength) hidden in satellite and aerial images (Angelliaume et al., 2017; Bianchi et al., 2020). Appendix A summarizes representative studies in the sub-field of spill detection, characterization, and monitoring based on chronological order.

RS uses hyperspectral/multi-spectral image data and SAR image data, obtained by optical sensors and microwave sensors, respectively (Balogun et al., 2020; Cantorna et al., 2019; Marghany, 2015). Generally, hyperspectral/multi-spectral imagery provides a suitable resolution

and wavelength band for terrestrial evaluation, and SAR imagery has a desirable spatial coverage and range of frequency (or wavelength) for offshore waters (Balogun et al., 2020; Giri et al., 2011; Marghany, 2015).

3.4.1 Hyperspectral and Multi-Spectral Imagery

Hyperspectral imaging is a procedure of using a spectrometer to collect images of the same target at different wavelengths (Fingas and Brown, 2014, 2018). While hyperspectral imaging continuously captures emitted radiation of the complete spectral bandwidth for each pixel of an image, multi-spectral imaging collects reflectance of a limited number of wavelengths (Angelliaume et al., 2017). Different wavelengths related to visible, infrared, and ultraviolet parts of hyperspectral images can be evaluated to extract their corresponding reflectance. The extracted features can be quantified and aggregated to indices for oil spill detection and characterization, such as fluorescence, rotation-absorption, hydrocarbon, vegetation health, narrowband vegetation, and normalized difference infrared/water indices (Angelliaume et al., 2017; Ozigis et al., 2019).

Various ML techniques, such as linear discriminant analysis, decision tree, random forest, and support vector machine (SVM), were frequently trained based on optical image-derived indices for detection/monitoring of different objects of interest (Balogun et al., 2020; Giri et al., 2011; Ozigis et al., 2019; Topouzelis and Psyllos, 2012; Trujillo-Acatitla et al., 2022). In addition to oil spill detection/monitoring, hyperspectral imaging combined with ML techniques can also be used to characterize oil slicks by estimating their properties based on various indices, such as the spectral information divergence index (Chang, 2000).

3.4.2 Synthetic-Aperture Radar Imagery

Microwave sensors, used in SAR imagery, emit electromagnetic radiation into the surface of interest and receive backscatters as the reflection of the radiation (Y. Li et al., 2014; Marghany, 2020). The presence of oil slicks damps short gravity and capillary waves on the seawater's

surface, resulting in low backscatter response that can be detected by evaluating SAR image data (Bianchi et al., 2020; Ma et al., 2014; Marghany, 2015). However, biogenic slicks (e.g., algae blooms and weed beds) and other natural phenomena (e.g., low-wind areas, currents, eddies, ships' wakes, rain cells) may also create similar patterns recognized as look-alikes (Angelliaume et al., 2017; Bianchi et al., 2020; Chen et al., 2017). Generally, the most influential parameter affecting contrast (backscatter response) of the oil slicks is wind (Ivshina et al., 2015; Li et al., 2022; Mera et al., 2017; Zeng and Wang, 2020).

Multi-polarized SAR imagery is another advancement that enhances the robustness and discrimination power of OSDCM tools (Chen et al., 2017; Marghany, 2015). Polarization is caused by the oscillation of a transverse wave in the perpendicular direction to the main path of waive motion (Avadhanulu, 1992). There are horizontal, vertical, and circular/elliptical polarizations. For example, "Vertical-Horizontal" representing a vertical transmit mode and a horizontal receive mode (Staples and Touzi, 2014). This type of cross polarization has a high capacity to identify all characteristics of a target object. While Horizontal-Horizontal polarization and Vertical-Vertical polarization are most suited for identifications of ships and oil spills, respectively, dual-polarized modes can be used to detect both (Bianchi et al., 2020; Staples and Touzi, 2014; Yekeen et al., 2020). Incidence angle, which is an angle between the emitted ray and the perpendicular direction to the point of incidence, is another parameter that can affect the quality of SAR images (Marghany, 2020). Vertical-Vertical polarization is less dependent on the incidence angle than other modes, and thus it can generate brighter and higher-quality images even in large incidence angles. This advantage makes it suitable for oil spills and look-alikes identifications (Yekeen et al., 2020). Furthermore, considering the type of sensor's antenna for SAR imagery, SAR X-band is more appropriate for oil spill detection due to the homogeneous pattern of oil slicks (more substantial wave damping) within its desirable frequency and bandwidth (Angelliaume et al., 2017; Li et al., 2022).

3.4.3 Computational Techniques in OSDCM

Evaluation of reflectance intensity values in RS image data is not sensitive enough for discrimination of oil slicks from look-alikes; therefore, it is necessary to implement computational techniques (particularly AI/SC) to process RS image data (Xu et al., 2014). These techniques can intelligently detect oil slicks based on extracted features of images that can also be used to define additional indices for discrimination.

3.4.3.1 Technique Evolution in OSDCM

To improve the evaluation of oil spill image data, some traditional techniques such as linear discriminant analysis, multiple linear regression, k-means clustering, and logistic regression techniques were mostly used during 2000 to 2005 (Xu et al., 2014). In the following years, more advanced AI/SC-based classification tools, such as fuzzy clustering, metaheuristic optimization, and neural network (NN)-based tools, were used as enhancements (Guo and Zhang, 2014; Marghany, 2015; Topouzelis and Psyllos, 2012; Xu et al., 2014). Some of the techniques, such as hybrid evolutionary optimization algorithms, were coupled with NNs to enhance the oil spill detection accuracy. For example, the accuracy could be increased by minimizing the information difference between manually annotated and output-segmented images (Jia et al., 2019). Annotation is the identification (labeling) of objects of interest in training dataset of images, which are mostly oil slicks in this field.

Eventually, to enhance the discrimination ability of the abovementioned techniques in OSDCM, DL models with a considerable number of training parameters have been developed in recent years (Ahmed et al., 2023; Bianchi et al., 2020; Zeng and Wang, 2020). These advanced models were also combined with image preprocessing techniques to eliminate unnecessary image background data or normalize extracted features for better classification (Cantorna et al., 2019; Yekeen et al., 2020). Application of DL in OSDCM seems a promising approach to enhance both accuracy and robustness of the AI-based tools.

3.4.3.2 Image Segmentation Process using AI/SC-Based Tools

Three main steps should be considered in the application of intelligent techniques for processing oil spill-related satellite/aerial images, including:

- Initial segmentation to separate potential areas of interest (dark spots) from the background of an image;
- Feature extraction from the separated segments, which is the main step of the detection; and
- Performing classification based on extracted vectors of features/indices (Mera et al., 2017; Xu et al., 2014).

Four categories of features can be defined: geometrical features corresponding to the shape of oil slicks (e.g., area and perimeter of slicks); physical features defining wave backscattering in the physical space (e.g., mean and standard deviation of contrast ratio of slicks); contextual features linking objects with the surrounding area (e.g., the number of adjacent dark spots); and textural features presenting characteristics of the surface of the object (e.g., slicks continuity) (Mera et al., 2017; Topouzelis and Psyllos, 2012; Xu et al., 2014). Among them, shape features (a subset of geometrical category) have been introduced as one of the most practical features due to its extraction simplicity (Guo and Zhang, 2014).

Accordingly, different indices can be defined based on extracted features to facilitate OSDCM. VV (i.e., vertical-vertical) channel intensity, entropy, alpha angle, degree of polarization, ellipticity, pedestal height, co-polarized phase difference, conformity coefficient, correlation coefficient and coherence coefficient are extensively used indices for characterizing SAR image data (Angelliaume et al., 2017; Jagdish and Jerritta, 2020; Marghany, 2015). For example, entropy is one of the widely used indices in evaluating the SAR images of Deepwater Horizon spill (Minchew et al., 2012). It has been frequently considered to identify the oil properties because of the positive relationship between the values of entropy and oil layer thickness and its capacity to discriminate mineral oil slicks and biogenic slicks (Chen et al., 2017; Marghany, 2015). It is suggested to use

quad-polarized data (containing four different modes of polarization's data) to monitor entropy variation (Chen et al., 2017; Minchew et al., 2012). The degree of polarization and conformity coefficient are recognized as two other import indices in oil spill classification. Specifications of clean water and oily water were determined based on these indices, and they were also used to train AI-based tools to identify oil spills (Chen et al., 2017; Y. Li et al., 2014).

Extracted features can also be directly used for OSDCM. Some studies combined several basic features to define a new feature that can technically be considered an index (Mera et al., 2017; Xu et al., 2014). Therefore, feature and index are used interchangeably In some studies, but index is more commonly used to characterize vegetation affected by oil spills (Balogun et al., 2020; Khanna et al., 2013; Ozigis et al., 2019). For example, Ozigis et al. (2019) defined eight vegetation health indices based on spectral parameters, such as RGB (red, green, blue) colour values, near infrared, and shortwave infrared bands These indices were used to monitor oil spill contaminated plants (Ozigis et al., 2019). Indices considering the effects of moisture and shortwave infrared band were the most influential indices in the classification process because plant growth processes highly depend on water contents (Balogun et al., 2020).

It is usually essential to use some statistical feature selection techniques, such as principal component analysis, to determine the influential features in oil spill detection (Ma et al., 2014). Feature selection techniques can be classified into three general groups: filters, wrappers, and embedded. All these techniques evaluate the sensitivity of detection tools to extracted features and reduce the number of effective features to enhance the robustness (Lassalle et al., 2020). While filters select the influential features based on the overall patterns in input data such as inputs' correlation, two other techniques work based on the performance of ML techniques. Thus, wrapper and embedded techniques, selecting influential features through classification processes, are usually considered more effective than filters (the simplest technique) with shorter computation time and less overfitting issues (Mera et al., 2017). Some studies have also investigated synergistic effects of features' combinations in oil spill detection (Topouzelis and

Psyllos, 2012). In general, there is a trade-off between the detection accuracy and robustness of a computation-based tool regarding the number of features considered (Ma et al., 2014; Mera et al., 2017; Xu et al., 2014). Also, a decrease in the number of influential features may reduce computation costs (Guo and Zhang, 2014). It should be mentioned that overfitting is the process of developing a model too closely fitted on limited training data. An overfitted model accurately predicts output values for its limited training dataset at the expense of poor performance on new sets of data (lacking robustness).

Xu et al. (2014) extracted many features from a set of preprocessed SAR images for oil spill detection. A preprocessing step, including a logarithmic transform and a normalization process, was done on the features extracted from the dataset due to a significant variation among values of features. These features were used for training and testing different statistical and ML-based tools, including multilayer Perceptron ANN, bootstrap-aggregated tree-based tools, generalized additive model, and SVM (Xu et al., 2014). The performance of the tools was compared based on the area under the curve of the receiver operating characteristic (ROC) metric, which is widely used in oil spill binary classifications (Cantorna et al., 2019; Xu et al., 2014). ROC depicts variation of true positive rate based on the rate of false positive. Area under the ROC curve of 1 represents 100% correct prediction. It should be noted that having a higher false-negative rate in binary classifiers is much more undesirable than having a high false-positive rate (Xu et al., 2014), because the former is prone to classify oil slicks as look-alikes. However, a lower false-positive rate may lead to a considerable reduction in on-site observation costs required to validate the tool's outcome (Mera et al., 2017).

Meta et al. (2017) conducted an extensive study on oil spill detection using manually annotated images from a training dataset. They extracted numerous features and employed various feature selection techniques to determine the most influential ones. The study compared the performance of several techniques, including correlation-based feature selection, consistency-based filter, information gain, relief-F, and recursive feature elimination. Following that, various combinations

of the identified influential features were used to train SVM-based tools for oil spill detection (binary classification). Notably, wind-related features, which were previously considered important for oil spill detection, were found to have no significant impact on the classification performance. This was most likely due to a manual annotation of the initial dataset that had already considered the effects of low-wind areas (Mera et al., 2017).

3.4.3.3 Deep Learning Application to Enhance OSDCM

Most oil spill detection tools used binary statistical classifiers to detect dark spots in images (Bianchi et al., 2020; Guo and Zhang, 2014; Pelta et al., 2019). For this purpose, classifiers should be trained based on influential features extracted from a manually annotated training dataset of oil spill images. Although some efforts were conducted to perform automatic annotation by defining an adaptive threshold of brightness (Ma et al., 2014), this may lead to a considerable number of misclassifications, especially for thin oil slicks (Yekeen et al., 2020; Zeng and Wang, 2020). Furthermore, the feature extraction process of binary classifiers, mostly using geometrical and physical features, has not been systematically defined. The last step of oil spill classification in these techniques also generates a considerable number of false-positive detections (Fingas and Brown, 2014; Yekeen et al., 2020).

New DL-based oil spill detection tools have been recently developed to address deficiencies of previously mentioned models in feature extraction and detection accuracy (Kang et al., 2022). These tools can automatically extract influential features from a training dataset of images (Zeng and Wang, 2020). However, DL models require a significant number of cases for training due to a large number of weights and parameters used (Chen et al., 2017; Yekeen et al., 2020; Zeng and Wang, 2020). In general, DL models have been classified into four architectural groups, including deep belief network, auto-encoder, convolutional neural network (CNN) and recurrent neural network (Yekeen et al., 2020). CNNs are the most frequently used architecture for image segmentation due to their superior segmentation performance during different ImageNet Large

Scale Visual Recognition Challenges. These annual competitions are designed to assess object detection and image classification techniques (Zeng and Wang, 2020). In comparison to ordinary NNs, CNNs differ in several aspects, such as the number of hidden layers, the application of convolution functions between layers, the determination of weights/parameters during the training process, and the use of activation functions (Aghdam and Heravi, 2017; Cantorna et al., 2019; Kang et al., 2022; Zeng and Wang, 2020).

Convolution is a linear operation implemented in CNNs for feature extraction. A convolution operation is performed on two input arrays (or tensors) within a convolutional layer by integrating their element-wise product after reversing and shifting one of the array (Kang et al., 2022; Oppenheim et al., 1999). The selection of appropriate weights in different layers of DL models is the most critical part of the training process. Some optimization algorithms have been suggested for weight tunning of DL models because traditional gradient descent-based approaches are less effective when dealing with a large number of layers (Cantorna et al., 2019; Chen et al., 2017). The gradient descent-based approaches calculate derivatives of training and validation loss functions concerning selected weights, which will be negligible when there are too many hidden layers. Alternatively, DL models can be used to develop a joint distribution between annotated image data and other sample data through the use of deep belief networks. This can be accomplished by finding training weights for multiple restricted Boltzmann machines, which are neural perceptrons that feature bi-directionally connected neurons arranged in layers. Having the resulted joint distribution, it becomes possible to map features to annotations (i.e., labels) based on the definition of joint probability and segment images. Additionally, stacking auto-encoders are another type of deep neural network that utilize multiple stacks of neurons to perform encoding and decoding procedures on inputs from preceding layers, thereby enhancing the efficacy of weight tuning for the model (Chen et al., 2017).

A pre-trained DL model can be used to facilitate object segmentation in OSDCM. To achieve this, DL models trained on widely-used image datasets, such as ImageNet and Common Objects in

Context (COCO), have been employed as baseline models to determine some weights and parameter values (Ghorbani and Behzadan, 2020; Jiao et al., 2019; Yekeen et al., 2020). This approach may reduce the required computation time and data for training a new DL model. Bianchi et al. (2020) developed a semantic segmentation technique that outperformed other DL models with promising oil spill detection performance. The performance of their DL model was compared with a basic version of U-net, which is a CNN initially developed for biomedical image segmentation, and DeepLabv3+, which is the third version of a semantic segmentation model initially developed to detect objects and animals in images (Bianchi et al., 2020).

3.5 Descriptive Analysis of Spill Management and Response Optimization

MSOM is an intrinsically multiple-stage operation, including the detection of spilt oil, the formation of decision-making teams and risk assessment, selection of suitable response method(s), dispatching response teams for clean-up, and making potential complementary decisions (Huang et al., 2020; Moroni et al., 2016; Ye et al., 2020). After detection of an oil spill, effective and efficient response activities should be carried out to contain or, better yet, eliminate the adverse impacts of spilt oil. ML techniques can be used to help select the most appropriate response method for any given oil spills based on influential attributes (Hu et al., 2020; Mohammadiun et al., 2021a). The response methods can also be optimized by setting different objective functions, such as maximizing the response efficiency (e.g., maximize oil recovery and minimize time and/or cost) and minimizing the environmental impacts of oil spills (e.g., based on equivalent costs) (Li et al., 2012; P. Li et al., 2014b; You and Leyffer, 2011). The application of SC techniques in response optimization (e.g., reducing response time and cost) has been frequently studied.

Figure 3-5 illustrates the general stages involved in deploying response teams following the detection of an oil spill, as used in the Canadian west coasts (WCMRC, 2021a). These include the Notification Stage, which involves informing duty officer(s) of the oil spill incident, the Mobilization Stage for deploying the required response resources and personnel, and the

Response Stage for initiating the response operation. Within this framework, the tools developed based on ML techniques and optimization algorithms can be particularly valuable during the initial incident assessment and decision-making processes, which encompass the Mobilization Stage and a portion of the Response Stage (Figure 3-5). These tools have the potential to assist decision-makers and response practitioners in responding to incidents promptly and efficiently.

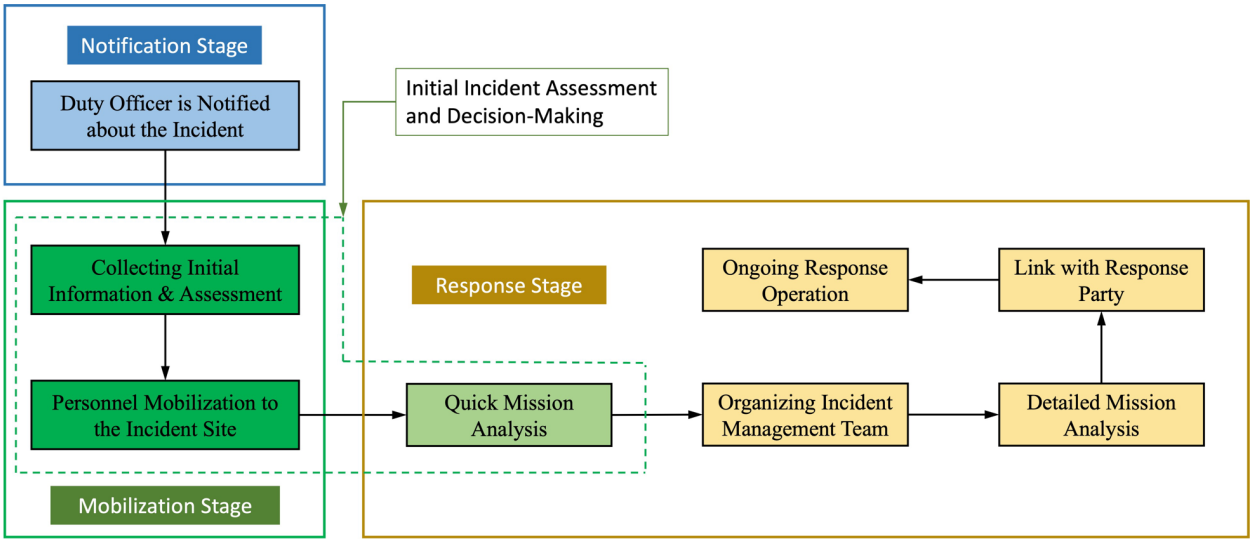


Figure 3-5. Simplified depiction of identification and response team mobilization processes in Western Canada (WCMRC, 2021a)

Appendix B summarizes representative studies in the sub-field of spill management and response optimization based on chronological order. The reviewed studies mainly focused on OSRM selection and optimization using SC-based tools; however, oily waste management, and oil spill probability estimation and trajectory prediction are also assessed. Development of effective OSRM selection tools was mainly done using regression analyses, Bayesian network (BN)/influence diagram, and various types of NNs (Helle et al., 2011; Li et al., 2013; Mohammadiun et al., 2021a). Optimization of a response procedure considering the dynamic nature of oil spills was also studied using mixed integer linear/non-linear programming, CBR, GA, and DL (Huang et al., 2020; Kim and Kim, 2020; P. Li et al., 2014b; You and Leyffer, 2011).

3.5.1 Oil Spill Response Selection and Risk Analysis

Various environmental- and oil characteristics-related attributes, such as wave height and type of spilt oil, have been identified important in OSRM selection (Hu et al., 2020; Polaris, 2009). It is essential to consider significant interactions of influential attributes in OSRM selection and process optimization. Additionally, sources of oil spill risk, such as human errors, and influential uncertainties in MOSM have also been evaluated using mathematical and AI techniques (Cao et al., 2013; Ye et al., 2020)

3.5.1.1 Response Selection

Response to an oil spill consists of a series of dynamic, time-sensitive, multifaceted, and complex processes subject to various constraints and challenges. The response effectiveness is largely reliant on the accuracy of spill information, availability of response resources, and decisions and actions made for the response operation (Das and Goerlandt, 2022). A spill response strategy often starts with the identification of appropriate response methods that are suitable for site-specific conditions. Many models, policies, and regulations have been in place to facilitate oil spill response; however, inadequate decision support may be one of the major challenges that limit the response efficiency (Lee et al., 2015). Selecting a suitable method for a particular oil spill response could be a complex process, which calls for effective tools to facilitate decision-making. The following evaluation of OSRM selection studies indicated that there is still an urgent need for a tool that can help select appropriate response methods by considering their operational suitability.

Best Available Technology (BAT) standards are being advocated to evaluate the effectiveness and technical/economic constraints of response technologies, especially in cold and harsh environments (Hout et al., 2014). However, BAT only applies to mechanical oil spill response equipment. Net environmental benefit analysis (NEBA) has also been developed to help response practitioners evaluate oil spill responses by determining whether additional environmental

damage could be caused by response actions (IPIECA, 2005). NEBA can provide decision-makers with information for determining the suitable response for a specific spill, based on the analysis of environmental trade-offs that may occur as a result of using various response methods (Buber and Koseoglu, 2022; Lee et al., 2015). The assessment is an integral part of contingency plans, offering several advantages, such as extended timeframes for data collection and analysis, consideration of a wide range of environmental factors and stakeholder involvement (Lee et al., 2015). NEBA compares the performance of response methods based on environmental impacts but places less emphasis on the operational feasibility of response methods (Coelho et al., 2013). Overall, various available decision support systems advocated for oil spill response method selection (e.g., NEBA, Consensus Ecological Risk Assessment, Methodology for Environmental Risk Analysis, and Spill Impact Mitigation Assessment) mainly work based on the impacts of oil spills on the environment and natural habitats over time (Afenyo et al., 2020; Wenning et al., 2018). To integrate uncertainty analysis into decision-making using available models, Sajid et al. (2020) performed a Monte Carlo-based ecological risk assessment of oil spills in the Arctic, and they quantified the damage of spills and restoration cost of ecological response measures (Sajid et al., 2020). Aune et al. (2018) performed NEBA in which the impacts on seasonality and spatio-temporal distributions of Arctic species posed by different oil spill response methods were assessed (Aune et al., 2018).

Helle et al. (2015) considered a total number of 55 variables within a BN-based tool to study the costs of MOSM practices, including MCR, shoreline clean-up, as well as waste storage transportation and treatment. The marginal and conditional probabilities of different variables were determined based on literature review, numerical models, and expert judgment. The environmental damage, estimated via a survey, was considered as another type of cost. The benefit of a practice was considered as a reduction in the expected costs of an incident. Annual expected values, which can be obtained by multiplying the cost of an incident by the number of annual incidents, were used to calculate all costs and benefits. Two decision variables:

purchasing a new response vessel and implementing an early warning system to avoid tankers collision, were considered in this study. Although a warning system was less effective in reducing environmental costs, it was considered as the more suitable method due to much lower implementation costs (Helle et al., 2015).

As mentioned, CBR is also used in OSRM selection. Conventional CBR techniques usually work based on simple Euclidean distance metrics to retrieve similar cases to a new introduced case, which is not sufficient for decision-making in complex and uncertain processes like oil spill response (Huang et al., 2020). It is suggested to use CBR in combination with other computational techniques to find the most similar previously happened cases into new oil spill incidents (Mata et al., 2011; Zacharias et al., 2021).

To include uncertainties analysis in MOSM using CBR-based tools, Huang et al. (2020) considered development of oil spill scenario trees within a hybrid DL and CBR tool. Scenario tree is also a technique to classify similar cases into the same group to be used later for decision-making in new incidents. The constructed tree included different branches, each of them contained a set of vectors defining an incident scenario. Branches of the tree were split based on the most significant factor controlling development of the tree. To this end, different influential attributes in an oil spill incident were categorized into three groups: hazard, exposure, and human behaviour describing various scenarios of oil spills. While the human behaviour category (e.g., selected response procedure) was used to develop branches of the oil spill scenario tree, exposure and hazard categories were used to determine oil spill scenario vectors. Each oil spill scenario was defined based on different attributes of exposure and hazard in a set of vectors, such as spilt oil specifications and wave height for hazard and remoteness of spill location and threatened species for exposure. After constructing the scenario trees, a hybrid DL and CBR tool was developed to maximize the effectiveness of OSRM in terms of impact mitigation. The impact was determined based on the amount, toxicity, and flammability of spilt oil and other factors related to response vessels and sea conditions. Several mapping functions (estimating the risk

based on attributes) were used to quantify the impacts of oil spills. The input to the hybrid tool was a vector of a new spill incident scenario, and the output was the effectiveness (reduction in impact values) of an OSRM, calculated by the DL module of the tool (Huang et al., 2020).

Mata et al. (2011) developed a tool based on a multi-agent system and trained it using previous oil spill data for predicting the coverage area and location of new oil slicks. While this research was a coupled OSRM selection and RS, it primarily focused on the former part. SAR image data of an incident were used to generate an initial dataset, and then influential attributes defining different oil slicks were extracted using principal component analysis. After preparing the initial dataset, it was possible to find the most similar oil slick cases to any new incidents using growing cell structures (GCS) coupled with a CBR approach. GCS is a structured arrangement of cells determining adjacent cells' locations based on their closeness to each other (similarity). Here, each cell was an oil spill incident, and GCS was used to arrange different cases stored in CBR based on their similarity. A coupled GCS-CBR can find the most similar previously happened incident to a new incident. After implementing the GCS-CBR, an ANN was trained based on similar cases and selected influential attributes for analyzing a new incident. If the obtained results from the analysis of the new incident were satisfying, the new incident would be added to the initial dataset to enrich samples (Mata et al., 2011).

3.5.1.2 Risk and Uncertainty Analyses

Human errors have been considered as an important factor causing oil spill incidents (Huang et al., 2020; Ye et al., 2020; Zhang et al., 2019). Ye et al. (2020) initially collected information about influential risk factors and experts' judgement. Risk factors covered influential human error factors in oil spill incident occurrence, such as improper operational actions during oil explorations. Experts' judgment was also used to estimate the impacts and importance of identified human errors on oil spill incidents using linguistic variables. Then, a fault tree analysis was performed on the data considering the occurrence rates of influential factors (i.e., low, moderate, high) to

estimate the occurrence of the ultimate top-event: oil spill incident. Afterwards, human factors analysis and a classification system were used to categorize influential factors for Fuzzy-TOPSIS analysis; the categorized factors can be ranked for preventing new oil spill incidents (Ye et al., 2020).

It is worth noting that fuzzy-TOPSIS comprises: representation of factors and decisions determined by experts; fuzzification and weight assignment of the decisions; aggregation of fuzzified decisions based on assigned weights; calculation of positive and negative ideal solutions (the best and the worst solutions); and ranking decisions (e.g., preventive practices) based on their distance to ideal solutions (Ye et al., 2020).

Zhang et al. (2019) identified different risk factors of a collision between an icebreaker and assisted ship based on available reports and experts' judgment. Then, the factors were classified by human factors analysis and a classification system into five levels of different human errors. A statistical analysis of 17 incidents was also conducted to determine the frequency of different factors' occurrence in each incident. Finally, a fault tree-based tool was developed using the mentioned frequency analysis and classified factors. The model drew a connection between different risk factors to develop a tree-like structure, presenting significance and interactions of factors to be considered in management plans (Zhang et al., 2019).

There are different techniques for tackling influential uncertainties in computational modeling, such as the analytical technique, first-order second-moment technique, fuzzy-based techniques, and sampling-based techniques (Morio and Balesdent, 2016; Tung et al., 2006). The last group (also known as Monte Carlo (MC) simulations), which is introduced as one of the most flexible and robust uncertainty assessment techniques, has been widely used in real-world engineering problems (Kroese et al., 2014; Mohammadiun et al., 2018). In oil spill management, MC simulations have been used to generate random values for influential variables (e.g., properties and locations of spilt oil and environmental variables) in the estimation of occurrence probability (Cao et al., 2013), response method selection (Das and Goerlandt, 2022; Li et al., 2013), oil spill

vulnerability estimation in shorelines (P. Li et al., 2014a), and skimmer's deployment optimization (P. Li et al., 2014b).

3.5.2 Oil Spill Response Optimization

Most previous studies defined an optimization problem considering variant characteristics of oil spills over time and used classic optimization approaches (gradient-based approaches) to solve the problem. Mixed integer linear/non-linear programming is the most frequently used technique for problem modeling (Garrett et al., 2017; Li et al., 2019; Zhong and You, 2011). This approach is an appropriate choice to model a simplified OSRM optimization problem by discretizing the formulated oil spill optimization by a selected time step. The applied time step takes the temporal variations of different effective parameters, such as oil spill viscosity, into account. While all effective parameters (e.g., spilt oil characteristics) should be considered unchanged during each step (P. Li et al., 2014b), the time step can be selected based on the desirable accuracy, computational capacity, and temporal resolution of available data, such as the oceanic data (Grubesic et al., 2017; Li et al., 2012).

Optimization of skimmer's deployment in MCR (Grubesic et al., 2017; P. Li et al., 2014b; Ye et al., 2019), resource investment for different infrastructures and response methods (Garrett et al., 2017; Li et al., 2019), and MCR, ISB, and UCD implementation process (You and Leyffer, 2011; Zhong and You, 2011) have been considered as the main objectives of the previous optimization studies. Table 3-2 depicts various decision variables, objective functions, and constraints of optimizations considered in the previous studies.

Some of these studies evaluated the effects of oil weathering processes by applying empirical algebraic equations and solving ODEs to model the main fate and transport effects (Li et al., 2012; P. Li et al., 2014b; You and Leyffer, 2011; Zhong and You, 2011). Alternatively, several studies used oil spill fate and transport simulators to consider the effects of weathering processes and oil

spill trajectories more comprehensively within the optimization (Grubesic et al., 2017; Kim and Kim, 2020).

Grubesic et al. (2017) optimized the time and costs of response procedure considering oil slicks' trajectory through a mixed integer programming (MIP)-based model. Locations of oil slicks, updated based on the desired time interval, were determined by an oil spill simulator (i.e., BLOSOM). Subsequently, the time and cost required to dispatch response vessels into the locations were estimated in this study. The capacity of response vessels (as the main constrain) was determined based on skimming rate, pumping rate, storage capacity, and containment boom length. Decision variables were the number of dispatched vessels and the sequential selection of oil slicks for cleaning up (Grubesic et al., 2017).

Ye et al. (2019) developed an optimization model to maximize the recovered oil as a top priority of MOSM. A hypothetical case study, characterized by predefined volumes and locations of oil slicks as well as a known response capacity, was developed for model demonstration. The model outperformed the shortest distance approach (SDA), which is an approach targeting the nearest oil slick to a response vessel one after another, concerning both response time and the volume of recovered oil. Different empirical equations were considered in their study for the skimmer's net oil recovery rate, oil slick thickness, evaporation, dispersion, and emulsification. They assumed that a response vessel assigned to an oil slick has the sufficient capacity to collect all spilt oil (Ye et al., 2019).

Garrett et al. (2017) used MIP for the scenario-based optimization of response resources in the Arctic. In their study, response resource allocation and deployment, as two important components, were considered to define different groups of constraints in optimization shown in Table 3-2. Resources allocation for budget and amount of purchased materials as well as infrastructure investment addressing storage capacity and response capability options (e.g., refueling and vessel sitting) were considered for optimization (Garrett et al., 2017).

Table 3-2. Decision variables, objective functions, and constraints used in OSRM optimization

Reference	Decision variable	Objective function	Major constraint
(You and Leyffer, 2011)	Scheduling MCR, ISB, UCD, and shoreline protecting boom operations	Minimizing response cost	Maximum length, durability, and deployment rate of booms; minimum slick thickness for ISB; maximum capacity and number of MCR, ISB, and UCD systems; as well as local regulations
(Zhong and You, 2011)	Scheduling MCR, ISB, and UCD operations and required transportation	Minimizing response time and cost	Oil slick properties and the clean-up target; boom capability for coastal protection; and maximum capacity and number of MCR, ISB, and UCD systems
(Li et al., 2012)	Scheduling different skimmers used for oil removal	Maximizing spilt oil removal	Maximum number of skimmers and regression equations for recovery rate of skimmers
(Grubesic et al., 2017)	The number of dispatched MCR response vessels and target slicks	Minimizing response time or cost	Maximum capacity of response equipment; and the clean-up target
(Garrett et al., 2017)	Resource allocation to different stockpiles and infrastructure investments	Minimizing the entire time for resource allocation and response operation	Response resource availability and capacity; minimum required and due time of response operations; transportation capacity; and safety requirements
(Ye et al., 2019)	Scheduling of different skimmers' application on various oil slicks	Maximizing spilt oil removal	Effects of evaporation and dispersion on the volume of spilt oil; as well as slick thickness variation based on the oil volume

Various tools have also been suggested for management and optimization of oily waste generated from response operations in marine environments. Hosseinipooya et al. (2022) estimated the volume of liquid and solid oily wastes generated from MCR operations through system dynamic modeling. They also used a response surface methodology to investigate the optimal values of influential input parameters of the system dynamic model to maximize oil recovery in an oil spill incident in Western Canada. Sea temperature and the arrival time of response vessels were the most significant factors affecting oil recovery and oily waste generation (Hosseinipooya et al., 2022).

Chen et al. (2021) minimized the management cost of waste generated from shoreline clean-up operations by finding the optimal allocation of oily waste to landfill sites and treatment facilities. They considered different levels of risks of constraint violation within an inexact inventory theory-

based optimization tool. Their tool considered two separate stages of waste collection and destocking for oily waste flow from temporary and intermediate storage sites to target waste management facilities (Chen et al., 2021).

Saleem et al. (2022) developed a life cycle assessment-based model to select the best management strategies for solid and liquid oily waste. The selection was made based on life cycle impact categories including climate change, eutrophication, human toxicity, and marine ecotoxicity. Combinations of various solid and liquid oily waste treatment/disposal methods including chemical demulsification, landfilling, incineration, decanting, and reuse were considered as waste management strategies in their study. Among them, chemical demulsification exhibited the highest environmental impact due to its significant toxicity (Saleem et al., 2022).

3.6 Summary

Effective MOSM is critical in minimizing the significant negative impacts of oil spills on the ecosystem and society. MOSM is a multi-stage and time-sensitive practice that should be well planned before the occurrence of any spill incidents. Effective OSRM selection and optimization of response and waste management operations are the most significant parts of MOSM. In order to facilitate decision-making tasks to appropriately manage an oil spill response, it is highly beneficial to implement computation-based tools. These tools can be used at different MOSM components, from OSDCM to OSRM selection and process optimization.

Artificial intelligence, soft computing, and mathematical modeling are computational techniques that may significantly enhance effective MOSM by identifying subtle patterns among the influential attributes in decision-making, tackling relevant uncertainties, considering more realistic variables and constraints in modeling and decision-making, and optimizing the response process. However, insufficient samples in oil spill incident databases are identified as one of the main challenges in developing effective computation-based tools for MOSM. Developing realistic hypothetical scenarios and enriching real-world datasets may improve the performance of various MOSM

tools. Furthermore, response to an oil spill and the management of its generated oily waste, that can be a bottleneck in the clean-up operation, are intricately connected. The integrated management of both response efforts and waste is essential to achieve an effective MOSM. Through an integrated management approach, MCR and OWM can be streamlined, ensuring efficient and environmentally sound practices throughout the entire process.

In this direction, this study used several promising computational techniques, including various ML techniques, mathematical models, and optimization algorithm to facilitate OSRM selection and response process and waste management optimization. To this end, some realistic hypothetical case studies were also developed based on relevant documents and experts' judgment.

Chapter 4 – Oil Spill Response Method Selection using Fuzzy Decision Tree

A version of this chapter has been published in Marine Pollution Bulletin, entitled “Selection of Oil Spill Response Method in Arctic Offshore Waters: A Fuzzy Decision Tree Based Framework.” https://doi.org/10.1016/j.marpolbul.2020.111705.

4.1 Introduction

Many ML tools have been used for data mining, such as artificial neural networks, decision trees, and evolutionary algorithms. Such tools have received increasing interests and efforts in addressing the complexity and dynamics of oil spill response operations and seeking optimal solutions for risk assessment and decision-making (Akinyokun and Inyang, 2013; Cai et al., 2009). Some techniques, such as fuzzy neural network (Akinyokun and Inyang, 2013; Cao and Fan, 2013) and predictive Bayesian model (Davies and Hope, 2015; Helle et al., 2015; Lu et al., 2019), have been developed and applied to assess oil spill risks and characterize their patterns for decision-making in oil spill responses. However, limited studies have focused on using ML techniques for facilitating decision-making in the selection of suitable oil spill response methods, in particular for the Arctic/Sub-Arctic offshore oil spills.

The frequently used response methods including MCR, ISB, and UCD can also be applied in Arctic and Sub-Arctic environments. However, these areas face challenges related to inadequate response equipment for managing large-scale incidents. There is also a lack of waste management facilities in the Arctic that are certified to handle hazardous materials. Consequently, oily waste needs to be stored locally and subsequently transported to southern regions for proper disposal/treatment (WWF-Canada, 2017). For the effective implementation of OSRMs in these regions, substantial improvement in response infrastructure and resources within the Arctic and Sub-Arctic areas are needed. Nonetheless, the development of suitable OSRM selection tools considering the operational suitability of response methods can assist response practitioners in making informed choices and enhancing contingency plans.

The main purpose of this chapter is to develop FDT-based tools to help oil spill response practitioners select suitable response methods for offshore oil spills in Arctic and Sub-Arctic regions. The tools are capable of predicting the field operational feasibility of response methods for oil spill incidents. Expert judgment on the suitability of three commonly used response methods including MCR, UCD, and ISB was obtained for hypothetical oil spill cases, while various influential attributes were considered, such as environmental conditions, logistical demand, and characteristics of spilt oil. Fuzzy logic was integrated into a decision tree algorithm to account for uncertainties originated from expert judgment on the influential attributes as a result of incomplete information and subjective perception of information. Moreover, LR was integrated into FDTs to improve the prediction and enhance decision-making.

4.2 Method

The model developed for the OSRM selection tool, which was briefly introduced in the Proposed Framework and Research Methodology chapter, is elaborated in the following.

4.2.1 Model for OSRM Selection Tool

As shown in Figure 4-1, the FDT-based tool consists of four main steps, including (i) selection of influential attributes, (ii) acquisition of oil spill response data, (iii) developing FDTs and performing tree pruning, and (iv) identifying the most suitable response method for oil spill case. In step (iii), fuzzy Interactive Dichotomizer 3 (ID3) algorithm was used to develop FDTs, and LR models were developed in step (iv) to enhance the prediction performance of FDTs.

In the context of oil spill response especially in the Arctic, the response objective can be defined as cleaning up spilt oil in the marine environment effectively and rapidly as well as minimizing waste generation during the cleanup process. Linked to the response objective is the need to select suitable response methods, which would directly affect the type and volume of waste generated (Polaris, 2009).

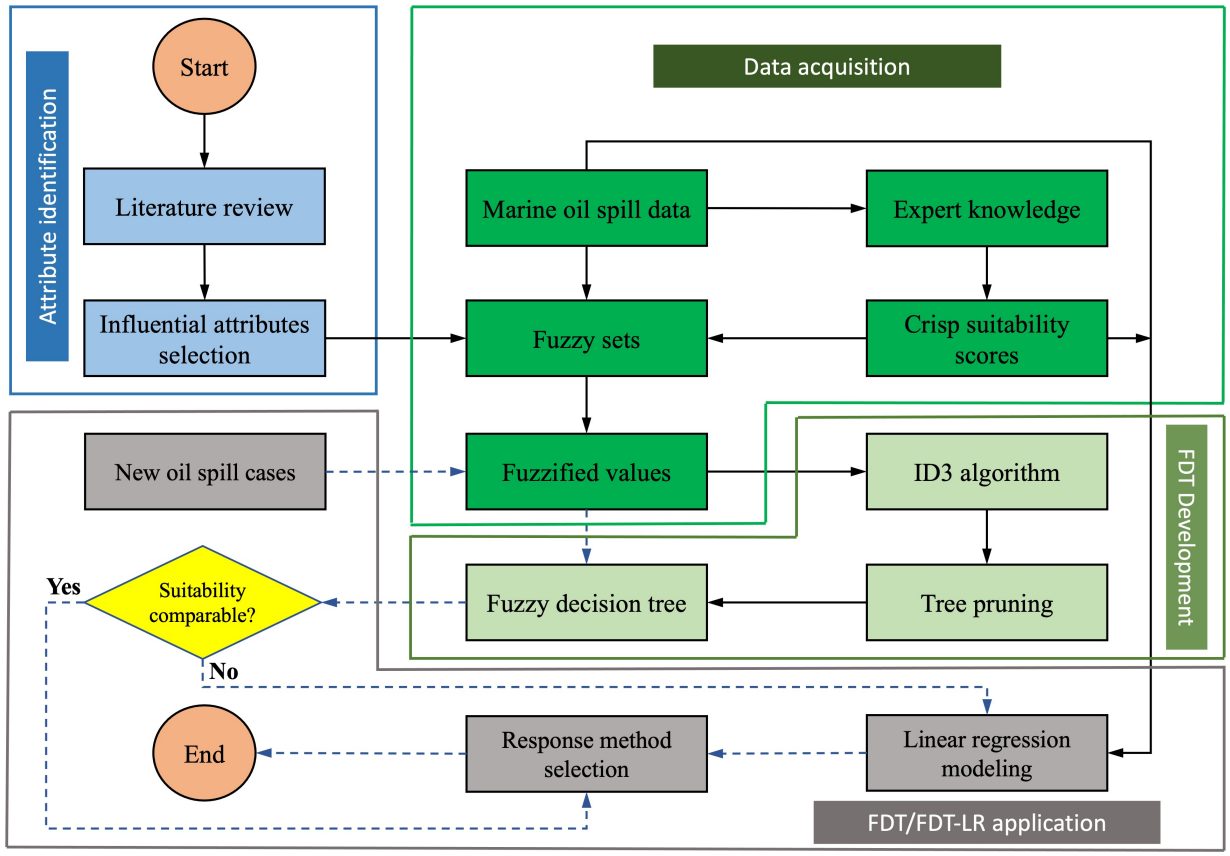


Figure 4-1. The model of FDT-based tool for selection of suitable oil spill response method
Note: arrows with dashed lines indicate the flow of using fuzzy decision tree developed for processing new oil spill data

The decision on response method selection is greatly dependent on various attributes, such as the nature of the spilt oil, oil slick characteristics, and environmental conditions. Attributes that affect the operational effectiveness and efficiency of three commonly used response methods were selected based on the review of recent articles, reports, and guidelines for oil spill response in the Arctic (Lee et al., 2015; NRC, 2014; Polaris, 2009; Potter et al., 2012; Wilkinson et al., 2017). The influential attributes in this study were selected for demonstrating the proposed model, and they can be modified to accommodate different judgments.

The suitable conditions regarding the selected attributes for effective application of the three response methods are summarized in Table 4-1. Generally, MCR is effective when treating relatively thick oil slicks under calm environmental conditions (e.g., low levels of winds, wave heights, and surface currents) (Polaris, 2009). Also, the presence of floating sea ice in Arctic

regions may interfere with MCR equipment such as boom, skimmers, and pumps (Abdelnour et al., 2001; Potter et al., 2012). MCR often generates a relatively large volume of oily wastes, and suitable transportation, storage, and treatment facilities on the scale needed often difficult to find in the remote regions (Wilkinson et al., 2017).

ISB can remove a large amount of oil from the sea surface within a short time. It can also be applied under conditions that involve a considerable amount of broken sea ice; however, it requires the oil slick to be thick enough and not-fragmented to maintain continuous controlled combustion (Polaris, 2009; Potter et al., 2012; Wilkinson et al., 2017). Strong wind and wave may break up oil slicks and disrupt the continuous combustion. Unlike ISB, UCD would be preferred when the oil slick is thin. Strong wind, wave, and surface current also work in favor of UCD as the turbulence can facilitate the dispersion of oil droplets in water column. Moreover, dispersants are reported to be more effective in treating light oil than heavy/weathered oil (Wilkinson et al., 2017). Both ISB and UCD essentially generate little waste, and thus they are preferred for response operations in remote areas such as Arctic regions where waste management may be challenging. The effects of attributes on the cleanup efficiency of the three response methods are illustrated in Figure 4-2.

Table 4-1. Important attributes and conditions favored by different oil spill response methods

	Attribute					
Method	A1: Type of spilt oil	A2: Oil slick thickness (mm)	A3: Oil slick continuity	A4: Wave height (m)	A5: Floating sea ice	A6: Remoteness
MCR	-[a]	> 0.1	-	< 1.0	Yes[b]	High[c]
UCD	Light oil	0.02-1.0	-	0.2-3.0	Yes	Low
ISB	-	> 2	Required	< 1.0	No	Low

[a] "-" indicates "no significant effect"
[b] "Yes" indicates that the suitability of method is adversely affected by floating sea ice
[c] "High" indicates that the suitability of method is highly sensitive to remoteness of response location
Note: MCR-mechanical containment and recovery; UCD-use of chemical dispersants; ISB-in-situ burning

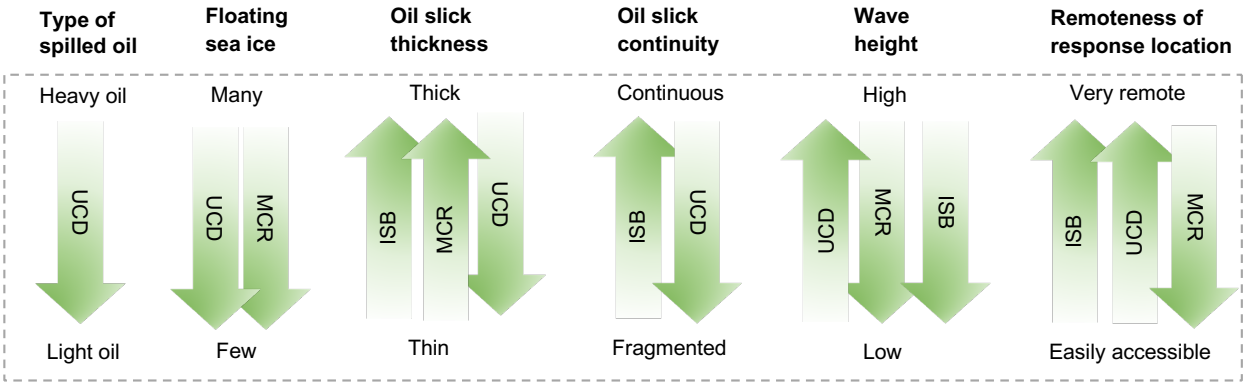


Figure 4-2. Effects of attributes on the operational suitability of oil spill response methods

Note: The direction of arrows indicates increased suitability. MCR-mechanical containment and recovery; UCD-use of chemical dispersants; ISB-in-situ burning.

4.2.2 Fuzzification of Attributes

Evaluating attributes through human judgment is an inherently fuzzy process as different decision makers have different perceptions on the attributes. Decision makers often use linguistic variables to describe attributes due to incomplete information and subjective insights as an amalgamation of personal knowledge, experience, and preference (Hu et al., 2019c). Using linguistic variables unavoidably introduces uncertainties into the evaluation results. Fuzzy logic-based techniques are capable of handling vagueness and imprecision in human insights by using fuzzy sets (Hu et al., 2019a). In fuzzy set theory, fuzzy set *A* of the universe of discourse *X* is defined by membership function $\mu_A(x): X \rightarrow 0,1$. For any element *x* of universe *X*, $\mu_A(x)$ equals the degree to which *x* is an element of set *A*. If *x* is totally in *A*, then $\mu_A(x)$ = 1; if *x* is not in *A*, then $\mu_A(x)$ = 0; if *x* is partially in *A*, then $\mu_A(x)$ takes a value between 0 and 1, reflective of the concept of partial truth (Hu et al., 2018). The most commonly used membership functions are triangular, trapezoidal, piecewise linear and Gaussian functions because they are easily prepared and computationally fast.

In this study, triangular membership functions were used to present linguistic variables for the attributes. As shown in Table 4-2, three linguistic variables, including “low” (L), “moderate” (M), and “high” (H) were used to describe the attributes according to their definitions. Fuzzy numbers

were also assigned for each attribute. As shown in Figure 4-3, the three linguistic variables are presented in fuzzy sets defined by fuzzy numbers (x_1, x_1, x_2), (x_1, x_2, x_3), and (x_2, x_2, x_3), respectively. The degree of membership ($\mu_A^x, \mu_A^x \in [0,1]$) for element x ($x \in [x_1, x_3]$) to fuzzy set A (x_1, x_2, x_3) is calculated:

$$\mu_A^x = \begin{cases} 0; & x \leq x_1 \\ \dfrac{x - x_1}{x_2 - x_1}; & x_1 < x \leq x_2 \\ \dfrac{x_3 - x}{x_3 - x_2}; & x_2 < x < x_3 \\ 0; & x \geq x_3 \end{cases} \qquad \textbf{Equation 4-1}$$

The suitability level of three response methods for a specific response case is also presented in fuzzy numbers on a scale from 0 to 10, including L: (0, 0, 5), M (0, 5, 10), and H (5, 10, 10), where a higher value of fuzzy number indicates higher suitability. The linguistic variables, fuzzy membership functions, and fuzzy numbers used in this study were developed based on our knowledge and the available literature (Hu et al., 2020), and they can be adjusted to suit different decision makers' preferences.

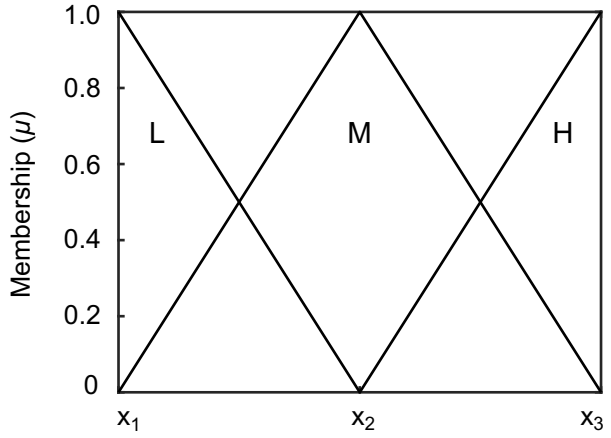


Figure 4-3. Graphic illustration of linguistic variables presented in three fuzzy sets

Table 4-2. Definitions and linguistic variables of attributes

Attribute	Definition (unit)	Linguistic variable		
		L	M	H
A1: Type of spilt oil	API gravity (°)	(45, 45, 30)	(45, 30, 15)	(30, 15, 15)
A2: Oil slick thickness	Thickness of oil slick (mm)	(0.1, 0.1, 1.5)	(0.1, 1.5, 3)	(1.5, 3, 3)
A3: Oil slick continuity	The percentage of area of the largest slick to the total area covered by spilt oil (%)	(30, 30, 60)	(30, 60, 90)	(60, 90, 90)
A4: Wave height	Wave height measurement (m)	(0, 0, 0.75)	(0, 0.75, 1.5)	(0.75, 1.5, 1.5)
A5: Floating sea ice	The percentage of area covered by sea ice to the total area covered by spilt oil (%)	(0, 0, 10)	(0, 10, 20)	(10, 20, 20)
A6: Remoteness of response location	Distance between the response location and the closest waste storage facility (km)	(100, 100, 300)	(100, 300, 500)	(300, 500, 500)

Note: API-American Petroleum Institute; L-low; M-moderate; H-high

4.2.3 Oil Spill Data Acquisition

Due to a lack of detailed information about historical oil spill incidents in the Arctic, hypothetical yet realistic oil spill cases were developed to demonstrate the application of the proposed FDT tool. Seventy hypothetical oil spill cases were randomly generated based on six attributes. Crisp values within the defined value ranges (Table 4-2), were randomly generated for each attribute. This randomization process followed a uniform probability distribution during sample generation. Each hypothetical case is characterized by the combination of the crisp values of six attributes. Three crisp suitability scores, with a value in the range of 0 to 10, were assigned to three response methods for each hypothetical oil spill case, based on expert judgment. Three academic researchers with experience in marine pollution prevention and spill response were consulted to obtain the crisp suitability scores based on the crisp values of different attributes. These consulted experts, with relevant work experience ranging from 2 to 10 years, reached a consensus on the assigned suitability scores after a series of meetings. Furthermore, the finalized dataset underwent an overall review and received endorsement during informal discussions with several other oil spill response experts at the International Oil Spill Science Conference 2022 in Halifax,

Canada. Table 4-3 provides the statistical parameters of the randomly generated attribute values and crisp suitability scores assigned by the experts. The crisp attribute values of hypothetical oil spill cases and the suitability scores of response methods are listed in Appendix C.

For example, the first oil spill incident in this dataset is characterized by the following randomly generated attributes within their associated reasonable range (see Appendix C): A1-Type of spilt oil = 40°; A2-Oil slick thickness = 0.6mm; A3-Oil slick continuity = 86%; A4-Wave height = 1.2m; A5-Floating sea ice = 3%; and A6-Remoteness of response location = 272km. Considering these attribute values to describe the first incident in this dataset, experts eventually assigned suitability scores of 4, 2, and 9 to MCR, ISB, and UCD, respectively. This indicates that UCD is the most appropriate method, while ISB is the least suitable for responding to this specific incident.

The crisp attribute values (C_A) and suitability scores (C_S) were fuzzified using Equation 4-1. The fuzzified attribute values and suitability levels are presented as $\tilde{C}_A$ and $\tilde{C}_S$, respectively. Although the data used in this study are hypothetical, the data assessment would still be similar when real-world data become available.

Table 4-3. Summary statistics of hypothetical oil spill data

Parameters	A1	A2	A3	A4	A5	A6	MCR	ISB	UCD
Min.	16	0.3	30	0.1	0	101	0.5	0.7	0
Q1	21	0.7	43	0.3	2	155	2	2	1.5
Median	29	1.4	59	0.6	4	254	4.5	4	3
Q3	36	2.4	75	1.1	10	406	6.5	5.5	5.5
Max.	45	3.1	88	1.5	20	499	10	10	10
STD	8.8	0.9	18	0.4	5.2	126	2.7	2.6	2.6

Note: MCR-mechanical containment and recovery; UCD-use of chemical dispersants; ISB-in-situ burning; STD-standard deviation; Q1-First quartile; Q3-Third quartile A1-Type of spilled oil; A2-Oil slick thickness; A3-Oil slick continuity; A4-Wave height; A5-Floating sea ice; A6-Remoteness of response location

4.2.4 Building Fuzzy Decision Trees

Decision trees, visualized in tree-like structures, are predictive models derived from the supervised data mining process (Quinlan, 1986). In a typical decision tree shown in Figure 4-4, decisions can be obtained through case classification (K1-K3, or decision classes) based on the

levels of relevant example attributes (E1-E4) or criteria. Each case classification is a split of the tree through an attribute, and each split creates a tree branch. The split is called a node and the first split (e.g., E1 in Figure 4-4) is called the root of the tree. When the splitting is terminated, the last nodes in branches are called terminal nodes or leaves of the tree (e.g., K1-K3 in Figure 4-4). Terminal nodes and the root node are connected by intermediate nodes at different hierarchies. An “if-then” rule is also created for each branch (Mistikoglu et al., 2015). A new case can be classified into the appropriate decision class using the established rules. In the present study, each case is an oil spill incident, defined by values of six influential attributes of Table 4-2.

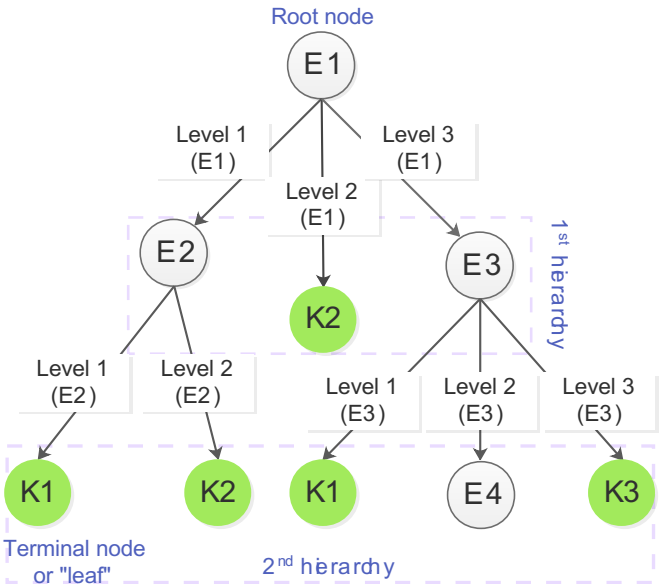


Figure 4-4. Structure of a decision tree

4.2.4.1 Fuzzy Interactive Dichotomizer 3 Algorithm

Many inductive algorithms have been developed for constructing decision trees, such as ID3, Classification and Regression Trees (CART), and Supervised Learning in Quest (SLIQ) (Altay and Cinar, 2016; Kotsiantis, 2013). These algorithms are heuristic techniques capable of generating near-optimal decision trees. ID3 is one of the most commonly used decision tree induction algorithms, which usually measures entropy, as an indicator of information gain/loss, to evaluate the discrimination power of attributes. The attribute giving the greatest information to a decision is selected as the root node of a decision tree. The greatest information gain is reflected

by the largest decrease in entropy (Altay and Cinar, 2016; Quinlan, 1986). The mathematical description of ID3 algorithm is provided in the following.

Let S be a training set of cases in which each case is represented by several attributes $A = \{A_1, \dots, A_i\}$. Each attribute A_i is associated with n levels: $A_i = \{A_i^1, \dots, A_i^n\}$. Based on the consideration of cases' information on all attributes (i.e., inputs), the cases are classified into several classes $K = \{1, \dots, k\}$ as decisions (i.e., outputs). The entropy of set S can be calculated:

$$E(S) = \sum_{k=1}^{K} -f(k)\log_2 f(k) \qquad \textbf{Equation 4-2}$$

where $E(S)$ represents the entropy of S and $f(k)$ is the relative frequency of appearance of class k in S:

$$f(k) = \frac{|k|}{|S|} \qquad \textbf{Equation 4-3}$$

where $|k|$ and $|S|$ represent the cardinality of class k and set S, respectively. In other words, $|k|$ indicates the total number of cases classified into k, and $|S|$ indicates the total number of cases in S. If all cases in S are in the same classification, then $|k| = |S|$ (i.e., $E(S) = 0$), indicating that there is no need for splitting S. Let T_i be the subset of cases generated from splitting S considering attribute A_i. Since A_i is associated with n levels, T_i can also be classified into subsets t based on the n levels of attribute A_i. The total number of t equals n. The information gain (IG) for attribute A_i is calculated:

$$IG(S, A_i) = E(S) - \sum_{t \in T_i} f(t)E(t) \qquad \textbf{Equation 4-4}$$

$$f(t) = \frac{|t|}{|S|} \qquad \textbf{Equation 4-5}$$

$$E(t) = \sum_{k=1}^{K} -f(t_k)\log_2 f(t_k) \qquad \textbf{Equation 4-6}$$

$$f(t_k) = \frac{|t_k|}{|t|}$$ **Equation 4-7**

where $t = \cup_{k=1}^{K} t_k$ and t_k represents the subset of cases both in subset *t* and class *k*. The iteration will identify the attribute with the greatest IG as the next split node with *n* branches. A new set-S_n will be generated for each branch and new IG will be calculated for rest attributes. A decision tree will be constructed once IG of all attributes are evaluated through the iteration process.

An FDT can be developed using the fuzzy ID3 algorithm (Yuan and Shaw, 1995). Let $\tilde{S}$ be a fuzzy training set of cases. The cases are classified into *k* classes as fuzzy subsets $\tilde{k}$. The relative frequency *f*(*k*) for class *k* can be written as:

$$f(k) = \frac{M(\tilde{k})}{\sum_{k=1}^{K}(\tilde{k})}$$ **Equation 4-8**

$$M(\tilde{k}) = \sum_{x \in S} \mu_{\tilde{k}}(x)$$ **Equation 4-9**

where $M(\tilde{k})$ is the cardinality of $\tilde{k}$ and $\mu_{\tilde{k}}(x)$ is the membership value of case *x* to class *k*. Each attribute A_i is described by *n* linguistic terms $(A_i^1, ..., A_i^n)$ to account for cognitive uncertainties. Splitting $\tilde{S}$ at attribute A_i will result in subsets $\tilde{T}_i$. The relative frequency of class *k* in subset $\tilde{t}$ ($\tilde{t} \in \tilde{T}_i$) is calculated:

$$f(\tilde{t}_k) = \frac{M(\tilde{t}_k \cap \tilde{k})}{M(\tilde{t}_k)}$$ **Equation 4-10**

where $\tilde{t} = \cup_{k=1}^{K} \tilde{t}_k$ and $M(\tilde{t}_k \cap \tilde{k}) = \sum_{x \in S} \min(\mu_{\tilde{t}_k}(x), \mu_{\tilde{k}}(x))$. In fuzzy ID3 algorithm, the entropy of $\tilde{S}$ is calculated:

$$E(\tilde{S}) = \sum_{k=1}^{K} -f(k) \log_2 f(k) = \sum_{k=1}^{K} -\frac{M(\tilde{k})}{\sum_{k=1}^{K}(\tilde{k})} \log_2 \left(\frac{M(\tilde{k})}{\sum_{k=1}^{K}(\tilde{k})} \right)$$ **Equation 4-11**

The *IG* after splitting $\tilde{S}$ at attribute A_i is calculated:

$$IG(\tilde{S}, A_i) = E(\tilde{S}) - \sum_{\tilde{t} \in T_i}^{n} f(\tilde{t}) E(\tilde{t})$$ **Equation 4-12**

where

$$E(\tilde{t}) = \sum_{k=1}^{K} -f(\tilde{t}_k) \log_2 f(\tilde{t}_k) = \sum_{k=1}^{K} -\frac{M(\tilde{t}_k \cap \tilde{k})}{\sum_{k=1}^{K} (\tilde{t}_k \cap \tilde{k})} \log_2 \left(\frac{M(\tilde{t}_k \cap \tilde{k})}{\sum_{k=1}^{K} (\tilde{t}_k \cap \tilde{k})} \right)$$ **Equation 4-13**

After each iteration, the truth level $P(\tilde{t}_k, \tilde{k})$ of classifying cases within the branch into each class is calculated:

$$P(\tilde{t}_k, \tilde{k}) = \frac{M(\tilde{t} \cap \tilde{k})}{M(\tilde{t})}$$ **Equation 4-14**

$P(\tilde{t}_k, \tilde{k})$ is the truth level of class *k* on the branch including set $\tilde{t}$. If the truth level is equal or larger than a predetermined threshold value (*β*), then the branch is terminated as a leaf. Otherwise, the next attribute with the greatest *IG* is investigated to split the branch until the termination criterion is met or no attribute is left. All cases in a leaf are classified into the class with the highest truth level. In this study, the value of *β* was set at 0.85 as a suggested threshold in the literature (Altay and Cinar, 2016).

4.2.4.2 Building Decision Trees

Within the model developed for OSRM selection tool, the fuzzy ID3 algorithm was applied to the acquired oil spill data through two approaches. In the first approach, one FDT was developed for evaluating all three oil spill response methods: fuzzified attribute values and normalized crisp suitability scores were used as inputs and outputs, respectively, to train the FDT. The values of crisp suitability scores were normalized by dividing by ten. The normalized suitability scores are in the value range of 0 to 1, which can be processed the same as membership values in Equation 4-9. The FDT developed through the first approach is named as FDT-AP1. For new attribute information, FDT-AP1 will lead the information discrimination to one suitable response method.

In the second approach, a decision tree was developed for each response method. The fuzzified attribute values of a spill case and the corresponding fuzzified suitability level of each response method for the same case were used as inputs and outputs for FDT training, respectively. For a new spill case, each FDT would generate three suitability levels for each response method, respectively. By comparing the suitability levels, the most suitable response method will be identified. However, the suitability of two or more response methods may be classified into the same class for a new oil spill case, and it could be difficult to select the most suitable response methods. To address this difficulty, LR analysis was performed to predict the crisp suitability scores of the response methods identified with the same suitability level. LR was initially used due to limitation in the number of available cases to develop a complex regression model (e.g., polynomial regression) at each terminal node as well as the high computational efficiency of LR analyses. However, potential of some additional regression analyses will also be investigated for suitability score calculation in the next chapter. LR models were developed at each terminal node of FDTs when data were available. The combined use of FDTs and LR analysis is named as FDT-LR.

To account for the uncertainty of selection of training and testing datasets, 10,000 iterations were performed using the fuzzy ID3 algorithm through the two approaches (FDT-AP1 and FDT-LR). In each iteration, fifty oil spill cases were randomly sampled and used as the training data for developing FDTs, while the rest twenty cases were used as the testing data set. The prediction accuracy was calculated as the percentage of the number of correct predictions to the total number of predictions.

4.2.4.3 Decision Tree Pruning

Each FDT developed can be expressed by a set of “if-then” rules. Generally, it is more desirable to reduce the number of rules without losing the prediction power for a robust decision support tool. This can be done by applying a tree pruning algorithm to FDTs. Tree pruning can reduce the number of nodes of a decision tree, ultimately avoiding the overfitting problem. In this study,

Minimum Error Pruning was used to prune the FDTs. In this technique, a decision tree is pruned back to the point where error rate is minimum. The error rate (*Er*) is compared between pruned and unpruned trees.

$$Er = \frac{m - m_c + k - 1}{m + k} \qquad \textbf{Equation 4-15}$$

where *m* is the number of cases that satisfy the conditions of the related non-terminal node, m_c is the maximum number of elements that belong to the same class, and *k* is the number of classes. If pruning a node does not increase the error rate, the pruning should be accepted. The pruning can provide an appropriate balance between the number of rules and the prediction performance of an FDT. The pseudocode developed for performing FDT development, tree pruning, and LR analysis shows as follows:

Algorithm: Fuzzy Decision Tree-Linear Regression (FDT-LR)
Input: fuzzy input ($\tilde{C}_A$), fuzzy output ($\tilde{C}_S$), crisp input (C_A), crisp output (C_S)
Training Applying fuzzy ID3 on $\tilde{C}_A$, $\tilde{C}_S$ to create a FDT Pruning FDT to create $\mathrm{FDT_{PR}}$ Classifying training data by $\mathrm{FDT_{PR}}$ For *i*∈ each branch Extraction of C_{A_i} , C_{S_i} for final attributes in each branch Development of linear regression model for each branch ($\mathrm{LR_{C_{Si}}}$) End Output: $\mathrm{FDT_{PR}}$ and $\mathrm{LR_{C_S}}$
Testing Classifying testing data by $\mathrm{FDT_{PR}}$ For *i*∈ each class Extraction of C_{A_i} for final attributes in each branch Computation of C_{S_i} by $\mathrm{LR_{C_{Si}}}$ in each branch End Output: C_S

4.3 Results and Discussion

For the initial evaluation of the simple FDTs developed by the ID3 algorithm (i.e., constructing a single tree without regression analysis), their performance was assessed on three commonly

used classification datasets with a wide range of cases and attributes. These include Iris Dataset widely used in data mining and two published cancer datasets (Dua and Graff, 2019; Ludwig et al., 2015). The obtained prediction accuracies of normal and pruned FDTs, reported in Table 4-4, were desirable indicating the promising performance of developed decision trees in the classification problems. The table provides the maximum amounts of training and testing accuracy, number of rules, and computation time values obtained from 10,000 iterations in each dataset. It also tabulated the number of cases, attributes, and classes in each dataset. A detailed description of the datasets can be found in the linked references in the table.

Table 4-4. Results of the simple FDT-based tool applied to multiple datasets

Dataset		Max. Train[a]	Max. Test[b]	Average[c]	Time[d]	Case	Attribute	Class
Iris	Normal	97, 74, 33[e]	85, 100, 35	89.1, 88.4, 34	0.012	150	4	3
	Pruned	97, 74, 17	85, 100, 9	89.4, 89.2, 17	0.012			
Colon Cancer	Normal	100, 64, 13	91, 100, 13	88.3, 75.9, 10	0.223	62	2000	2
	Pruned	98, 74, 7	91, 100, 11	86.2, 75.3, 5	0.224			
Ovarian Cancer	Normal	100, 98, 9	99, 100, 9	98.7, 97.8, 13	3.094	253	15154	2
	Pruned	100, 98, 9	99, 100, 7	98.5, 96.9, 8	3.397			

[a] The maximum of training accuracy (%) obtained from 10,000 iterations with its corresponding values of testing accuracy (%) and number of rules
[b] The maximum of testing accuracy (%) obtained from 10,000 iterations with its corresponding values of training accuracy (%) and number of rules
[c] The average amounts of training (%), testing (%), and number of rules in 10,000 iterations
[d] The average computation time (s) on 10,000 iterations
[e] The first and second numbers are training and testing accuracy percentages, respectively, and the third one is the number of rules

Results of applying the proposed structure for FDT-based OSRM selection tool to the hypothetical oil spill dataset are provided and discussed in the following.

4.3.1 Visualization of Decision Trees used in OSRM Selection

The unpruned FDT-AP1 with the highest prediction accuracy on the training data and its pruned version are shown in Figure 4-5. The unpruned FDT-AP1 (Figure 4-5a) contains 19 attribute nodes, whereas the highest information gain can be obtained at A2 regarding response method selection. The FDT determines that ISB should be applied to a spill case where the oil slick thickness is at the “H” level. After A2, the FDT branches at A6, indicating that MCR is suitable only when the remoteness of response location is “L”. A total of 37 terminal nodes as leaves are linked to the attribute nodes at six hierarchies, resulting in a total of 37 “if-then” fuzzy rules. It should be noted that in Figure 4-5a, several branches are connected to the same leaves for a clearer presentation. Two terminal nodes are characterized by “no data” because of the limitation of the given data. UCD is associated with the highest occurrence frequency (15 times) among the terminal nodes, followed by MCR (13 times) and ISB (5 times), suggesting that UCD has relatively less dependence on the selected attributes. However, UCD is greatly affected by A1 and A5. The information discrimination power of attributes follows an order of A2, A6, A1, A3, and A4/A5.

After the pruning, the FDT (Figure 4-5b) exhibits a much simpler structure, in which only five attribute nodes and 11 terminal nodes left. The hierarchy of attributes in terms of information gain is similar to that of the unpruned tree, where the attributes are in the order of A2, A6, A1, and A3. Attributes A4 and A5 are omitted from the pruned tree because the omissions do not increase the error rate of the decision tree. The prediction performances of different FDTs are compared in Section 4.3.2 Evaluation of Decision Trees. In the real-world application of the pruned tree, an “if-then” rule example could be “if attributes A2, A6, A1, and A3 are found at their corresponding moderate levels, then ISB could be a suitable response method.”

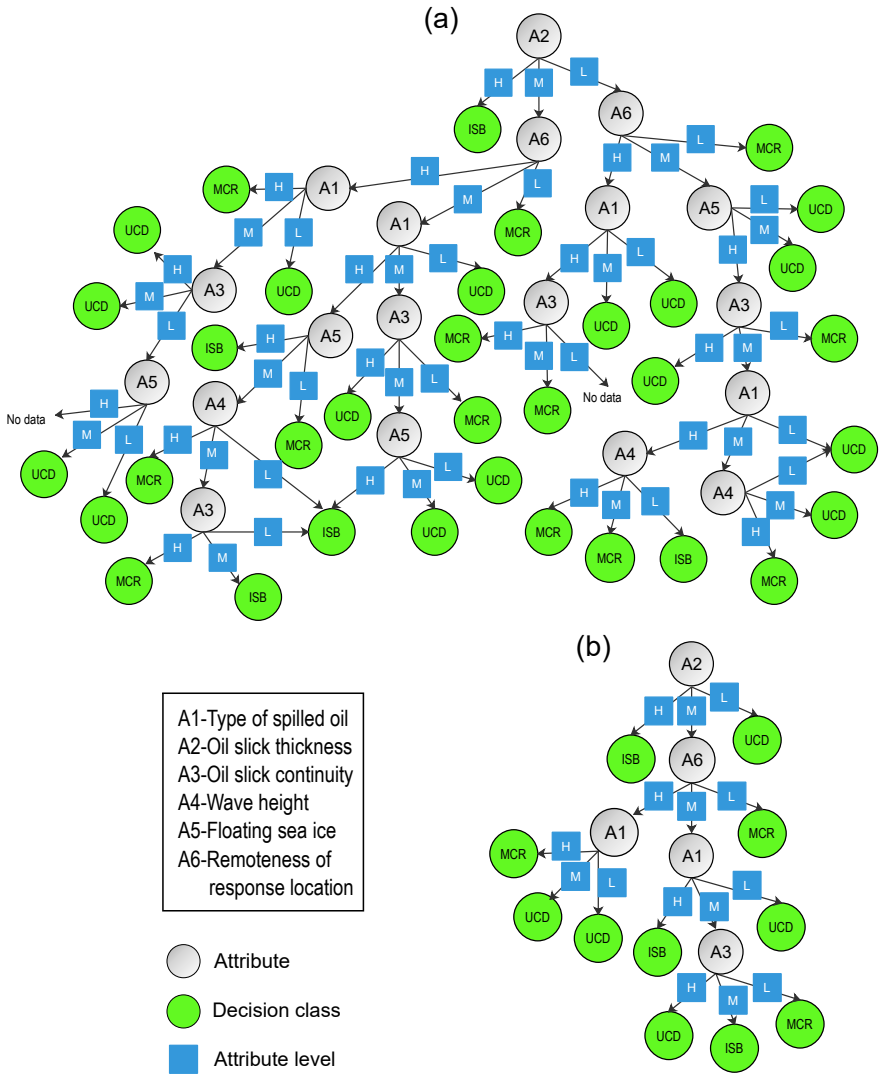


Figure 4-5. Decision tree (FDT-AP1) for the selection of suitable oil spill response methods: (a) unpruned and (b) pruned versions

As shown in Figure 4-6, three pruned FDTs were also developed for evaluating the suitability of three response methods separately through the FDT-LR approach. After pruning, three attributes and seven terminal nodes are found in the decision tree for MCR (Figure 4-6a). Attributes "remoteness of response location", "wave height", and "floating sea ice" play important roles in determining the suitability of MCR. A major problem of applying MCR is that the recovered material mainly comprises oiled ice and water, yet has to be stored safely and disposed of. Suitable storage/disposal facilities often do not exist in the Arctic (Wilkinson et al., 2017). Hence, the distance between waste storage/disposal facilities and the response location plays a critical

role in the application of MCR. It has also been reported that ice coverages as low as 10% could negatively affect towed-boom technologies (DeCola et al., 2006).

As shown in Figure 4-6b, attributes A5, A1, A2, and A4 have a great influence on the suitability of UCD. It can also be seen that the presence of floating sea ice, heavy oil, and thick oil slick are not favored in UCD. The same adverse impacts of these attributes on the effectiveness of chemical dispersants were reported. For example, the presence of sea ice could dampen the mixing energy brought by the waves, resulting in a reduced oil dispersing rate in the water column (Spring et al., 2006). It has also been reported that dispersants become less effective in treating heavy oil with a high viscosity (10,000 cP) (Lessard and DeMarco, 2000).

For ISB, oil slick thickness and continuity, as well as wave height, are critical attributes determining its operational suitability (Figure 4-6c). The minimum ignition thickness of fresh crude oil was reported to be 1 mm, while a thicker slick (2–5 mm) is required for successful ignition of weathered oil on water (Potter and Buist, 2008). Brandvik et al. (2010) reported that ISB can eliminate 50%-90% of spilt oil in open water with 70%–90% ice coverage. The adverse effects of floating sea ice on ISB were also determined insignificant by the FDT. The deductive logic of the developed trees generally is consistent with the effects of attributes on the effectiveness and efficiency of oil spill responses reported in the literature (Polaris, 2009; Wilkinson et al., 2017). An application instance could be “if A6, A5, A4, A1, and A2 are identified at the corresponding low levels, then the suitability levels of MCR, UCD, and ISB will be determined high, moderate, and low, respectively; thus, MCR will be considered as the most suitable response method in this situation.”

In the real-world application of the decision trees developed through the FDT-LR approach, the suitability levels may be determined at the same level for two or even all three response methods. For instance, in a response situation where A6, A1, A2, and A4 are identified at the “M” level, and A5 is at the “L” level, then moderate suitability will be determined for the three response methods using the FDTs shown in Figure 4-6, and the results are not helpful for the final method selection.

LR models are developed at each terminal node of the pruned trees to enhance the decision-making process.

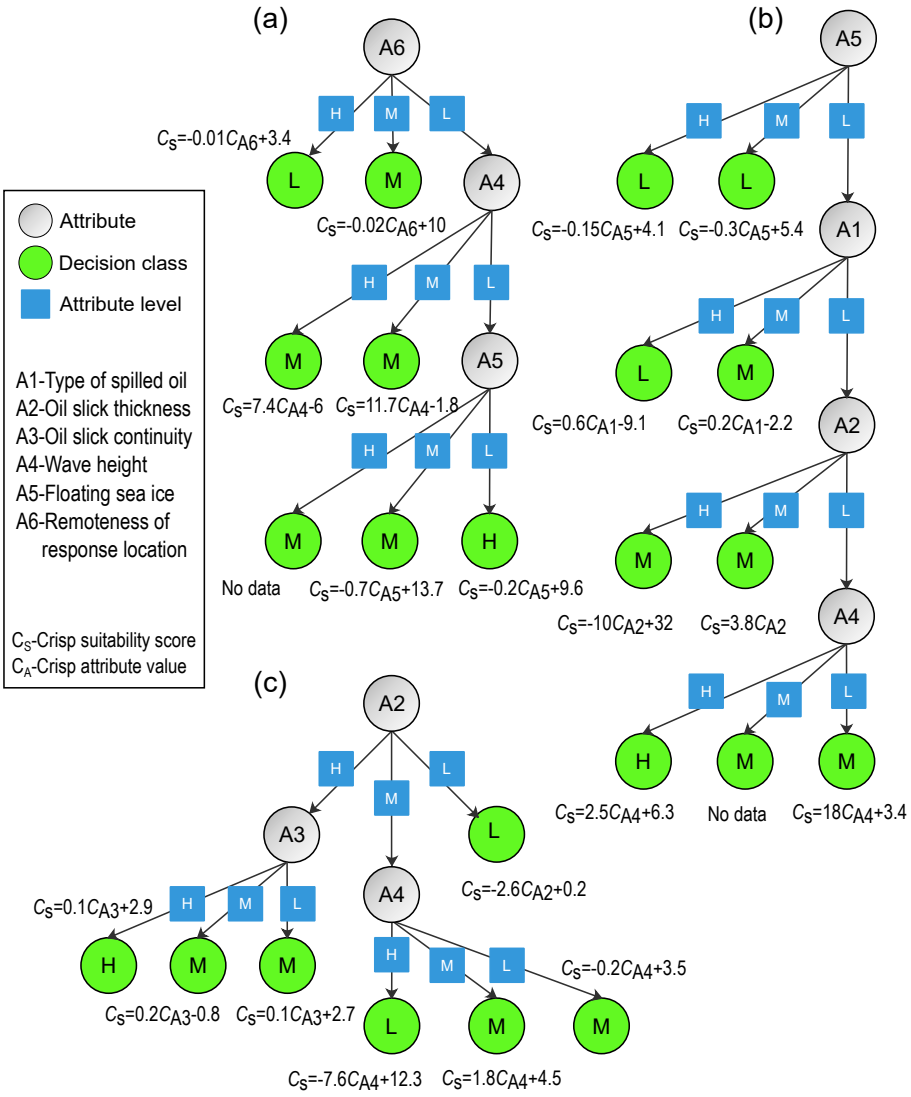


Figure 4-6. Fuzzy decision trees (pruned) for (a) mechanical containment and recovery (MCR), (b) use of chemical dispersant (UCD), and (c) in-situ burning (ISB)

A crisp suitability score will be generated at each terminal node based on the crisp value of the attribute immediately connected to that node. The response method with the highest crisp suitability score will be considered most suitable; however, when the suitability levels are identified "L" for all three response methods, then a deeper investigation on the combined use of response

methods and/or method enhancement is needed. For example, using chemical herding agents and fire-resistant booms to increase the thickness and continuity of oil slick to facilitate ISB can be considered when the suitability of using individual ISB is low (Bullock et al., 2017; Potter et al., 2012; Rojas-Alva et al., 2019). Additionally, when multiple response methods have similar suitability levels "M" or "H" and closely matched suitability scores, a further investigation might be required to explore the potential for combining these response methods.

4.3.2 Evaluation of Decision Trees

The prediction performances of FDTs developed through FDT-AP1 and FDT-LR approaches are summarized in Table 4-5. As can be seen, both the maximum and average prediction accuracies of the FDT-LR approach are higher than those of unpruned and pruned FDT-AP1. On average, roughly 10% and 16% increases in prediction accuracy on training and testing data sets, respectively, can be achieved by using FDT-LR. The average number of rules generated from the FDT-LR approach is smaller than that generated from the unpruned FDT-AP1. However, the computation time of the FDT-LR approach is two times longer than that of unpruned FDT-AP1. Nevertheless, the computation time is short enough to be neglected in real-world applications.

Table 4-5. Prediction performance of decision trees developed through two approaches

Approaches	Decision trees	Max. accuracy-training data (%)[a]	Max. accuracy-testing data (%)[b]	Average performance (%)[c]	computation time (sec)[d]
1	FDT-AP1 (unpruned)	(84, 41)	(91, 29)	(65, 57, 43)	0.014
	FDT-AP1 (pruned)	(86, 19)	(86, 21)	(69, 55, 14)	0.013
2	FDT-LR[e]	(98, 25)	(100, 21)	(79, 75, 26)	0.028

[a] Results are presented in (maximum prediction accuracy percentage from iterations on training data, the number of "if-then" rules obtained from the same iteration)
[b] Results are presented in (maximum prediction accuracy percentage from iterations on testing data, the number of "if-then" rules obtained from the same iteration)
[c] Results are presented in (average prediction accuracy percentage from iterations on training data, average accuracy percentage on testing data, average number of rules)
[d] The computation was performed using a personal computer with a CPU of Intel® Core™ i7-8700K and 32 GB of RAM (DDR4-2400 MHz)
[e] Three fuzzy decision trees in combination with LR modeling

The distributions of prediction accuracies of 10,000 iterations using different FDTs are shown in Figure 4-7. There is no significant difference between the prediction accuracies of the pruned and unpruned FDT-AP1. Using the FDT-LR approach can significantly increase the prediction accuracies. Therefore, the FDT-LR approach has better potential to be used in real-world response method selections. The prediction accuracy could be further increased when more training data becomes available.

As shown in Figure 4-8, in the FDT-LR approach, most of the terminal nodes (> 66% out of 10,000 iterations) can be reached within the first three hierarchies (including the root node). For MCR and UCD, more than 40% of outcomes were obtained at the 2nd hierarchy of the corresponding FDTs, while more than 40% of suitability assessment was determined for ISB at the 1st hierarchy, suggesting that identifying the operational suitability of ISB for a new oil spill case could be easier than the other two methods. In the unpruned FDT-AP1, the final decisions were predominately determined at the 5th hierarchy of the tree due to the complexity of tree structures. It is more difficult to generate comprehensive fuzzy rules required for the prediction of every possible condition by using one FDT. The large number of rules generated in the FDT from the FDT-AP1 approach could be the cause for the comparatively low prediction accuracy and robustness of this approach.

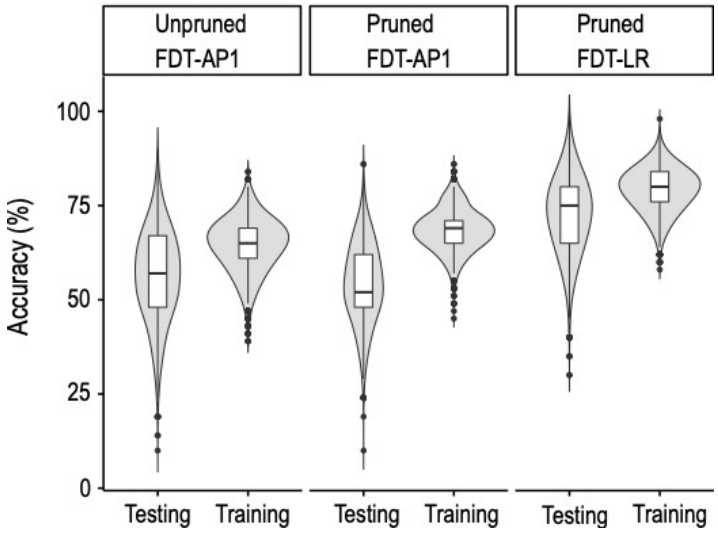


Figure 4-7. Prediction performance of decision trees developed through two approaches

The influences of attributes on the suitability of the three response methods are summarized in Table 4-6. About 95.2% of attributes in the 1st hierarchy of 10,000 iterations generated for MCR were found to be A4. The information discrimination power of attributes decreased from the top to the bottom in an FDT. A higher information discrimination power of an attribute suggests that it has a stronger influence on the decision-making process.

Table 4-6. Distribution of attributes identified in different hierarchies of decision trees (10,000 runs)

Tree	MCR		ISB		UCD	
hierarchy	Attribute	Percent	Attribute	Percent	Attribute	Percent
Root	A6	100	A2	100	A1	54.6
1st	A4	95.2	A3	55.1	A1	45.0
2nd	A5	92.5	A5	39.1	A4	45.7
3rd	A5	55.4	A1	54.7	A5	25.2
4th	A2	62.1	A6	44.7	A3	36.4
5th	A1	34.7	A6	51.2	A6	43.5

Note: A1-Type of spilt oil, A2-Oil slick thickness, A3-Oil slick continuity, A4-Wave height, A5-Floating sea ice, A6-Remoteness of response location

In 10,000 FDTs developed for MCR, all root nodes were identified to be A6, indicating that spill response location had the most influence on the suitability of MCR. A6 was followed by A4 and A5, which represent external environmental conditions such as wave height and presence of floating sea ice. In the FDTs developed for ISB, A2 was the most determining attribute affecting the application of ISB. After this, A3 and A5 also played a critical role in the selection of ISB. In the FDTs developed for UCD, A1 was associated with the predominant influence as roughly half of A1 nodes were found in the root and the 1st hierarchy of FDTs. The influences of critical attributes on the suitability of the three response methods also show the same patterns as those illustrated in the FDTs (Figure 4-6).

The OSRM selection tool can be used to facilitate the screening of suitable response methods for offshore oil spill incidents. The prediction accuracy of the decision support tool can further be improved by big data obtained during the contingency planning of oil spill response.

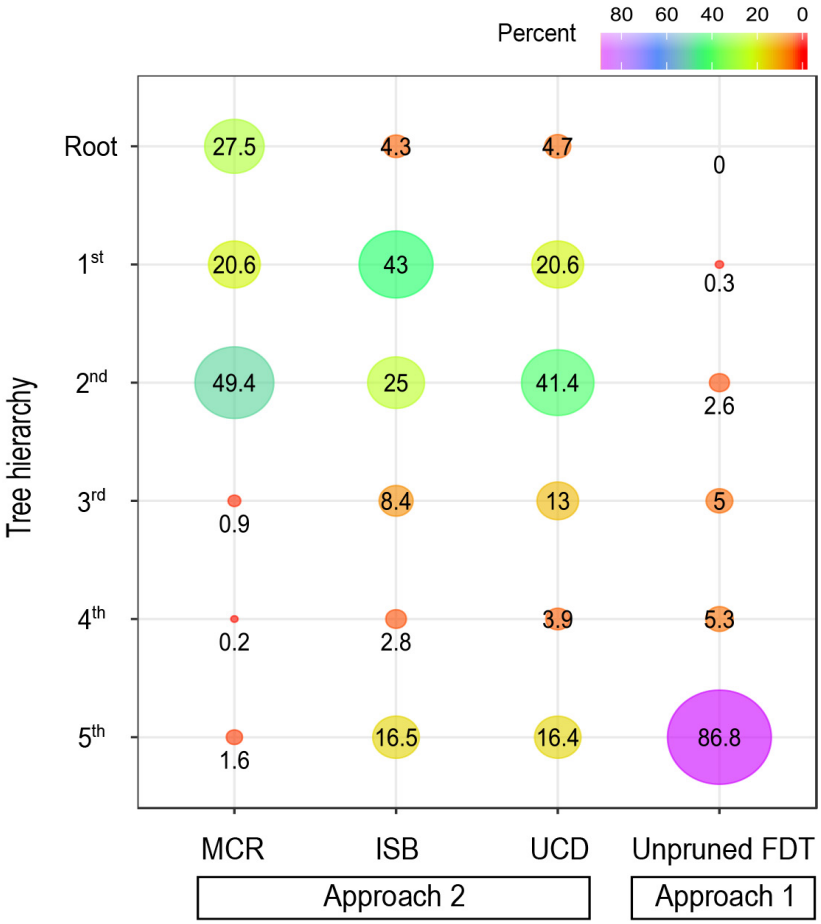


Figure 4-8. Distribution of terminal nodes reached at different hierarchies of decision trees (a total of 10,000 iterations)

4.4 Summary

This chapter proposed an FDT-based model for a decision support tool to help response practitioners select response methods suitable for oil spills, particularly in Arctic/Sub-Arctic regions. The developed tool combines the advantages of ML, fuzzy logic, and LR to assess the suitability of response methods based on the given information of influential attributes. Attributes that influence the operational suitability of offshore oil spill response methods were identified first, and hypothetical oil spill cases were developed based on the randomly generated attribute data.

Expert judgments on the operational suitability of the three most commonly used response methods, including MCR, ISB, and UCD, were obtained. Fuzzy logic was used to reduce the uncertainties resulting from subjective human judgments and linguistic evaluation of attributes. Using the fuzzified suitability judgments and linguistic variables as the training data set, decision trees were developed through ML and tree pruning algorithms.

The results show that the FDTs can be successfully generated using the developed model. The FDT constructed through the FDT-AP1 approach had a much more complex structure (i.e., more attribute and terminal nodes) than the FDT-LR approach; however, the FDT-LR approach was associated with higher prediction accuracies for both training and testing data sets. After training, the FDT-LR approach can generate a maximum of 100% accurate predictions for testing oil spill cases. The developed model represents a screening-level decision support tool for facilitating the selection of Arctic/Sub-Arctic offshore oil spill response methods. The tool can be enhanced by including more regional-specific data regarding influential attributes and expert judgments. This tool can be used in the oil spill contingency planning to collect data for training FDTs, and the trained FDTs can be used to provide rapid decision support for time-sensitive oil spill responses in the region of concern. By following the similar model, specific FDT-based tools can be developed for other offshore oil spill response practices.

Chapter 5 – Optimization of Fuzzy Decision Tree-Based Tools

A version of this chapter has been published in Knowledge-Based Systems, entitled “Optimization of Integrated Fuzzy Decision Tree and Regression Models for Selection of Oil Spill Response Method in the Arctic.” https://doi.org/10.1016/j.knosys.2020.106676.

5.1 Introduction

As discussed before, DT is a well-structured, supervised technique that helps improve the decision-making process in a transparent and intelligible manner, as compared to some “black-box” decision support tools (Altay and Cinar, 2016; Janikow, 1998; Jin et al., 2014; Quinlan, 1986; Yeung et al., 2001). DTs can be simply evaluated by users based on the explicit rules generated within the construction of a tree-like structure. Fuzzy logic can be conveniently integrated with DTs to form fuzzy decision trees reducing the epistemic uncertainties in expert judgments (Crockett et al., 2006; Hu et al., 2019c). While maximizing both training and testing accuracies of FDTs are important, restricting the number of “if-then” rules is another critical aspect to avoid overfitting, as discussed in the previous chapter and literature (Altay and Cinar, 2016; Fan et al., 2011; Wang et al., 2001). In this direction, a model was suggested in the previous chapter to select a suitable OSRM in Arctic/Sub-Arctic waters consisting of an integrated FDT with an LR analysis. In this chapter, OSRM selection tools, developed in the previous Chapter, underwent enhancements aimed at improving their predictive capabilities. This improvement was achieved through the application of a novel model that incorporates a multi-objective optimization algorithm in the development of various FDTs.

Recent studies focused on the construction of efficient DTs by the application of heuristic algorithms and the development of more effective splitting criteria. Karabadji et al. (2017) used the genetic algorithm to enhance the prediction of a DT over testing datasets. Their approach generated more accurate DTs compared to conventional methods of DT construction and some evolutionary-based models (Karabadji et al., 2017). Wang et al. (2017) developed a new entropy information measure and DT construction approach to improve prediction accuracy and escape

from local optima in the construction process (Wang et al., 2017). Yan et al. (2020) integrated DT with some binarization techniques and also defined a new splitting criterion to improve the performance of DTs in multi-level classifications dealing with a greater number of outcome classes (Yan et al., 2020).

These integrated models improved the prediction power of a simple DT to some extent, but these complicated methodologies do not thoroughly satisfy the requirement for a reliable and applicable tool for real-world problems (Hu et al., 2020). In some challenging decision-making situations, such as the selection of oil spill responses, additional techniques may be used in combination with DTs to improve the prediction performance (Hu et al., 2020; Wenning et al., 2018).

To enhance prediction power (including accuracy and robustness), non-dominated sorting differential evolution (NSDE), a multi-objective metaheuristic optimization algorithm, is integrated into FDTs with various regression analyses (FDTRs) in the present chapter. Through the selection of the optimal combination of training and testing datasets to construct an FDTR, the prediction accuracy of an FDTR can be increased and the number of "if-then" rules can be reduced to improve robustness. NSDE is an enhanced version of differential evolution for solving multi-objective optimization problems based on non-dominated sorting criteria. It has been recognized as an excellent meta-heuristic optimizer with respect to convergence and diversity of optimal solutions (Mohammadiun et al., 2018; Yazdi et al., 2018, 2017b). Its unique mutation and cross-over operators improve the diversity and convergence of NSDE in comparison with other widely used multi-objective evolutionary algorithms, such as the non-dominated sorting genetic algorithm and non-dominated sorting harmony search (Yazdi et al., 2018). Moreover, two other regression analyses, including second-order non-linear regression (NLR) and GPR, are used to enhance the prediction capacity of FDTRs. NLR can be a potential method for improving prediction accuracy due to its capacity to consider the interrelation of attributes. Also, GPR is recently developed regression analysis with satisfying performance reported in other fields of engineering (Alizadeh et al., 2019; Rasmussen and Williams, 2006).

As an extension of the previous tool, the objective of this chapter is to develop an optimization-based fuzzy decision tree approach to select oil spill responses in the Arctic by optimizing two objective functions simultaneously. A complementary model comprising a multi-objective optimization algorithm, FDT, and various regression analyses is proposed to select the most appropriate combination of training and testing datasets for constructing FDTRs. The effects of four different information measures calculating the information discrimination power of attributes are examined on the prediction capacity of FDTRs. The hyper volume indicator (HVI) is used to compare the performance of different optimized tools in obtaining optimal solutions. The proposed model is used to predict the most suitable response methods for the previously developed hypothetical oil spill dataset to demonstrate its applicability.

5.2 Method

A coupled optimization and fuzzy decision tree with regression analyses (optimization-based FDTR) model was developed to enhance the selection of suitable oil spill response methods in harsh environments. The developed model is discussed in detail within the following sub-sections.

5.2.1 Model for Optimization-Based FDTR Selection Tool

As shown in Figure 5-1, this model consists of three main steps, including (i) oil spill dataset preparation, (ii) FDTR tools development and optimization process, and (iii) statistical evaluation and best-performing tool selection.

Step (i): Hypothetical oil spill data explained in the previous chapter was also used within the present model. As mentioned in Chapter 4, the dataset including 70 oil spill incidents and suitability scores of three response methods for each spill event were elicited from experts. Likewise, this oil spill dataset will be grouped into a training dataset and a testing dataset in this model.

Step (ii): The oil spill data were used to develop FDTs based on four different measures of information gain (IG), deviance (DEV), GINI impurity (GINI), and misclassification error (ME). In

addition, three regression analyses including LR, NLR, and GPR were combined with the FDTs to enhance their prediction performance. In this step, FDTR tools were also optimized by identifying the optimal combination of training and testing datasets. The optimization can maximize the robustness and accuracy of the FDTR tools.

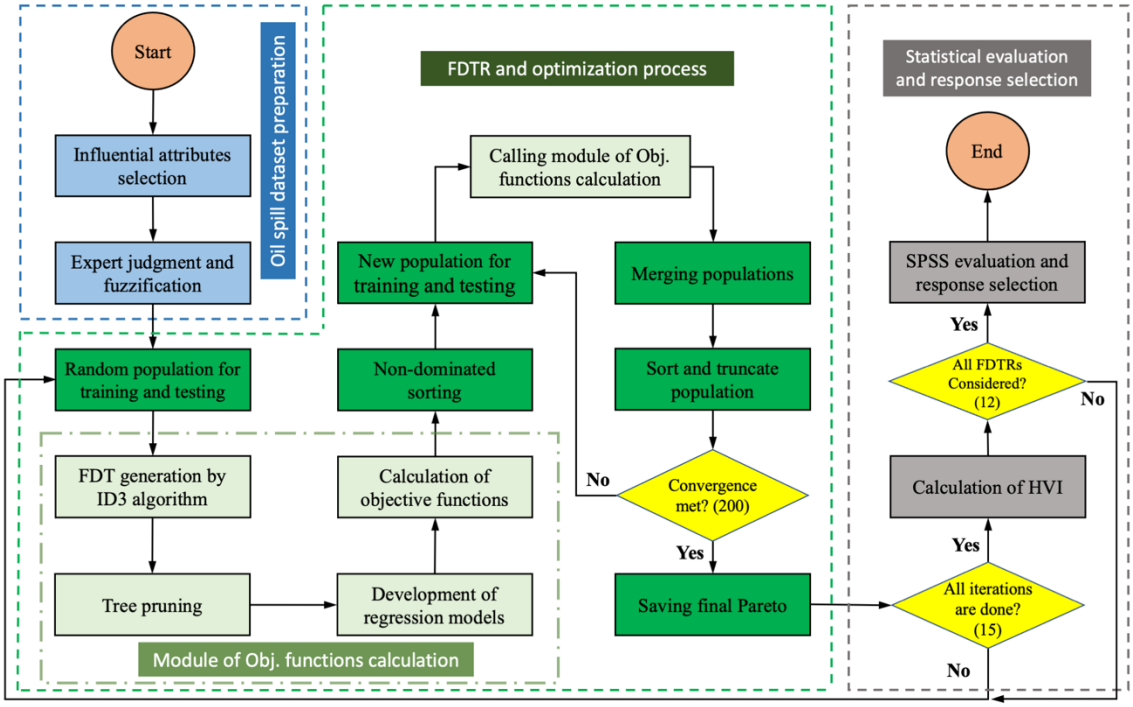


Figure 5-1. The developed optimization-based FDTR model for oil spill response selection

Step (iii): The prediction performance of the optimized FDTR tools was compared using the HVI. The best-performing FDTR tool was selected to predict suitable response methods for any given new oil spill incidents.

5.2.2 Oil Spill Dataset Preparation

The first step in the proposed model is considered to develop training and testing datasets. The preparation of oil spill data to develop FDTs have been explained in Section 4.2 Method of the previous chapter. Previous chapter also explained detailed process of FDT development using ID3 algorithm. As mentioned, a dataset of 70 hypothetical oil spill incidents characterized by six influential attributes was developed. The attributes included the type of spilt oil, floating sea ice

coverage, oil slick thickness, slick continuity, wave height, and remoteness of response location. Each of the 70 hypothetical incidents (cases) was defined by the random generation of six numbers for the influential attributes within their logical ranges. Then, in case of different oil spill incidents, a suitability score was assigned to each response method (ISB, USD, MCR) based on the experts' judgment.

5.2.3 FDTR Development and Optimization

The second step of the proposed model begins with the development of FDTs based on a randomly selected training dataset. Similar to the OSRM selection tool, 50 oil spill incidents out of 70 were selected as the training data, and the remaining 20 events were used as testing data to evaluate the prediction accuracy of the developed FDTs. Pruning and integration of FDTs with regression analysis are the next steps to generate efficient FDTRs. The developed FDTRs with promising prediction accuracy can be used later to predict suitable response methods for new oil spill cases. The developed FDTRs were then optimized for two objective functions related to prediction accuracy and robustness simultaneously. The prediction accuracy of an FDTR was determined using both training and testing datasets. As mentioned, the robustness of an FDTR depends on the number of rules generated throughout an FDTR generation process. As the number of rules decreases, the tool robustness increases (Altay and Cinar, 2016; Fan et al., 2011; Wang et al., 2001); however, the decreased number of rules may result in a decrease in prediction accuracy. Therefore, it is important to obtain an optimal trade-off between robustness and prediction accuracy. For this purpose, two conflicting objective functions (f_{n1} and f_{n2}) were defined to optimize the combination of training and testing data (decision variables) among all 1.619×10^{17} possible combinations (50 combinations of 70 oil spill incidents):

$$f_{n1} = Min\{(1 - Ac_{tr}) + (1 - Ac_{ts}) + |Ac_{tr} - Ac_{ts}|\} \quad \textbf{Equation 5-1}$$

$$f_{n2} = Min\{\sum_{i=1}^{3} Nr_i\} \quad \textbf{Equation 5-2}$$

Subject to:

$$S \cup T = DS = \{DS_j | j = 1 \; to \; 70\}$$ **Equation 5-3**

$$S \cap T = \emptyset$$ **Equation 5-4**

where Ac_{tr} and Ac_{ts} are prediction accuracies of FDTRs on training and testing data, respectively. The prediction accuracy, in the range of 0 to 1, is calculated by dividing the number of correctly classified incidents by all number of incidents in training and testing datasets separately. Nr_i is the number of rules obtained from three FDTRs of MCR (i = 1), ISB (i = 2), and UCD (i = 3); S, T, and DS are training set, testing set, and the total set of oil spill incidents, respectively. *DS* consists of all 70 oil spill incidents and its members are defined as DS_j (j=1 to 70). It should be noted that, only feasible sets of decision variables satisfying the abovementioned constraints (Equation 5-3 and Equation 5-4) was considered as the candidate optimal solutions during the entire optimization process.

The values of prediction accuracies (Ac_{tr} and Ac_{ts}) and the number of rules (*Nr*) depend on the selection of training and testing datasets (decision variables) and can be obtained from FDTRs. By minimizing the first objective function (Equation 5-1), the prediction error of FDTRs on both training and testing datasets will be minimized. Also, the difference in prediction accuracies of training and testing datasets will not be significant due to the consideration of their absolute difference in the first objective function. In addition to the first objective function, minimizing the second objective function (Equation 5-2) will enhance the robustness of FDTRs by reducing the number of rules while still maintaining acceptable prediction accuracies considered in the first objective.

Due to the discrete nature of the search space of this evaluation and the procedure required for calculating the objective functions, metaheuristic algorithms are best suited to solve this type of optimization problems.

5.2.3.1 Optimization Iterative Process

NSDE was used to optimize the objective functions within the second step of the proposed model. The parameter setting of NSDE is selected based on the recommendation of previous studies (Mohammadiun et al., 2020; Yazdi et al., 2018) (Table 5-1).

In this metaheuristic optimization process, the objective function values were minimized within an iterative process by increasing the number of generations. Each generation is a population of candidate optimal solutions in each iteration of the optimization. Populations are specific sets of decision variables, here considered as combinations of training and testing datasets. Thus, each member of the population is a selected combination of training and testing datasets resulting in specific values of objective functions. The objective functions of each member of the population can be calculated based on the prediction accuracies and the number of rules obtained from FDTRs constructed, based on each combination of training and testing datasets. Within this optimization algorithm, two vectors called “donor vector” and “trial vector” were constructed for each generation of the optimization process. Then, the objective functions corresponding to the elements of the trial vector (in the present generation) were compared with the amounts of objective functions in another vector, called vector of objectives, obtained from a previous generation. To this end, a random population was produced in the first generation, and then operators of the optimization process were utilized to modify the population members throughout different generations in order to minimize the values of objective functions. Optimization operators were considered the mathematical process defined by a combination of three randomly generated vectors and two weighting coefficients, including scaling factors and cross-over probability (Table 5-1).

Therefore, the optimization started with generating a random population of decision variables, which were specific combinations of training and testing datasets. The randomly selected datasets were used to develop FDTRs first, and then differential evolution operators were applied

to find a new population with better objective function values, based on Equation 5-5 to Equation 5-10 (Storn and Price, 1997).

Table 5-1. The setting of parameters for NSDE

Parameter	Value
Maximum number of populations	400
Number of new populations	400
Maximum number of generations	200
Cross-over probability	0.7
Scaling factors (lower and upper bands)	0.2 - 0.8

The overall formulation of a minimization problem in case of an objective function $f_n : DS \subseteq R^D \rightarrow R, DS \neq \emptyset$ is as follows to find x^* (optimal decision variables):

$$x^* \in DS \rightarrow f_n(x^*) \leq f_n(x), \forall x \in DS \quad \textbf{Equation 5-5}$$

$$f_n(x^*) \neq -\infty \quad \textbf{Equation 5-6}$$

In the differential evolution algorithm, with the selection of $N \geq 4$ as the number of populations, the objective vector can be written as follows for optimization of a function with *D* real parameters (*G* is the number of generations):

$$x_{i,G} = \left[x_{1,i,G}, x_{2,i,G}, \dots, x_{D,i,G}\right];\ i = 1,2, \dots, N \quad \textbf{Equation 5-7}$$

In order to optimize each parameter, the lower and upper bounds of variables (shown by x^{Lb} and x^{Ub}, respectively) should be determined as:

$$x_j^{Lb} \leq x_{j,i,1} \leq x_j^{Ub}$$
$$j = 1,2,\dots,D \quad \textbf{Equation 5-8}$$

Regarding the abovementioned ranges, the initial values of each parameter can be randomly generated (for the first generation). Then, every *N* vectors of parameters will undergo a set of optimization operators. To this end, three random vectors $x_{r1,G}$, $x_{r2,G}$ and $x_{r3,G}$ will be produced for each variable $x_{i,G}$. Subsequently, scaling factors *FS*, with random values in a predefined range (Table 2), will be utilized to generate the donor vector based on Equation 5-9:

$$v_{i,G+1} = x_{r1,G} + FS(x_{r2,G} - x_{r3,G})$$ **Equation 5-9**

Next, the trial vector will be constructed using components of objective and donor vectors based on cross-over probability *CR*:

$$u_{j,i,G+1} = \begin{cases} v_{j,i,G+1}, & rand_{j,i} \leq CR \\ x_{j,i,G}\ , & rand_{j,i} > CR \end{cases}$$ **Equation 5-10**

where $rand_{j,i}$ is a random value in [0,1]. Regarding this equation, components of the donor vector are substituted in the trial vector to produce a new generation of optimization (*G*+1). Then, vectors of $x_{i,G}$ and $u_{i,G+1}$ will be compared, based on their values in single-objective optimization or using non-dominated sorting criteria in multi-objective optimization, to keep the most efficient solutions with respect to the maximum number of populations. Following this comparison, $x_{i,G+1}$, which is the next generation of candidate solutions, will be determined. This process will be repeated to satisfy the convergence/stopping criterion of the optimization process, which can be a predefined number of generations (Table 5-1).

Non-dominated sorting criteria, which are widely used in multi-objective optimization problems, are a set of rules ranking the suitability of population members considering values of multiple objective functions simultaneously, rather than just a single objective. These criteria work based on the relative values of objective functions and the population density in the search space of feasible solutions (the number of optimal results on the Pareto fronts found by optimization) (Yazdi et al., 2017b, 2017a). Based on these criteria, solutions that cannot be dominated by other potential candidates and that are also located in low-density regions of a Pareto front (dispersed solutions) have a higher probability of being selected as the optimal results (Yazdi et al., 2017b, 2017a).

As shown in Figure 5-1, after the first step, a random population of different training and testing datasets was generated. Based on the random population of different training and testing datasets, the objective functions of each member (Equation 5-1 and Equation 5-2) of this population were calculated by generating an FDT using the ID3 algorithm, FDT pruning, and the

development of one of the regression analyses at the terminal nodes of the FDT. The process of objective function calculation is illustrated in “Module of Obj. functions calculation” of Figure 5-1. In addition to explanations provided in the previous chapter, complementary information to develop FDTRs are provided in the following sections.

After determination of objective functions, population members were ranked based on non-dominated sorting criteria. Then, by application of differential evolution operators to generate the trial vector (Equation 5-7 to Equation 5-10), a new population of potential solutions was generated and the updated objective functions were calculated. In the next step, by merging the new and previous populations, as well as non-dominated sorting and truncating based on the maximum number of populations (Table 5-1), the best potential candidates of optimal solutions were kept. The aforementioned process was iterated for a predefined number of optimization generations, specifically 200, as outlined in Table 5-1. This particular number of generations served as the convergence criterion for the optimization process and was determined through a sensitivity analysis that will be further expounded upon in the sub-section entitled "Pareto fronts of optimized tools." When this convergence criterion has been satisfied, the final Pareto front, showing the relative variation of two objective functions, was identified as the most efficient group of solutions.

5.2.3.2 Fuzzy Decision Tree with various Regression Analyses

In Step 2, integrated FDT-regression models were also developed through several processes, including fuzzification of attributes, generation of FDTs based on different information measures, pruning, and integration with regression analyses. The process explained in the previous chapter were used to develop FDTs. Alternatively, the information discrimination power of attributes was evaluated through four information measures: IG (used in the previous chapter), GIN, ME, and DEV in the present model (Kahraman and Kabak, 2016). To improve the prediction power, the developed FDTs incorporated three types of regression analyses to generate FDTRs, in contrast to the single LR used in the previous chapter. Figure 5-2 provides an illustration of an FDTR

developed in this chapter along with simplified processes of the “Module of Obj. functions calculation” depicted in Figure 5-1.

Similar to the previous chapter, attributes are represented as nodes in the decision tree, which is formed through splitting. The hierarchy of the decision tree places attributes in different levels based on their importance. For instance, in the assumed FDTR shown in Figure 5-2, attribute A4 is the most significant and is considered as the root of the decision tree. Attributes A5 and A1 follow as the next most important attributes in the first hierarchy of the decision tree. In the present model, three FDTRs were developed for the three oil spill response methods separately. A series of “if-then” rules can be established based on each decision tree for predicting the suitability level of the related oil spill response method. Then, based on the predicted suitability levels, the most appropriate response method can be selected for a new oil spill case. In the shown example (Figure 5-2), 11 “if-then” fuzzy rules can be generated from this decision tree diagram.

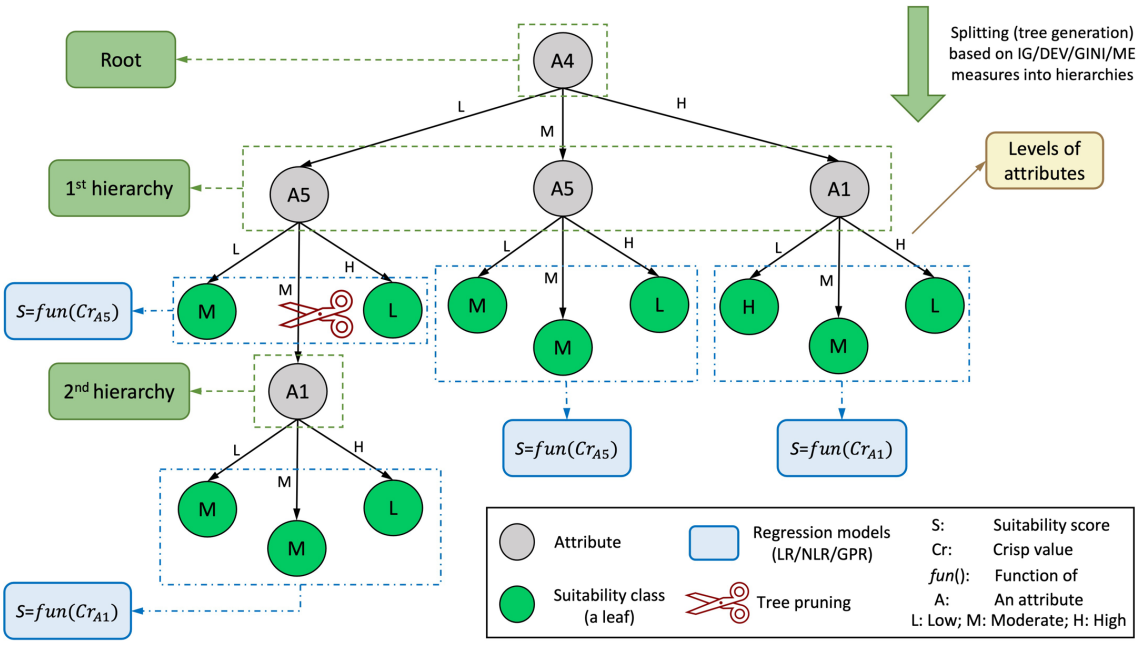


Figure 5-2. A typical FDT with various regression analyses (FDTR) and information measures

Here, three types of regression analysis, including LR, NLR, and GPR, were used to develop regression analyses at terminal attributes. Since the information discrimination power of attributes was evaluated based on one of the four information measures (i.e., DEV, GINI, ME, and IG) and

also three types of regression analysis were developed at each terminal attribute, the total number of 12 optimization-based FDTR tools were developed in each evaluation. Different information measures such as IG, DEV, GINI, and ME can be determined using formulas provided in Table 5-2.

Table 5-2. Different information measures to construct a decision tree

Information Measure	Formulation	Comments
Information Gain (IG)	$IG(S,A_i) = E(S) - \sum_{t\in T_i}^{n} f(t)E(t)$ where $f(t) = \frac{\|t\|}{\|S\|}$, $E(t) = \sum_{k=1}^{K} -f(t_k)\log_2 f(t_k), f(t_k) = \frac{\|t_k\|}{\|t\|}$	here, $t = \bigcup_{k=1}^{K} t_k$ in which t_k shows a case belonging to both subset *t* and output class *k*. The attribute with the highest IG will be selected.
GINI Impurity (GINI)	$GINI(t) = 1 - \sum_{k=1}^{K}[f(t_k)]^2$, $GINI(S,A_i) = \sum_{t\epsilon T_i}\frac{\|t\|}{\|S\|}GINI(t)$	$f(t_k)$ is the relative frequency of class *k* in node *t* of the tree. An attribute with a lower GINI Impurity has more influence in the event classification.
Misclassification Error (ME)	$ME(t) = 1 - max_k[f(t_k)]$ $ME(S,A_i) = \frac{\|Missclassified\ by\ A_i\|}{\|S\|}$	the lower the misclassification error, the better contribution of an attribute in the selection of the final method.
Deviance (DEV)	$DEV(t) = -2\sum_{k=1}^{K}\{n_{tk}[\ln f(t_k)]\}$	in which n_{tk} is the number of cases at node *t* and class *k*. Likewise, because deviance depicts the level of deviation and subsequently heterogeneity in a class, an attribute leading to the smallest amount of deviance will be selected as the most influential attribute.

As mentioned, three types of regression analyses were also compared in the present model. In LR analysis, a linear first-order equation ($y=a+bx_r$) was developed as a regression analysis between the crisp values of an attribute immediately before a leaf and the suitability scores of each response method. In NLR regression analysis, a second-order equation ($y=a+bx_r+cx_r^2$) was considered for the regression analysis. In both formulas, y and x_r are dependent (output) and independent (input) variables of regression analysis, respectively; and a, b, and c are constants of the equations.

GPR is a kernel-based probabilistic regression analysis considering the output values (y) as a function of input variables (x_r) and a noise value (ε). The noise value may be estimated by a normal distribution (N_d) as $\varepsilon \sim N_d(0, \sigma^2)$, in which σ^2 is the error variance. Considering the general form of the regression analysis as $y = h(x_r)^{Tr}\beta + g(x_r)$, formulation of this method is as Equation 5-11 to Equation 5-17 (Alizadeh et al., 2019; Rasmussen and Williams, 2006). It should be noted that h is called basis function, which maps input data into a different dimensional space, and $g(x_r)$ is the Gaussian process (GP) function, which has a Gaussian distribution for any finite variables r. The value of $g(x_r)$ can be defined by the mean value of data ($m_e(x_r)$) and covariance (cov) or kernel function (kr). Also, β is a vector of regression weights determined based on available observed values in a specific branch of an FDT, and Tr represents the transpose operator.

$$g(x_r) \sim GP\big(m_e(x_r), kr(x_r, x_r')\big) \quad \textbf{Equation 5-11}$$

$$m_e(x_r) = EV\big(g(x_r)\big) \quad \textbf{Equation 5-12}$$

$$kr(x_r, x_r') = EV\big[\big(g(x_r) - m_e(x_r)\big) \times \big(g(x_r') - m_e(x_r')\big)\big] \quad \textbf{Equation 5-13}$$

$$kr = \begin{bmatrix} cov(x_{r_1}, x_{r_1}) & cov(x_{r_1}, x_{r_2}) & \cdots & cov(x_{r_1}, x_{r_r}) \\ cov(x_{r_2}, x_{r_1}) & cov(x_{r_2}, x_{r_2}) & \cdots & cov(x_{r_2}, x_{r_r}) \\ \vdots & \vdots & \vdots & \vdots \\ cov(x_{r_r}, x_{r_1}) & cov(x_{r_r}, x_{r_2}) & \cdots & cov(x_{r_r}, x_{r_r}) \end{bmatrix} \quad \textbf{Equation 5-14}$$

EV is the expected value. The probability (P) of response y can be modeled by the following vector-form equation:

$$P(y|g, X) \sim N_d(y|H\beta + g, \sigma^2 I) \quad \textbf{Equation 5-15}$$

where for r observations:

$$X = \begin{bmatrix} x_{r_1}^{Tr} \\ x_{r_2}^{Tr} \\ \dots \\ x_{r_r}^{Tr} \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_r \end{bmatrix}, H = \begin{bmatrix} h(x_{r_1}^{Tr}) \\ h(x_{r_2}^{Tr}) \\ \dots \\ h(x_{r_r}^{Tr}) \end{bmatrix}, g = \begin{bmatrix} g(x_{r_1}) \\ g(x_{r_2}) \\ \dots \\ g(x_{r_r}) \end{bmatrix} \quad \textbf{Equation 5-16}$$

In this formulation, the kernel function estimates the effects of variable x_{r_j} on the response of variable x_{r_i}. Different formulations can be assumed to represent kernel function using several

kernel/hyperparameters, including the length scale parameter (σ_l) and the signal standard deviation (σ_f). In the present study, the most frequently used function, squared exponential kernel, was utilized (Rasmussen and Williams, 2006):

$$kr(x_r, x_r') = \sigma_f^2 \exp\left(-\frac{(x_r - x_r')^2}{2\sigma_l^2}\right), Note: (x_r - x_r')^2 = (x_{r_i} - x_{r_j})^{Tr}(x_{r_i} - x_{r_j})$$

Equation 5-17

Considering the presented formulation, it is possible to define a GPR by determination of σ, σ_l, σ_f, and a vector of β, all determined within the training process.

5.2.4 Statistical Evaluation and Response Method Selection

To improve the optimization process of each tool, given the random nature of evolutionary algorithms, the process was repeated 15 times. This decision was based on a sensitivity analysis demonstrated in the subsection entitled "Pareto fronts of optimized tools." The HVI was considered to evaluate the statistical specifications of different tools. In the third step of the proposed model, 12 different FDTR tools were evaluated, and for each tool, 15 iterations were performed for 400 populations within 200 optimization generations (Table 5-1). Therefore, a total of 180 tools were evaluated (15×12), including 60 iterations for each regression analysis and 45 iterations for each information measure. After this evaluation, it is possible to select the most appropriate decision support tool from obtained Pareto fronts. Subsequently, the best tool can be used for the selection of the most suitable response method in case of new oil spill incidents.

5.2.4.1 Hyper Volume Indicator Calculation

HVI is one of the most reliable indicators to compare the performance of different multi-objective evolutionary algorithms and obtained Pareto fronts concerning both convergence and diversity criteria (Auger et al., 2009; Beume et al., 2009; Yazdi et al., 2018). A Pareto front that yields the highest HVI surpasses other Pareto fronts in the ability to discover global optima while considering variations of multiple objectives simultaneously (Fleischer, 2003). For a two-objective optimization, this indicator measures the area representing dominated solutions by a specific set

of optimal results on a Pareto front. The HVI concept has been illustrated in Figure 5-3 for a general two-objective optimization problem, where $(\overline{f_{n1}}(z_{n+1}), \overline{f_{n2}}(z_{n+1}))$ is a single reference point determined based on the extreme amounts of normalized objective functions ($\overline{f_n}$) in a set of *no* optimal solutions. Also, *z* represents the variables required to calculate the values of any objective functions.

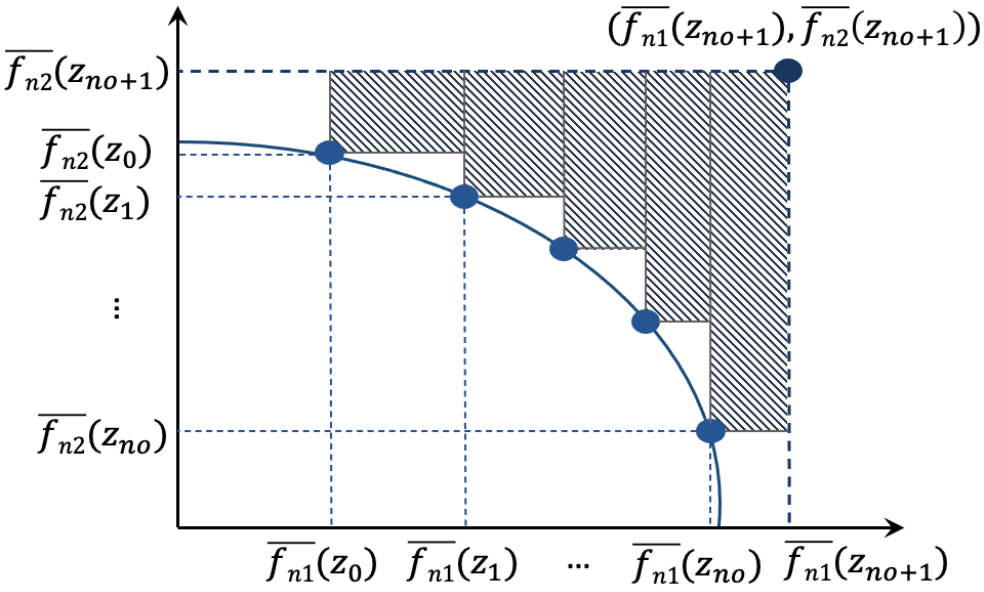


Figure 5-3. The area considered in the calculation of HVI (Yazdi et al., 2018)

To calculate this area, two objective functions were first normalized using their maximum amounts among all 180 considered tools, and then the following formulation was utilized (Yazdi et al., 2018):

$$HVI \cong \sum_{i=0}^{no} \left[\overline{f_{n2}}(z_{no+1}) - \overline{f_{n2}}(z_i)\right] \times \left[\overline{f_{n1}}(z_{i+1}) - \overline{f_{n1}}(z)\right] \quad \textbf{Equation 5-18}$$

5.2.4.2 Statistical Evaluation

The Kruskal-Wallis non-parametric test (Kruskal and Wallis, 1952) was selected to assess HVI variations in different tools due to the non-normal distribution of independent variables of the statistical evaluation (HVI values). The *p*-values obtained from this test were considered as a measure of the significance of the difference. Also, statistical parameters including mean values

as well as the standard deviations, 95% confidence levels, minimum, and maximum of the obtained mean values were reported to evaluate the HVI variation in different tools. All statistical evaluations of obtained results were performed using SPSS™ (version 23).

Kruskal-Wallis test is the non-parametric equivalent of one-way ANOVA test for the detection of stochastic difference between two or more independent samples. In this test, by ignoring the group assignment, data are sorted and ranked from lowest to highest values. Afterwards, the samples are returned to their corresponding groups, and then the sum of data ranks within a group is calculated. The test statistic is calculated by the following formula (Kruskal and Wallis, 1952):

$$H = \frac{12}{N_{ob}(N_{ob}+1)} \sum_{i=1}^{n_g} \left\{ \frac{{R_i}^2}{n_i} - 3(N_{ob}+1) \right\} \qquad \textbf{Equation 5-19}$$

Here, N_{ob}, n_g, n_i, R_i are the total number of observations, number of groups, sample size of a particular group, and sum of ranks for each group, respectively. Considering the chi-square distribution of *H* with degree of freedom of (number of groups - 1), the *p* value is calculated.

5.3 Results and Discussion

The following sections presents the outcomes of the proposed model in terms of improving the prediction power of OSRM selection tools.

5.3.1 Tool Enhancement by Optimization

To compare the previously developed tool using information gain and LR (IG-LR) and tools developed through the present optimization-based approach, the maximum and the average amounts of prediction accuracies on training and testing datasets as well as the number of rules are compared in Table 5-3. To compare, two best-performing examples of the optimized tools, including GPR-GINI and LR-GINI, are shown in Table 5-3.

In the IG-LR, 10,000 random combinations of training and testing datasets have been considered. The present approach uses an optimization process to find more efficient tools, and also

evaluates the effects of different regression analyses and information measures on the tools' prediction power.

It is evident that the optimization process led to a significant reduction in the number of rules, thereby increasing the tools' robustness. In addition, there was an improvement in the average accuracies of both testing and training processes. Although there was a slight decrease in the maximum prediction accuracies for both testing and training data, the corresponding testing accuracy (in case of the maximum of training data, MxTr) and the related training accuracy (in case of the maximum of testing data, MxTs) have seen significant improvements. Furthermore, the optimized tools decreased the average total number of rules by approximately 57%. This reduction was attributed to a 21%, 59%, and 70% decrease in the average number of rules for MCR, ISB, and UCD decision trees, respectively, compared to unoptimized IG-LR tools. The optimization process considers both accuracy and the number of rules criteria, which contributes to the enhancement of the decision support tools. The present model takes into account both training and testing accuracies through the first objective function (Equation 5-1), and simultaneously minimizes the difference between them. Therefore, the obtained optimized tools demonstrate superior performance in terms of both accuracy and robustness.

Table 5-3. Comparison of IG-LR (from previous chapter) and the optimization-based FDTRs

Method	Max. accuracy – training data (MxTr) in %	Max. accuracy – testing data (MxTs) in %	Average performance in %
IG-LR (without optimization)	(98, 70, 25)	(74, 100, 21)	(79, 73, 26)
Optimization-based FDTR (GPR-GINI)	(92, 90, 17)	(88, 95, 15)	(81, 83, 11)
Optimization-based FDTR (LR-GINI)	(90, 85, 15)	(90, 95, 19)	(80, 82, 11)

Note: The first number in the bracket is training accuracy, the second number is the corresponding testing accuracy, and the third one is the total number of rules.

5.3.2 Pareto Fronts of Optimized Tools

Figure 5-4 and Figure 5-5 depict the typical Pareto fronts demonstrating the convergence trend in various generations of optimization and the final Pareto fronts (generation 200) in different iterations, respectively. GPR-GINI, being the best-performing tool, was chosen for presenting these variations, while other developed tools showed similar patterns to GPR-GINI. As shown in Figure 5-4, the final optimal solutions remained relevantly unchanged from generation 150 to 200, hence 200 was chosen as the stopping criterion for the optimization process (i.e., the maximum number of optimization generations). Improvement in values of both objective functions (i.e., movements of Pareto fronts towards the center point) is observable by an increase in the number of generations in Figure 5-4. Moreover, Figure 5-5 reveals that a significant number of identical optimal solutions were obtained in 15 different iterations of GPR-GINI optimization, and a predictable range of variation was observed among the iterations. As mentioned earlier, this insignificant variation of Pareto fronts was expected because of utilizing the random parameters during several steps of the optimization process in evolutionary algorithms, such as the random generation of the first population as well as two random parameters used in Equation 5-9 and Equation 5-10.

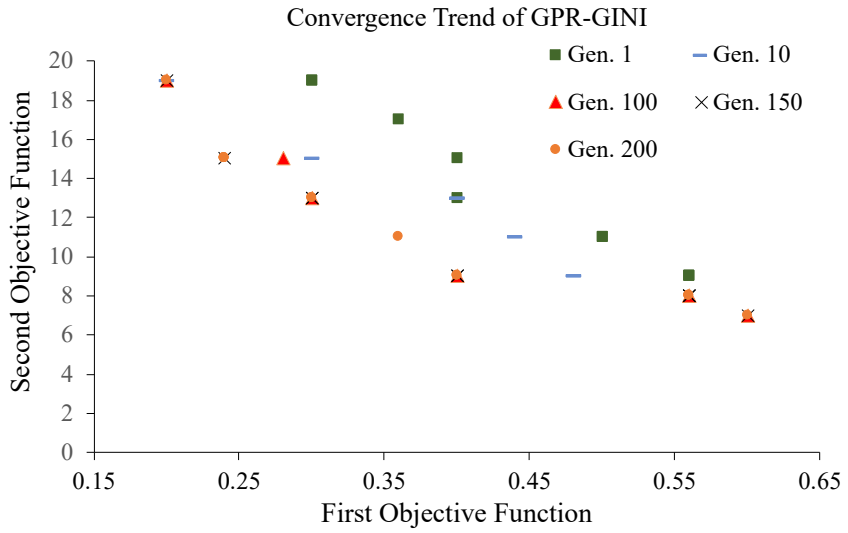


Figure 5-4. Pareto fronts of different generations in one of the iterations of the GPR-GINI tool

The range of variation depicted in Figure 5-5, delineated by schematic lower and upper bands, is adequately accounted for by the optimal solutions obtained through 15 iterations. Since these solutions are mostly identical or within an acceptable range and considering the computational resource available, 15 iterations of coupled models were deemed sufficient.

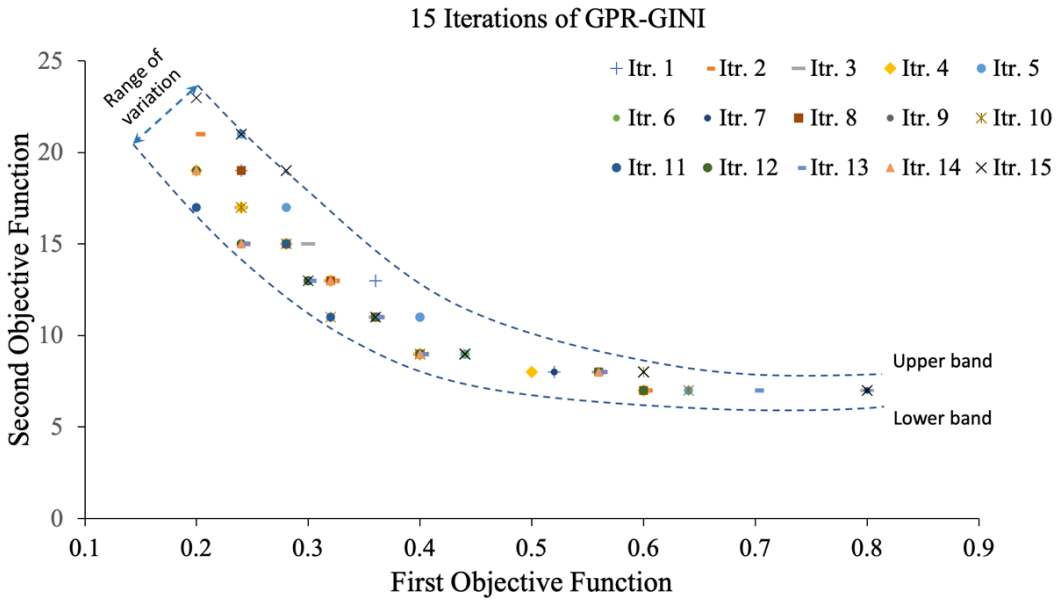


Figure 5-5. Final Pareto fronts (generation No. 200) of different iterations in GPR-GINI

Table 5-4 to Table 5-6 show a statistical summary of HVIs calculated for different tools. The tables contain the HVI assessments calculated from 15 iterations of each tool, resulting in a total of 180 HVIs.

Based on the *P*-values in these tables, there is a significant difference between each pairwise comparison of the developed tools as well as between those of regression analyses and information measure techniques. The *P*-values indicate that the detected differences in mean values of HVI are not random. Moreover, Table 5-4 depicts that GPR-GINI, NLR-GINI, and LR-GINI showed the best performance based on the mean HVI values. This is also verified by the results shown in Table 5-5 and Table 5-6 illustrating that GPR and GINI have the best mean values of HVI among regression analyses and information measures, respectively. Alternatively, the required computation time of each regression analysis reported in Table 5-5 shows that GPR

is the most time-consuming regression analysis among the tools. About 3-4 minutes are required for each iteration of tools working with GPR, which resulted in approximately 11 hours for all 200 generations, compared with about 2 hours needed for tools with LR and NLR.

After optimization, the FDT developed using GINI as the information measure outperformed those developed using other information measures concerning HVI values. Furthermore, every regression analysis in combination with GINI demonstrated better performance than other combinations. Additionally, in terms of regression analyses, GPR was marginally better than NLR, while LR was the least-performing regression analysis due to its simplicity.

Table 5-4. Statistical parameters of normalized HVIs

Method	No. of iterations	Mean	Std. deviation	95% CI for mean		Minimum	Maximum
				Lower bound	Upper bound		
GPR-GINI	15	0.6703	0.0137	0.6628	0.6779	0.6512	0.6893
NLR-GINI	15	0.6695	0.0165	0.6604	0.6786	0.6464	0.7053
LR-GINI	15	0.6597	0.0184	0.6495	0.6699	0.6261	0.6864
GPR-IG	`15	0.6318	0.0145	0.6237	0.6398	0.6119	0.6532
GPR-DEV	15	0.6292	0.0160	0.6204	0.6380	0.6106	0.6512
LR-IG	15	0.6243	0.0166	0.6150	0.6335	0.5990	0.6770
GPR-ME	15	0.6237	0.0123	0.6169	0.6306	0.6129	0.6420
NLR-IG	15	0.6236	0.0118	0.6170	0.6301	0.6093	0.6436
NLR-DEV	15	0.6235	0.0211	0.6118	0.6352	0.5950	0.6681
NLR-ME	15	0.6219	0.0164	0.6128	0.6310	0.6003	0.6612
LR-DEV	15	0.6198	0.0119	0.6132	0.6264	0.6086	0.6473
LR-ME	15	0.6149	0.0097	0.6096	0.6204	0.5990	0.6357
Total	180	0.6343	0.0242	0.6308	0.6379	0.5950	0.7053
P-Value (Kruskal-Wallis)	**0.000**						

Although the most complex and non-linear analysis (i.e., GPR) displayed better performance in prediction, its complexity resulted in considerably longer computation time than other, simpler tools. Nevertheless, it should be noted that once the best tool is identified using the proposed model, it can be readily employed for subsequent evaluations when choosing an OSRM for new

cases. In other words, the application of the developed tools does not entail substantial computational costs since they are comprised of a limited number of "if-then" fuzzy rules. The proposed model can be applied to any other datasets to find the most accurate and robust tool for the prediction.

Table 5-5. Evaluation of normalized HVIs based on the analysis of each regression analysis

Reg. method	No. of iterations	Mean	Std. deviation	95% CI for mean		Min.	Max.	Computation time (s)[a]
				Lower bound	Upper bound			
GPR	60	0.6388	0.0232	0.6328	0.6448	0.6106	0.6893	198.6
NLR	60	0.6346	0.0261	0.6279	0.6414	0.5950	0.7053	35.1
LR	60	0.6297	0.0228	0.6238	0.6356	0.5990	0.6865	39.3
Total	180	0.6343	0.0242	0.6308	0.6379	0.5950	0.7053	-
P-Value	**0.047**							

[a] Reported for in each generation and executed on a personal computer with a CPU of Intel® Core™ i7-8700K and 32 GB of RAM (DDR4-2400 MHz)

Table 5-6. Evaluation of normalized HVIs based on the analysis of each information measure

Information measure	No. of iterations	Mean	Std. deviation	95% CI for mean		Minimum	Maximum
				Lower bound	Upper bound		
GINI	45	0.6665	0.0167	0.6615	0.6715	0.6261	0.7053
IG	45	0.6265	0.0146	0.6221	0.6309	0.5990	0.6770
DEV	45	0.6242	0.0168	0.6191	0.6292	0.5950	0.6681
ME	45	0.6202	0.0134	0.6162	0.6242	0.5990	0.6612
Total	180	0.6343	0.0242	0.6308	0.6379	0.5950	0.7053
P-Value	**0.000**						

Figure 5-6 shows one set of Pareto fronts used to evaluate the normalized HVIs related to the combinations of GPR and FDTs developed using the four information measures. As can be seen,

the HVI determined for the GINI method was the largest, indicating that GINI was the best-performing information measure for the development of FDTs. Subsequently, its Pareto front dominated others, too.

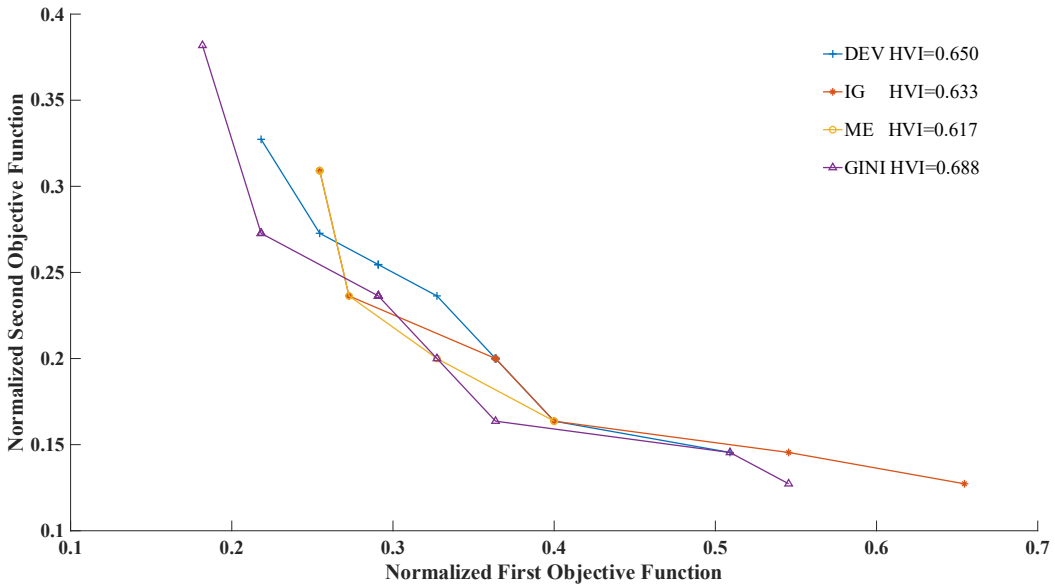


Figure 5-6. Typical normalized HVI variations for one of the GPR iterations

The performance of GPR-GINI tools with the largest HVI value is shown in Figure 5-7. Because two conflicting objective functions were considered in the optimization, the decreased number of rules enhanced the robustness of the tool but compromised the prediction accuracy. As can be seen, the obtained Pareto front of the optimization process can help decision-makers select the most appropriate tool to determine a suitable response method. For example, by reducing the number of rules from 21 to 15, the decline in the prediction accuracy of the tool was negligible. Therefore, a tool with the first and second objective function values of 0.24 and 15, respectively, can be considered an appropriate choice. Also, other optimal solutions are available to be chosen based on decision-makers' preference for robustness and prediction accuracy. It should be noted that each optimal GRP-GINI tool (each circle) has its specific combination of training and testing datasets that cannot be dominated by other potential results (other dataset combinations) with respect to the defined objective functions.

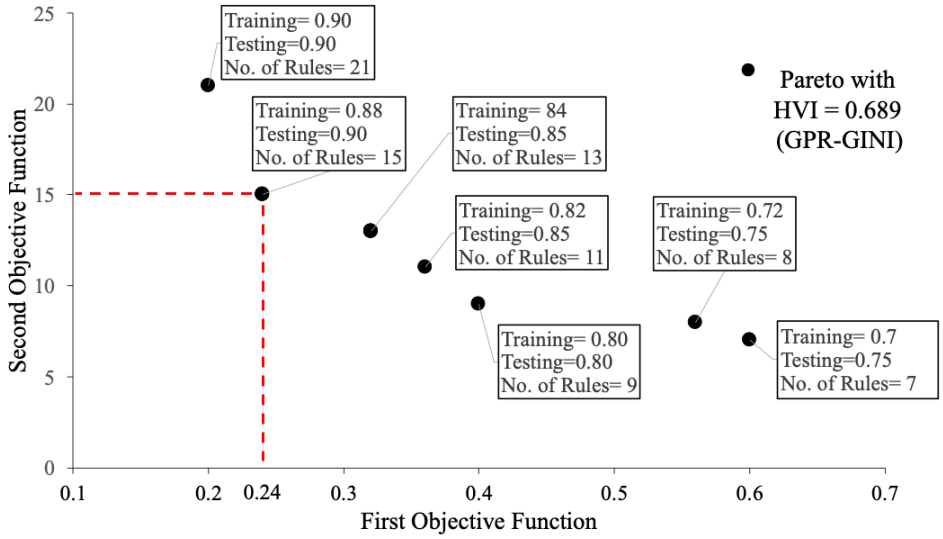
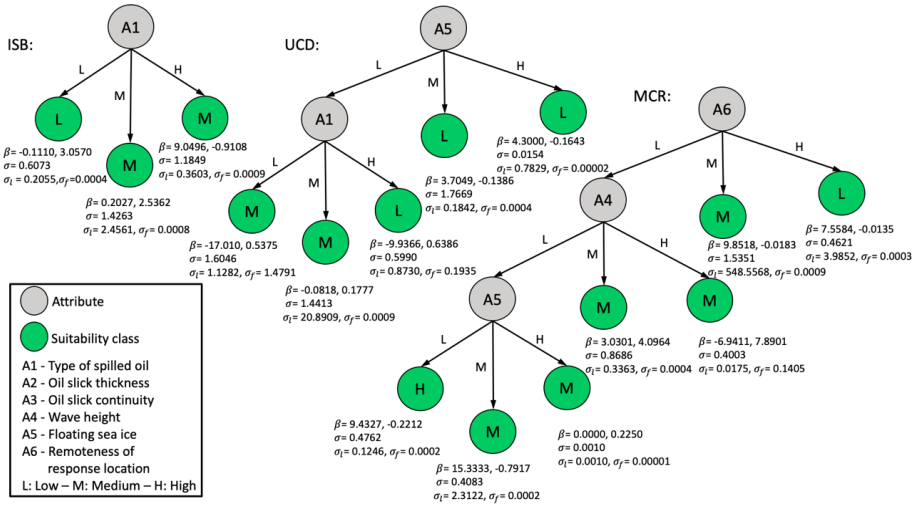


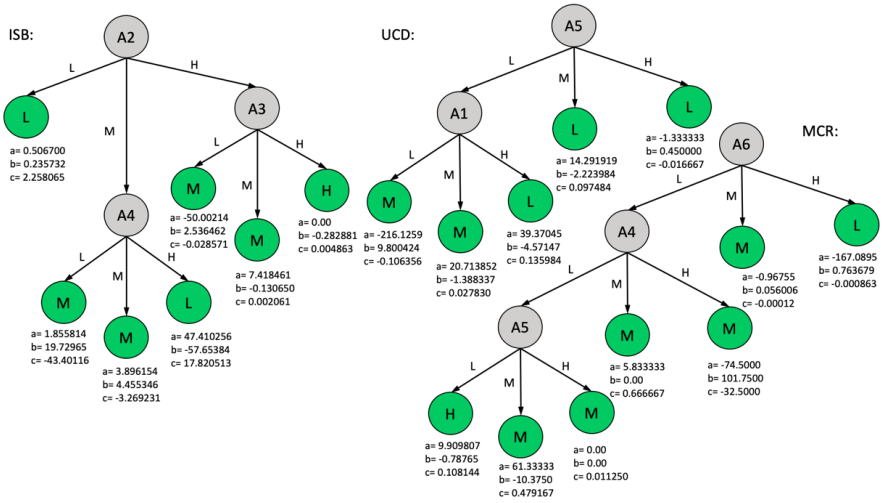
Figure 5-7. Final Pareto front for GPR-GINI tool with the highest HVI

5.3.3 Visualization of Optimized FDTRs

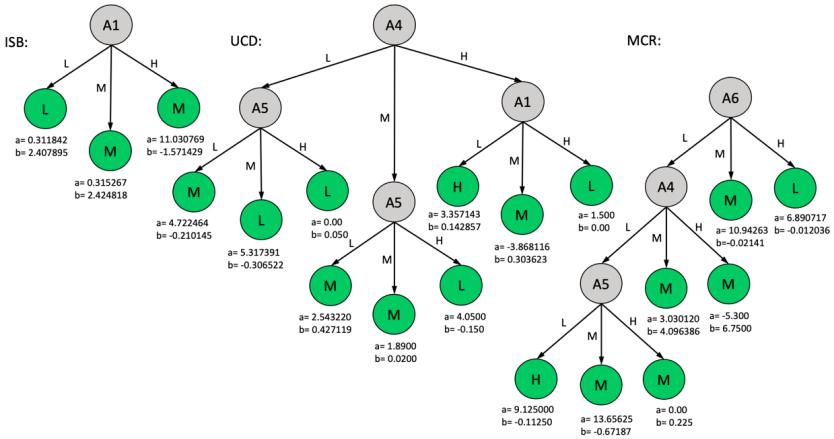
Figure 5-8 shows the visualized FDTRs obtained for each oil response method using the combination of GINI information measure with GPR, NLR, and LR (three tools with the highest mean values of HVI). The effects of influential attributes on the suitability of three response methods (Figure 4-2) can also be seen in the developed FDTs. For example, in the determination of MCR suitability, the remoteness of response location is a crucial attribute that was selected as the root of different FDTs. As the remoteness of response location increases, the suitability of the MCR method decreases from H to L, which is consistent with the overall relations defined in Figure 4-2. Based on the utilized regression analyses, three suitability scores can be obtained for the three oil spill response methods (MCR, ISB, and UCD) from the three FDTs, enabling selection of the most suitable response if two response methods are in a same and higher suitability class than the third response method. Therefore, after selecting the most suitable tool based on HVI values, three related FDTRs of that tool can be used to select the most appropriate response method based on FDTs and the associated suitability scores.



a. GPR-GINI tool



b. NLR-GINI tool



c. LR-GINI tool

Figure 5-8. One set of FDTRs developed based on GINI information measure and different regression analyses

5.4 Summary

In this chapter, NSDE optimization algorithm was applied to FDTR tools for selecting the most suitable oil response method in cold waters. To this end, two objective functions, minimizing prediction error and minimizing number of rules, were simultaneously optimized by consideration of training and testing dataset members as decision variables. While these FDTR tools were developed using different combinations of regression analyses (GPR, NLR, and LR) and information measures (GINI, IG, DEV, and ME), HVI was used as an indicator to compare performance of the optimized FDTR tools based on variations of two objective functions.

The results demonstrated that the optimization not only increased the overall prediction accuracy of FDTR tools, but also significantly improved their robustness compared to unoptimized tools developed in the previous chapter. On average, prediction accuracy on the training and testing datasets increased by 3% and 14%, respectively, after optimization. Moreover, the robustness of FDTRs was notably enhanced by reducing the number of rules generated within an FDT (about 57% reduction on average) compared to unoptimized tools. Among all FDTRs, the GPR-GINI tool has the highest performance according to the HVI values. In addition to enhancements in the predication power, this developed coupled optimization-FDTR model also enables decision-makers to select the most appropriate combination of training and testing datasets based on a trade-off between the prediction accuracy and robustness of the tool. This can potentially result in a more efficient oil spill response, achieved by utilizing a robust OSRM selection tool chosen based on a desirable performance level.

Chapter 6 – Comparison of Different Machine Learning Techniques for OSRM Selection

A version of this chapter has been published in Journal of Hazardous Materials, entitled "Evaluation of Machine Learning Techniques to Select Marine Oil Spill Response Methods under Small-Sized Dataset Conditions." https://doi.org/10.1016/j.jhazmat.2022.129282.

6.1 Introduction

Decision tree-based tools showed a desirable performance, considering an acceptable prediction power and a transparent decision-making process by constructing a tree-like structure for decision support. Like many other ML applications, developing ML-based tools for OSRM selection requires appropriate datasets of oil spill incidents. As mentioned, a significant lack of real-world oil spill datasets has been reported frequently, particularly for the cold and harsh environments like the vast Arctic and Sub-Arctic regions (Hu et al., 2020; Ye et al., 2020, 2019). The small number of training and testing cases (i.e., data points and/or vectors) has also been identified as a common barrier to developing reliable ML-based tools in different fields of study (Alizadeh et al., 2019; Mohammadiun et al., 2021b; Vamathevan et al., 2019; Zodiatis et al., 2016). While it seems impossible for researchers to acquire enriched, high-quality datasets of real-world oil spill incidents in the immediate future, it is worth finding ML techniques that perform well under data constraints. In this chapter, several other promising ML techniques were used to develop effective tools for OSRM selection under dataset constraints while comparing their performance with FDT-based tools (developed and optimized in the previous two Chapters). Three types of ANNs with different training algorithms, including gradient descent (ANN-GD), Levenberg Marquardt (ANN-LM), and Bayesian regularization (ANN-BR) were evaluated for OSRM selection. In addition, GPR and SVR were two other regression-based ML techniques evaluated as an independent tool for OSRM selection in this chapter.

ANN is a commonly used ML technique with a desirable performance level in a wide range of problems (Berardi et al., 2008; Mohammadiun et al., 2018; Tabesh et al., 2009). Other more

recently developed ML techniques, such as GPR and SVR, showed promising results under data constraints in various research fields (Alizadeh et al., 2019; Jiang et al., 2022; Shirzad et al., 2014; Zeng et al., 2020). FDTs' structure is somehow different from the abovementioned regression-based ML techniques (i.e., ANNs, SVR, and GPR), but it is possible to compare their performance based on the prediction power (Altay and Cinar, 2016). All these techniques depicted beneficial performance on smaller-size datasets compared to some other ML techniques, such as deep learning that usually needs large datasets (Ghomi et al., 2020; Mohammadiun et al., 2021a).

The performance of tools is appraised based on prediction accuracy and other important statistical metrics. Recommendations are provided here to enhance the performance of ML-based tools developed on small datasets. As mentioned before, the performance of regression-based ML techniques was also compared with previously developed optimized FDTs (OFDTs). The OFDTs were developed here using the GINI information measure in combination with LR, NLR, and GPR analyses and optimized by NSDE, as described in the previous chapter. The average performance of optimized FDTs with LR, NLR, and GPR reported in this chapter for OFTDs.

6.2 Method

The model developed for performance comparison of various ML techniques, which was briefly introduced in the Proposed Framework and Research Methodology chapter, is elaborated in the following.

6.2.1 Model for Performance Evaluation of MLs for OSRM Selection

The performance of various ML techniques was compared following the model presented in Figure 6-1. The previously developed dataset of oil spill incidents was also used within this model. In another word, similar to the previous models presented in Chapters 4 and 5, influential decision-making attributes were first identified within the hypothetical oil spill dataset stage, and random values were generated concerning their reasonable ranges of variation. Then, experienced decision-makers assigned suitability scores to OSRMs through each oil spill scenario, defined by

values of influential decision-making attributes. The generated hypothetical dataset was then fuzzified based on the values reported in Table 4-2. In the next step, 70% and 30% of cases in the dataset were selected for the training and testing of ML-based tools, respectively (i.e., identical to the split ratio used for FDT-based tools). It should be mentioned that the sensitivity of each technique to different sizes of training and testing datasets is also compared in this chapter.

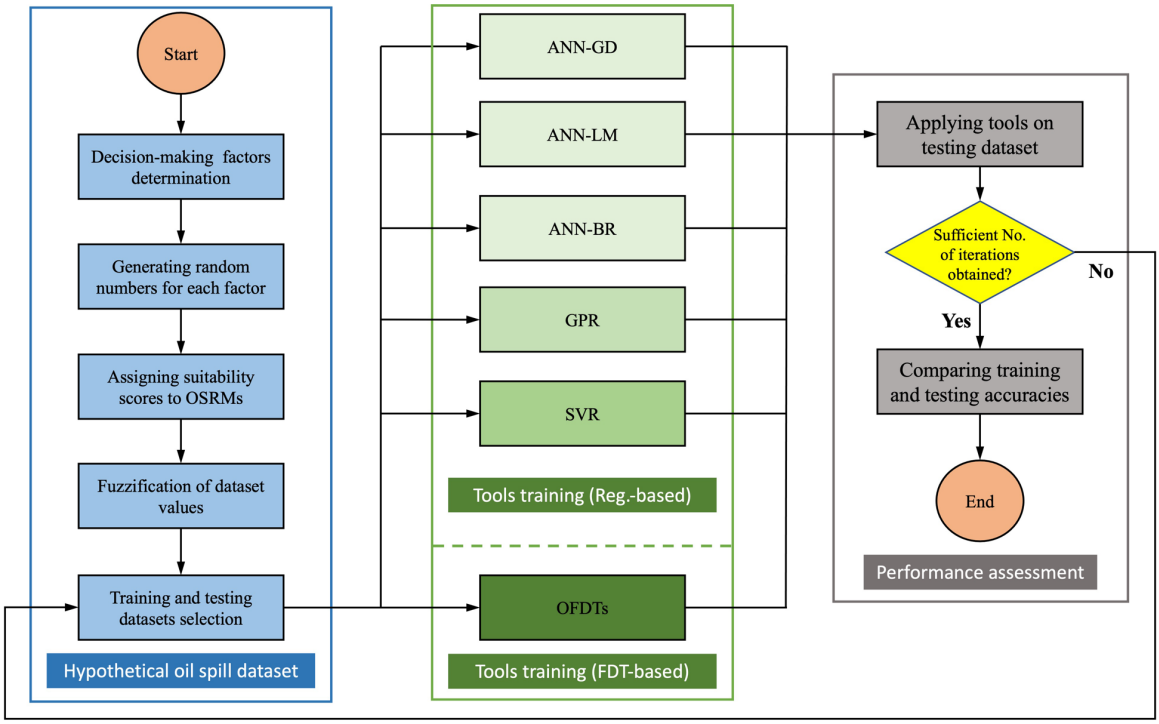


Figure 6-1. The model to evaluate the performance of different ML techniques in the OSRM selection

Different ML-based tools were developed based on the training dataset to select the most appropriate OSRM by predicting and comparing their assigned suitability scores in the tools' training stage. In this stage, the training accuracy of each tool was calculated. Then, the trained tools were applied to testing datasets for measuring testing accuracy. Frequencies of correct selection of OSRMs on training and testing datasets were reported as the training and testing prediction accuracies, respectively. The training and testing process by different dataset samples repeated for a sufficient number of iterations, large enough to reach almost fixed average values of prediction accuracies. ANNs, GPR, and SVR-based tools were executed 10,000 times to

consider the randomness effects of various processes in their development (e.g., random selection of training and testing datasets).

As seen, the previously developed optimized FDTs in Chapter 5 was also incorporated within the model. The FDT-based tools were developed based on the ID3 algorithm. They were also an integration of three FDTs with regression analyses at terminal nodes, assigning suitability scores to each OSRM. Furthermore, a metaheuristic optimization algorithm was also connected to integrated FDTs and regression analyses to enhance tools' prediction accuracy and robustness by selecting the optimal training and testing datasets (i.e., decision variables) to develop OFDTs. Within the developed model in this chapter, the prediction accuracies of regression-based ML tools (i.e., ANNs, SVR, and GPR) were also compared with FDT-based one (i.e., OFDTs).

The performance of the tools was compared based on prediction accuracies and well-known metrics, including Pearson correlation coefficient ($-1 \leq r \leq +1$) and mean squared error ($0 \leq e \leq 1$) for regression-based ML techniques. Pearson correlation coefficient and mean squared error between target values (T_v) and predicted values (P_v) on all training cases (n) can be calculated by Equation 6-1 and Equation 6-2, respectively (Benesty et al., 2008).

$$r = \frac{\sum_{i=1}^{n}(T_v - \overline{T_v}) \times (P_v - \overline{P_v})}{\sqrt{\sum_{i=1}^{n}(T_v - \overline{T_v})^2} \times \sqrt{\sum_{i=1}^{n}(P_v - \overline{P_v})^2}} \qquad \text{Equation 6-1}$$

$$e = \frac{1}{n} \times \sum_{i=1}^{n} \left(T_{v_i} - P_{v_i}\right)^2 \qquad \text{Equation 6-2}$$

where $\overline{T_v}$ and $\overline{P_v}$ are averaged values of T_v and P_v, respectively. While ± 1 are the ideal values for r representing the normalized covariance of variables, the smaller values (ideally 0) are more desirable for e measuring estimation error values (Benesty et al., 2008; Chai and Draxler, 2014). It should be mentioned that the FDTs' classification process clusters training cases within different branches of a tree, split based on the influential decision-making attributes. Subsequently, the number of training cases and variables for regression equation development in integrated FDTs

is less than in regression-based ML techniques. Thus, FDT-based and regression-based tools have just been compared based on the prediction accuracy (and not regression error evaluation criteria).

6.2.2 Regression-Based Machine Learning Techniques

While GPR has already discussed in Section 5.2.3.2 Fuzzy Decision Tree with various Regression Analyses, the remaining regression-based ML techniques including ANNs and SVR are briefly introduced in the following.

ANN is widely used in solving two types of problems, pattern recognition and regression analysis (i.e., function approximation). As mentioned in the third chapter, ANNs imitate neural network connections of human brains to discover a prediction pattern among input and target values. These networks encompass several layers of neurons receiving input values and aggregating them based on weights and biases to estimate target values (Priddy and Keller, 2005; Shanmuganathan, 2016). The number of layers and neurons may affect the performance of ANNs. Generally, problems with a higher degree of non-linearity and complexity require larger numbers of layers and neurons. In this study, the number of neurons and hidden layers were determined by sensitivity analyses, which were performed using the trial-and-error approach.

ANN weights and biases are equivalent to coefficients and intercepts used in LR models, respectively. Weights and biases are determined through a training process aiming to minimize a loss function. The function estimates the difference between target and predicted values. As depicted in Equation 6-2, the mean squared errors (*e*) between T_v and P_v is the most widely used loss function, also considered in the present study for all ANNs' development (Livingstone, 2008). In this equation, P_{v_i} is a function of weight and bias variables' vector (*X*).

ANN training process is an optimization problem that can be solved by various training algorithms. Training algorithms exhibit different performances in ANN construction, depending on the complexity and type of the initial problem and the number of training cases. Generally, training

algorithms initialize the optimization problem from random values of weights and biases to minimize the sum of squares of errors corresponding to all inputs (i.e., loss function defined by Equation 6-2). These values will be updated by calculating a vector of adjustment (ΔX) to reduce the error (*e*). One of the stopping criteria in this process is the maximum of epochs (i.e., the number of times when all inputs are presented to the network), considered as 1000 in the development of all ANNs in the present study. Also, the Previous studies widely discussed details of ANNs (Hagan et al., 2002; Livingstone, 2008; Shanmuganathan, 2016). Three types of backpropagation training algorithms to construct multi-layer feedforward ANNs (i.e., networks with no loops or cycles of information movements) were evaluated in this study, as discussed below. Backpropagation algorithms calculate gradients and Jacobians (i.e., derivatives of loss functions), required to update weights and biases, in the backward direction through the network. In this study, three types of widely used training algorithms in ANN including Gradient decent, Levenberg Marquardt, and Bayesian regularization are considered.

Gradient decent training algorithm determines weight and bias values based on the gradient descent optimization. This algorithm updates weight and bias values within an iterative process by considering the steepest descent directions in the function variation (Ruder, 2016). The steepest descent directions are identified by calculating the negative gradient of the loss function at the current iteration point, which corresponds to the derivatives with respect to the current weight and bias values. In this algorithm, a learning rate parameter is defined to be multiplied with the negative gradient values, specifying the range of variation for weights and biases in each iteration. The learning rate controls the step size taken during each update and influences the conflicting criteria of stability and convergence rate of the algorithm (i.e., reliability and speed of the algorithm in finding optimal solutions) (Hagan et al., 2002). The general formula to adjust the vector of weight and bias values (*X*) is:

$$\Delta X = -L_r \times \frac{de(X)}{dX} \qquad \text{Equation 6-3}$$

where ΔX, L_r and $e(X)$ are the vector of adjustment, learning rate, and loss function (Equation 6-2), respectively. Based on the desired trade-off between the stability and convergence rate, the initial learning rate assumed as 0.001 in this study while it can increase up to 10^{10}. This rate has the flexibility to be adjusted, depending on fluctuations in the loss function (i.e., greater values in case of an increase in loss function). Hagan et al. (2002) provided a discussion about L_r selection (Hagan et al., 2002). Furthermore, the used activation function for hidden layers of ANNs was the hyperbolic tangent sigmoid activation function.

Levenberg Marquardt training algorithm interpolates between gradient descent and Gauss-Newton non-linear least-squares algorithms to address their slow convergence and divergence issues in finding optimal values, respectively (Kelley, 1999). It defines an adaptive coefficient (μ) through Newton's technique to switch between the gradient decent (for larger values of μ) and Gauss–Newton (for smaller values) in updating vector of weight and bias values as Equation 6-4.

$$\Delta X = [J^{Tr}(X)J(X) + \mu I]^{-1} J^{Tr}(X)e(X) \qquad \text{Equation 6-4}$$

where *J*, *Tr*, and *I* denote the Jacobian matrix (i.e., matrix of first-order partial derivatives) of the loss function, matrix transpose operator, and identity matrix, respectively. $e(X)$ also returns error values corresponding to all inputs, according to Equation 6-2.

The loss function is calculated to apply this algorithm, and its Jacobean matrix is constructed. Then, weights and biases are updated by solving Equation 6-4 to recalculate the loss function. The Levenberg-Marquardt algorithm also involves tuning an adaptive coefficient by comparing the new values of the loss function with the previous ones to iteratively repeat the process (Hagan and Menhaj, 1994).

Levenberg-Marquardt and gradient descent algorithms have both validation processes in the development of an ANN based on a training dataset. Within the validation process, they consider the validation partition of cases to terminate tuning weights and biases in order to avoid overfitting (Burden and Winkler, 2008). Datasets with a limited number of cases are more susceptible to overfitting due to insufficient members in training, testing, and validation sub-sets (partitions).

Bayesian regularization algorithm works similarly to Levenberg-Marquardt in terms of updating weight and bias values (i.e., it also uses Equation 6-4). However, it aims to generate a more generalized and robust type of ANN by incorporating regularization into the training process. This algorithm introduces a diagonal matrix of penalty weights into the conventional loss function, regularizing the solution for weights and biases (Foresee and Hagan, 1997). The additional penalty matrix helps prevent overfitting by penalizing inflated weights and biases, particularly for parameters with high collinearity. The regularization process is efficiently controlled by implementing Bayes' theorem, which facilitates optimal termination of the training process once a sufficient number of non-zero parameters have been obtained for prediction. There is no need to conduct a separate validation process within this algorithm, which may lead to a more robust ANN that is less susceptible to overfitting with smaller datasets. The algorithm assumes a Gaussian prior distribution for weights, biases, and training data. Then, weights and biases are updated to maximize the posterior probability distribution while minimizing the regularized loss function (Burden and Winkler, 2008; Foresee and Hagan, 1997). The posterior probability distribution is constructed using Bayes' theorem (conditional probability) based on the prior density of weights and biases and the likelihood function, representing the initial knowledge about weight and bias values and probability of data occurring after considering input data, respectively.

SVR is another supervised learning technique successfully applied to various ML problems for function approximation (Alizadeh et al., 2019). SVR typically assumes a linear equation between target values and input data, incorporating a vector of parameters (X) that is determined through an iterative training process. This vector of parameters, sometimes referred to as coefficients, is analogous to weights and biases needed to be determined in ANNs. SVR also allows to consider a non-linear relationship by applying a non-linear mapping function (Alizadeh et al., 2019; Schölkopf and Smola, 2002).

Equation 6-5 is used to calculate the accuracy of a trained model in SVR by defining a deviation threshold (error tolerance) from the observed data. Then, Equation 6-6 is minimized as a loss function to determine the vector of parameters.

$$Ac(P_v, T_v) = \begin{cases} 0, & |P_v - T_v| \leq \varepsilon \\ |P_v - T_v| - \varepsilon, & |P_v - T_v| > 0 \end{cases} \quad \text{Equation 6-5}$$

$$e(X) = \frac{1}{2}\|X\|^2 + C\sum_{i=1}^{n} Ac\left(P_{v_i}, T_{v_i}\right) \quad \text{Equation 6-6}$$

subject to $|P_v - T_v| \leq \varepsilon$

where P_v, T_v, n, and ε are predicted outputs, target values, number of training cases, and error tolerance, respectively. C is a weighting coefficient used to balance the regression equation's complexity and model error. Lagrange duality theorems can be applied to minimize the dual form of Equation 6-6 as a quadratic programming problem and determine the vector of parameters. Using this theory, another optimization problem (i.e., dual form of the primal problem) can be defined, which is easier to solve and lead to the identical optimal values to the primal problem. Detail of this method and its minimization process can be found in the reported studies (Alizadeh et al., 2019; Schölkopf and Smola, 2002). In the present research, a linear mapping function was considered to develop the regression equation and the quadratic programming optimization approach was used to minimize the dual form of Equation 6-6.

6.3 Results and Discussion

The performance of regression-based ML techniques is compared based on prediction accuracies and several metrics for regression error evaluation, including correlation coefficient, absolute error, and mean squared error. Prediction accuracies of decision tree-based tool is also compared with those of regression-based ML techniques. Additionally, tools' sensitivity to the ratio of training and testing datasets are assessed.

6.3.1 Comparison of Prediction Accuracies

Testing accuracy is the most significant parameter in comparing different tools because it represents the prediction power on new cases. Therefore, the comparison was mainly focused on the testing accuracy; however, training accuracy was also evaluated for the developed tools.

To determine the optimal architecture for ANNs, the average prediction accuracy, correlation coefficients, and mean squared errors obtained from 50 iterations of ANN-LM on testing data were compared. This number of iterations was considered based on the preliminary sensitivity analysis and was used to initiate the determination of optimal architecture. As shown in Figure 6-2a, a 12-neuron network was identified as the best-performing architecture for the present problem.

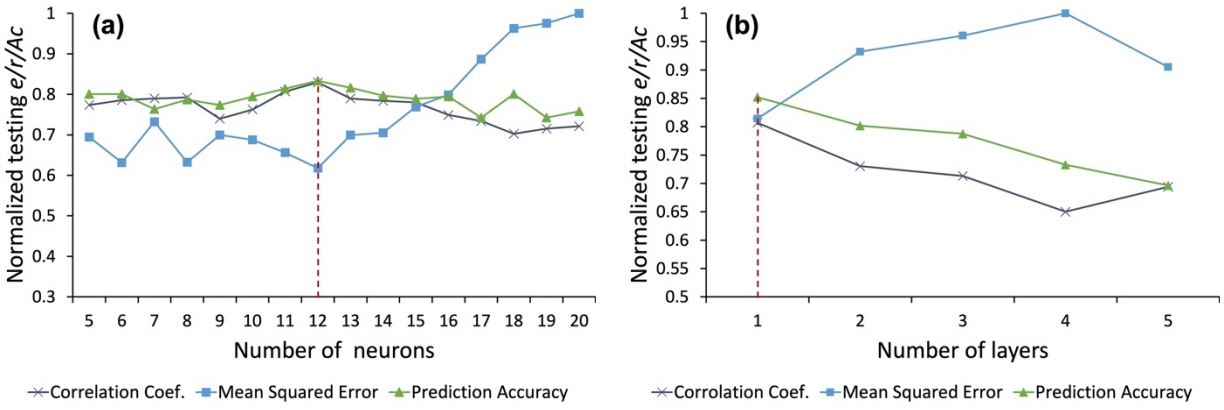


Figure 6-2. Sensitivity analysis to the number of (a) neurons and (b) layers of ANNs

Note: Mean squared error (*e*), correlation coefficient (*r*), and prediction accuracy (*Ac*) were normalized against the maximum obtained values of each metric.

When the number of neurons was 12, ANNs with one hidden layer outperformed others with more than one hidden layer (Figure 6-2b). This is reasonable because the degree of non-linearity and complexity of the present problem are not significantly high. Therefore, one hidden layer with 12 neurons was used to develop ANN-based tools.

Table 6-1 illustrates average training and testing accuracies from 10,000 iterations of various tools. Required computation time (i.e., CPU time) is also reported for each tool. GPR obtained the most desirable performance among regression-based ML techniques considering both testing

and training accuracies. SVR also attained comparable results to GPR, showing their better compatibility with small training datasets than ANNs. The probabilistic nature of GPR, inferring a probability distribution over all potential values for regression parameters, led to better performance on a wide range of dataset sizes compared to other techniques.

These two techniques generally require tuning a considerably smaller number of prediction parameters than ANN and therefore showed better prediction performance on small datasets. ANN-GD and ANN-LM did not attain appropriate prediction accuracies, especially on testing datasets. However, both training and testing accuracies and the convergence rate of ANN-LM were better than ANN-GD due to utilizing an algorithm working faster and more reliable to find optimal values of weights and biases in this problem. Defining an adaptive coefficient to interpolate between gradient descent and Gauss-Newton algorithms enhanced ANN-LM performance in finding optimal values of weights and biases compared to ANN-GD.

Table 6-1. Average prediction accuracies and computation time for different techniques

Technique		Training accuracy (%)	Testing accuracy (%)	CPU time (s)
ANN	GD	70.80	60.20	2.160
	LM	82.54	68.84	0.472
	BR	90.01	77.98	9.658
SVR		86.39	80.30	0.016
GPR		93.49	80.93	0.047
OFDTs		84.57	85.60	12.625[a]

Note: Average values of 10,000 iterations are reported for accuracies and CPU time of all regression-based tools; Average values of 15 iterations for 200 generations of optimization are reported for accuracies of OFDTs.

[a] Required time for one generation of optimization is reported for OFDTs' CPU time. All tools were executed on a personal computer with a CPU of Intel® Core™ i7-9700 and 16 GB of RAM (DDR4-2667 MHz)

Although ANN-BR outperformed two other ANN techniques, it still has a lower testing accuracy than GPR and SVR. Unlike two other ANNs, there is no need to consider a validation subset of data in ANN-BR because of implementing Bayes' theorem. Bayesian regularization algorithm

ameliorated the overfitting issue of ANNs, while a small dataset of cases is available, and enhanced the testing accuracy. As discussed, this is mainly due to skipping the validation process by implementing this algorithm. Previous studies in the field of computer science also reported a desirable performance of ANN-BR against overfitting (Burden and Winkler, 2008).

In the previous chapter, three regression analyses were integrated with FDTs, including LR, NLR, and GPR (Hu et al., 2020; Mohammadiun et al., 2021a). These regression analyses could develop a relationship between OSRMs' suitability scores and cases' crisp values of an immediate influential decision-making attribute before a terminal node. As mentioned before, results of the optimized version of FDTs have also been compared with regression-based tools in this chapter. Considering prediction accuracy, OFDTs outperformed other tools in terms of testing accuracy due to the application of a multi-objective metaheuristic optimization algorithm to enhance both prediction accuracies and robustness of FDTs. It is also possible to connect optimization algorithms with other ML techniques to enhance prediction accuracy. However, unlike the number of rules in FDTs, there is no specific criterion for other techniques to optimize the robustness and reliability of their models.

OFDTs and ANN-BR were the most time-consuming tools in the training process. However, the computation time values of all tools were in a reasonable range that would not hinder their applicability. Additionally, after training a tool, it can be used in the future to instantly predict outcomes in a new case. It is worth noting that OFDTs are generally less susceptible to overfitting compared to ANN-BR. This characteristic arises from the fact that in this problem, ANN-BR tends to achieve higher training accuracy but considerably lower testing accuracy, resulting in a noticeable gap between these two values. Conversely, OFDTs exhibit the desirable quality of having both training and testing accuracies that are close and satisfactory.

Figure 6-3 illustrates a boxplot of obtained testing accuracies from all iterations. While ANNs have the largest ranges of variation and mostly smaller mean and median values, OFDTs and GPR show a smaller variation range and larger mean and median values of testing accuracy.

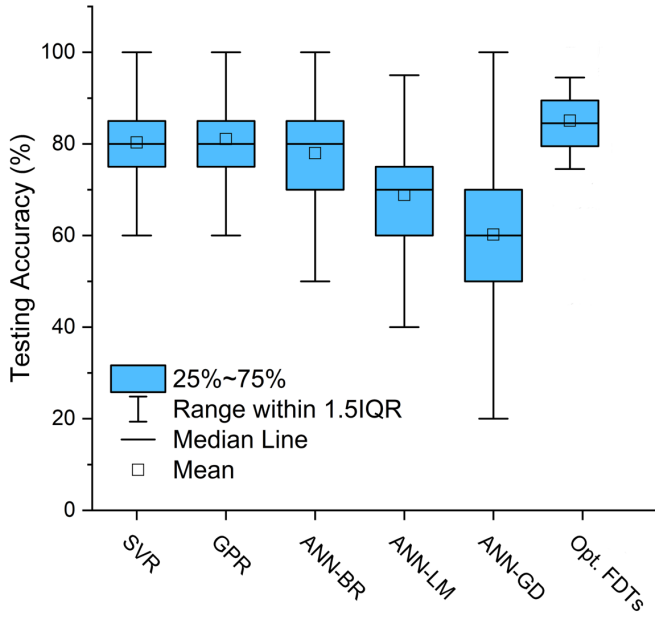


Figure 6-3. Boxplots of obtained testing accuracies through all iterations of tools

Generally, a smaller range of variation represents a more stable (reliable) tool for prediction. SVR also presents comparable results with GPR but with a smaller mean value. Based on this figure, ANN-BR depicts better prediction accuracy and stability than two other ANNs.

6.3.2 Error Evaluation in the Regression-Based Techniques

For further evaluations, *e* and *r* values are also reported in case of ML techniques performing regression analyses on all cases of the dataset (i.e., ANNs, SVR, and GPR). Likewise, *e* and *r* values obtained from 10,000 iterations of regression-based ML techniques (Table 6-2) are consistent with the previous observations. From this perspective, GPR has the highest correlations and lowest mean squared error for both training and testing datasets resulting in better predictions. While ANN-BR has better performance on training datasets than SVR, the latter outperformed considering testing datasets. Unlike ANNs, the desirable performance of SVR on testing datasets may prove its less susceptibility to overfitting on datasets with smaller sizes. Based on the values in this table, ANN-GD is the worst-performing tool, which is also consistent observation with the data distribution illustrated in Figure 6-3.

Table 6-2. Average absolute correlation coefficients and prediction errors in regression-based ML techniques

Technique		Training *r* [a]	Testing *r*	Training *e* [b]	Testing *e*
ANN	GD	0.720	0.555	0.046	0.076
	LM	0.865	0.689	0.019	0.046
	BR	0.954	0.845	0.008	0.025
SVR		0.919	0.867	0.011	0.018
GPR		0.976	0.884	0.003	0.016

[a] Pearson correlation coefficient
[b] Mean squared error in models' prediction
Note: Average values of 10,000 iterations over three OSRMs are listed

Figure 6-4 also shows distributions of correlation coefficient values on testing data for 10,000 iterations in all regression-based ML techniques. The width of the violin graph depicts the density of data with respect to correlation coefficient values in a specific area.

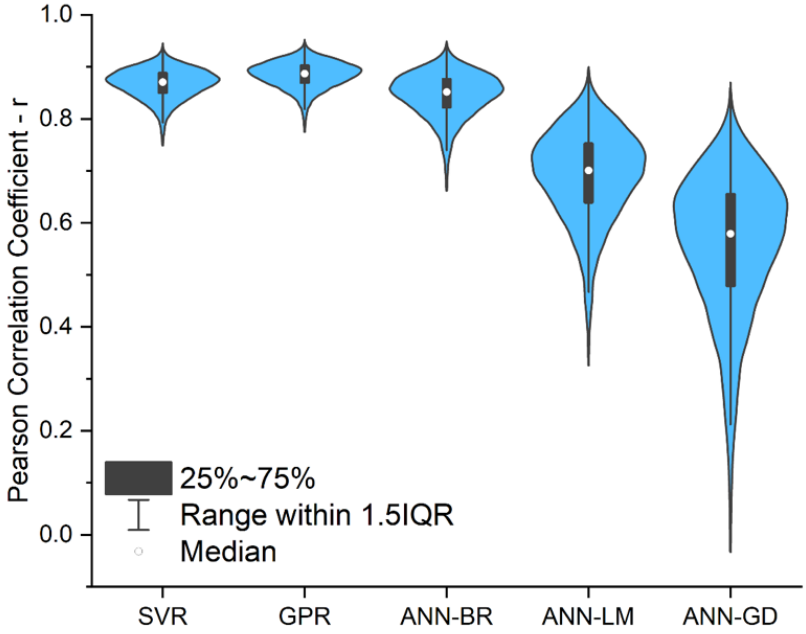


Figure 6-4. Distribution of testing correlation coefficients for regression-based ML techniques

As can be seen, GPR outperformed others (i.e., larger *r* values) and has the most stable prediction performance (i.e., smaller range of vertical variations). Also, ANN-BR considerably performed

better than two other ANNs. Again, ANN-GD is the worst-performing tool with a large variation range of correlation coefficients (i.e., instability in the prediction) which are also considerably less than other tools. Similar performance trends were also observed in mean squared error distributions.

Table 6-3 shows the performance assessment of different regression-based tools on a specific oil spill incident in a testing dataset. Absolute errors between OSRMs' suitability scores assigned by experts and calculated by different tools as well as the most suitable OSRM selected by the tools are tabulated. After considering all influential attributes, experienced decision-makers selected ISB as the most suitable OSRM for this incident. This oil spill incident happened in a highly remote location (i.e., a large value of A6), has a sufficient slick thickness for combustion (A2>1mm), and is not significantly fragmented (i.e., not a relatively large value for A3). Other values of influential decision-making attributes of this specific incident are also given in Table 6-3.

Table 6-3. Performance of regression-based tools on a specific spill incident in testing dataset

Technique		Absolute error in MCR	Absolute error in ISB	Absolute error in UCD	Average absolute error	Models' prediction
ANN	GD	5.74	2.10	2.02	3.28	MCR
	LM	3.92	0.38	0.73	1.68	MCR
	BR	1.69	1.31	0.96	1.32	ISB
SVR		0.77	0.75	0.54	0.69	ISB
GPR		0.90	0.30	0.80	0.67	ISB

Note: Values of influential decision-making attributes for this incident were A1=30, A2=1.6, A3=59, A4=0.5, A5=3, and A6=460.
The most suitable OSRM in the hypothetical dataset was ISB.
MCR - Mechanical containment and recovery; ISB - In-situ burning

MCR is usually not an appropriate response method for incidents in highly remote areas. ISB is a suitable choice that is not sensitive to the Remoteness attribute, but it requires sufficient oil slick thickness (Table 4-1). Based on Table 6-3, while GPR, SVR, and ANN-BR selected the correct OSRM for this incident (i.e., ISB), ANN-LM and ANN-GD incorrectly selected MCR. Smaller error

values in suitability scores estimation led to a more reliable OSRM selection tool. Considering the average absolute error values, GPR shows the best performance among all regression-based tools.

6.3.3 Sensitivity Analysis to the Number of Cases

Finally, to conduct a sensitivity analysis on prediction performance in relation to the ratio of training and testing datasets size, the number of cases in the training dataset was altered from 40% to 80% of the total dataset size. The rest of the cases were used for testing through the previously mentioned iterative process in each case. The minimum, maximum, mean, and standard deviation (STD) of average training and testing accuracies on all iterations are shown in Table 6-4 for each tool. Considering the performance on the testing dataset, OFDTs, GPR, and SVR achieved larger values of prediction accuracies while having smaller values of STDs, representing a more accurate and more stable prediction performance, respectively.

Table 6-4. Variations of statistical metrics of prediction accuracies with the size of the training dataset

Technique		Minimum (%)		Maximum (%)		Mean (%)		STD[a] (%)	
		Training	Testing	Training	Testing	Training	Testing	Training	Testing
ANN	GD	66.06	52.27	71.36	61.47	69.31	57.71	1.91	3.26
	LM	79.38	61.27	83.49	70.29	82.07	66.54	1.40	3.21
	BR	80.52	70.49	91.68	79.04	86.96	75.62	4.08	3.10
SVR		86.21	77.90	88.26	80.46	87.03	79.49	0.77	0.97
GPR		93.02	78.26	96.02	80.93	94.41	80.13	1.13	1.06
OFDTs		82.00	80.63	88.15	87.78	84.76	85.14	1.50	1.66

[a] Standard deviation

Almost a similar trend can also be seen in their performance on the training dataset. Considering ANN-based tools, ANN-BR has the highest values of minimum, maximum, and mean. However,

it is still sensitive to the size of the training dataset concerning the range of variation for prediction accuracies (i.e., relatively high values of STDs).

The comprehensive evaluation of various metrics revealed that OFDTs and GPR techniques are most suited for the present problem with insufficient training cases. These techniques are less susceptible to overfitting and can provide reliable accurate prediction for response method selection. As mentioned, FDT is also desirable for developing ML-based tools considering the transparent decision-making process. Unlike ANNs, FDTs can generate explicit rules reasoning the decision-making process and make the process understandable to the decision makers. It should be noted that the best-performing tool identified can be used to promptly select the most suitable response method for new oil spill incidents with given information about the six influential attributes.

6.4 Summary

Applying intelligent decision support tools may considerably facilitate OSRM selection and enhance the response practitioners' preparedness. However, insufficient data may impede the development of the tools for OSRM selection and many other real-world applications. While the ideal solution is to acquire enriched and relevant datasets for developing robust ML-based tools, such datasets may not always be easily available. In such cases, employing advanced computational techniques capable of mitigating the impacts of data limitations and performing effectively under constrained data conditions becomes a viable alternative.

This chapter evaluated the capability of different types of ML techniques, including various ANNs, GPR, and SVR, to develop reliable, intelligent decision support tools under data limitations. The prediction power of these techniques was also compared with optimized decision-tree based tools for OSRM selection. To this end, these techniques were concisely introduced, especially with respect to the sources of their distinct performance within their formulations. Then, ML-based tools were trained on a hypothetical dataset of oil spill incidents, containing values of influential

decision-making attributes and corresponding OSRMs' suitability scores in cold waters. The performance of all tools was comprehensively assessed in relation to prediction accuracies on training and testing datasets as well as significant statistical metrics. Evaluation of the results shows that OFDTs and GPR outperformed other techniques in terms of prediction accuracies and reliability of the tools. By applying these techniques, approximately 30% and 40% enhancements have been respectively achieved for training and testing accuracies compared to the worst-performing model (ANN-GD). The enhanced performance on the relatively small-size dataset indicates a considerable capacity of OFDTs and GPR to avoid overfitting on a limited number of training cases. Moreover, the implementation of Bayesian regularization within ANN-BR has also improved the prediction power of ANNs. This algorithm reduced the risk of ANN's overfitting on limited data, causing 29% and 12% improvement in the testing accuracy of ANN-GD and ANN-LM, respectively.

Chapter 7 – Integrated optimization of oil spill response and oily waste management operations

A version of this chapter is under revision for publication in the Journal of Hazardous Materials, entitled “Integrated Optimization of Marine Oil Spill Response and Liquid Oily Waste Management using Mathematical Programing and Evolutionary Metaheuristic Techniques.”

7.1 Introduction

After selecting an OSRM, optimizing the response process becomes highly beneficial in order to attain an efficient response level. As mentioned, after an oil spill incident, there is a narrow window of opportunity to effectively reduce the adverse impacts of spilt oil through a well-organized and timely oil spill response operation. In most response operations, the first step is to contain floating oil slicks to prevent them from spreading and reaching nearby shorelines because the cleanup of contaminated shorelines can generate a large volume of solid oily wastes. Among OSRMs, MCR is the ideal response method that can remove spilt oil containing toxic components from the marine environment. In this chapter we optimized the involved process in the mechanical removal of spilt oil in marine environment as the most frequently used and preferable respone method.

There is an inextricable link between MCR and OWM operations, and disregarding this linkage would lead to inefficient or even ineffective oil spill response management. Optimization of MCR and OWM operations is in great demand as those may considerably improve oil recovery efficiency and reduce operation time and costs. Simplified forms of these operations have already been optimized individually in previous studies.

By adopting an integrated management approach, the response and waste management operations can be streamlined, ensuring effective oil spill response management. In addition, efficient management of the entire operation is a highly complex task even for experienced decision-makers with access to a well-established system of communication between available response assets. There is a lack of comprehensive decision support tools that can holistically consider the operational details of both MCR and OWM for maximum benefit. In particular, the

following practical considerations have not been investigated through the lens of integrated MCR and OWM operations:

1. The operational considerations in MCR such as the competitive skimming process of several vessels deployed to the same oil slick, as well as the storage capacity and variable skimming rates of vessels.
2. The operational considerations in OWM such as offshore temporary storage capacity, available shoreline storage capacity, transportation capacity, treatment infrastructure, and scheduling constraints.
3. Multi-objective optimization of time, cost, and recoverable oil through an integrated MCR and OWM.
4. Testing optimization models in a comprehensive case study of oil spill response and waste management operations.

To address these gaps, an innovative decision support tool was developed in the present chapter working as a multi-agent system to minimize the response cost and time as well as the volume of weathered oil, considering the detailed operational procedures in MCR and OWM. The system includes three main components of metaheuristic multi-objective optimization, oil weathering process, and MCR and OWM operational agents.

7.2 Method

The integrated optimization of MCR and OWM operations is a complex problem that involves multiple operational stages with interconnected activities. Figure 7-1 illustrates a network representing the general concept and components of the oil spill response problem. The components include oil slicks, vessels, mini barges, big barges, trucks, and treatment facilities. The problem is divided into two main stages: offshore and onshore. During the offshore stage, the response operation begins with deploying skimming vessels to boomed oil slicks and ends with offloading collected oily wastewater into the big barges located near the shoreline. Vessels

equipped with skimmers should be dispatched to collect the oil contained by booms and store the collected oil-water mixture in their internal storage tanks. Mini barges, providing additional storage capacity, are responsible for transporting oily wastewater from the vessels to the big barges. For a large oil spill incident, skimming vessels should be kept close to oil slicks to accelerate the oil removal process. In the onshore stage, trucks transport the oily wastewater from the big barges to treatment facilities.

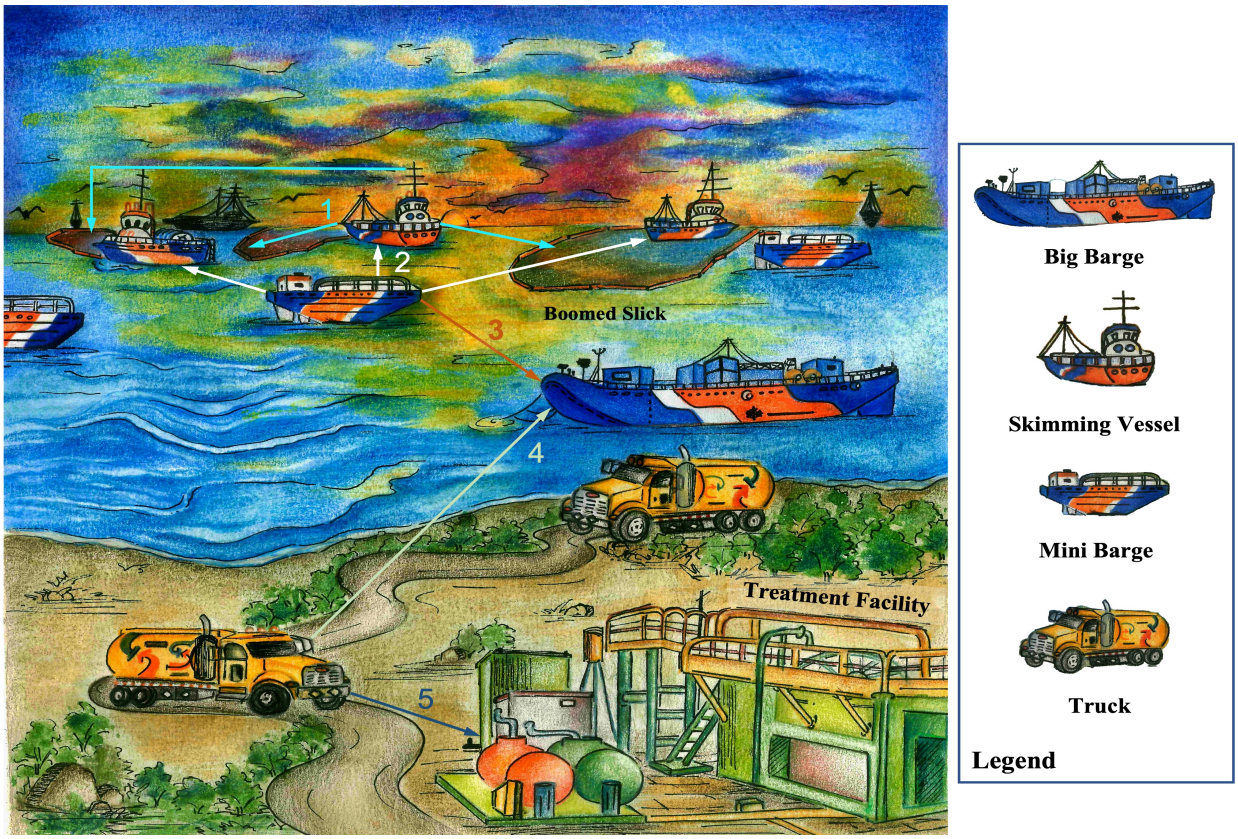


Figure 7-1. Network of marine oil spill response and liquid oily waste management operations

As shown in Figure 7-1, the entire operation can be modeled as an assignment-based scheduling problem. One critical aspect of this problem is the assignment of mobile response assets (i.e., vessels, mini barges, and trucks) in the offshore and onshore stages over time. This includes the assignment of (1) skimming vessels to oil slicks, (2) mini barges to skimming vessels, and (3) mini barges to big barges in the offshore stage, as well as assigning (4) trucks to big barges and (5)

trucks to treatment facilities in the onshore stage. Defining binary integer variables is necessary to solve this assignment-based scheduling problem.

Another important aspect is the time needed to recover a unit volume of spilt oil. The recovery of spilt oil is affected by skimming rate and oil slick thickness that is varying over time with oil weathering processes. The volume of recovered oil over time also depends on the available storage capacity of skimming vessels. These considerations demand to define continuous (non-integer) variables. Because both binary integer and continuous variables are necessary, the research problem is mathematically modeled using MIP (Sharif et al., 2022).

7.2.1 Model for Integrated Optimization of MCR and OWM

A decision support tool was developed based on the multi-agent system concept to optimize integrated MCR and OWM operations as illustrated in Figure 7-2. The system encompasses three main agents: (i) multi-objective optimization agent, (ii) mathematical model (or operational agent), and (iii) oil weathering process agent.

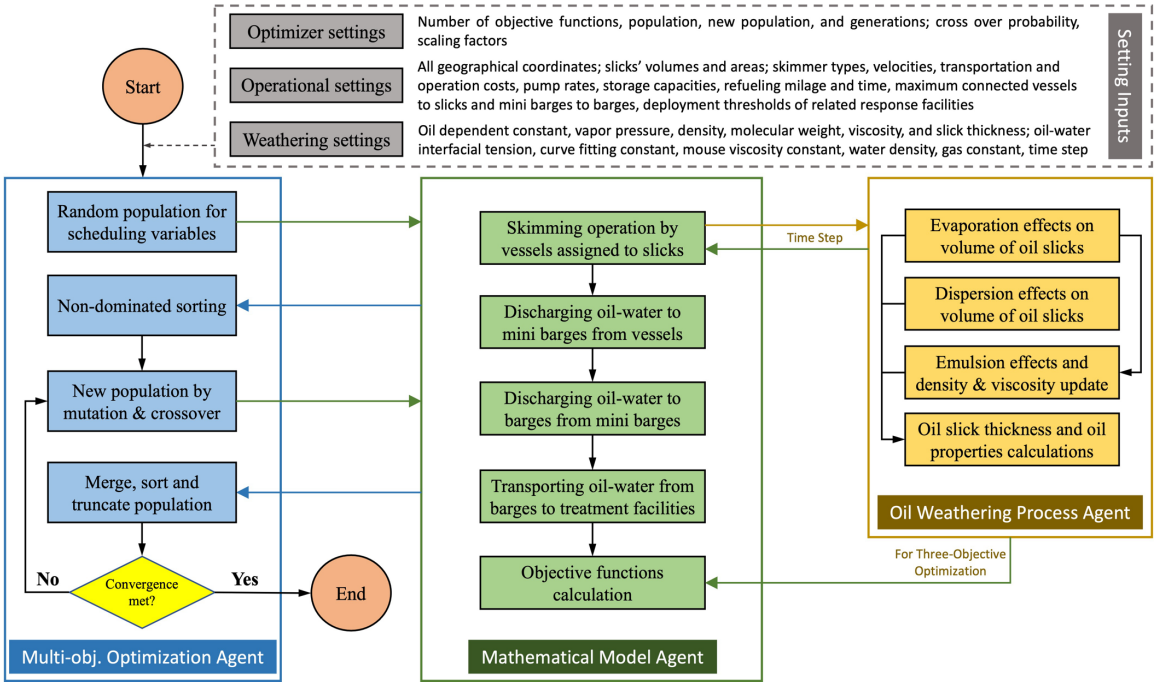


Figure 7-2. Model for optimization of oil spill response and oily waste management operations

The multi-objective optimization agent uses a soft computing technique (i.e., NSDE) to efficiently optimize the complex and interdependent processes in MCR and OWM. The mathematical model agent formulates the asset assignment problem based on MIP. In particular, NSDE is used to find efficient solutions for the developed MIP-based model. The oil weathering processes are also essential to determining the characteristics of spilt oil over time for developing effective response strategies.

The tool begins with the collection of data related to the MCR and OWM operations and oil weathering processes, which is indicated as setting inputs for optimizer, operational, and weathering agents in Figure 7-2. To streamline the multi-agent system, a sophisticated code is developed in MATLAB-R2022b^{TM}, facilitating timely and automated integration of the three agents. The associated pseudocode is provided in Appendix D.

7.2.1.1 Multi-Objective Optimization Agent

Similar to Chapter 5, the NSDE algorithm was used to optimize multiple objective functions in the optimization agent (Figure 7-2). The objective functions include minimizing the time, cost, and volume of weathered oil during the response operation. As seen before, solutions that could not be dominated by other candidates based on the values of objective functions and locations in low-density regions of the feasible search space have a higher probability of being selected as the optimal results in NSDE. In each iteration of optimization (i.e., a generation), a population of candidate optimal solutions is generated, where each member of the population represents a specific scheduling of all available mobile assets for the entire operation. The details of NSDE and relevant mathematical equations were already provided in Chapter 5. In the present chapter, the NSDE code is modified based on parallel processing requirements to be effectively used to solve a large-scale and time-sensitive optimization problem. The number of populations were taken as 1500 determined based on the size of the optimization problem.

As Figure 7-2 shows, the first step is input setting. The optimizer settings include the number of objective functions, population, new population, and generation, as well as cross-over probability

and scaling factors. The optimizer then generates a random population for decision variables to schedule mobile assets, which are linked to the mathematical model agent. The optimizer receives values of objective functions from the mathematical model agent to rank the solutions based on non-dominated sorting criteria. The optimization agent then generates new population using mutation and crossover operators and pass it to the mathematical model agent to calculate the values of associated objective functions. After calculation, the previous and new populations are merged and sorted in the optimization agent and then truncated based on the maximum number of population. The process will continue until the stopping criterion is met to reduce the objective function values through different generations. The stopping criterion consists of two conditions: a predefined maximum number of iterations (i.e., 200) and the absence of any changes in the optimal solutions obtained within the past ten generations after at least 75 generations have been formed.

7.2.1.2 Mathematical Model Agent

The optimization problem for marine oil spill response and OWM operations is formulated based on MIP. It is worth noting that this MIP-based model is solved by the optimization agent using a metaheuristic technique (i.e., NSDE) due to the significant size and complexity of the problem, which also includes considering time as a continuous variable in assets deployments. The following mathematical formulation employs small and capital letters to distinguish between decision variables and parameters, respectively. Different sets and indices were used to mathematically model the problem. Specifically, the following sets are used: I ($\forall i$ = 1, 2,…, OSN) for the oil slicks, J ($\forall j$ = 1, 2,…, VSN) for the available response vessels, K ($\forall k$ = 1, 2,…, MBN) for the mini barges, L ($\forall l$ = 1, 2,…, BBN) for the big barges, M ($\forall m$ = 1, 2,…, TRN) for the transportation trucks, N ($\forall n$=1,2,…, TFN) for the treatment facilities. It should be mentioned that all decision variables and parameters are listed in Appendix E.

When a vessel j is deployed, a unit operation cost (VOC_j) is imposed accounting for personnel, maintenance, depreciation, etc. The rest of the components in both offshore and onshore stages

are also associated with operation costs. Therefore, MOC_k and BOC_l are the unit operation costs of using mini barge *k* and big barge l in the offshore stage, respectively. TOC_m and TRC_n are the unit operation costs of truck *m* and treatment facility *n* in the onshore stage, respectively. Additionally, the unit transportation costs of response vessel *j* (VTC_j), mini barge *k* (MTC_k), and truck *m* (TTC_m) are separately considered in the model. It is also possible to merge transportation and operation costs where it is desirable, as considered in the present study for trucks.

The mathematical model also considers the distance between various components of the network (e.g., a vessel and an oil slick). This distance is calculated based on the coordinates of each component, which are continuously updated over time. The model uses the following variables: $(X_i, Y_i), (X_j, Y_j), (X_k, Y_k), (X_l, Y_l), (X_m, Y_m), (X_n, Y_n)$, representing the coordinates of oil slick *i*, vessel *j*, mini barge *k*, big barge l, truck *m*, treatment facility *n*, respectively. In this evaluation, the location of the one of the big barges is selected as the origin of a Cartesian coordinate system. It is important to note that the locations of oil slicks, big barges, and treatment facilities are assumed to be fixed. Taking the distance between vessel *j* and oil slick *i* as an example, their distance can be formulated as $\overrightarrow{dis}(j,i) = \sqrt{(X_i - X_j)^2 + (Y_i - Y_j)^2}$ in the offshore stage. However, the assignment of trucks is simulated based on the actual driving distance between big barges and treatment facilities in the onshore stage. By continuously updating the coordinates of each component, the model can calculate the distance between relevant components.

To minimize the overall response time, the continuous time variables required for performing a task, refueling a mobile response asset, and travelling between the components of the network are considered. Their associated variables are listed in Table 7-1.

During the response operation, decisions are made to assign mobile assets to their relevant targets and maybe to refuel these assets before being dispatched to their next destination. These require the use of binary decision variables listed in Table 7-2.

Table 7-1. List of time related variables

No.	Temporal variables	Definition
1	$T_{ij}, i \in I, j \in J$	The time required for vessel *j* to collect spilt oil from slick *i* calculated based on skimming capacity of the vessel *j* (VSK_j).
2	$\vec{T}_{ij}, i \in I, j \in J$	The time needed for vessel *j* to reach oil slick *i*.
3	$VRD_j, j \in J$	The time required for refueling vessel *j* before moving towards another oil slick.
4	$\vec{T}_{jk}, j \in J, k \in K$	The time required for mini barge *k* traveling to vessel *j*.
5	$\vec{T}_{kl}, k \in K, l \in L$	The time required for mini barge *k* traveling to big barge *l*.
6	$T_{jk}, j \in J, k \in K$	The time required for mini barge *k* to receive oily wastewater from vessel *j* calculated based on the pumping rate of the mini barge *k* (MPR_k).
7	$T_{kl}, k \in K, l \in L$	The time required for mini barge *k* to offload oily wastewater to big barge *l* calculated based on the pumping rate of the big barge (BPR_l).
8	$MRD_k, k \in K$	The time required for refueling mini barge *k*.
9	$\vec{T}_{lm}, l \in L, m \in M$	The time required for truck *m* traveling to big barge *l*.
10	$\vec{T}_{mn}, m \in M, n \in N$	The time required for truck *m* traveling to treatment facility *n*.
11	$T_{lm}, l \in L, m \in M$	The waiting time for truck *m* until oily wastewater is transferred from big barge *l* into that truck calculated based on the pumping rate for transferring wastewater from the big barge (BPR_l).
12	$T_{mn}, m \in M, n \in N$	The time required for truck *m* to offload oily wastewater to treatment facility *n* calculated based on the pumping rate of the treatment facility (TPR_n).
13	$TRD_m, m \in M$	The time required for refueling truck *m* before moving towards treatment facilities or big barges.

Table 7-2. List of binary decision variables

No.	Binary decision variables	Definition
1	$\delta_{ij}, i \in I, j \in J$	1 if oil slick *i* is skimmed by assigned vessel *j*, and 0 otherwise.
2	$\delta_{jk}, j \in J, k \in K$	1 if oily wastewater is transferred from vessel *j* to assigned mini barge *k*, and 0 otherwise.
3	$\delta_{kl}, k \in K, l \in L$	1 if mini barge *k* is assigned to offload into big barge *l*, and 0 otherwise.
4	$\delta_{lm}, l \in L, m \in M$	1 if oily wastewater is loaded from big barge *l* to assigned truck *m*, and 0 otherwise.
5	$\delta_{mn}, m \in M, n \in N$	1 if the load of truck *m* is sent into treatment facility *n*, and 0 otherwise.
6	$\psi_j, j \in J$	1 if vessel *j* needs to be refueled, and 0 otherwise.
7	$\psi_k, k \in K$	1 if mini barge *k* needs to be refueled, and 0 otherwise.
8	$\psi_m, m \in M$	1 if truck *m* needs to be refueled, and 0 otherwise.

Here, optimization is conducted using two approaches: one aims to minimize the entire time and cost of response operations within a two-objective optimization, while the other approach incorporates a third objective function within a three-objective optimization. The additional objective focuses on minimizing the volume of weathered oil, which is determined using the oil weathering process agent. This agent updates rates of evaporation and dispersion in each time step and calculates the volume of weathered oil based on these parameters along with the initial volume of oil slick *i* (OSV_{0_i}). In this direction, the volume of weathered oil is also affected by scheduling of skimming vessels changing the slick thickness by collecting oil. It should be noted that minimizing the weathered oil is equivalent to maximizing the volume of recovered oil. Using the defined parameters and decision variables, the following objective functions and constraints are formulated:

*f*1: Minimization of the overall operation and transportation costs:

$$
\begin{aligned}
f_1 = \min \Bigg\{ \sum_{i\epsilon I} \sum_{j\epsilon J} \Bigg(& \left(VOC_j \cdot T_{ij} + VTC_j \cdot \overrightarrow{dis}(j,i) \right) \cdot \delta_{ij} \\
& + \sum_{k\epsilon K} \Bigg(\left(MOC_k \cdot T_{jk} + MTC_k \cdot \overrightarrow{dis}(k,j) \right) \cdot \delta_{jk} \\
& + \sum_{l\epsilon L} \Bigg(\left(BOC_l \cdot T_{kl} + MTC_k \cdot \overrightarrow{dis}(k,l) \right) \cdot \delta_{kl} \\
& + \sum_{m\epsilon M} \Bigg(\left(TOC_m \cdot T_{lm} + TTC_m \cdot \overrightarrow{dis}(m,l) \right) \cdot \delta_{lm} + \sum_{n\epsilon N} \Big(TRC_n \cdot TSC_m \\
& + TTC_m \cdot \overrightarrow{dis}(m,n) \Big) \cdot \delta_{mn} \Bigg) \Bigg) \Bigg) \Bigg) \Bigg\}
\end{aligned}
$$

Equation 7-1

f_2: Minimization of the total operation and transportation time:

$$f_2 = min\left\{\sum_{i\epsilon I}\sum_{j\in J}\left(VRD_j\cdot\psi_j + \left(T_{ij} + \frac{\overrightarrow{dis}(j,i)}{VSS_j}\right)\cdot\delta_{ij}\right.\right.$$
$$+\sum_{k\epsilon K}\left(MRD_k\cdot\psi_k + \left(T_{jk} + \frac{\overrightarrow{dis}(k,j)}{MBS_k}\right)\cdot\delta_{jk}\right.$$
$$+\sum_{l\epsilon L}\left(MRD_k\cdot\psi_k + \left(T_{kl} + \frac{\overrightarrow{dis}(k,l)}{MBS_k}\right)\cdot\delta_{kl}\right. \qquad \text{Equation 7-2}$$
$$+\sum_{m\epsilon M}\left(TRD_m\cdot\psi_m + \left(T_{lm} + \frac{\overrightarrow{dis}(m,l)}{TRS_m}\right)\cdot\delta_{lm} + \sum_{n\epsilon N}TRD_m\right.$$
$$\left.\left.\left.\left.\left.\cdot\psi_m + \left(T_{mn} + \frac{\overrightarrow{dis}(m,n)}{TRS_m}\right)\cdot\delta_{mn}\right)\right)\right)\right)\right\}$$

f_3: Minimization of the total volume of evaporated and dispersed (i.e., weathered) oil:

$$f_3 = min\left\{\sum_{i\in I}(FE + FD)\cdot OSV_{0_i}\right\} \qquad \text{Equation 7-3}$$

The constraints of the optimization problem and definitions of the rest of parameters are listed in Table 7-3. The implementation of the optimization agent with two-objective functions (Equation 7-1 and Equation 7-2) is demonstrated under OPA-II and its implementation with three-objective functions (Equation 7-1, Equation 7-2, and Equation 7-3) is referred to as OPA-III in this chapter. Moreover, the outcomes of these approaches are compared with SDA (i.e., shortest distance approach). SDA deploys response assets by considering the minimum distance between each available asset and the target while using the least number of required vessels, mini barges, and trucks. The scheduling of available assets in SDA also considered all the constraints listed in Table 7-3, employing the mathematical model and oil weathering process agents.

It should be noted that the time required for oil slick removal is just a portion of the entire operation time considered in f_1. Alternatively, f_3 primarily relies on the time required for oil slick removal and the selection of vessels and slicks for the skimming operation.

Table 7-3. Constraints of the MCR and OWM optimization problem

No.	Equation	Remark
1	$\sum_{j \in J} VSC_j \cdot \delta_{ij} = OSV_{0_i} \cdot [1 - (FE + FD)], \forall i \in I$ where VSC_j is the storage capacity of vessel *j*.	The total volume of oil skimmed by all assigned vessels *j* from oil slick *i* plus the volume of weathered oil is equal to the initial volume of the oil slick.
2	$1 \leq \sum_{j \in J} \delta_{ij} \leq ODV_i, \forall i \in I$ where ODV_i is the maximum number of concurrently deployable vessels to slick *i*.	The assigned number of vessel(s) to oil slick *i* can vary between one and a predefined number of vessels for that slick.
3	$\sum_{i \in I} \delta_{ij} \leq 1, \forall j \in J$	Each vessel is assigned to at most one oil slick at a time.
4	$\sum_{i \in I} \vec{T}_{ij} \cdot VSS_j \cdot \delta_{ij} \leq VFT_j, \forall j \in J$ where VFT_j is the millage can be travelled with a full tank of vessel *j* and VSS_j is the average speed of vessel *j*	For each vessel *j*, the constraint ensures that it can only be assigned to an oil slick *i* if it has enough fuel for the required travel duration.
5	$\sum_{j \in J} \psi_j \leq VNR_j$ where VNR_j is the maximum number of concurrently refueling vessels.	The number of concurrently refueling vessels cannot be larger than a predefined number of vessels (VNR_j).
6	$\sum_{j \in J} \delta_{jk} \leq 1, \forall k \in K$	A mini barge *k* can receive oily wastewater from at most one response vessel *j* at a time.
7	$\sum_{j \in J} \vec{T}_{jk} \cdot MBS_k \cdot \delta_{jk} \leq MPT_k, \forall k \in K$ where MPT_k is the maximum millage that can be travelled with a full tank of mini barge *k* and MBS_k is the average speed of mini barge *k*.	Each mini barge *k*, can only be assigned to a vessel *j* if it has enough fuel for the required travel distance.
8	$\sum_{l \in L} \vec{T}_{kl} \cdot MBS_k \cdot \delta_{kl} \leq MPT_k, \forall k \in K$	Each mini barge *k* can only be assigned to a big barge *l* if it has enough fuel for the required travel distance.
9	$\delta_{jk} - \delta_{ij} \leq 0, \forall i \in I, \forall j \in J, \forall k \in K$	Each mini barge *k* can only receive oil from a vessel *j* if it is assigned to that vessel and the vessel *j* has already been assigned to an oil slick *i*.
10	$VSC_j \cdot \delta_{jk} \leq MSC_k, \forall k \in K, \forall j \in J$ where MSC_k is the storage capacity of mini barge *k*.	Each mini barge *k* can only receive oil up to its storage capacity. **Note:** The smaller value between oil stored in the vessel and available storage capacity of a mini barge could be

No.	Equation	Remark
		transferred to that mini barge based on a decision by the optimizer.
11	$\sum_{k \in K} \psi_k \leq MNR_k$ where MNR_k is the maximum number of concurrently refueling vessels.	The number of concurrently refueling mini barges is at most a predefined number of mini barges (MNR_k).
12	$(BSC_l + \sum_{m \in M} TSC_m \cdot \delta_{lm}) \geq \sum_{k \in K} MSC_k \cdot \delta_{kl}, \forall l \in L$ where BSC_l is the storage capacity of big barge *l*, TSC_m is the storage capacity of truck *m*, and MSC_k is the storage capacity of mini barge *k* (the volume of oil transferred from mini barge *k* to big barge *l*).	The total oil collected by the assigned mini barges does not exceed the storage capacity of big barge *l*. The sum of the big barge's storage capacity and the volumes of oil transferred to the assigned trucks must be greater than or equal to the volumes of oil transferred from the assigned mini barges to that big barge. **Note:** When the volume of stored oil in mini barges or trucks is a fraction of their storage capacities, the precise amount of stored oil is considered for calculations.
13	$\sum_{k \in K} \delta_{kl} \leq BCM_l, \forall l \in L$ where BCM_l is the maximum number of concurrently connectable mini barges.	Each big barge *l* can be loaded by at most a predefined number of mini barges at the same time.
14	$\sum_{m \in M} \delta_{lm} \leq BCT_l, \forall l \in L$ where BCT_l is the maximum number of concurrently connectable trucks.	Each big barge *l* can load at most a predefined number of trucks at the same time.
15	$\sum_{l \in L} \vec{T}_{lm} \cdot TRS_m \cdot \delta_{lm} \leq TPT_m, \forall m \in M$ where TPT_m is the maximum millage that can be travelled with a full tank by truck *m* and TRS_m is the average speed of truck *m*.	Each truck *m* can only be assigned to big barge *l* if it has enough fuel for the required travel distance.
16	$\sum_{n \in N} \vec{T}_{mn} \cdot TRS_m \cdot \delta_{mn} \leq TPT_m, \forall m \in M$	Each truck *m* can only be assigned to treatment facility *n* if it has enough fuel for the required travel distance.
17	$\sum_{k \in K} MSC_k \cdot \delta_{kl} \geq BAT_l \cdot \delta_{lm}, \forall l \in L, \forall m \in M$	To initiate the offshore stage operations, the volume of stored oily wastewater in big barge *l* should be equal to or greater than the volumetric activation threshold for discharging (BAT_l). **Note:** When the volume of stored oil in mini barges is a fraction of their storage capacities, the precise amount of stored oil is considered for calculations.

No.	Equation	Remark
18	$VOL_{lm} \leq TSC_m \cdot \delta_{lm}, \forall l \in L, \forall m \in M$	The volume of oil loaded by each big barge *l* to each truck *m* (VOL_{lm}) should be less than or equal to the maximum storage capacity of each truck *m*.
19	$\sum_{m \in M} \psi_m \leq TNR_m$ where TNR_m is the maximum number of concurrently refueling trucks.	The number of concurrently refueling trucks is at most a predefined number of trucks (TNR_m).
20	$\sum_{m \in M} \delta_{mn} \leq TDT_n, \forall n \in N$	Each treatment facility *n* can accept oily wastewater from at most a predefined number of trucks (TDT_n) at the same time.
21	$\sum_{m \in M} TSC_m \cdot \delta_{mn} \leq TFC_n, \forall n \in N$	The total transferred oily wastewater into a treatment facility *n* must be equal to or less than its treatment capacity (TFC_n).

7.2.1.3 Oil Weathering Process Agent

As mentioned before, the weathering processes can affect the behavior of the spilt oil, including its thickness and viscosity. Considering the weathering processes is necessary in development of effective oil spill response strategies, as it can help responders anticipate changes in the behavior of spilt oil and make informed decisions about how to contain and recover it (Nazir et al., 2008). Evaporation, dispersion, and emulsification are among the most influential process affecting the spilt oil in marine environment. The weathering processes are complex which could be considered within a dynamic system. The mentioned weathering processes were incorporated in the developed multi-agent system through the following formulations:

- Evaporation effects on the volume of oil slicks:

The evaporation rate of an oil slick is influenced by multiple factors, including wind velocity (*U*) in m/s, temperature (*T*) in K, oil vapor pressure (*P*) in Pa, oil molecular weight (*M*) in g/mol, oil density (ρ_m) in kg/m^3, gas constant (*R*) in m^3·Pa/mol·K, oil slick volume (*OSV*) in m^3, oil slick area (*OSA*) in m^2, and time (*t*) in hr. Time is measured here from the start of the response operation, when all mobile assets are situated at their initial locations reported in Appendix E. The volume of fraction of oil disappeared in an oil slick due to evaporation (dimensionless) can be expressed as (Li et al., 2012):

$$FE = 1 - e^{\left(-1.5\times10^5\times\frac{T\times U^{0.8}}{M^2}\times\frac{P\times M}{\rho_m\times R\times T}\times\frac{OSA}{OSV}\right)\times t} \quad \text{Equation 7-4}$$

Considering an appropriate time step (taken as 1 hr in this study), the volume of each oil slick is updated, and *FE* is calculated at each time step. Then, the evaporated volume of each oil slick is calculated by multiplying *FE* at each time step with the initial volume of oil slick. It should be noted that ρ_m is also variable and can calculated based on evaporation and emulsification (Equation 7-7). The initial value is the parent oil density (ρ_{m1}) at the beginning of the operation that can be used to calculate the initial volume of fraction of oil disappeared due to evaporation.

- Dispersion effects on the volume of oil slicks:

The dispersion is affected by *U*, *t*, *OSV*, *OSA*, interfacial tension of oil and water (S_t) in dyne/m, and oil viscosity (μ_m) in cP. The volume of fraction of oil disappeared in an oil slick due to dispersion (dimensionless) can be expressed as (Li et al., 2012):

$$FD = 0.11 \times \frac{(U+1)^2}{1 + 50\sqrt{\mu_m} \times S_t} \times \frac{OSA}{OSV} \times t \qquad \text{Equation 7-5}$$

Considering the time step, the volume of each oil slick is updated, and *FD* is calculated at each time step. Then, the dispersed volume of each oil slick is calculated by multiplying *FD* at each time step with the initial volume of oil slick. It should be noted that μ_m is also variable and can calculated based on evaporation and emulsification (Equation 7-8). The initial value is the parent oil viscosity (μ_{m1}) at the beginning of the operation that can be used to calculate the initial volume of fraction of oil disappeared due to dispersion.

- Emulsion effects and density and viscosity update:

The emulsification can affect the density and viscosity of the spilt oil and accordingly other weathering processes. The emulsion effects can be expressed in terms of the fractional water content (*FW*) using the following equation based on curve fitting constant (K_a) and mousse viscosity constant (K_b) (Li et al., 2012; Sarhadi zadeh and Hejazi, 2012):

$$FW = K_b \left[1 - e^{\left(-K_a / K_b (U+1)^2 t\right)}\right] \qquad \text{Equation 7-6}$$

Then, the following equations can be used to update the oil density and viscosity values over time (Guo and Wang, 2009; Li et al., 2012):

$$\rho_m = FW \times \rho_w + (1 - FW) \times (\rho_{m1} + K_b \times FE) \qquad \text{Equation 7-7}$$

$$\mu_m = \mu_{m1} \times e^{(K_c \times FE)} \times e^{\left(\frac{2.5 \times FW}{1 - K_b \times FW}\right)} \qquad \text{Equation 7-8}$$

where ρ_w is water density in kg/m^3, and $1 \leq K_c \leq 10$ is a constant depending on the type of spilt oil.

Oil slick thickness can affect both recovery rate of a skimmer and properties of the spilt oil. The thickness of the slick is estimated here based on the volume and initial surface area of the slick. Within a coupled system of equations, the changes in the slick's volume are calculated with respect to weathering processes mentioned above and skimming process.

7.3 Hypothetical Case Study

After consulting oil spill responders in the western coasts of Canada, a hypothetical yet realistic case study was developed based on the Western Canada Operational Response Plans (WCMRC, 2017, 2020, 2021b). An oil spill responding organization in this area is currently certified to respond to up to 10,000 tons of spilt oil and intend to enhance their response capacity to 20,000 tons of spilt oil in Canadian West Coasts, concerning the risk assessment performed for Trans Mountain Expansion Project in Salish Sea, British Columbia, Canada (WCMRC, 2021b). Figure 7-3 illustrates the selected hypothetical incident of a tanker collision as a case study. This scenario was chosen based on considerations of available response capacity and risk assessment in the specific area under investigation. The incident location was deliberately assumed to be about 30 km away from the shoreline to emphasize the impact of available storage capacity on the response operation, which in turn leads to a longer operation time.

The hypothetical incident releases approximately 10,000 tons of heavy crude oil near the entrance of Juan de Fuca Straight in Salish Sea. Accordingly, it is presumed that the total number of seven oil slicks was boomed within an approximate radius of 15 km. To respond to this incident, six skimming vessels, 20 mini barges, two big barges, 25 trucks, and two treatment facilities were assumed in total, considering the available response capacity in the study area (WCMRC, 2021b). The location of Big Barge 2 is selected as the center of a Cartesian coordinate system to determine relative locations of different assets. All other required input and assumptions are listed in Appendix E. Inputs were chosen based on available documents (Ensor, 2018; Li et al., 2012; OTA, 2019; WCMRC, 2017, 2020, 2021b; Ye et al., 2021) as well as extensive consultation with

oil spill responders in the study area. The initial locations (i.e., the coordinates at time zero of the optimization) of response assets are also tabulated in Appendix E. It is assumed that all the mobile assets will return near their closest big barge at the end of the operation when no oil slicks are left and vessels, mini barges, trucks, and big barges are empty. The associated times and costs of this process were added to the overall time and cost of the operation.

In order to simulate the process of oil slick removal, certain assumptions were made. It was assumed that the volume associated with each oil slick represents the total volume of the oil-water mixture needed to be collected and transported to treatment facilities, and also subjected to weathering processes, as previously assumed in similar studies (Li et al., 2012; P. Li et al., 2014b; Ye et al., 2021, 2019). Any potential decanting and treatments required for the collected oily wastewater would be carried out at the treatment facilities. It is also assumed that a fuel dock is available on-site for refueling the mobile assets used in offshore stage. Expenses related to the required boom vessels, landing crafts, and work boats are not considered in calculating the cost for the purpose of simplification. Costs associated with non-durable/consumable equipment such as personal protective equipment and sorbents are considered in a lump sum. It is also assumed that the two treatment facilities are qualified to receive industrial hazardous oily wastewater.

Since the skimming rates may vary due to change in oil slick volume (Li et al., 2012), it is presumed that each vessel can be equipped with two available skimmers with variable skimming rates depending on the oil slick thickness. The skimming capacity of vessel *j* (VSK_j) working on oil slick *i* at each time step can be calculated based on the following regression equation for these skimmers, developed by previous studies (Li et al., 2012):

$$VSK_j = max\{5, \quad VNS_{j1} \times ORR_1 + VNS_{j2} \times ORR_2\} \qquad \text{Equation 7-9}$$

$$= max\left\{5, \quad VNS_{j1} \times \left[0.01437 \times \left(10^3 \times \frac{OSV_i}{OSA_i}\right)^2 + 0.01602 \times \left(10^3 \times \frac{OSV_i}{OSA_i}\right) + 0.05829\right] + VNS_{j2} \times \left[-0.00791 \times \left(10^3 \times \frac{OSV_i}{OSA_i}\right)^2 + 0.84975 \times \left(10^3 \times \frac{OSV_i}{OSA_i}\right) + 0.19929\right]\right\}$$

where *ORR*, *VNS*, *OSV*, and *OSA* are oil recovery rate of skimmers (m^3/hr), number of skimmers, volume of oil slick (m^3) and area of oil slick (m^2), respectively. The minimum skimming capacity of 5 m^3/hr is assumed for each skimmer since complete removal of spilt oil is needed.



Figure 7-3. An illustration of the hypothetical oil spill incident considered for optimization

7.4 Results and Discussion

The tool developed in this chapter was tested on the case study to demonstrate its performance in scheduling the response assets. Initially, the developed mathematical model and oil weathering process agents were used to assign mobile assets based on SDA. Subsequently, the optimization agent, along with two other agents, was employed to schedule the response assets while minimizing two objective functions, f_1 and f_2 in OPA-II. Then, a three-objective optimization has also been performed to minimize all f_1, f_2, and f_3 simultaneously in OPA-III.

7.4.1 Two-Objective Optimization

Figure 7-4 illustrates the variation in the overall operation time and cost in solutions over the final Pareto front of OPA-II as well as the solution obtained from SDA. The OPA-II Pareto offers

decision-makers a range of optimal solutions from which they can select the most desirable one based on their preferences to response time and cost. As observed in Figure 7-4, the best-performing optimal solution in terms of operation time, resulted in a 13.6% reduction in time compared to SDA, but increased the cost slightly (i.e., 0.7%). Alternatively, the best-performing solution in terms of operation cost reduced the cost by 1.4% and the operation time by 8.4%. Figure 7-4 also presents one of the candidate solutions over the Pareto selected for further evaluations for balancing the low operational cost and time.

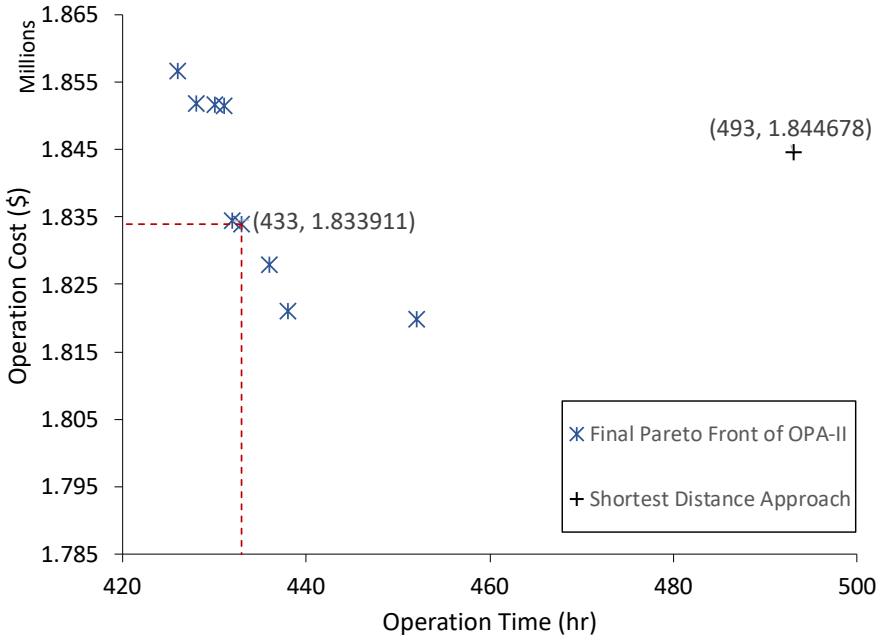


Figure 7-4. Variation of time and cost in the final Pareto front of OPA-II and SDA

Table 7-4 compares the total costs of different assets as well as the entire operation cost, time, and volume of weathered oil in SDA and OPA-II. While OPA-II exhibited superior performance in terms of entire operation cost and time, SDA demonstrated better results in terms of the volume of weathered oil with a reduction of about 32% compared to OPA-II. This led to a higher recovery of spilt oil in SDA. OPA-II demonstrated a shorter overall operation time than SDA, which includes the time required for oily wastewater transportation to big barges and treatment facilities in addition to the time needed for oil slick removal. OPA-II effectively reduced the overall operation time by optimizing the scheduling of the available assets during the entire operation. However,

OPA-II resulted in a longer duration for eliminating oil slicks from the marine environment, leading to a larger volume of weathered oil.

Table 7-4. Overall comparison of SDA and selected OPA-II solutions

		Vessels	Mini barges	Big barges	Trucks	Treatments
SDA	Operation ($)	469,168.8	40,611.9	300,460.0	0.0	334,762.6
	Transportation ($)	10,363.6	88,005.6	0.0	601,305.0	0.0
OPA-II	Operation ($)	451,946.8	40,241.2	287,445.0	0.0	333,120.5
	Transportation ($)	10,223.3	92,378.7	0.0	618,555.0	0.0
Difference	Operation (%)	-3.7	-0.9	-4.3	0.0	-0.5
	Transportation (%)	-1.4	5.0	0.0	2.9	0.0
Overall Difference	Cost (%)	-0.6				
	Time (%)	-12.2[a]				
	Weathered oil (%)	32.0[b]				

[a] Entire durations were 433 hr and 493 hr in OPA-II and SDA, respectively.
[b] Volumes of weathered oil were 273 m^3 and 206 m^3 in OPA-II and SDA, respectively.

Table 7-5 to Table 7-8 provide a detailed analysis of SDA and OPA-II concerning the cost and workload associated with each mobile asset and treatment facility throughout the entire operation. Coefficients of variation (CVs) are included in Table 7-5 to Table 7-7 to provide insights into the variation and distribution of cost and workload across different assets.

Table 7-5. Comparison of operational cost and workload of vessels in SDA and OPA-II

Vessel No.	SDA		OPA-II	
	Total cost ($)	Total skimmed oil (m^3)	Total cost ($)	Total skimmed oil (m^3)
1	12,2463.8	3,231.0	13,3814.4	2,683.1
2	3,0631.0	1,486.7	27,936.2	1,478.8
3	10,0887.6	2,007.7	91,105.9	1,607.4
4	104,893.4	1,668.6	74,779.6	1,974.9
5	98,825.3	2,095.5	103,087.6	1,985.8
6	21,831.5	905.0	31,446.4	1,596.9
CV	0.49	0.37	0.49	0.21

SDA indicates larger values of CV in terms of the total skimmed oil by vessels, as well as the cost and volume of oily wastewater transported by mini barges and trucks. These exhibit an uneven distribution of workload compared to OPA-II with considerably smaller CV values. In contrast, OPA-II tends to use all available mobile assets through a more balanced distribution of workload

among them. This approach shows a more efficient use of mobile assets considering all stages of the operation, resulting in a reduction of the overall operation time.

Table 7-6. Comparison of different mini barges in SDA and OPA-II

Mini Barge No.	SDA		OPA-II	
	Total cost ($)	Total transported oily wastewater (m^3)	Total cost ($)	Total transported oily wastewater (m^3)
1	16,369.5	1,080.0	12,560.3	760.0
2	19,024.9	1,187.7	12,918.7	720.0
3	11,751.7	1,120.0	8,450.1	730.0
4	12,022.4	1,138.6	7,431.2	676.8
5	11,540.6	1,040.0	8,431.0	740.0
6	6,306.5	714.7	4,865.5	543.1
7	6,596.1	748.5	6,128.7	706.9
8	6,034.4	680.0	5,634.1	617.4
9	5,137.8	560.0	5,131.4	560.0
10	4,466.6	480.0	4,920.9	560.0
11	4,808.4	520.0	5,660.7	600.0
12	4,300.8	480.0	3,593.5	390.0
13	2,755.1	205.0	5,761.4	478.8
14	2,278.4	240.0	4,798.1	530.0
15	5,943.0	320.0	7,702.7	459.0
16	3,931.5	360.0	5,711.7	480.0
17	2,094.2	240.0	3,291.3	360.0
18	1,402.6	80.0	11,009.6	634.9
19	528.9	40.0	5,357.3	420.0
20	1,324.2	160.0	3,261.7	360.0
CV	0.78	0.65	0.41	0.22

The discrepancy between CV values of SDA and OPA-II are particularly noticeable in case of truck scheduling, as indicated in Table 7-7. Truck scheduling significantly affected operation time due to the large driving distances between big barges and treatment facilities. The evaluation of the results revealed that deployed trucks in SDA travelled a total distance of 364,970.0 km during the entire operation, whereas in OPA-II, the equivalent value was 354,820.0 km (i.e., about 2.8% less mileage). Scheduling of trucks optimized by OPA-II ensures that all trucks are efficiently used, leading to the minimum time required for wastewater transportation. While refueling times for vessels and mini barges were insignificant, it accounted for approximately 2.5% of total transportation time for trucks due to these considerable travelled distances. However, it is

important to note that if a fuel dock was not available on-site, the associated refueling times for mobile assets in the offshore stage could potentially cause significant delays in the operation.

Table 7-7. Comparison of different trucks in SDA and OPA-II

Truck No.	SDA		OPA-II	
	Transportation cost ($)	Total transported oily wastewater (m^3)	Transportation cost ($)	Total transported oily wastewater (m^3)
1	270.0	10.0	27,315.0	450.0
2	555.0	10.0	24,705.0	430.0
3	43,605.0	710.0	25,065.0	400.0
4	43,920.0	710.0	22,455.0	420.0
5	44,820.0	730.0	19,710.0	330.0
6	43,020.0	700.0	18,690.0	320.0
7	42,345.0	700.0	23,355.0	410.0
8	42,165.0	690.0	24,390.0	370.0
9	46,700.0	860.9	23,820.0	480.0
10	44,000.0	840.0	24,660.0	458.0
11	39,555.0	710.0	20,715.0	350.0
12	36,225.0	690.0	24,450.0	410.0
13	26,280.0	550.0	22,605.0	390.0
14	23,145.0	520.0	19,800.0	330.0
15	26,480.0	653.6	25,140.0	435.0
16	26,180.0	630.0	25,080.0	525.0
17	24,920.0	570.0	27,440.0	573.8
18	19,700.0	435.0	24,360.0	465.0
19	13,460.0	285.0	25,460.0	465.0
20	6,740.0	150.0	28,100.0	495.0
21	3,120.0	105.0	24,680.0	510.0
22	2,160.0	75.0	25,580.0	525.0
23	720.0	30.0	28,380.0	540.0
24	1,220.0	30.0	28,460.0	570.0
25	0.0	0.0	34,140.0	675.0
CV	0.73	0.66	0.13	0.19

Table 7-8. Treatment operation comparison in SDA and OPA-II

Treatment No.	SDA		OPA-II	
	Operation cost ($)	Received oily wastewater (m^3)	Operation cost ($)	Received oily wastewater (m^3)
1	174,650.0	4,990.0	174,825.0	4,995.0
2	160,112.6	6,404.5	158,295.5	6,331.8

According to the data presented in Table 7-8, both SDA and OPA-II used almost the entire capacity of Treatment Facility 1. This can be associated with the favorable proximity and better

road access of Treatment Facility 1 to the big barges, particularly Big Barge 1, which is closer to the spill location. Treatment Facility 2 received the remaining oily wastewater due to its larger capacity and lower unit operation cost compared to Treatment Facility 1.

Figure 7-5 depicts the variations in the volume of oil slicks in two different approaches. It is evident that OPA-II (Figure 7-5b) requires more time to completely remove all the oil slicks from the water compared to SDA (Figure 7-5a). Figure 7-5 also demonstrates a more even distribution of vessels among the slicks in OPA-II. In the beginning of the operation, all the vessels were allocated to oil slicks 2, 6, and 7 in SDA.

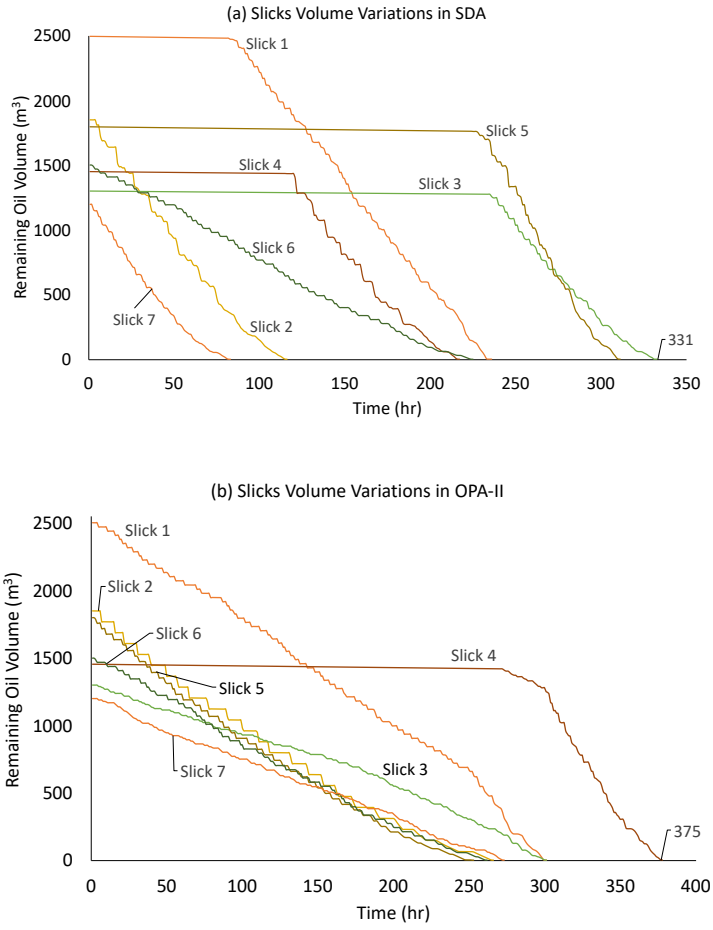


Figure 7-5. Variations in remaining volume of oil slicks in SDA and OPA-II

As the operation progressed and after the elimination of Slick 7 in SDA, its associated vessels were reassigned to Slick 1. Also, vessels assigned to Slick 2 at the start of the operation were

sent to Slick 4 after eliminating Slick 2. Slicks 3 and 5 were the last two slicks collected in SDA. There were also delays in the commencement of skimming for subsequent slicks due to the transportation time required for vessels to move between slicks and the potential time needed for offloading vessels to mini barges.

In OPA-II, all six available skimming vessels were deployed to deal with six different slicks. Throughout the operation, all slicks were continuously being skimmed, except for Slick 4. Slick 4 remained unattended until near the end of the operation. Lastly, three vessels were assigned to Slick 4, concerning the maximum number of vessels that could be concurrently dispatched to this particular slick, as mentioned in the input parameter table provided in Appendix E.

Figure 7-6 presents a comparison of the change in oily wastewater stored in the big barges between SDA and OPA-II. The volumes of stored oily wastewater are increased up to 10% of the capacity of each big barge, which serves as the activation threshold for initiating the loading of oily wastewater from big barges into trucks. After surpassing the 10% of the total capacity of big barges, fluctuations in the volume of stored oily wastewater are observed in both SDA and OPA-II due to the simultaneous delivery of oily wastewater from mini barges to the big barges and its transfer to trucks.

In terms of Big Barge 2, there is a balance between the input and output of oily wastewater in both SDA and OPA-II, resulting in relatively stable volumes of stored oily wastewater. However, Big Barge 1 received a greater volume of oily wastewater in both approaches compared to Big Barge 2 because it was closer to the oil slicks. Additionally, the input to Big Barge 1 exceeded the output for approximately two-thirds of the operation time, indicating an accumulation of stored oily wastewater. In SDA, the stored oily wastewater in Big Barge 1 reached a peak volume of 1,845 m^3, representing an approximately 37% increase compared to that in OPA-II.

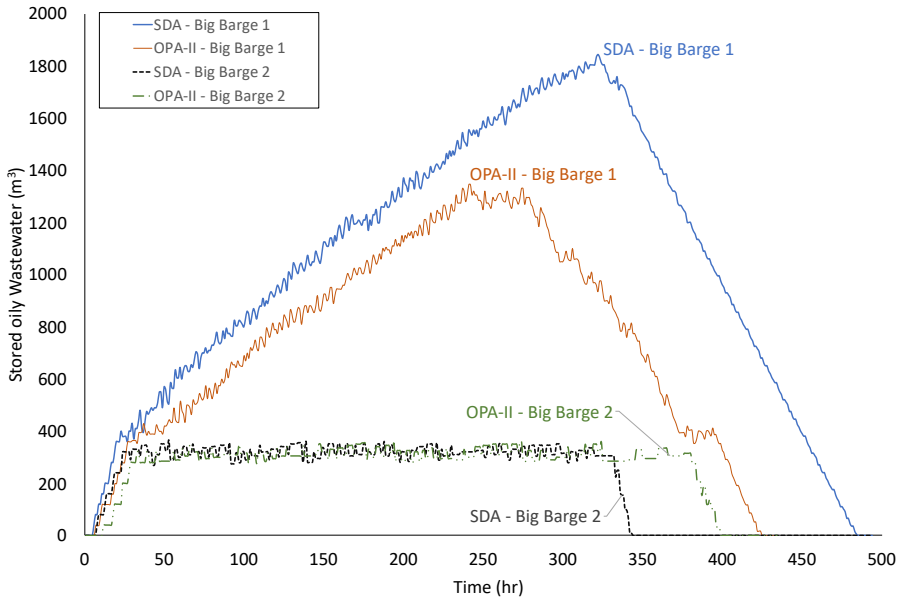


Figure 7-6. Volume of stored oily wastewater in big barges in SDA and OPA-II

There is also a noticeable delay between the peak values of stored oily wastewater in Big Barge 1 for the two approaches. The delay can be justified by considering the remaining oil slicks near the completion of approximately two-thirds of the operation. At that time, OPA-II had only Slick 4 remaining (Figure 7-5b), while SDA still had Slicks 3 and 5 (Figure 7-5a). As a result, when the declining phase of stored oily wastewater distribution in Big Barge 1 commenced in OPA-II (i.e., its falling limb in Figure 7-6), there was a reduced inflow of oily wastewater due to the presence of only one slick (i.e., Slick 4). In contrast, Big Barge 1 continued to receive significant amounts of oily wastewater from mini barges originating from two slicks (i.e., Slicks 3 and 5) in SDA. This inflow exceeded the capacity of available trucks to offload the big barge, resulting in a delay in the occurrence of peak volume of stored oily wastewater in SDA.

7.4.2 Three-Objective Optimization

In order to find an appropriate trade-off between overall operation time, cost, and oil recovery, the third objective function was considered in the optimization. This additional objective function aims to minimize the volume of weathered oil (i.e., maximize the volume of oil for possible recovery) calculated using the oil weathering process agent.

Figure 7-7 visualizes the final three-dimensional Pareto front (i.e., all optimal solutions) obtained from the OPA-III. The values of the objective functions in two potential solutions from this Pareto are also illustrated in this figure. In one of the potential solutions, the volume of weathered oil is almost identical to that of SDA, measuring 206 m^3. However, there is a reduction of approximately 9.5% in operation time and 0.6% in cost compared to SDA (see Figure 7-4 and Figure 7-7), showing the desirable performance of the optimization process.

The other potential solution represents the best-performing solution in terms of oil spill recovery. It demonstrates a reduction of approximately 9.2% in the volume of weathered oil compared to SDA. Figure 7-8 depicts the variation of weathered oil in this solution along with selected solution in OPA-II and SDA. The reduction in the volume of evaporated and dispersed oil in OPA-III led to the highest value of oil recovery achieved within the shortest time, indicating a quicker removal of oil slicks.

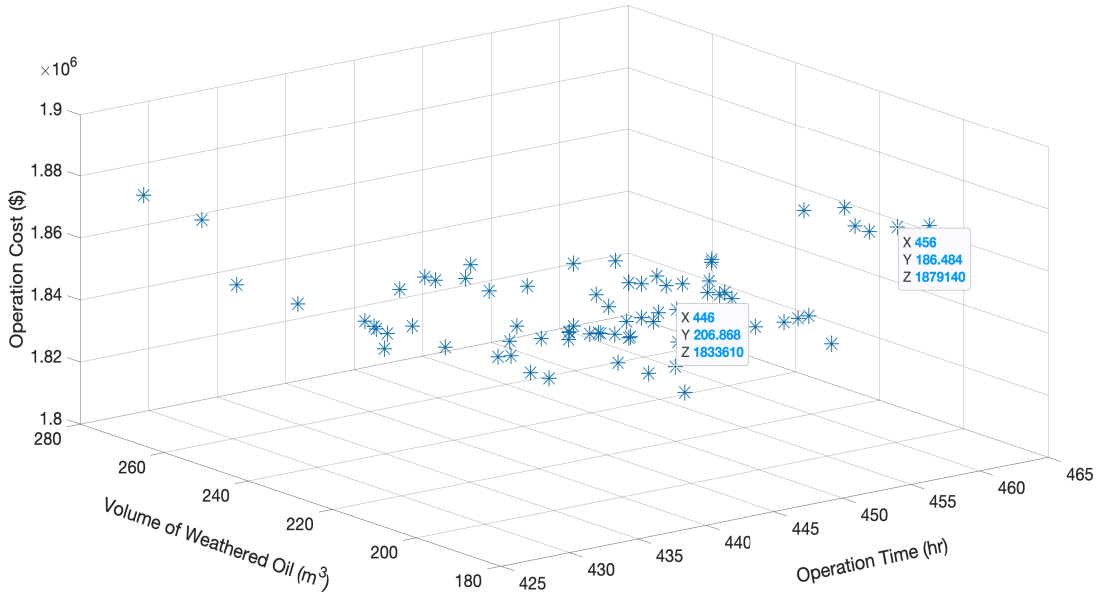


Figure 7-7. Final Pareto front of three-objective optimization (OPA-III)

The best-performing solution of OPA-III in terms of oil spill recovery experiences a 7.5% reduction in operation time and a 1.9% increase in cost compared to SDA. Considering the interdependence

of the objective functions, the selection of the optimal solution requires decision-makers to consider their priorities.

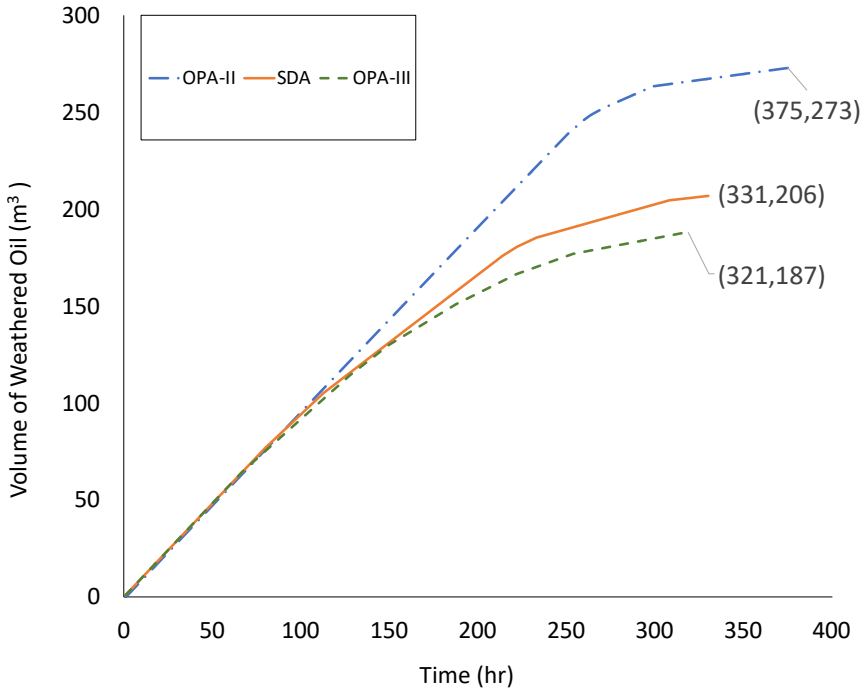


Figure 7-8. Volume of weathered oil in three different approaches

It is worth noting that the vessels are equipped with different types and numbers of skimmers, and the area and thickness of the boomed oil slicks vary (as listed in Appendix E). Also, the efficacy of different skimmer types depends on the thickness of the oil slick (Equation 7-9), which itself varies over time. Therefore, having the appropriate type of skimmers operational at the right time can accelerate the process of oil spill removal. Additionally, parameters such as scheduling of mini barges also notably impact the rate of oil recovery from the slick through providing vessels with the required storage capacity. For further evaluations, Figure 7-9 and Figure 7-10 compare the oil slick removal over time. Figure 7-9 shows the variations in the volume of remaining oil in different slicks based on the OPA-III solution with the best oil spill recovery. In Figure 7-10, the variations of the remaining oil volume over time and the volume of skimmed oil in different time steps are specifically presented for Slick 3 on which SDA has the poorest performance. This figure

also illustrates the number and specifications of the skimming vessels assigned to Slick 3 in each approach.

As can be seen in Figure 7-9, OPA-III demonstrated a skimming vessel assignment pattern similar to that of SDA. However, it used the storage and skimming capacity of vessels as well as the storage capacity of mini barges in a more effective way (more like to OPA-II) to collect the spilt oil. As a result of this improved efficiency, OPA-III achieved a faster elimination of the spilt oil, surpassing SDA and OPA-II by approximately 9% and 14%, respectively. However, this accelerated oil recovery came at the expense of an increased overall operation time and cost in OPA-III in comparison with OPA-II. Specifically, the operation time was extended by 5.4%, and the cost increased by 2.5% compared to OPA-II. These results also align with the observations regarding the completion time of operations associated with each network component, as shown in Table 7-9. OPA-II significantly reduced operation time for the scheduling of trucks between big barges and treatment facilities, although it resulted in longer operation times for the slick, vessel, and mini barge components. In contrast, OPA-III achieved time savings in all components of the network, though to a lesser extent in truck scheduling compared to OPA-II. Based on Table 7-9, the slick removal operations were completed first. Subsequently, the mini barges received the stored oily wastewater from the last active vessels, and they all returned to the nearest big barge. Obviously, the arrival time of the mini barges and vessels depended on their location and speed during the final deployment. Since the trucks were the last assets to return to the closest big barges, their completion time aligned with the end of the operation.

According to Figure 7-10, SDA shows the poorest performance in collecting Slick 3, as it assigns Vessels 1 and 2 to this slick after 234 hours from the start of the operation. It is important to note that Vessel 1 is a high-speed vessel with relatively excellent capabilities, including three skimmers and a large internal storage capacity. On the other hand, Vessel 2 has only two skimmers and the lowest storage capacity among all the vessels. The combination of five skimmers in SDA

results in the highest volume of skimmed oil at different time steps compared to the other two approaches.

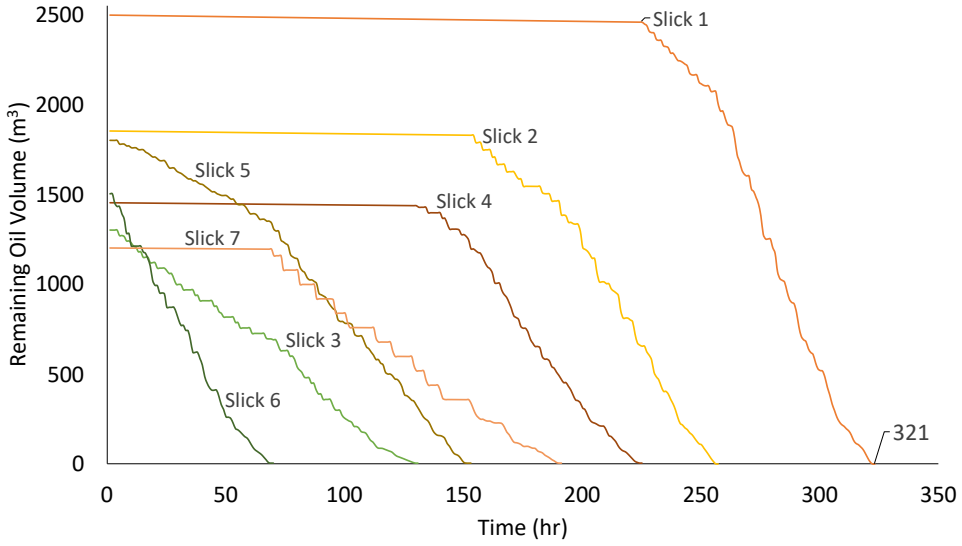
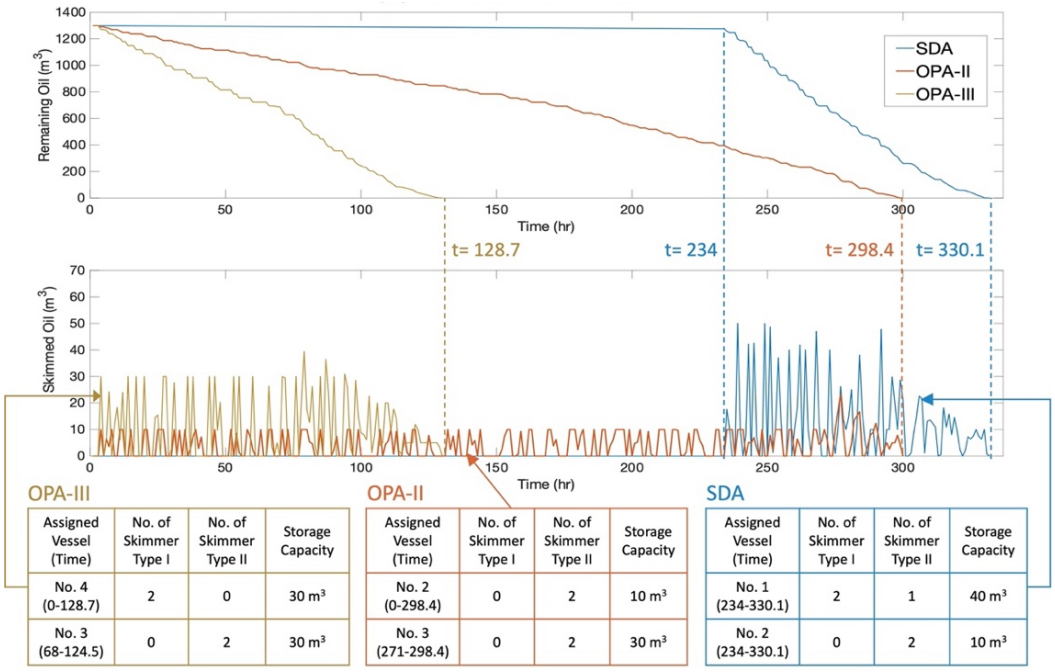


Figure 7-9. Oil slicks removal in OPA-III



OPA-III

Assigned Vessel (Time)	No. of Skimmer Type I	No. of Skimmer Type II	Storage Capacity
No. 4 (0-128.7)	2	0	30 m³
No. 3 (68-124.5)	0	2	30 m³

OPA-II

Assigned Vessel (Time)	No. of Skimmer Type I	No. of Skimmer Type II	Storage Capacity
No. 2 (0-298.4)	0	2	10 m³
No. 3 (271-298.4)	0	2	30 m³

SDA

Assigned Vessel (Time)	No. of Skimmer Type I	No. of Skimmer Type II	Storage Capacity
No. 1 (234-330.1)	2	1	40 m³
No. 2 (234-330.1)	0	2	10 m³

Figure 7-10. Variations of skimmed and remaining oil volumes in Slick 3 in different approaches

In contrast to SDA, OPA-II assigned Vessel 2 from the beginning of the operation. Furthermore, Vessel 3 equipped with the same type of skimmers joined the efforts to remove Slick 3 after 271

hours since the start. This sequential scheduling from the start of the response allowed for a smooth and continuous removal of the oil slick throughout the entire operation.

Table 7-9. Completion time (hr) for operations related to each component in different approaches

	SDA	OPA-II	OPA-III	Difference compared to SDA (%)	
				OPA-II	OPA-III
Slick	331	375	321	13.3	-3.0
Vessel	338	386	328	16.3	-1.2
Mini barge	337	393	332	16.6	-1.5
Big barge	484	424	447	-14.0	-9.3
Truck	493	433	456	-12.2	-7.5
Treatment	488	429	451	-13.0	-8.5

Alternatively, OPA-III adopted a different approach for eliminating Slick 3. It deployed Vessels 3 and 4, which were equipped with different types of skimmers compared to each other and had relatively high storage capacities among all available vessels. This vessel assignment from the start of the operation enabled the fastest removal of Slick 3 (i.e., 128.7 hr) among the three approaches.

Overall, the evaluation of the results indicated that OPA-II proves to be an effective approach for optimizing the distribution of workload among available assets, outperforming other approaches in terms of reduced overall operation time and cost. Alternatively, SDA is a desirable approach in terms of oil spill recovery compared to OPA-II but by compromising the entire operation time. OPA-III is the most suitable approach to consider all former influential parameters in scheduling of available resources. OPA-III can find an appropriate trade-off between conflicting objective functions of time, cost, and oil spill recovery. The solutions obtained from both OPA-II and OPA-III can be considered by decision-makers for effective management of the operation based on their priorities.

7.5 Summary

Management of marine oil spill response operations is a complex and time-sensitive task, given the serious consequences of large-scale oil spill incidents and the numerous factors that must be considered for deploying available response assets efficiently. While MCR is the most common response method for removing spilt oil from the marine environment, its effectiveness can be compromised by response capacity limitations in the affected areas, such as available storage and transportation capacities of oily wastewater and the availability of fleets of response vessels. Accounting for all the influential factors in decision-making is a highly demanding task even for experienced oil spill response practitioners. Consequently, there is a dire need to develop an integrated optimization-based tool that can efficiently utilize accessible response resources to manage both MCR and the generated oily wastewater.

In this chapter, an innovative decision support tool was developed working as a multi-agent system to minimize the operation time and cost of MCR and OWM as well as the volume of weathered oil through multi-objective optimizations. The tool was applied to a large hypothetical oil spill incident in Western Canada. Using the developed tool, it was initially possible to schedule available assets using SDA whilst considering all operational details to compare its results with optimized solutions. Following that the scheduling of available assets was optimized through different multi-objective optimization approaches. The evaluation of the results indicated that the comprehensive optimization-based tool presented in this study led to a reduction of the entire operation time and cost as well as an increase in the volume of recoverable oil. Two-objective optimization resulted in a 13.6% (67 hr) reduction in entire operation time and 1.4% ($24,800) reduction in operation cost compared to SDA. The three-objective optimization also reduced the volume of weathered oil by about 9.2% in comparison with SDA, suggesting that more spilt oil could be recovered.

According to the results, OPA-II is recommended for minimizing overall operation time and cost. For a more holistic optimization that considers time, cost, and spilt oil recovery, OPA-III proves to

be the most suitable choice. Additionally, the Pareto fronts obtained from the optimization process provides decision-makers with multiple optimal scheduling strategies that can be selected for maximum efficiency. By efficiently utilizing available resources, the developed decision support tool can reduce the negative environmental and social impacts of oil spill incidents and minimize response operation costs.

Chapter 8 – Conclusions

8.1 Introduction

The effective management of marine oil spill incidents is essential to mitigate the catastrophic consequences, especially in harsh and fragile marine environments. MOSM is a complex practice affected by various factors, such as characteristics of spilt oil and environmental conditions. Oil spill detection, characterization, and monitoring; risk evaluation; response selection and process optimization; and waste management are the major components of MOSM demanding timely decision-making. Adopting and applying robust computational techniques, in particular AI and SC techniques may considerably facilitate decision-making processes in different components of MOSM.

This study examined the role of advanced computational techniques in MOSM and aimed to develop several AI/SC-based decision support tools to assist in decision-making for oil spill response and waste management. To this end, a framework was designed in order to:

- perform content analysis to identify available gaps in the application of intelligent computational techniques in MOSM and collect relevant data for developing decision support tools for its key components;
- develop multiple intelligent OSRM selection tools to timely select response methods based on operational suitability using capable AI/SC techniques that mimic expert judgment; and
- conduct multi-objective optimization of integrated oil spill response and waste management operations to minimize the negative environmental impacts and associated response costs.

In this direction, a variety of AI/SC techniques, including multiple ANNs, advanced regression-based ML models, FDTs, and NSDE, were explored to develop decision support tools for MOSM. The findings of this study revealed that FDT-based tools, combined with regression analyses for OSRM selection, offer a transparent and effective strategy for response practitioners to select the most suitable response method. These tools generate easily interpretable "if-then" rules that

decision-makers can use for OSRM selection, offering a profound understanding of the ranking of attributes' influence.

Furthermore, integrating multi-objective metaheuristic optimization into the development of FDTRs, significantly improved their predictive power. Optimized FDTRs, along with GPR, outperformed other evaluated techniques for OSRM selection in this study, particularly under data constraints.

Alternatively, the integrated optimization of MCR and OWM operations through a sophisticated multi-agent system led to a notable reduction in overall operation time, cost, and the volume of weathered oil when compared to conventional resource scheduling techniques. The application of the developed tools of present study has the potential to reduce the adverse environmental impacts of oil spill incidents while simultaneously minimizing the associated operational costs of response efforts.

8.2 Originality and Contributions

AI and SC are among the most effective computational techniques for supporting decision-making. It is necessary to perform a comprehensive study focusing on the application of these techniques in MOSM. This research thoroughly evaluated the essential role of intelligent computational techniques in MOSM and develop several AI/SC-based decision support tools to facilitate marine oil spill response.

Applying ML techniques in OSRM selection can reduce the reliance on experienced decision-makers and generate accurate, immediate selection under complex conditions due to a wide range of influential attributes. The ML-based tools can consider the subtle patterns in the OSRM selection process that cannot be easily discerned by human brains. Furthermore, a flexible metaheuristic optimization algorithm used within a sophisticated multi-agent system for MCR and OWM coordination can further augment the capacity to create more comprehensive scheduling

tools. This can be accomplished by considering the intricate operational procedures of response operations and optimizing response efficacy, without the need for excessive simplifications.

The developed tools in this study could also be beneficial to oil spill response practitioners in training and contingency planning as the frequency of oil spill incidents has decreased. The novelty of the present work lies in:

- Conducting an in-depth evaluation of computational techniques in MOSM and identifying knowledge gaps for future research needed to enhance oil spill response operations;
- Developing fuzzy-based oil spill response selection tools to assist response practitioners in selecting suitable OSRMs for offshore oil spills under uncertainty;
- Developing practical decision support tools for timely OSRM selection in remote and harsh environments;
- Evaluating and comparing the performance of various advanced AI/SC techniques for OSRM selection;
- Applying innovative FDT-based tools and optimizing them for OSRM selection, mimicking experts' cognition of the operational suitability of response methods;
- Developing state-of-the-art techniques to enhance the prediction power of available AI/SC-based decision-making tools;
- Integrating the optimization of MCR and liquid oily waste management operations to minimize the time and cost of response tasks and maximize oil recovery;
- Considering the detailed operational procedures of both MCR and OWM operations within a multi-agent system, avoiding extensive simplifications; and
- Optimizing the spill response process and oily waste management in Canadian West Coasts using advanced optimization techniques to enhance response and waste management efficacy.

8.3 Limitations

Several challenges and limitations were encountered during this research. The main limitation in developing the OSRM selection tools was the lack of real-world datasets of oil spill incidents containing information of influential attributes and effective response methods implemented, particularly oil spill incidents in Arctic waters. Several promising AI/SC techniques were used to develop robust and accurate OSRM selection tools; however, other capable techniques can also be integrated into the suggested models to improve the prediction power of these tools. Furthermore, the influential attributes and their definitions and levels used in this study cannot encompass all factors affecting OSRM selection. Also, the spill cases are hypothetical and the judgments on the suitability of three response methods were subjectively made.

The lack of oil spill incident data also affected the integrated optimization of MCR and OWM. A hypothetical oil spill scenario was developed in the integrated optimization of MCR and OWM to overcome the data constraint. Although operational details of MCR and OWM were considered in the optimization, some simplifications were still assumed. Some considerations in MCR and OWM were not included in the integrated optimization, including oily solid waste generated from response operations; the precise locations of refueling facilities for vessels, mini-barges, and trucks; the detailed use of certain response assets like booms, sorbents, and work boats; advection effects on the transport of spilt oil; potential impacts of high and low tide conditions, and availability of incineration and oil refinery facilities or onsite decanting technologies for waste management.

It should be mentioned that the lack of real-world datasets may cause uncertainties affecting the accuracy of the mentioned decision support tools. Nevertheless, the developed models and associated tools can be used in future studies in case of more reliable data become available.

8.4 Recommendations

To address the limitation of this research and current gaps in the application of intelligent computational techniques in MOSM, following recommendations are made:

- One major challenge in the development of OSDCM tools is the lack of sufficient usable satellite/aerial images of oil spill incidents. The application of DL models for semantic/instance segmentation requires a considerable number of samples. Although the requirement can be lowered using pre-trained DL models and data augmentation techniques, there is still a significant need for image datasets with sufficient samples.
- Metaheuristic optimization algorithms have a great potential to be used for selecting the best combination of features in developing OSDCM tools. This can be done by defining appropriate objective functions of the accuracy and number of features. To this end, metaheuristic algorithms may be used to find the optimal combination of training datasets for better performance in the two objective functions.
- Lack of oil spill incident data and significant dependence on experts' judgment are frequently identified as shortcomings in the development of OSRM selection/management and risk estimation tools. Real-world oil spill incident data should be recorded and associated influential attributes should be evaluated to address the data-related shortcomings.
- Although the attributes and hypothetical oil spill cases for OSRM selection tools were developed subjectively in this study, they can be modified by different decision-makers to suit specific cases differed by response equipment requirements, expert knowledge, and local environmental conditions. Some other attributes affecting the selection of OSRM could be enumerated as operational cost, workplace safety, primary and secondary environmental risks, and societal impacts.
- Overall, it is encouraged to generate additional hypothetical data for training the OSRM selection tools to consider the features of the local environment. The sufficiently trained tools can be used by oil spill response practitioners in training process and contingency planning. Even by generating localized, hypothetical yet realistic oil spill data, response

practitioners can train the tools to enhance its prediction performance and increase its usefulness in the context of local oil spill response.

- A holistic computation-based framework can be developed for MOSM considering the usefulness of multiple AI/SC-based tools in oil spill detection and monitoring, risk assessment and contingency planning, adaptive response method selection, and response and waste management process optimization. Various modules/sub-modules of this framework can be mutually affected by their neighbours and dynamically update each other as parts of a multi-agent system.
- Compared to previously developed optimization models, the decision support tool of the present study for MCR and OWM optimization considers significantly fewer simplifications to maximumly simulate the real-world problems. But future studies can incorporate other operational details, such as the management of solid oily waste and the detailed use of additional response assets like booms, sorbents, and work boats. Moreover, the oil weathering process agent of the tool can be improved through the application of more comprehensive oil spill fate and transport simulators that also consider advection and biodegradation effects.
- In future studies, it would also be beneficial to optimize the number and specifications of available response assets in the MCR and OWM. Particularly, the multi-objective optimization can also focus on the number of skimming vessels, mini barges, big barges, and trucks, and consider their associated specifications such as storage capacity, skimming capacity, and other relevant factors.
- Appropriate uncertainty assessment techniques (e.g., the Monte Carlo simulation technique) can also be incorporated into the developed multi-agent system to account the potential effects of input parameters' variations in the real-world situations.

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Appendices

Appendix A – Summary of Reviewed Articles in ODCM

Spill detection, characterization, and monitoring papers

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
(Giri et al., 2011)	Development of a monitoring and classification framework for mangrove forests, which are one of the most vulnerable and sensitive plants to oil spills, using RS	- Decision tree - Aerial and satellite optical images - See5 was used to generate the DT	- Providing variations in mangrove forest cover during a long-term period in Louisiana, US - Developing a conceptual framework for monitoring and impact assessment on mangrove forests - Comparing the performance of models working based on aerial photographs and satellite images on mangroves detection	The developed decision tree can classify objects of interest into three groups of mangrove, terrestrial non-mangrove, and water. Oil impacts on mangroves and their recovery rate can be assessed using this framework
(Topouzelis and Psyllos, 2012)	Evaluation of different feature combinations for oil spill detection based on RS and DTs	- Decision tree forest - SAR image data	- Selection of the most influential features' combination in oil spill detection using SAR image data and binary classification - Application of enhanced version of decision tree for oil spill detection - Comparison of different AI-based models and selected features based on obtained accuracies	The total number of 25 features are extracted from 24 images, annotated manually, for training DTs. To reduce the bias of a DT to the selection of a training dataset, a set of different DTs are generated using randomly selected datasets voting for the classification
(Khanna et al., 2013)	Assessment of oil spill impacts on coastal saltmarsh and evaluation of the recovery process using DT	- Decision tree to classify images based on vegetation and angle indices - Linear regression for evaluation of shoreline recovery with time - Airborne visible infrared imaging spectrometer (AVIRIS) imagery - Automated image registration technique as	- A list of influential vegetation indices and their formulation were provided - Effects of oil spills on sensitive vegetation shorelines and species were evaluated - DT can classify the affected area into different regions of concern to prioritize spill management procedure	- One part of the study is about training and validation of DTs for the classification of images. Images were classified into six different groups, including water, soil, dry plant material or non-photosynthetic vegetation (NPV), photosynthetic vegetation, oiled soil and oiled NPV. Also, two groups of oil-contaminated and non-contaminated vegetations were evaluated - The other part of the study is about the evaluation of vegetation indices for oil-contaminated and non-contaminated shorelines to assess the effect of spilt oil

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
		a preprocess aligning images - ArcGIS for calculations of land distance to shorelines - ANOVA and Tukey-HSD test for statistical comparison of contaminated or non-contaminated shorelines		on the vegetation density and the recovery process
(Guo and Zhang, 2014)	Selection of influential geometrical (shape) features for oil spill detection based on SAR image data and ANN	- Differential evolution feature selection (DEFS) to determine discriminative features - An ANN and a deep-learning technique for classification of features - SAR image data	- Nine important shape features in image processing have been identified - The most discriminative parameters pertaining to features have been determined - Enhancing accuracy and sensitivity of the binary classification by reducing the number of influential features' parameters using DEFS	A total of 95 eigenvalues calculating shape features were considered: most of them from Elliptical Fourier descriptors capturing low frequent shape contents. DEFS was used to determine the most influential sets of decision supports based on their discrimination power in oil spill detection of an ANN
(Ma et al., 2014)	Oil spill detection based on SAR image data and AI-based models	- multilayer perceptron (MLP) and radian basis function (RBF) ANNs - SAR image data - Principal Component Analysis (PCA)	- Comparing MLP and RBF performance - Selection of influential features using PCA	Although RBF is a faster model, MLP outperforming based on the robustness criterion.
(Xu et al., 2014)	Comparing the performance of different statistical and ML models in oil spill detection using SAR image data	- MLP ANN - Bootstrap-aggregated tree-based techniques - Generalized additive model - SVM - Penalized linear discriminant analysis - k-fold cross-validation technique to train models	- Evaluation of a considerable number of spills from 2004 to 2008 - Addressing the imbalance problem in contaminated and non-contaminated pixels - Providing variation ranges of statistical parameters for different features - Performing log-transformation and normalization of datasets as the preprocessing step led to better predictions	- Bootstrap-aggregated tree-based techniques, including bagging and bundling, outperformed other evaluated models owing to the robustness and stability of their sampling technique - ANN showed the lowest prediction performance among the models

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
		- R software for programing - Kruskal-Wallis and Wilcoxon rank tests for statistical evaluation		
(Marghany, 2015)	Development of an evolutionary optimization model to automatically detect oil spills from SAR image data	- SAR image data - Genetic Algorithm (GA)	- Training a GA using fully polarimetric SAR image data to desirably discriminate between oil spills and look-alikes boundaries - Performing automatic segmentation through a heuristic optimization problem	The use of GA can help increase the accuracy of the prediction of oil spill areas
(Juahir et al., 2017)	Discrimination between different types of oil-based on spill fingerprints and classification models	- Six Sigma (6σ), which is a process and production improvement technique - Multivariate analysis techniques including principal component, cluster, and discriminant analyses (to classify oils) - ANOVA - Gas chromatography/flame ionization detector - Gas chromatography/mass spectrometry (used to identify the origin of oil spill samples experimentally) - JMP 10 and XLSTAT 2015 (for statistical analysis)	- Accelerating the complex process of oil spill classification with the desired accuracy - It is possible to determine the primary source of an oil spill in a study area	- The 6σ technique has been integrated with chemometric approaches to classify oil spills based on their chemical compounds obtained experimentally - Data reduction technique was used to identify the variable with the most discrimination power - Clustering techniques were used to generate homogeneous groups of spill sources - ANOVA was used to detect significant variation between different types of oil
(Chen et al., 2017)	Evaluation of DL models' performance in oil spill detection using polarimetric SAR image data	- Stacked autoencoder (SAE) DL model - Deep belief network (DBN) DL mode - SVM	- Sufficient explanations were provided for the construction and structure of appropriate DL networks - The performance of DL models was compared with conventional models (such	- DL models outperform conventional classification techniques, especially when smaller datasets are available due to lack of sufficient data to tune parameters in the latter

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
		- ANN - LIBSVM (a library to apply SVM)	as SVM and ANN) on the identification of oil spills on small to large datasets	- SVM outperformed ANN on small datasets due to the application of structural risk minimization led to classification margin maximization
(Mera et al., 2017)	Evaluation of different Feature Selection (FS) techniques to define efficient indicators for oil detection using SAR image data	- Correlation-based feature selection, consistency-based filter, Information Gain, Relief-F, and Recursive Feature Elimination for SVM (as five techniques of FS) - CMOD5 model to reduce low-wind effects - SAR image data - MATLAB and OpenCV to develop the model	- Performance of different FS techniques was evaluated - Several features to discriminate oil slicks and look-alikes were identified - A considerable number of oil spill and look-alikes cases that occurred from 2007 to 2011were evaluated - Details of influential features for oil spill detection were provided - The developed model has been uploaded on GitLab	- A link for the developed model: https://gitlab.citius.usc.es/cograde/feature-selection-in-SAR-images - FS techniques were used to eliminate unnecessary quantitative and qualitative features and select influential ones Sensitivity, specificity, precision, and Cohen's kappa coefficient were used as classification performance metrics. True Positives and Negatives are oil spills and look-alikes, respectively - Geometrical features have been reported as the most influential group
(Pelta et al., 2019)	Detection and monitoring of oil spill incidents using airborne hyperspectral Remote Sensing and Machine Learning under uncontrolled conditions	- Hyperspectral optical images - ANOVA to select more influential bands - Partial least square to reduce prediction variables (bands) in ML - Linear discriminant analyses (LDA) as the binary classifier - ENVI software and Python	- Detection of trace amount of oil in a real-world condition two years after a spill - This research developed its own spectral absorption threshold of petroleum hydrocarbon by ML	ML identified the most appropriate petroleum hydrocarbon spectral range to detect oil spills from aerial images after a long time
(Jiao et al., 2019)	Development of an automatic oil spill detection and evaluation model working based on enhanced unmanned aerial vehicle (UAV) images	- Deep convolutional neural network (DCNN) - developed by VGG16-a pre-trained network on the dataset of ImageNet - UAV images - Otsu algorithm was used for interface removal	- Automatic, real-time monitoring of oil transportation systems reduced the workload and costs of oil spill detection, especially in long-distance pipelines - The performance of different conventional AI-based models and the developed DL model was compared	- Using Faster Regional-Convolutional Neural Network (R-CNN) as the backbone of the DL model, an augmented image dataset was processed to detect oil spill traces. Mean Square Error of classification in dataset's samples was minimized as the loss function of the model

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
				- Otsu algorithm, determining the maximum variance (the best threshold) between regions of interest and background in images, was used to identify oil spills more distinctively and remove the interface of the region of interest and the background
(Cantorna et al., 2019)	Enhancing the oil spill detection capacity of a DL model; the enhanced model can segment SAR image data even for small-scale spills	- Convolutional neural network (CNN) DL model - Logistic regression algorithm - Clustering algorithms (classic clustering k-means, fuzzy c-means, and spatially constrained kernelized fuzzy clustering c-means) - CUDA platform to implement CNN	- The oil spill detection performance of different image processing and segmentation models was compared - An appropriate discussion was provided for tuning of the CNN DL model and enhancement of its identification power	- After the acquisition of images, preprocessing was performed to reduce the number of unnecessary pixels, and then different machine learning and neural network-based models can be trained, calibrated, and tested on the dataset to detect oil spills - The best performing model in this study was CNN by reducing the number of false-positive detection and obtaining 0.95 % as the mean precision
(Jia et al., 2019)	Improving oil spill segmentation performance of pulse coupled NN using a hybrid heuristic optimization for tunning the parameters	- Hybrid optimization algorithm including thermal exchange and seagull optimization algorithms (SOA) - UAV and satellite images - Pulse-coupled neural network - Wilcoxon rank sum test was used for statistical comparison	- The SOA performance was enhanced using temperature exchange capacity - Comparisons were made between the performance of the developed hybrid evolutionary algorithm and other ordinary metaheuristic algorithms in segmentation enhancement - Providing both quantitative and visual qualitative comparisons between different algorithms	- Cross entropy was selected as the objective function of the optimization of model parameters, which can minimize the information difference between probability distributions of manually annotated and output segmented images - The developed optimization algorithm trains the network by defining the weight of each neuron, and determination of a dynamic threshold for each iterative process that generates a series of the temporal pulse from intensities of each input pixel
(Ozigis et al., 2019)	Detection of oil-contaminated lands in a multi-landcover area using an ML models	- Random forest (RF) classifier - Optical images of operational land Imager (OLI) from Landsat 8	- Consideration of separate classification processes for individual landcover enhanced the classification performance	Eight different vegetation health indices have been used for oil spill detection from OLI satellite images. The indices were defined based on spectral characteristics, including RGB color values as well as near

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
		- EnMap Box to perform RF	- Suggesting influential variables and indices to detect oil-contaminated lands from OLI images - Combining spectral band and vegetation indices to obtain suitable classification parameters	and shortwave infrared bands. The Random Forest algorithm has been used for supervised classification based on developed indices
(Ryu et al., 2019)	A report of the journal special issue on application of RS and GIS in coastal environments	For the oil spill detection part: - ANN - CNN - SAR - Satellite optical and hyperspectral images	- This special issue contains the application of UAV, SAR, spectrometry etc. for the detection of various phenomena, including oil spills - Various case studies have been considered in Kuwait, China, Western parts of North America, Greenland, and the Arctic ocean	A total number 52 paper has been published in this issue considering the application of RS and GIS for gathering and evaluation of data
(Ghorbani and Behzadan, 2020)	Small-scale oil spill semantic (binary classification) and instance segmentation	- CNN (pre-trained models of VGG16 and Mask R-CNN) - UAV and satellite images	- Application of two recently developed DL models for both semantic and instance segmentation - Consideration of various types of images to check the robustness of the model	- VGG16 has a 16 convolutional layers network structure and is pre-trained on the ImageNet dataset - Mask R-CNN's network structure is Resnet-50, and is pre-trained on the COCO dataset - For R-CNN, average precision of 61% and recall of 70% have been obtained (maybe due to inaccurate labeling)
(Jagdish and Jerritta, 2020)	Oil spill detection and characterization in SAR image data of various incidents using machine learning and signal processing techniques	- k-nearest neighbour (KNN) algorithm - Wavelet analysis	- Evaluation of a wide range of samples (images) related to oil spills in different locations - Providing general data (such as the volume of spilt oil) for different oil spill incidents	Different features of SAR image data, including RGB, spreading, complexity, standard deviation, entropy, ellipticity, intensity and correlation coefficient, have been classified and processed by KNN and wavelet analysis for detecting and characterizing oil spills
(Bianchi et al., 2020)	Developing a DL model working based on remote sensing for oil spill semantic segmentation (both detection and characterization)	- SAR image data - U-Net, which is a CNN DL model (used as the backbone of the developed model)	- Developing a highly accurate binary detection model - Extracting geometrical specifications and detailed properties of spilt oil slick using a separate NN trained by the initial dataset	- A total of 12 parameters have been annotated in the training dataset defining shape and some oil spills properties. These parameters were used later for training an NN to automatically identify the characteristics of oil spills

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
		- Kruskal-Wallis h-test used for statistical evaluation	- Evaluation of different loss-functions for network training to handle classification imbalance - Effects of satellite incidence angle on oil spill detection were evaluated - Providing sufficient explanations about the segmentation process in DL models	- The developed model outperformed the original version of U-Net and another DL model, DeepLabV3+, considering computation time and performance metrics - The variation of incidence angle of the satellite within the range of 30 to 45 degrees, which was suggested for oil spill detection, has no significant effects on the model performance
(Yekeen et al., 2020)	Application of instance segmentation DL model for oil spill detection and characterization	- Mask R-CNN (COCO pre-trained ResNet 101 as the backbone and pre-trained model for CNN) - SNAP was used for scaling, preprocessing, and labeling initial images - SAR image data - Supervisely™ was used as a platform to implement the model	- Improving the performance of the oil spill detection model working based on SAR image data. The model's performance has been compared with several previous models - Indirectly considering the correlation of relevant elements to oil spill evaluation such as ships and terrestrial areas by separately annotating them - Covering a wide range of elements in evaluations, including labeling ground truth mask of the spill, ship, look-alikes, and terrestrial regions	This study performed dB conversion, and speckle filtering (by SNAP) as the preprocess step to reduce noise affecting dark spot discrimination power
(Balogun et al., 2020)	Development of ML models to evaluate oil spill impacts on terrestrial vegetations and their recovery process using satellite optical sensor images	- RF - SVM - OLI images - EnMap Box software to develop ML models - McNemar's chi-square test to statistically compare the models	- Evaluation of the recovery process by comparing pre- and post-oil spill multispectral images - Selection of the most significant indices based on statistical evaluations (*p*-values) for both vegetation and wetlands	The number of 16 vegetation health indices has been defined based on different features extracted from OLI images to evaluate oil spills' effects on different vital plant processes. Also, elements of interest have been classified by ML models into four different groups of contaminated/non-contaminated vegetation/wetlands. An enhancement in classification accuracy may facilitate the evaluation of recovery rate in oil-contaminated vegetation

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
(Zeng and Wang, 2020)	Development of a sufficiently deep CNN for oil spill detection using SAR image data	- VGG-16 and OSCNet DL models (as the basic architecture) - SAR image data - Data augmentation to increase available training samples	- Development of an original DL model rather than using pre-trained networks - Addressing the imbalance of datasets by extensive processing during image segmentation - Performing sensitivity analyses on activation function, optimization algorithm, initial learning rate, and batch size - Loss functions of the augmented and non-augmented datasets were compared - Comparison of manual and automatic feature extractions (Comparison of DL and NN)	Pixel level (rather than patch-level) DL classifiers such as RED-Net has been suggested as a high potential DL for future studies on oil spill detection
(Zacharias et al., 2021)	Enhancing identification of oil spill sources using fate and transport simulation and stochastic evaluation	- Spill, Transport, and Fate Model (STFM) - Monte Carlo simulation - DT	- Integrating an uncertainty assessment technique, ML technique, and fate and transport simulator to reduce uncertainties of the oil spill source identification in a time-reverse evaluation - Finding the most probable incident scenario based on the probabilistic assessment	The developed approach is appropriate for post-incident evaluation rather than real-time oil spill assessment
(Trujillo-Acatitla et al., 2022)	Developing ML models for spill detection and characterization using multi-spectral image data	- KNN - RF - DT - SVM - RBF - Multi-spectral image data - Multiple statistical analyses on image data based on ANOVA	- Determining the variable concentration of spilt oil in water to monitor its degradation rate - Considering reflectance intensity of different materials, such as plastic, soil, and vegetation, within the oil spill detection process by ML techniques - Introducing the most influential features (i.e., wavelength bands) in oil spill detection and characterization	- KNN outperformed other ML techniques in terms of oil spill detection accuracy. About 20% variation was observed in the oil spill concentration in seawater - To enhance the accuracy and robustness of ML techniques, it is suggested to increase the number of spectral response data for model training under various environmental conditions and types of oil
(Abdallah and Chantsev, 2022)	Oil spill risk assessment of growing transportation activities based on oil spill fate and transport simulations	- GNOME for trajectory modeling - Automated Data Inquiry for Oil Spills (ADIOS2) for weathering processes modeling	- Scenario-based evaluation of spilt oil trajectory for two hypothetical cases helping contingency plan development - Evaluating the effects of physical and chemical processes on spilt oil over time	- The developed model for the studied area could be further evaluated and validated using RS and oceanic data related to the case study - A sensitivity map of the affected area and the likelihood of various incidents could

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
				also be incorporated into the risk assessment - Instantaneous release of complete cargo tanks after a tanker collision or grounding is less likely these days
(Li et al., 2022)	Oil spill detection using ML and SAR image data obtained from a shipborne x-band antenna	- SVM - Gray-Level Co-occurrence Matrix (GLCM) - Fuzzy C-Means (FCM) - SAR image data	- Incorporating complementary pre- and post-processing steps (coordinate system transformations on images) with ML to enhance oil spill detection - Using economical shipborne radar technology for oil spill detection - Using GLCM and SVM to identify suitable monitoring areas to detect oil spill - Using FCM to distinguish oil films and water waves following the monitoring area determination	The model works well on thick oil slicks. It is suggested to train and test it on various oil slick thicknesses and types in order to enhance the robustness of the model.
(Kang et al., 2022)	Appraisal of different ML techniques coupled with various feature extraction and post-processing techniques for oil spill detection and characterization	- 2DCNN - Auto-decoder (AE) DL model - Spectral Former (SF) DL model - Multiscale Adaptive Sparse Representation (MASR) - Locality-Constrained Sparse Representation Classifier (LSRC) - Hyperspectral image data	- Evaluating oil spills' extent, type, and thickness based on hyperspectral image data - Comparing the performance of multiple MLs (DL and signal processing-based models) in ODCM on several case studies	- The models working based on the Structural Profile feature extraction technique outperformed others in terms of noise reduction and robustness - There is a lack of a sufficient number of training and testing samples of high-quality images

Appendix B – Summary of Reviewed Articles in Spill Management and Response Optimization

Spill management and response optimization papers

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
(Mata et al., 2011)	Developing a system, combining different methods for data communication and prediction of oil slick specifications based on records of previous incidents	- Multi-agent modeling was used to develop the basic system - SAR image data (develop the initial dataset) - Fast Iterative Kernel Principal Components Analysis was used to reduce dataset variables and select influential ones for training - CBR was used to select the most similar cases to a new incident - A growing radial basis NN was trained for prediction based on similar cases and influential variables	- The system is able to analyze oil spill incident specifications (e.g., area, location, and distance to a coast) to enrich samples required for NN training and enhance its applicability. For instance, the system can download real-time meteorological and oceanic information of the study area - The developed model is compatible with cell phones or personal digital assistants (PDAs), which are influential devices in remote areas - A multiple user interaction capacity was provided for the system due to the implementation of SOAP protocol facilitating communication between different elements	- The developed system (CROS) is an integrated platform connecting metrological data, a database of previous oil spills information, and a NN model to select a group of available oil slicks that are similar to a new case for training the NN -The trained NN can accurately predict the area and possible location of the new slick - The system works like an enhanced combination of a clustering technique and an ANN
(Helle et al., 2011)	Development of a probabilistic model for evaluation of the effectiveness of oil spill responses considering ecological aspects	- Bayesian network (BN) - SPILLMOD 2.0 and Geoinformatica were used to develop probabilistic distributions for species exposure to oil and inshore wave height, respectively - Hugin software was used to construct a BN	- Taking ecological variables into account to identify vulnerable aquatic species in the area. - Effects of spilt oil and response methods have been evaluated on the vulnerable aquatic species - A considerable number of variables were evaluated through the BN to determine response efficiency - Providing reliable resources to extract required data for most of the variables in the study area	- BN included in this review due to significant influence of its generalized version on AI (Boutilier, 2005) - Various influential variables of the oil spill response operation are considered by using their conditional probability distributions as inputs to a BN. Effects of the selection of random samples for some variables and selected values for decision variables can be evaluated using this interconnected network to determine the efficiency of decisions in the reduction of death of different aquatic species

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
(You and Leyffer, 2011, 2010)	Optimization of oil spill response considering the fate and transport processes for maximizing response efficiency	- Mixed integer non-linear programing - Mixed integer linear programing was used first to obtain a suitable initial condition of variables for non-linear programing - Finite element method was used to discrete differential equations - CPLEX 12.2 was used to solve linear programing - Semi-empirical equations for fate and transport - DICOPT and KNITRO were used to solve non-linear programing	- Ordinary differential equations were provided and solved in case of major components of fate and transport processes - An integrated optimization and simulation problem was clearly formulated and solved - Use of a pragmatic formulation for booms considering potential damages and deployment rates - Considering simplified constraints (thickness for ISB and regulations for UCD) for determining the suitability (selection) of an oil spill response method	In this study, an optimization problem has been formulated, considering semi-empirical differential equations for the weathering process, to enhance the efficacy of response strategy based on response time and total costs
(Li et al., 2012)	Optimization of skimmers' allocation in oil spill responses considering the oil weathering process	- Mixed integer nonlinear programming model - MATLAB and LINDO API were used to solve the optimization problem	- Performing separate optimizations with and without the consideration of the weathering process to show its influence on skimmers' efficiency - Defining an optimization problem by the integration of the weathering models and regression models of skimmers' efficiency	A time step is selected to discrete an optimization problem designed based on influential variables of skimmers and the weathering process (a multiple-stage dynamic programming problem)
(Cao et al., 2013)	Estimation of spill occurrence probability in terrestrial systems considering model selection uncertainties	- Monte Carlo simulation - Bootstrap re-sampling technique for quantification of samples estimations' accuracy generated by MC - Maximum likelihood technique was used to identify suitable	- The North America Industry Classification System was used to consider possible spill locations (sources) - Uncertainties evaluation of developed probability distributions to determine their accuracy in the representation of variables - Evaluation of correlation between variables to assure using independent ones	A long-term historical dataset of benzene was collected, and probabilistic distributions of different variables, including location code, spill mass, and interevent time were fitted. Through an MC-base framework, the randomly generated samples based on the probabilistic distributions were statistically evaluated for a simulation time frame of 10 yrs.

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
		probabilistic distributions for data - MATLAB was used for the implementation		
(Li et al., 2013)	OSRM selection using a neural network-based model working on fuzzified features of response methods and oil spill incidents	- Fuzzy set theory - Monte Carlo simulation - Adaptive resonance theory (ART)	- Integration of ARTMap NN with fuzzy set theory to address practical complexities and uncertainties in the selection of response method - The model is able to consider fuzzified parameters, such as temperature, wave, wind, oil viscosity, and slick thickness in the selection	In this study, random values were assumed for influential features in a hypothetical case study, and these values were fuzzified using triangular membership functions. The same fuzzification process was performed for the assumed suitable ranges of four different responses. Monte Carlo simulation was used to generate different values in these ranges of responses, and then the developed model selected an appropriate response method based on these values and influential features
(Verma et al., 2013)	A scenario-based optimization of response facility locations and their stockpiles of response equipment considering oil spill frequency	- A two-stage stochastic recourse model was used to consider uncertainties through single objective optimization - CPLEX 12.4.2 was used to solve both linear and non-linear optimization problems	- Two separate stages, including the first investments stage and the following response stage, were considered in an optimization model which has real-world application demonstrations - Consideration of environmental costs and annual frequency of spills among influential parameters of the model	In this research, different zones are considered around sought coast of Newfoundland, Canada, with different oil spill frequencies. Also, the effects of the oil spill locations and sea conditions were considered through a probabilistic model used on long-term data in this area. Then, a single objective optimization problem was defined to solve logistical issues of oil spills
(P. Li et al., 2014a)	Intelligent classification of shoreline areas based on oil spill vulnerability considering various meteorological, environmental, and marine traffic conditions	- Monte Carlo simulation to address uncertainty - ART classification - MATLAB grid data to spatially interpolate data (such as wave height and oil spill frequencies) - Mann-Whitney, Kruskal-Wallis, and Jonckheere–Terpstra nonparametric	- The classification criteria of the model were generated automatically - Connecting the model with Monte Carlo simulation to address influential uncertainties of features such as the wave height - Considering interrelations between different influential features defining vulnerability index	A number of different uncertain and deterministic features have been considered to define an oil spill vulnerability index. Then, a MC-based model is used for unsupervised classification of near shoreline areas based on the values of the mentioned features

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
		tests to statistically compare index values		
(P. Li et al., 2014b)	Optimization of device allocations and operational time in mechanical recovery responses considering uncertainties due to the oil weathering process	- Monte Carlo simulation - Mixed integer nonlinear programming model - MATLAB with LINDO API to solve the optimization problem	- Considering effects of the oil weathering process and uncertain parameters in the optimization - Comparing deterministic and uncertain results - Considering the variation of viscosity of spilt oil by time through the main optimization process - A comprehensive evaluation of the obtained results has been provided, especially for the effects of variations in uncertain parameters on optimal scenarios	In this study, some regression equations have been developed for three different skimmers based on the variation of oil spill thickness with time, calculated using the remaining oil volume in each time step and the initial area of spilt oil. Then, the deployment order of these skimmers as well as their operational time have been optimized using multiple-stage (considering time steps) nonlinear programing while taking weathering process into account by empirical equations. Slick area, wind speed, and temperature were considered as uncertain features in this study to be handled by Monte Carlo simulation by assuming a normal distribution for them
(Helle et al., 2015)	Developing a probabilistic cost-benefit analysis model to determine the most influential proactive or active OSRMs	- Influence diagram, a generalized version of BN in decision-making - Cost-benefit analysis - Hugin Researcher 7.6 software	- This approach can help decision-makers to determine whether an investment in oil spill response infrastructures is rational and which response method is more beneficial - The model considers influential uncertain parameters in an explicit manner	- Influence diagram included in this review due to its significant influence on AI (Boutilier, 2005) - A total number of 55 variables were considered in a BN-based model to study the costs of MOSM practices, including MCR, shoreline clean-up, as well as waste storage transportation, and treatment. The marginal and conditional probabilities of different variables were determined based on literature review, numerical models, and expert judgment. Results showed that preventive measurements (early warning systems) are more efficient than mitigative measures, and waste treatment cost is also an important parameter in the selection of response strategy

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
(Mokhtari et al., 2015)	Estimation of oil spills occurrence probability based on ships' routes and distance to coastal areas and exploration facilities	- Generalized linear model (GLM) - R software to implement the model	- Considering effects of exploration and transportation decision supports on oil spill occurrence - Estimation of oil spill probabilities without using a surveillance system - Sensitivity analysis was used to determine the most influential factors in oil spill occurrence	Regression analysis was used to estimate the rate of oil spills based on the influential cause of incidents. Marine traffic was considered as the main proxy of oil spill occurrence
(Encheva, 2015)	Development of a decision-making tool to select an efficient alternative to manage ship-generated wastes	Grey theory (a mathematical technique)	Suggesting influential attributes in ship-generated waste management, including the availability of facilities to receive, treat, and dispose of wastes as well as green energy consumption capacity	- Grey theory deals with uncertainties that always exist between two conditions: one with fully available information (white) and the other with a total lack of information (black). The model work with mapping a universal set with two functions determining lower and upper membership functions - In this model, a set of different influential attributes were selected that can be defined by linguistic variables, then a discrete weighting structure was used to define weights of attributes. Following that, the degree of possibility for each alternative was estimated using Grey theory to select the best one. So, each alternative can be ranked based on uncertain attributes
(Zodiatis et al., 2016)	Evaluation of an integrated decision support model for detecting and evaluating (e.g., trajectory prediction) oil spill incidents in the Mediterranean Basin	Reviewing an available multi-model system for oil spill management	- Details of oceanographic and meteorological forecasting systems in the Mediterranean Sea, such as types of data, temporal and spatial resolutions, were provided - Different oil spill simulation models were introduced, including an open-source one	MEDESS-4MS is a model integrating different oil spill simulation models with real-time harmonized meteo-oceanographic datasets and prediction models, as well as feeding real-time satellite SAR image data. This model can be used to predict the trajectory and characteristic of detected oil slicks to facilitate a response process
(Moroni et al., 2016)	Developing an electronic nose-based	- ANN (Kohonen Self-Organizing Map)	- Application of E-Nose elements on the moored buoys to enhance their	- Two separate ANNs have been trained by laboratory benchmark datasets to

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
	buoy, evaluating air samples near the water surface for oil spill detection and water quality monitoring	- A sensor-equipped buoy - MATLAB to train ANNs	performance in carbohydrate detection in water - Performing laboratory and field tests to evaluate the performance of the buoy based on accuracy and response time	identify the concertation (Acc.: 88%) and type of the hydrocarbon (Acc.: 81%) sampled from the water surface - There is a bidirectional connection (using a SIM card and GSM modem) between the buoy and Marine Information System (MIS) to store and analyze real-time data on the server and also receive configurations for the buoy
(Grubesic et al., 2017)	Development of a coupled numerical and optimization model to find efficient allocations of oil spill response resources	- Blowout and spill occurrence model (BLOSOM) was used for spill modeling - Classic optimization problems were solved using Gurobi Optimizer in Python	Consideration of trajectory and characteristics of the oil spill, being updated with time, in the coupled model led to more reliable outcomes	- Objective function was assumed as minimizing dispatching time of response vessels considering available resources constraints - The outputs (location of oil plumes) of BLOSOM for each day of response were used as the input of the optimization model
(Garrett et al., 2017)	Policy optimization for timely investment in response infrastructures and resources in the Arctic	- Mixed integer linear programing - CPLEX 12.6 to solve the optimization problem	- Considering negative effects of missed deadlines (e.g., as a result of lack of resources in remote areas) to mitigate the response procedure - Evaluation of the model by experienced experts in the field of oil spill response in Arctic offshore waters - Consideration of two separate components for resource allocation and response deployment, in a chronological pattern - Providing a list of practical constraints and weighted variables that should be considered during the response, categorized through two previously mentioned groups - Providing a dataset of oil spill incidents in the Arctic	- This paper answers higher-level questions pertaining to the optimization of oil response policies as well as the importance and effects of deadlines - Different scenarios of time, resource and budget allocation have been evaluated
(Akyuz and Celik, 2018)	Oil spill risk assessment and mitigation considering	- Failure mode and effects analysis (FMEA) was used to identify and	- Integration of fuzzy sets with FMEA to enhance the robustness and accuracy of the risk assessment	In this study, the fuzzification of linguistic parameters related to the risk estimation process of oil spills has been done along

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
	uncertainties of influential attributes in decision-making	prioritize risk factors of oil spills - Fuzzy set was used to handle uncertainties resultant from the use of linguistic variables	- Providing mitigative and practical strategies to reduce oil spill risk	with FEMA's application to determine the ranking of influential factors. To this end, the number of 125 fuzzy rules were generated based on severity, occurrence rate, and level of detectability of oil spills pertaining to tanker collisions. These rules can be used for risk assessment of oil spills and determination of risk factors ranking
(Zhang et al., 2019)	Risk estimation of icebreaker and assisted ship incident in Arctic waters	- Human factors analysis and classification system for classification of collision risk factors - A fault tree (FT)-based model - Easy Draw (V2.19) software to perform a quantitative risk assessment	- Consideration of 28 different sources of collision risks caused by human errors - Four experts with considerable relevant work experience in Arctic environments were asked to classify collision factors - Presentation of a tree-like structure showing order and interaction of collision risk factors, and performing a quantitative risk assessment based on this model to determine contribution degree of each factor in incidents	A quantitative risk assessment has also been performed based on the definition of minimum cut set technique in FT model and determination of importance degree of each factor in incident contribution
(Ye et al., 2019)	Evolutionary optimization of mechanical OSRMs considering weathering effects	- Particle swarm optimization (PSO) - Multi-agent modeling - NetLogo was used as the implementation platform	Interactions of various weathering processes including evaporation, dispersion, and emulsification with a mechanical response method were considered through an optimization process	Agent-based modeling has been performed considering three different layers of optimization, interaction, and behaviour layers. The output of the empirical-based simulation model has been given to the optimization algorithm through an iterative process
(Li et al., 2019)	Optimization of required resources to respond to large-scale oil spills considering real-world constraints	- Linear programing (LP) - YALMIP (a toolbox in MATLAB) to call CPLEX	- An appropriate discussion has been provided on practical considerations and constraints that in the optimization of the oil spill response operation - Considering an estimation of emergency materials required for each scale of oil spills in the formulation, to be more realistic	A single objective and simple optimization problem has been defined, mostly emphasizing sailing time and distance to reach the point of interest
(Huang et al., 2020)	Development of a hybrid emergency oil spill management	- CBR - Deep Q-network algorithm (combining	- Considering rescuing vessels and crews as well as protection of resources as two	Deep Q-Network integrates reinforcement learning (an action and reward system) and deep neural network-based models,

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
	model working based on historical spill records	reinforcement learning and CNN) to implement the model - k-means clustering algorithm	categories of response in addition to clean-up procedure - Consideration of hazard and exposure elements in the definition and selection of oil spill scenarios	and defines a reward for each action that can be selected based on previous observations. It works by considering the temporal sequences of observations and actions and try to maximize rewards summation
(Janati et al., 2020)	Development of an ANN surrogate model based on a multiphase Lagrangian simulator to predict oil spill trajectory	- Back propagation ANN to be trained to generate a surrogate model that will be running instead of the simulator - A multiphase Lagrangian numerical simulator considering water and oil phases	- Two-phase simulation was performed, and the outcomes were validated by experimental data - Sufficient accuracy can be achieved with a significant reduction in computation time by using a trained ANN instead of a numerical simulator	In this research, a previously developed and validated numerical model was used in a considerable number of cases (with different channel geometry, oil spill specifications, and environmental conditions) to create a dataset for ANN training. Then, two separate ANNs were developed for the prediction of oil spill spread concerning advection effects of currents or waves
(Ye et al., 2020)	Development of a fuzzy multi-criteria decision support tool to determine the most influential human errors in oil spill management (prevention and response)	- Technique for order preference by similarity to ideal solution (TOPSIS) - Fuzzy sets analysis to address vagueness - Fault tree analysis to consider the interactions of different sources of error (in early stages of a spill) to determine their simultaneous effects on the oil spill occurrence rate	- Finding the most important sources of human errors in the offshore oil spill management - Performing a sensitivity analysis to find effects of different individual judgement on the ranking of risky factors - Reducing vagueness and uncertainties by integrating fuzzy numbers with TOPSIS	In this research, a considerable number of human error sources (factors) were evaluated by experts to find their effects on oil spill incidents and management. Triangular fuzzy membership functions were used to define different linguistic variables. Then, obtained expert judgments were aggregated using the Fuzzy-TOPSIS technique to determine the rank of the influence of human error sources
(Kim and Kim, 2020)	Integrating a deep NN-based model with a GA for optimization of skimmer assignments	- GA - Deep NN	About 61% reduction in computation time has been achieved by the application of the surrogate model instead of the simulation model	- DNN is trained, based on the results of a simulation model for oil spills' fate and transport, to use within the optimization process of skimmers application - Two scenarios of minimizing work hours of oil skimmers by reallocation and minimizing the amounts of total allocations

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
				(use of skimmer capacity) were considered
(Sai et al., 2020)	Conceptualization of an intelligent buoy that can be used for spilt oil recovery	- Internet of Things (IoT) - ML models - Various sensors to digitalize environmental and oil spill conditions	- Suggestion of a concept of intelligent buoys for timely oil spill response - ML models are considered to evaluate data collected from different sensors- Evaluation of required material for the buoy to work properly under extreme weather conditions - Consideration of two different sources of energy for the buoy	- A buoy has been designed here that can absorb the oil spills from the surface of the water and evaluate various information about oil spills using different sensors, including temperature, gyroscope, ambient light, vibration, pH, infrared, hydrocarbon detection (Moroni et al., 2019) and color sensors. ML technique and data mining models are considered in this intelligent system capable of communication using marine environment networks MANET (Mobile Ad-hoc Network) - Some oil tankers can be equipped with sufficient dispersant and recovery buoys to use them in emergency conditions
(Hu et al., 2020)	Development of an intelligent fuzzy decision tree-based decision support tool for the selection of suitable OSRMs in the Arctic	- Fuzzy decision tree (FDT) - Linear regression (LR) - MATLAB™	- Facilitating the OSRM selection process by the integration of FDT and LR - Consideration of three widely used oil spill responses - Providing a detailed description for influential attributes of response selection in the Arctic - The developed framework could expedite the decision-making process while considering vagueness and uncertainties	In this research, influential attributes in the selection of OSRMs in harsh environments have been determined, and a realistic hypothetical dataset of oil spill incidents has been developed based on these attributes and their suitable response methods. Fuzzy logic was considered to reduce vagueness in the decision-making process, and various types of decision trees were trained and tested on this dataset to select the most appropriate response method. The model facilitates the decision-making process in complicated real-world conditions
(Mohammadiun et al., 2021a)	Optimization of integrated FDTs-regression analyses models for optimal	- FDT - Non-dominated sorting differential evolution (NSDE)	- Enhancing prediction accuracy of FDT-based models for optimal selection of OSRMs - Optimization of the FDTs training process	The prediction accuracy of fuzzy decision support tools integrated with different regression analyses has been enhanced by optimization of these models based on training dataset selection. By considering

Reference	Objective(s)	Techniques and Models	Strength/Novelty	Remarks
	OSRMs selection in Arctic waters	- Gaussian process regression (GPR) - LR and non-linear regression (NLR) - Kruskal-Wallis test for statistical evaluation - MATLAB™	- Consideration of different information discrimination measures to develop FDTs - Investigating the prediction performance of combinations of FDTs with different regression analyses - Proposing an integrated framework selecting the most efficient model based on the Hyper Volume Indicator	separate objective functions of prediction accuracy and the number of rules of FDTs, the average prediction power and robustness of the models have been significantly increased
(Das and Goerlandt, 2022)	Response method ranking for Arctic incidents using a data-driven technique	- Bayesian Inference - Monte Carlo simulation - DT	- Generating a considerable number of hypothetical scenarios in the Arctic using a DT and Monte Carlo-based model - Reducing the number of influential attributes compared to their original scenario generators	- More realistic probability density functions can be considered for scenario generation in case of better data availability in the Arctic - Incorporation of oil spill fate and transport factors in data generation could also enhance the data reliability - Scenario generator needs a significant number of attributes as input parameters - There is still a need for additional data to enhance the robustness of the model

Appendix C – Hypothetical Dataset of Oil Spill Incidents used in OSRM Selection Tools

No.	A1	A2	A3	A4	A5	A6	MCR	ISB	UCD	No.	A1	A2	A3	A4	A5	A6	MCR	ISB	UCD
1	40	0.6	86	1.2	3	272	4.0	2.0	9.0	36	21	2.9	47	1.1	5	402	2.0	6.5	2.5
2	23	0.4	43	0.2	0	214	8.0	1.0	2.5	37	22	1.9	34	0.1	1	312	6.5	2.5	1.5
3	44	0.6	50	1.3	2	122	4.5	2.5	10.0	38	28	3.0	37	0.9	2	288	7.5	5.0	2.0
4	39	2.9	47	0.2	5	146	8.5	5.5	3.0	39	23	1.2	39	1.2	3	224	5.0	2.5	4.7
5	36	0.4	70	1.2	3	476	1.0	1.5	7.5	40	21	0.6	66	0.4	2	362	4.0	1.5	4.2
6	36	0.6	38	0.3	0	122	10.0	1.0	5.5	41	17	2.4	57	0.1	5	191	9.0	6.5	1.0
7	36	0.9	80	0.1	2	418	3.0	3.0	5.0	42	30	0.6	80	0.8	5	499	1.0	1.5	7.0
8	38	1.4	87	0.1	2	184	8.5	5.5	3.0	43	31	1.5	36	1.4	4	101	4.5	1.7	6.5
9	36	0.6	83	1.2	3	456	1.0	1.5	9.0	44	26	2.6	82	0.1	5	260	8.0	9.5	2.7
10	18	1.1	86	0.9	3	136	7.5	5.0	1.5	45	18	2.6	56	1.4	0	172	4.0	5.5	1.0
11	38	2.8	74	0.3	15	418	1.0	10.0	1.5	46	18	0.6	38	1.3	4	332	3.5	1.5	2.0
12	31	0.7	55	1.1	5	254	4.5	2.0	5.5	47	42	0.6	82	0.9	0	240	7.2	2.5	6.7
13	40	0.5	31	0.2	1	128	9.0	1.0	7.0	48	30	1.4	34	0.4	3	149	9.0	2.5	3.5
14	34	2.4	66	0.3	13	456	1.0	9.0	3.0	49	39	0.9	55	0.1	5	461	2.0	2.5	5.2
15	43	1.1	43	0.3	2	126	9.0	4.0	4.2	50	30	1.6	59	0.5	3	460	1.5	5.2	3.4
16	17	0.7	50	0.5	15	148	4.0	3.0	1.5	51	16	2.2	32	0.1	10	138	5.5	4.0	0.0
17	32	1.2	76	1.2	0	124	3.0	4.0	5.6	52	40	2.6	74	0.2	13	307	4.0	8.4	1.7
18	24	1.3	40	1.3	1	188	2.5	2.5	4.5	53	45	2.1	78	0.7	9	430	2.0	8.0	5.5
19	36	0.5	77	1.3	1	306	1.5	0.7	7.5	54	17	0.5	40	0.6	17	422	1.0	2.0	1.5
20	19	0.7	50	1.3	0	120	3.2	1.0	1.5	55	16	1.3	62	0.6	13	351	1.5	4.0	1.0
21	18	3.0	39	1.5	4	483	0.5	5.5	0.5	56	24	1.4	30	1.5	3	142	5.0	1.0	3.0
22	17	2.6	38	0.6	2	467	1.5	6.7	0.0	57	26	0.7	59	0.5	19	469	1.0	3.0	1.2
23	20	3.0	70	0.1	5	440	1.0	9.5	0.0	58	16	2.4	46	0.6	11	478	1.0	6.0	0.0
24	32	2.2	76	1.1	2	257	5.5	6.5	4.2	59	27	3.1	48	1.1	13	316	1.5	5.0	2.0
25	27	0.7	73	0.5	1	211	6.5	3.0	5.5	60	36	2.1	40	0.2	20	168	4.5	5.0	1.0
26	27	0.4	80	1	2	227	5.5	1.5	6.5	61	16	1.8	83	1	3	247	7.0	5.5	1.0
27	25	0.3	56	0.6	4	406	2.5	1.0	6.0	62	29	3.1	39	1.3	13	250	4.5	5.0	1.0
28	22	0.7	59	0.7	5	359	3.0	1.4	5.0	63	20	1.4	59	0.2	12	190	5.5	4.0	0.0
29	32	2.4	46	1	14	362	3.0	7.2	3.2	64	26	1.9	45	0.4	12	206	5.5	4.0	1.0
30	45	0.5	60	1.4	3	236	4.0	1.2	9.5	65	40	3.1	74	0.5	12	143	6.0	8.3	1.5
31	31	0.8	75	0.4	4	302	6.0	1.5	4.8	66	43	2.8	79	0.4	12	109	6.5	8.0	2.5
32	26	2.8	88	0.8	5	155	7.5	8.9	3.7	67	28	1.1	39	0.3	8	137	9.0	3.0	4.5
33	30	0.9	81	0.4	3	426	2.0	2.2	4.5	68	33	1.6	72	1	13	113	6.5	5.0	5.5
34	40	2.9	51	0.3	4	200	8.0	7.0	4.8	69	17	1.1	62	1	8	428	2.3	2.0	0.0
35	35	1.2	51	1.2	3	334	5.5	5.3	6.5	70	37	3.1	62	0.5	2	344	6.5	7.0	1.0

Note: MCR-mechanical containment and recovery; UCD-use of chemical dispersants; ISB-in-situ burning; A1-Spilt oil type; A2-Oil slick thickness; A3-Oil slick continuity; A4-Wave height; A5-Floating sea ice; A6-Remoteness of response location;

Appendix D – Pseudocode Developed for Optimizing MCR and OWM Operations

Algorithm: Integrated Multi-Objective Optimization of MCR and OWM

Inputs/Variables:

Slick's Number (OSN), Locations (OSL_i), Volumes (OSV_i), Areas (OSA_i), Maximum Number of Concurrently Deployable Vessels (ODV_i)

Vessel's Number (VSN), Initial Locations (VSL_j), Skimming Capacity (VSK_j), Number of Skimmers - Type 1 (VNS_{j1}); Number of Skimmers - Type 2 (VNS_{j2}); Speeds (VSS_j), Storage Capacities (VSC_j), Unit Operation Costs (VOC_j), Unit Transportation Costs (VTC_j), Miles per Full Tank (VPT_j), Refueling Durations (VRD_j), Maximum Number of Concurrent Refueling (VNR_j)

Mini barge's Number (MBN), Initial Locations (MBL_k), Storage Capacities (MSC_k), Speeds (MBS_k), Unit Operation Costs (MOC_k), Unit Transportation Costs (MTC_k), Pumping Rates (MPR_k), Miles per Full Tank (MPT_k), Refueling Durations (MRD_k), Maximum Number of Concurrent Refueling (MNR_k)

Big barge's Number (BBN), Locations (BBL_l), Storage Capacities (BSC_l), Operation Costs (BOC_l), Pumping Rates (BPR_l), Maximum Number of Concurrently Connectable Mini Barges (BCM_l), Volumetric Activation Threshold for Discharging (BAT_l), Maximum Number of Concurrently Connectable Trucks (BCT_l)

Truck's Numbers (TRN), Initial Locations (TRL_m), Storage Capacities (TSC_m), Speeds (TRS_m), Unit Operation Costs (TOC_m), Unit Transportation Costs (TTC_m), Miles per Full Tank (TPT_m), Refueling Durations (TRD_m), Maximum Number of Concurrent Refueling (TNR_m)

Treatment Facility's Number (TFN), Locations (TFL_n), Capacities (TFC_n), Unit Operation Costs (TRC_n), Pumping Rates (TPR_n), Maximum Number of Concurrently Deployable Trucks (TDT_n)

Oil Weathering Parameters: Wind Velocity (U), Temperature (T), Oil Vapor Pressure (P), Parent Oil Density (ρ_{m1}), Parent Oil Viscosity (μ_{m1}), Oil Molecular Weight (M), Interfacial Tension of Oil and Water (S_t), Curve Fitting Constant (K_a), Mousse Viscosity Constant (K_b), Oil Dependent Constant (K_c), Water Density (ρ_w), Gas Constant (R), Time Step (t_s)

Optimizer Parameters: Number of Objective Functions (NOF), Number of Populations (NOP), Number of New Populations (NNP), Number of Generations (NOG), Cross Over Probability (COP), Lower Bound of Scaling Factor (LSF), Upper Bound of Scaling Factor (USF), Maximum Length of Decision Variables' Vector (MLV)

Optimizer

Parallel For N_P =1 to NOP
- Randomly generating the vector of decision variables (VDV) with the length of MLV to schedule vessels, mini barges, and trucks
- Calculate the operation time and cost (i.e., NOF=2) using the next MCR & OWM and Weathering agents in two-objective optimization and the operation time and cost and volume weathered oil (i.e., NOF=3) in three-objective optimization

End
- Ranking the population based on non-dominated sorting criteria

For N_G =1 to NOG
  Parallel For N_P =1 to NNP
    - Apply mutation and cross over operators with respect to COP, LSF, and ISF on VDV to minimize operation time, costs, and/or volume of weathered oil obtained from the following agents (MCR & OWM and Weathering)
  End
  - Merging, sorting, and truncating the population based on NOP and non-dominated sorting criteria
End
Saving the final Pareto front and computation time

Weathering Processes – Evaporation and Dispersion Effects on Spilt Oil

For *i* =1 to OSN

- Update OSV_i with respect to evaporation based on t_s, T, U, M, P, ρ_{m1}, ρ_w, K_a, K_b, FE_p, R, P, previous OSV_i, and OSA_i
- Update OSV_i with respect to dispersion based on t_s, U, S_t, K_a, K_b, K_c, FE_p, previous OSV_i, and OSA_i
- Update oil viscosity with respect to emulsion based on t_s, U, μ_{m1} K_a, K_b, K_c, and FE_p
- Update oil density with respect to emulsion based on t_s, U, ρ_{m1}, ρ_w, K_a, K_b, and FE_p

End

- Oil weathering process agent will update oil slick specifications for each t_s during the entire response operation

MCR & OWM Operations based on Mathematical Model

Vessels to Oil Slicks – Skimming Spilt Oil

For *i* =1 to OSN

While $OSV_i \neq 0$ & $VSC_j \neq 0$

- Assign vessel(s) to oil slick *i* based on VDV generated by Optimizer agent, ODV_i, OSN, and available vessels (i.e., not already assigned)
- Check refueling requirement considering VPT_j, VNR_i and VRD_j
- Update VSL_j if assigned based on t_s, OSL_i, previous VSL_j, VSS_j, VPT_j, and VRD_j
- Update OSV_i based on remaining operation time, number of assigned vessels to slick *i*, OSA_i. previous OSV_i, VSK_j, VSC_j, and VNR_i
- Calculate the time, cost of the operation based on previous steps as well as VOC_j and VTC_j in case of assigned vessel(s)
- Vessel *j* will be discharged by a mini barge through the next component if $VSC_j = 0$

End

End

If $\sum_{i=1}^{OSN} OSV_i = 0$ & $\sum_{j=1}^{VSN} VSC_j = Max$ (i.e., the initial storage capacity at time zero)

For *j* =1 to VSN

- Vessel *j* will be sent to the nearest big barge
- Associated transportation cost with Vessel *j* will be calculated based on VTC_j, VPT_j, VNR_i, and VRD_j

End

End

Mini Barges to Vessels – Collecting Oily Waste from Vessels

For *j* =1 to VSN

If $VSC_j = 0$ & $MSC_k \neq 0$

- Assign mini barge *k* to vessel *j* based on the VDV, and MBN
- Check refueling requirement considering MPT_k, MNR_k, and MRD_k
- Update MBL_k if assigned based on t_s, VSL_j, previous MBL_k, MBS_k, MPT_k, MRD_k, and MNR_k
- Update VSC_j and MSC_k based on remaining operation time, previous VSC_j, previous MSC_k, and MPR_k
- Calculate the time and cost of the operation based on previous steps as well as MTC_k and MOC_k

End

- Vessel *j* will be assigned to the same slick *i* if $OSV_i \neq 0$, otherwise it can be assigned to another slick based on VDV prior to $\sum_{i=1}^{OSN} OSV_i = 0$
- Associated transportation costs with vessel *j* will be calculated based on VTC_j, VPT_j, VRD_j, and VNR_j after reassignment

End

If $\sum_{i=1}^{OSN} OSV_i = 0$ & $\sum_{j=1}^{VSN} VSC_j = Max$ & $\sum_{k=1}^{MBN} MSC_k = Max$

For *k* =1 to MBN

- Mini barge *k* will be sent to the nearest big barge
- Associated transportation cost with Mini barge *k* will be calculated based on MTC_k, MPT_k, MRD_k, and MNR_k

End

End

Mini Barges to Big Barges – Transferring Oily Waste to Onshore

For *k* =1 to MBN

If $MSC_k = 0$ & $BSC_l \geq$ Initial MSC_k

- Assign mini barge *k* to big barge *l* based on VDV, BCM_l, and BBN
- Check refueling requirement considering MPT_k, MNR_k, and MRD_k
- Update MBL_k if assigned based on t_s, BBL_l, previous MBL_k, MBS_k, MPT_k, MRD_k, and MNR_k
- Update BSC_l and MSC_k based on remaining operation time, previous BSC_l, previous MSC_k, BPR_l, and BCM_l
- Calculate the time and cost of the operation based on previous steps as well as MTC_k and BOC_l

End
- Mini barge *k* will be assigned again to vessel *j* based on VDV if $VSC_j = 0$, otherwise if $\sum_{j=1}^{VSN} VSC_j = Max$ & $\sum_{i=1}^{OSN} OSV_i = 0$ then it will stay near the big barge *l* (i.e., the closest big barge)
- Associated transportation costs with mini barge *k* will be calculated based on MTC_k, MPT_k, MRD_k, and MNR_k if reassigned

End

Trucks to Big Barges – Collecting Oily Waste from Big Barges

If $BSC_l < BAT_l$

For *m* =1 to TRN

- Assign truck *m* to big barge *l* based on VDV, BCT_l, and BBN
- Check refueling requirement considering TPT_m, TNR_m, and TRD_m
- Update TRL_m if assigned based on t_s, BBL_l, previous TRL_m, TRS_m, TPT_m, TRD_m, and TNR_m
- Update TSC_m and BSC_l based on remaining operation time, previous TSC_m, previous BSC_l, BPR_l, and BCT_l
- Calculate the time and cost based on previous steps as well as BOC_l, TTC_m, and TFC_n

End

End

Trucks to Treatment Facilities – Transferring Oily Waste to Wastewater Treatment

For *m* =1 to TRN

If $TSC_m = 0$ & $TFC_n \neq 0$

- Assign truck *m* to treatment facility *n* based on VDV and TFN
- Check refueling requirement considering TPT_m, TNR_m, and TRD_m
- Update TRL_m if assigned based on t_s, TFL_n, previous TRL_m, TRS_m, TPT_m, TRD_m, and TNR_m
- Update TSC_m and TFC_n based on remaining operation time, previous TSC_m, previous TFC_n, TPR_n
- Calculate the time and cost of the operation based on previous steps as well as TRC_n, TTC_m, and TOC_m

End

- Truck *m* will be assigned again to big barge *l* based on VDV, BCT_l, and BBN if $\sum_{l=1}^{BBN} BSC_l \neq Max$
- Associated transportation costs with truck *m* will be calculated based on TTC_m, TPT_m, TRD_m, TOC_m, and TNR_m if reassigned

End

If $\sum_{i=1}^{OSN} OSV_i = 0$ & $\sum_{j=1}^{VSN} VSC_j = Max$ & $\sum_{k=1}^{MBN} MSC_k = Max$ & $\sum_{l=1}^{BBN} BSC_l = Max$

- Truck *m* will return to the closest big barge
- Associated transportation costs with truck *m* will be calculated based on TTC_m, TPT_m, TRD_m, TOC_m, and TNR_m

End

- End of simulation for MCR & OWM Operations and Weathering Processes
- The total time and cost of the operation, and/or volume of weathered oil (objective functions) will be calculated and sent to the optimizer
- The optimizer will generate a new population of VDV and send it to other agents until NG=NOG (stopping criterion) or the absence of any changes in the optimal solutions obtained within the past ten generations, after at least 75 generations (convergence criterion).

Outputs:

- Final Pareto fronts
- Total operation time, cost, and/or volume of weathered oil in the selected optimal strategies
- Temporal distribution of skimmed oil for each oil slick and each vessel
- Temporal distribution of evaporated and dispersed oil for each oil slick
- Temporal distribution of transported oily wastewater to big barges and treatment facilities
- Details of vessel assignments
- Details of mini barge assignments
- Details of truck assignments

Appendix E – Input Parameters and Assumptions of Case Study of MCR and OWM Optimization

No.	Description	Nomenclature	Value
	Oil Slicks		
1	Number of Oil Slicks	*OSN*	7
2	Locations (x, y) **Note:** Listed values are in km and Big Barge 2 is selected as the center of the coordinate system	*OSL*	Slick 1: (-66, 8) Slick 2: (-62, 6) Slick 3: (-67, 7) Slick 4: (-65, 9) Slick 5: (-68, 5) Slick 6: (-75, 8) Slick 7: (-70, 11)
3	Volumes (m^3)	*OSV*	Slick 1: 2500 Slick 2: 1850 Slick 3: 1300 Slick 4: 1450 Slick 5: 1800 Slick 6: 1500 Slick 7: 1200
4	Areas (m^2)	*OSA*	Slick 1: 25000 Slick 2: 22000 Slick 3: 15000 Slick 4: 18000 Slick 5: 22000 Slick 6: 20000 Slick 7: 14000
5	Maximum Number of Concurrently Deployable Vessels **Note:** It may depend on the area of the slick and skimming techniques used	*ODV*	Slick 1: 5 Slick 2: 4 Slick 3: 2 Slick 4: 3 Slick 5: 4 Slick 6: 3 Slick 7: 2
	Skimming vessels		
6	Number of Vessels	*VSN*	6
7	Initial Locations (x, y) **Note:** half of the number of vessels were assumed to be located near the big barge 1 and the rest near big barge 2	*VSL*	Vessel 1 - Barkley Sentinel: (-80, 42) Vessel 2 - Eagle Bay Sentinel: (-80, 42) Vessel 3 - GM Penman: (-80, 42) Vessel 4 - Hecate Sentinel: (0, 0)

No.	Description	Nomenclature	Value
			Vessel 5 – Salish Sentinel: (0, 0)
			Vessel 6 - MJ Green: (0, 0)
8	Number of Skimmers - Type 1	VNS_1	Vessel 1: 2
			Vessel 2: 0
			Vessel 3: 0
			Vessel 4: 2
			Vessel 5: 2
			Vessel 6: 0
9	Number of Skimmers - Type 2	VNS_2	Vessel 1: 1
			Vessel 2: 2
			Vessel 3: 2
			Vessel 4: 0
			Vessel 5: 1
			Vessel 6: 2
10	Average Speeds of Vessels (km/hr) **Note:** 12 km/hr is equivalent to about 6 knots which is the planning speed for the vessels in the study area	*VSS*	Vessel 1: 20
			Vessel 2: 12
			Vessel 3: 20
			Vessel 4: 20
			Vessel 5: 12
			Vessel 6: 20
11	Storage Capacities (m^3)	*VSC*	Vessel 1: 40
			Vessel 2: 10
			Vessel 3: 30
			Vessel 4: 30
			Vessel 5: 79.5
			Vessel 6: 10
12	Unit Operation Costs ($/hr) **Note:** Operation cost is considered while a vessel is working to skim oil and discharge into a mini barge	*VOC*	Vessel 1: 750
			Vessel 2: 300
			Vessel 3: 750
			Vessel 4: 750
			Vessel 5: 750
			Vessel 6: 300
13	Unit Transportation Costs ($/km)	*VTC*	Vessel 1: 20
			Vessel 2: 15
			Vessel 3: 20
			Vessel 4: 20
			Vessel 5: 20
			Vessel 6: 15
14	Miles per Full Tank (km) **Note:** - Fuel consumption is variable based on the speed of vessel - 950 km is equivalent to about 515 nautical miles	*VPT*	Vessel 1: 1000
			Vessel 2: 2000
			Vessel 3: 1000
			Vessel 4: 1000
			Vessel 5: 2000
			Vessel 6: 1000
15	Refueling Duration (hr)	*VRD*	0.7

No.	Description	Nomenclature	Value
16	Maximum Number of Concurrent Refueling	*VNR*	2
	Mini barges		
17	Number of Mini Barges (type: TB4)	*MNB*	20
18	Initial Locations (x, y) **Note:** half of the number of mini barges were assumed to be located near the big barge 1 and the rest near big barge 2	*MBL*	10 mini barges: (-80, 42) 10 mini barges: (0, 0)
19	Storage Capacities (m^3)	*MSC*	40
20	Average Speeds of Mini Barges (km/hr)	*MBS*	4 mini barges: 20 6 mini barges: 18 10 mini barges: 14
21	Unit Operation Costs ($/hr) **Note:** Operation cost is considered while a mini barge is working to offload a vessel and discharge into a big barge	*MOC*	50
22	Unit Transportation Costs ($/km)	*MTC*	4 mini barges: 5 6 mini barges: 3 10 mini barges: 2
23	Pumping Rate to Load a Mini Barge (m^3/hr)	*MPR*	4 mini barges: 50 6 mini barges: 40 10 mini barges: 35
24	Miles per Full Tank (km)	*MPT*	4 mini barges: 1000 6 mini barges: 1200 10 mini barges: 1500
25	Refueling Duration (hr)	*MRD*	0.7
26	Maximum Number of Concurrent Refueling	*MNR*	2
	Big barges		
27	Number of Big Barges	*BBN*	2
28	Locations (x, y)	*BBL*	Big Barge 1 - Sentinel 302: (-80, 42) Big Barge 2 - Sentinel 303: (0, 0)
29	Storage Capacities (m^3)	*BSC*	Big Barge 1: 4000 Big Barge 2: 3500
30	Operation Costs ($/hr) **Note:** Operation cost is considered while a big barge is working to offload a mini barge and discharge into a truck	*BOC*	Big Barge 1: 415 Big Barge 2: 300
31	Pumping Rates to Load and Offload a barge (m^3/hr)	*BPR*	Big Barge 1: 20

No.	Description	Nomenclature	Value
			Big Barge 2: 25
32	Maximum Number of Concurrently Connectable Mini Barges Note: one mini barge at each side of the big barge	*BCM*	2
33	Activation Threshold for Discharging (% of capacity) **Note:** start offloading to trucks after this threshold	*BAT*	10
34	Maximum Number of Concurrently Connectable Trucks **Note:** depends on the number of tanks in big barge, available hose length, and dock space	*BCT*	Big Barge 1: 6 Big Barge 2: 6
	Trucks		
35	Number of Trucks	*TRN*	25
36	Initial Locations (x, y) **Note:** half of the number of trucks were assumed to be located in the dock near Big Barge 1 and the rest near Big Barge 2	*TRL*	12 trucks: (-80, 42) 13 trucks: (0, 0)
37	Storage Capacities (m^3)	*TSC*	12 trucks: 10 13 trucks: 15
38	Average Speeds of Trucks (km/hr)	*TRS*	12trucks: 75 13 trucks: 65
39	Unit Transportation Costs ($/km) **Note**: operation costs (*TOC*) are included in TTC for trucks	*TTC*	12 trucks: 1.5 13 trucks: 2
40	Miles per Full Tank (km)	*TPT*	12 trucks: 600 13 trucks: 800
41	Refueling Duration (hr)	*TRD*	0.33
42	Maximum Number of Concurrent Refueling	*TDT*	8
	Treatment facilities		
43	Number of Treatment Facilities	*TFN*	2
44	Locations (x, y)	*TFL*	Treatment 1: (35, 79) Treatment 2: (77, -11)
45	Total Capacities (m^3)	*TFC*	Treatment 1: 5000 Treatment 2: 8000
46	Operation Costs ($/$m^3$)	*TRC*	Treatment 1: 35 Treatment 2: 25
47	Pumping Rate to Offload a Truck in a Treatment Facility (m^3/hr)	*TPR*	Treatment 1: 50 Treatment 2: 70
48	Maximum Number of Concurrently Deployable Trucks **Note:** no limitation in this case study.	*TDT*	30
	Oil and Weathering Parameters		
49	Wind Velocity (m/s)	*U*	5
50	Temperature (K)	*T*	298
51	Oil Vapor Pressure (Pa)	*P*	10.4
52	Parent Oil Density (kg/m^3)	ρ_{m1}	832
53	Parent Oil Viscosity (cP)	μ_{m1}	31

No.	Description	Nomenclature	Value
54	Oil Molecular Weight (g/mol)	M	128.2
55	Interfacial Tension of Oil and Water (dyne/m)	S_t	2000
56	Curve Fitting Constant	K_a	2×10^{-6}
57	Mousse Viscosity Constant	K_b	0.7
58	Oil Dependent Constant	K_c	10
59	Water Density (kg/m^3)	ρ_w	1020
60	Gas Constant (m^3·Pa/mol·K)	R	8.314
61	Time Step (hr)	t_s	1

Appendix F – Artwork Usage Consent

October 8, 2023

To Whom It May Concern,

I am Arezoo Rahimi, holding a master's degree in Illustration from the University of Art, Tehran, Iran. I recently collaborated with Mr. Saeed Mohammadiun, a Ph.D. candidate in Civil Engineering at the University of British Columbia - Okanagan (UBCO), and his colleagues to create several paintings for various articles, including:

- S. Mohammadiun, G. Hu, A. Alavi Gharahbagh, J. Li, K. Hewage, and R. Sadiq, "*Evaluation of Machine Learning Techniques to Select Marine Oil Spill Response Methods under Small-Sized Dataset Conditions*" (2022) Journal of Hazardous Materials, Elsevier, Volume 436, 129282, DOI: doi.org/10.1016/j.jhazmat.2022.129282
- S. Mohammadiun, G. Hu, A. Alavi Gharahbagh, J. Li, K. Hewage, and R. Sadiq, "*Intelligent Computational Techniques in Marine Oil Spill Management: A Critical Review*" (2021) Journal of Hazardous Materials, Elsevier, Volume 419, 126425, DOI: doi.org/10.1016/j.jhazmat.2021.126425
- S. Mohammadiun, G. Hu, A. Alavi Gharahbagh, R. Mirshahi, J. Li, K. Hewage, and R. Sadiq, "*Optimization of Integrated Fuzzy Decision Tree and Regression Models for Selection of Oil Spill Response Method in the Arctic*" (2021) Knowledge-Based Systems, Elsevier, Volume 213, 106676, DOI: doi.org/10.1016/j.knosys.2020.106676
- S. Mohammadiun, A. Alavi Gharahbagh, E. Bakhtavar, G. Hu, J. Li, K. Hewage, and R. Sadiq, "*Integrated Optimization of Marine Oil Spill Response and Liquid Oily Waste Management using Mathematical Programing and Evolutionary Metaheuristic Techniques*" Under revision in the Journal of Hazardous Materials.

I confirm that I have been duly compensated by Mr. Saeed Mohammadiun for my work on these paintings. Therefore, all artwork created is Mr. Saeed Mohammadiun's intellectual property, and he has the right to use them as he sees fit, such as in his thesis. Additionally, I grant permission to UBCO to use these paintings in any form, such as reports and media releases on their website, specifically in relation to the aforementioned papers and thesis.

Kind regards,
Arezoo Rahimi



